# OECD ECONOMIC SURVEYS

## JAPAN

1997

**SPECIAL FEATURE**

- THE SUSTAINABILITY OF THE SOCIAL SECURITY SYSTEM

# OECD ECONOMIC SURVEYS

## 1996-1997

JAPAN

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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# BASIC STATISTICS OF JAPAN

## THE LAND

| | |
|---|---|
| Area (1 000 sq. km) | 377.8 |
| Cultivated agricultural land (1 000 sq. km, 1994) | 51.7 |
| Forest (1 000 sq. km, 1994) | 251.6 |
| Densely inhabited districts[1] (1 000 sq. km, 1990) | 11.7 |

| Major cities, October 1996 estimate (10 000 inhabitants): | |
|---|---|
| Tokyo (23 wards) | 796 |
| Yokohama | 332 |
| Osaka | 260 |
| Nagoya | 215 |
| Sapporo | 177 |
| Kobe | 142 |
| Kyoto | 146 |

## THE PEOPLE

| | |
|---|---|
| Population, October 1996 estimate (1 000) | 125 864 |
| Number of persons per sq. km in 1996 | 333 |
| Percentage of population living in densely inhabited district in 1990[1] | 63.2 |
| Net annual rate of population increase (1985-1996) | 0.4 |

| | |
|---|---|
| Labour force in per cent of total population, October 1996 | 53.8 |
| Percentage distribution of employed persons, 1996: | |
| Agriculture and forestry | 5.1 |
| Manufacturing | 22.3 |
| Service | 24.6 |
| Other | 48.0 |

## PRODUCTION

| | |
|---|---|
| Gross domestic product in 1996 (billion yen) | 500 314 |
| Growth of real GDP, 1981-1996 average (annual rate, per cent) | 3.2 |
| Gross fixed investment in 1996 (per cent of GDP) | 29.6 |

| | |
|---|---|
| Growth of real gross fixed investment, 1981-1996 average (annual rate, per cent) | 3.8 |
| Net domestic product of agriculture, forestry and fishery, at market prices, in 1994 (billion yen) | 7 500 |
| Growth of production in manufacturing 1981-1996 (annual rate, per cent) | 2.4 |

## THE GOVERNMENT

| | |
|---|---|
| Public consumption in 1996 (in per cent of GDP) | 9.8 |
| Current public revenue in 1995 (in per cent of GDP) | 31.9 |
| Government employees in per cent of total employment, 1996 | 8.2 |
| Outstanding long-term national bonds in per cent of GDP (FY 1996) | 47.9 |

| | House of Representatives | House of Councillors |
|---|---|---|
| Composition of Parliament, August 1997: | | |
| Liberal Democratic Party | 249 | 112 |
| New Frontier Party | 133 | – |
| Heiseikai | – | 59 |
| Social Democratic Party | 15 | 21 |
| Democratic Party | 52 | 23 |
| Others | 51 | 37 |
| Total | 500 | 252 |
| Last election: | Nov. 1995 | July 1995 |

## FOREIGN TRADE AND PAYMENTS

(1996, million yen)

| | |
|---|---|
| Commodity exports (fob) | 435 659 |
| Commodity imports (fob) | 344 693 |
| Services | –67 792 |
| Investment income | 58 180 |
| Current balance | 71 579 |
| Exports of goods and services in per cent of GDP | 9.9 |
| Imports of goods and services in per cent of GDP | 9.4 |

| | Exports | Imports |
|---|---|---|
| Percentage distribution: | | |
| OECD countries | 55.6 | 51.3 |
| *of which:* North America | 28.7 | 25.8 |
| Far East | 36.9 | 32.9 |
| Other | 7.6 | 15.8 |
| Total | 100.0 | 100.0 |
| Crude material and fuels (SITC 2, 3, 4) | 1.2 | 26.1 |
| Semi-manufactured goods (5, 6) | 18.1 | 17.9 |
| Machinery and transport equipment (7) | 69.5 | 25.1 |
| Other (0, 1, 8, 9) | 11.2 | 30.9 |
| Total | 100.0 | 100.0 |

## THE CURRENCY

Monetary unit: Yen

| Currency units per US$ average of daily figures: | |
|---|---|
| Year 1996 | 108.8 |
| September 1997 | 120.7 |

1. Areas whose population density exceeds 5 000 persons per sq. km.

*Note:* An international comparison of certain basic statistics is given in an annex table.

*This Survey is based on the Secretariat's study prepared for the annual review of Japan by the Economic and Development Review Committee on 15 September 1997.*

•

*After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 16 October 1997.*

•

*The previous Survey of Japan was issued in December 1996.*

# Assessment and recommendations

***With economic growth pausing in the first half of 1997...***

Japan's recovery from the drawn-out recession of the early 1990s accelerated in 1996, with output expanding by 3.5 per cent on average during the year. This acceleration was essentially driven by the strength of business investment, as firms continued their restructuring efforts to reduce costs and improve their competitiveness. At the same time, the combination of stronger export growth, following the significant depreciation of the yen's exchange rate, and the deceleration of imports, in part due to substantial destocking in the manufacturing sector, provided a further boost to activity. While accelerating in the first quarter of 1997, GDP fell unexpectedly by 2.9 per cent in the second quarter, as the strength of exports – which boosted the external current account surplus to around 2½ per cent of GDP – only partly offset the sharp fall in private demand. Some of that decline, mainly personal consumption and business investment, may be temporary, in the wake of anticipatory spending prior to the April increase in the consumption tax. Overall, output growth in the first half of 1997 was merely 0.8 per cent at annual rate, suggesting a pause in the recovery.

***... the current upswing remains weak by past standards***

As a result, the current recovery appears to be one of the slowest of the post-war period, averaging only 1¾ per cent since it began in late 1993. This is somewhat below the growth rate needed to close the output gap estimated by the OECD, so that, by the first half of 1997, the margin of

spare capacity in the economy was still relatively large and higher than in 1996. Moreover, with the recovery having drawn more people into the labour force, the pick-up in employment growth has not yet proved strong enough to reduce the unemployment rate which, at 3¼ per cent, remains high by Japanese standards. The resulting weakness in wage growth has helped keep underlying domestic prices stable, although the recent rise in the consumption tax has raised the reported rate of domestic inflation to just under 2 per cent by August 1997.

***Nonetheless, the recovery may regain some momentum...***

Some reacceleration of the economy can be expected during the second half of 1997. As suggested above, after the exceptionally weak second quarter, a rebound in the level of activity is likely in the rest of the year to the extent that consumers bring their spending back into line with incomes and businesses catch-up with their planned investment. Nonetheless, according to OECD projections, such a pattern of demand will probably only generate growth of less than 1 per cent for 1997 as a whole. In 1998, economic activity is expected to revive somewhat to about 2 per cent, as in the absence of additional tax increases (see below), the growth of disposable income should accelerate, thereby boosting personal consumption. Renewed growth of private demand, though, might be partially offert by continued weakness in public demand. In addition, the contribution of foreign trade to growth may slacken, as the effects of the weak yen on trade diminish and exports to South East Asia slow as a result of the financial instability in that region. However, the current account surplus is projected to rise moderately in 1998, due both to increased investment income and a rise in net exports. At the same time, the unemployment rate may fall only slightly despite expected growth in private-sector jobs during the projection period.

With unit labour costs falling, domestic inflation should slacken to below 1 per cent by the end of 1998.

***... if external risks do not materialise...***

There are some risks surrounding these projections. The recent volatility of output creates the possibility that the rebound in the Japanese economy may be faster than expected in the short term. On the other hand, the timing and strength of the rebound will depend on how business and consumer confidence reacts to the recent weakening of activity. Another significant risk, at the moment, is that the recent financial difficulties of some South East Asian countries could have a marked effect on Japanese business sentiment and export growth to that region. The deeper and longer these difficulties last, the more severe would be the impact on Japan's growth prospects, particularly if the economic slowdown spreads to the rest of Asia.

***... and provided that both monetary conditions remain easy for a while...***

The above outlook assumes a continued easy stance of monetary policy. The historically low short-term interest rates, which have stayed at around 0.5 per cent since September 1995, have played an important role in supporting the recovery. At the same time, as the prospects for budget deficit reduction improved, long-term interest rates also dropped to extremely low levels. Consequently, international differentials in both short and long-term interest rates widened further, contributing, through increased capital outflows, to the downward pressure on the yen in the exchange market. As the domestic price level first stabilised and then increased somewhat, estimated real rates, at the short end of the market, became negative. Together with the weaker exchange rate, this has resulted in exceptionally accommodating monetary conditions. With underlying inflation expected to remain practically flat, an early tightening of monetary policy does not seem to be warranted in the current economic environment. Once the

recovery becomes firmly established, with the output gap clearly falling, there would be a case for a progressive reduction in the extent of monetary ease.

***... and significant progress is achieved toward restoring financial stability***

By supporting economic activity, the maintenance of easy monetary conditions would also help restore the health of the banking system. Even though high operating profits, due in part to cost cutting efforts, have enabled banks to reduce the size of their unprovisioned non-performing loans, it will take some time before their bad loan problem is fully resolved. This process might be further complicated by the rise in bankruptcies and the recent slowdown in economic growth in Japan as well as by the impact, albeit uncertain, of the financial turmoil in South East Asia. Moreover, the financial situation of credit co-operatives and a few second-tier regional banks remains relatively weak, while some mutual life insurance companies are experiencing difficulties in meeting the projected returns offered to investors at a time of low yields on their portfolios. A prompt resolution of the problems facing the small and medium-sized financial institutions should be facilitated by the recent reform of the deposit insurance system, whose resources have been considerably strengthened. The newly-created financial supervisory agency, designed to supervise deposit-taking institutions and other private-sector financial enterprises, should also contribute to preventing the recurrence of such problems. It is essential, however, that this body be given sufficient resources to accomplish its supervisory function and ensure that the new capital adequacy legislation is administered transparently. Attention also needs to be given to strengthening the capital position of commercial banks.

***Given the sharp deterioration in public finances...***

The prolonged weakness of private demand and the series of economic packages designed to stimulate growth have resulted, in recent years, in a marked worsening in the government financial position. The general government deficit rose to an estimated 4 per cent of GDP in 1996, one of the largest in the OECD area. Moreover, the existence of a still large social security surplus – which is expected to fall progressively as the population ages – masks the extent of the fiscal deterioration. Indeed, the central and local government deficit has risen to nearly 7 per cent of GDP, resulting in a steep increase of gross general government debt to 86 per cent of GDP at the end of 1996. Although net debt has also risen, it remains relatively low (14 per cent of GDP) compared with other OECD countries. This is due to the accumulated social security assets, the income of which is designed to contribute to the payment of future pensions.

***... achieving fiscal sustainability over the medium term appears crucial...***

In the expectation of rising economic activity, the government has taken steps towards correcting the fiscal imbalance. It first raised indirect taxes, ended the temporary income tax rebate and reduced the level of public investment in the 1997 budget. This was followed by the announcement of a medium-term plan, starting in 1998, to reduce the central and local government deficit to 3 per cent of GDP (equivalent to about 1 per cent on a general government basis) by 2003. Beyond that date, the government intends to gear fiscal policy towards stabilising the level of gross public debt. One of the main features of the current plan is a further reduction in public investment. Specific targets, over the same period, have been formulated to restrain other categories of expenditure while there will also be a slight rise in tax revenues resulting from fiscal drag. However, given the estimated scale of the decline in the structural budget deficit in 1997 (2 per cent of GDP), the extent of fiscal tightening will be less in this period, so

reducing the adverse effect on demand. Nonetheless, flexibility will be needed when framing budget proposals in the years to come, so as to ensure that actions taken to achieve long-run fiscal sustainability are compatible with cyclical economic developments. In view of the downward risk on activity outlined above, care should be taken to avoid too rapid a withdrawal of government support to the economy in the short term.

***... the more so since ageing will put pressure on government spending***

One important aspect that fiscal consolidation needs to take into account is the rapid rise in pension and health expenditures that is expected in Japan in the years to come. Indeed, as discussed in the special chapter of this Survey, with the ageing of the population and the maturing of the pension system, retirement benefits are projected to rise from 10 per cent of national income in 1998 to nearly 16 per cent by 2025, a much larger increase than in the rest of the OECD area. In addition, with the number of elderly growing, government spending on health care for the aged may also increase sharply from 2 to 5 per cent of national income during the same period. As a result, assuming other public expenditure is held constant, total government outlays are projected to rise to just above half of national income by 2025 – 10 percentage points higher than the average of the past decade.

***Despite recent pension reforms...***

Such potential pressures on public finances are likely despite the introduction, in recent years, of several reforms in the area of both pensions and health care for the elderly. In 1994, the government decided to progressively raise the retirement age to 65 for the first-tier state pension, while leaving the age for the second-tier earnings-related pension at 60. The increase of benefits has been restrained by indexing them to the growth of earnings net of tax and social security contributions rather than to that of gross wages.

Finally, the government announced a phased increase of the contribution rate, to a peak of close to 30 per cent in 2025 – though following new population projections, the government now estimates that contributions may have to rise to around 34 per cent of basic earnings. In addition, to limit the expansion of health expenditure on the elderly, the daily hospital charge has been raised and a new social contribution to pay for long-term care for old infirm people will be introduced.

***... additional measures will be required in that area...***

However, further measures will be necessary to avoid breaching the government's long-term objective of limiting the sum of all taxes, social security contributions and the fiscal deficit (which is equivalent to total public spending less non-tax revenues) to half of national income (or currently 43 per cent of GDP). One option would be to lower pension spending by raising the retirement age to 65 for earnings-related pensions and accept some increase in contributions; another option would be to avoid increasing the level of contributions through reducing benefits. Reforms to raise the investment income received by the state pension schemes could also help in this regard. At present, most of the state funds are loaned at fixed rates to the government and used for programmes which finance unprofitable public enterprises and provide credit to selected sectors at below market rates. Private-sector investment funds, operating under freer asset allocation rules but subject to appropriate prudential oversight, might manage the social security resources more efficiently, provided they took into account the risks and rewards of different financial instruments. Moreover, to mitigate the pressure on health expenditure, the extent to which the elderly finance their own health care could be increased. At the same time, the cost of financing pensions should perhaps be distributed more equitably, both across and within generations. To this end, the govern-

ment could consider either reducing the large income tax deductions on pension incomes or limiting the full tax deductibility for pension contributions.

***... together with policies designed to raise participation and birth rates***

Over the longer term, the conflict between assuring the well-being of the elderly and limiting the level of the overall tax burden could in part be resolved by accelerating the deregulation of the economy (see below), so boosting real incomes, and by expanding the size of the labour force. This would require increasing participation rates by eliminating measures that discourage people from working. Phasing out the pension benefit reductions that are applied to working pensioners would help in this perspective, as well as removing aspects of the social security contribution systems that tend to limit the incentives of married women to work. However, such measures to boost the labour supply would not be sufficient to prevent the labour force from falling in the long run unless the fertility rate is raised from its current low level. Measures that reduce the cost to women of working and having families could be considered in this context. These might include better provision of maternity leave and improved child-care facilities for young children.

***These should be matched by a rationalisation of public spending...***

To keep a rein on public finances, such policies would need to be accompanied by a rationalisation of government expenditure, most particularly public investment. This category of spending, which, in Japan, is about three to four times higher than in other major OECD countries, appears to be allocated not just on the basis of obtaining the highest return on investment but also to support local activity. Thus, in implementing the cuts in investment already envisaged under the budget deficit-reduction programme, it will be important to thoroughly evaluate all projects, dropping those with the lowest returns, rather than merely delaying

the implementation of planned investments. Further privatisation would help in these regards.

***... and increased budget transparency***

At the same time, the transparency of budget decisions needs to be improved. Two areas are of particular concern here, as pointed out in previous OECD surveys: the presentation of government accounts and the operation of the so-called Fiscal Investment and Loan Programme (FILP). Regarding the first item, government financial statements, though detailed and specific, do not permit consolidation of the many different accounts nor do they distinguish between categories of spending and revenue that have very different economic effects. A step forward would thus be to present the government budget in a way that would enable monitoring, on a national accounts basis, of the chosen fiscal target – the central and local government deficit – as well as the internationally-comparable general government account. As for the FILP, there are two main challenges. One is to move those public corporations engaged in infrastructure investment on to the budget and rationalise their investments. Another is to increase the market orientation of related government financial institutions, which account for one-third of total lending in the economy. At present, loans allocated by the FILP through these institutions benefit selected sectors, such as housing and small businesses, generating considerable hidden subsidies. In the future, the performance of the economy would gain from allocating such funds to projects with the highest return.

***Continued emphasis on deregulation is also essential***

More generally, further improving economic performance will necessitate keeping the accent on structural reforms. Japan has made significant progress in deregulating key sectors of its economy in recent years. Most notably, the easing of the Large-Scale Retail Store Law has helped increase the number of larger, more efficient retail outlets,

thus reducing distribution costs. In the telecommunications sector, increased competition, combined with technological progress, has boosted demand and lowered prices. In addition, the liberalisation of controls on imports of gasoline has lowered its price significantly during the past two years. In total, according to estimates by the Japanese government, deregulation measures in major sectors of the economy may have resulted in gains to consumers and additional demand equivalent, respectively, to 1 and 1.6 per cent of GDP on average each year since 1990. For the future, studies by both the OECD and the Japanese government suggest that further deregulation of key sectors of the economy could boost GDP by as much as 6 per cent.

***Although recent structural initiatives constitute important steps...***

To achieve such gains, the government has recently announced significant initiatives, including a plan to reform the financial system by 2001. The first step of the plan is to completely liberalise foreign exchange transactions from April 1998, a measure that is likely to lead to changes in other areas. Brokerage commissions will be liberalised in two stages beginning in 1998, while competition in insurance premiums will be permitted. Other reforms will reduce segmentation among financial intermediaries and allow the introduction of new products. At the same time, pension funds will be given greater freedom to decide on asset allocation and employ investment advisory firms. Important measures in other sectors include reform of telecommunications, such as the restructuring of Nippon Telephone and Telegraph, which dominates the domestic market, and the elimination of entry barriers in the transport sector over the next three to five years. Moreover, the regulations in construction in city centres have been eased, while the government has widened the scope of activities permitted for private job placement firms. Finally, the ban on holding companies has been lifted, subject to some restrictions.

***... there is scope for further progress in this field***

Despite such progress, the scope of deregulation could be broadened and its pace accelerated in many areas. In particular, removing remaining entry barriers in several non-manufacturing sectors would be necessary. A more transparent system of allocating take-off and landing slots at major airports would enable new airlines to compete with existing carriers. In addition, phasing-out the Large-Scale Retail Store Law, which continues to limit the size and operating hours of stores, would help maintain the momentum towards a more efficient retail sector. Furthermore, a reduction in the high price of electricity could probably be achieved by separating the generation of power from its transmission. Also, as recommended in the previous OECD Survey of Japan, the trend toward increased turnover in labour markets would be facilitated by removing controls on temporary worker agencies as well as those on job placement firms. Finally, to be really effective, the decision to allow holding companies should be accompanied by the introduction of a well-designed consolidated taxation system for such structures. To help advance progress in this wide range of areas, the existence of an independant body overseeing the deregulation process would be beneficial.

***Summary***

*In sum*, the Japanese economy appears to have emerged from the deep recession of the early 1990s, helped by a substantial fiscal and monetary stimulus. However, a sustained upswing has yet to be achieved as the upturn paused in the first half of 1997 despite buoyant exports. The underlying momentum of the recovery might reappear subsequently provided that monetary conditions remain easy for a while and the consequences of the recent financial instability in South East Asia are limited. Greater strength of private demand would facilitate the reduction of Japan's budget deficit, which has increased sharply in recent years, without stalling the pace of the recovery. The need for

fiscal consolidation is accentuated by the rapid ageing of Japan's population, which is boosting public expenditures on pensions and health. Further reforms to limit the growth of such spending, combined with policies aimed at raising the birth rate and increasing the labour force participation of women and the elderly, would be beneficial. Such measures, coupled with further structural reforms to enhance productivity, would help assure favourable growth prospects over the longer term.

# I. Recent trends and prospects

In 1994, the Japanese economy began to recover from the prolonged period of slow growth that followed the speculative ‘‘bubble’’ of the late 1980s. So far, the upswing has proved to be relatively weak compared both with most projections and with past cycles and, though showing some signs of improvement, unemployment has remained high by Japanese standards. Following the reversal of the yen’s appreciation, which started in the middle of 1995, there was a pick-up in the growth of private-sector domestic demand but, despite sustained exports, this strength was not maintained in the second quarter of 1997, which registered a sharp drop in both personal consumption and business investment. This occurred in a context of falling public-sector demand as a result of a tightening of fiscal policy. The recovery may resume somewhat during the second half of the year as the temporary decline in private demand unwinds and the pace of budget deficit-reduction becomes more moderate, so boosting personal disposable income and consumption. However, there are risks surrounding this outlook. Most notably the recent financial difficulties of some South East Asian countries could have a more marked effect on Japanese business sentiment and export growth to that region. This chapter first reviews the main features of the current recovery and then presents the OECD’s economic projections for 1997 and 1998.

## A moderate recovery...

Economic activity gained momentum in 1996. With stronger domestic demand, the growth of economic activity was boosted to 3.5 per cent for the year as a whole (Table 1), despite a continued negative contribution from the foreign balance. For the first time since 1991, GDP growth exceeded the average of the other six major countries (Figure 1), reflecting the strong performance in the first

Table 1. **Demand and output**

Percentage change from previous period at 1990 prices

| | 1993 Per cent of total | 1994 | 1995 | 1996 | Seasonally-adjusted annual rates | | |
|---|---|---|---|---|---|---|---|
| | | | | | 1996 | | 1997 |
| | | | | | 1st half | 2nd half | 1st half |
| **Consumption** | **67.4** | **2.0** | **2.2** | **2.8** | **3.7** | **0.3** | **3.3** |
| Private | 58.3 | 1.9 | 2.0 | 2.8 | 4.0 | –0.2 | 3.9 |
| Public | 9.1 | 2.4 | 3.5 | 2.3 | 1.8 | 3.5 | –0.5 |
| **Gross fixed investment** | **30.1** | **–0.8** | **1.1** | **8.7** | **12.9** | **2.4** | **–9.5** |
| Public | 8.6 | 2.8 | 0.7 | 9.9 | 18.0 | –10.0 | –26.3 |
| Private residential | 4.9 | 8.5 | –6.4 | 13.6 | 28.8 | 9.3 | –14.6 |
| Private non-residential | 16.5 | –5.3 | 3.9 | 6.6 | 5.6 | 8.0 | 2.2 |
| **Stockbuilding** [1] | **0.2** | **–0.2** | **0.3** | **–0.1** | **0.0** | **–0.4** | **0.0** |
| **Total domestic demand** | **97.7** | **1.0** | **2.2** | **4.5** | **6.4** | **0.5** | **–0.8** |
| **Exports** | **11.4** | **4.6** | **5.4** | **2.3** | **–0.1** | **8.6** | **13.3** |
| **Total demand** | **109.1** | **1.3** | **2.5** | **4.2** | **5.6** | **1.4** | **0.7** |
| **Imports** | **9.1** | **8.9** | **14.3** | **10.5** | **10.8** | **3.1** | **0.7** |
| **GDP** | **100.0** | **0.6** | **1.4** | **3.5** | **5.1** | **1.2** | **0.8** |
| *Memorandum* | | | | | | | |
| Net exports [1] | – | –0.3 | –0.8 | –0.9 | –1.2 | 0.6 | 1.5 |
| Current external surplus/GDP [2] | – | 2.8 | 2.2 | 1.4 | 1.3 | 1.5 | 0.0 |

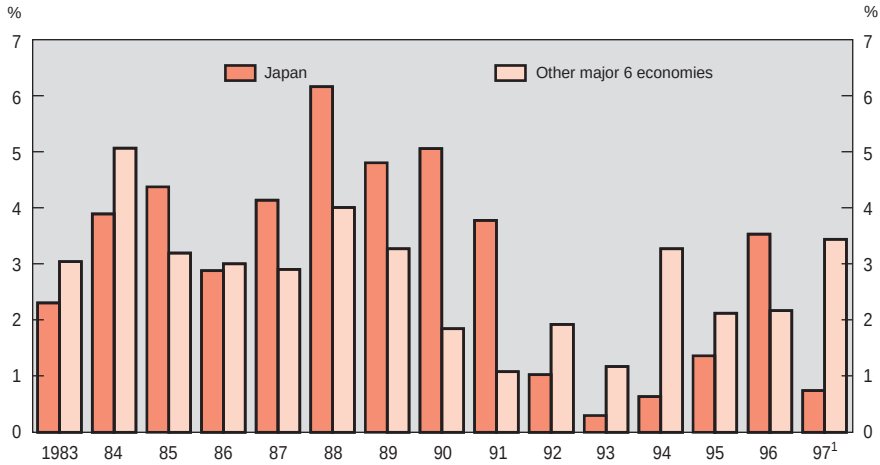
1. Contribution to GDP growth.
2. At current prices, per cent.

*Source:* EPA.

half of 1996. In the second half of the year, the pace of expansion slowed somewhat due to reduced public works spending and the slight fall in household consumption as personal expenditure came back into line with real incomes. Nevertheless, overall private demand remained buoyant thanks to the continued expansion of business investment and renewed growth of exports.

While the pace of the recovery accelerated at the beginning of 1997, its strength was not confirmed in the second quarter when the level of GDP dropped significantly. For the first half of the year as a whole, the jump in consumption spending – in anticipation of tax increases – was more than offset by a marked fall in government spending, a slowing in the growth of business investment and a fall in housing investment. As a result, domestic demand fell. Nonetheless, with export growth accelerating, GDP rose – but by less than ¾ per cent at an annual rate.

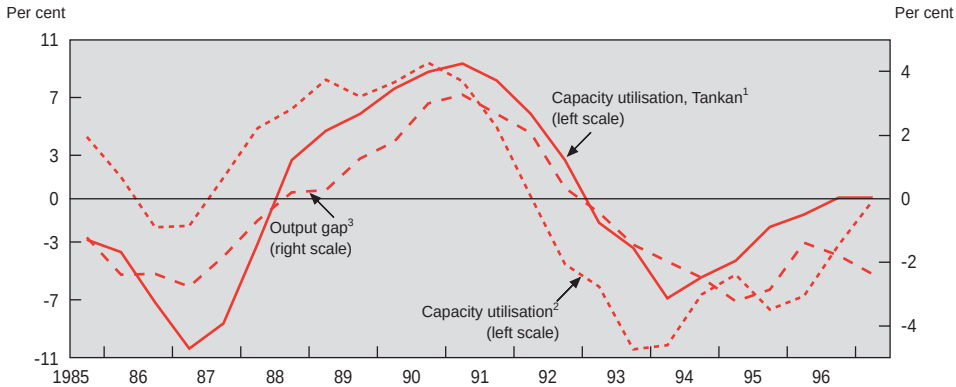
Figure 1. **REAL GDP GROWTH**




1. Figures are for the first half of 1997 and represent the growth from the second semester of 1996.
*Source:* OECD.

Overall, the current recovery appears to be one of the longest and slowest in the post-war period. During the past three and a half years, annual output growth has averaged only 1¾ per cent, with the economy showing few signs of autonomous momentum. Such a growth path is somewhat below that of potential output, which is currently estimated by the OECD at 2 per cent per year – down from 3¼ per cent between 1988 and 1993. This suggests that the output gap is slightly larger than when the recovery started. In contrast, a number of indicators point to the extent of spare capacity having been reduced (Figure 2). For instance, the official index of capacity utilisation in manufacturing has moved up and is now close to its average over the past fifteen years. In addition, business surveys such as the Tankan suggest that major companies have considerably reduced the extent of spare capacity over the past two years. Moreover, the size of the estimated output gap may be overstated as part of the recent rise in unemployment may reflect growing mismatches in the labour market. In addition, there appears to have been a slowdown in the growth of labour productivity in the past five years. However, participation rates continue to increase, more than offsetting the decline in the working-age population. Consequently, there still appears to be some degree of spare capacity in the economy which, so far, the recovery has been unable to eliminate.

Figure 2. **CAPACITY UTILISATION AND OUTPUT GAP**

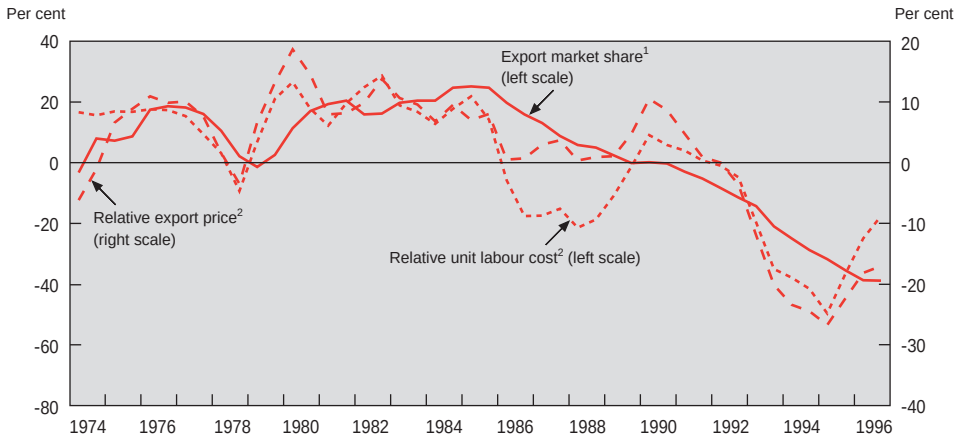



1. Difference between companies judging capacity to be insufficient and those considering it excessive.
2. Difference between the current level of the MITI capacity utilisation index and the average for the past 15 years.
3. Output gap as estimated by the OECD.

*Source:* Bank of Japan, MITI and OECD.

Figure 3. **INTERNATIONAL COMPETITIVENESS AND EXPORT MARKET SHARE**

Percentage deviation from period average




1. Ratio of actual export market share to its period average.
2. A fall in the series indicates a loss in competitiveness.

*Source:* OECD.

## ... driven by exports...

Economic growth has been essentially supported by exports, which have increased markedly since the middle of 1996, and, by the second quarter of 1997, were more than 14 per cent above their level of the same period a year previously. This reflects economic expansion abroad and the substantial depreciation in the value of the yen that started in the summer of 1995 (see Chapter II), which has improved Japan's international competitiveness to levels last seen in the second half of 1992 (Figure 3). Exports of capital goods, in particular, grew significantly, as well as shipments of cars in response to a decline in stocks in the United States and other areas. Indeed, sales of Japanese cars abroad increased significantly in the first half of 1997 after two years of decline.

Japanese trade has become further concentrated in other Asian countries, a development which has been supported by continued direct investment in these countries (Table 2). As a result, the ratio of overseas to domestic production rose once again (Table 3). Overseas production has become oriented to local sales, which are not necessarily substitutes for exports from Japan. This suggests that the exchange rate may become a less important factor in determining direct

Table 2. **Japanese direct investment abroad**

Yen billion, fiscal years

| | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|
| **Total** | 4 431 | 4 151 | 4 281 | 4 957 | 5 409 |
| **Manufacturing** | 1 304 | 1 277 | 1 443 | 1 824 | 2 282 |
| Electrical machinery | 236 | 313 | 273 | 519 | 734 |
| Transport machinery | 156 | 110 | 214 | 194 | 436 |
| Chemicals | 258 | 204 | 272 | 208 | 232 |
| **Non-manufacturing** | 3 081 | 2 845 | 2 798 | 3 039 | 3 012 |
| Finance and real estate | 1 264 | 1 432 | 1 226 | 1 109 | 1 576 |
| Services and commerce | 1 332 | 1 004 | 1 176 | 1 550 | 995 |
| Other | 485 | 409 | 395 | 381 | 442 |
| **Asia** | | | | | |
| Manufacturing | 400 | 418 | 540 | 781 | 747 |
| Non-manufacturing | 410 | 336 | 439 | 357 | 475 |
| **Other areas** | | | | | |
| Manufacturing | 904 | 859 | 903 | 1 042 | 1 535 |
| Non-manufacturing | 2 671 | 2 509 | 2 359 | 2 682 | 2 537 |

*Source:* Ministry of Finance.

Table 3. **Japanese manufacturing production overseas**

Percentage of domestic production

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|
| **All manufacturing** | 6.0 | 6.2 | 7.4 | 8.6 | 9.1 | 9.9 |
| Non-ferrous metals | 5.2 | 7.8 | 6.5 | 8.8 | 6.7 | . . |
| General machinery | 7.6 | 4.1 | 5.8 | 8.1 | 8.1 | . . |
| Electrical machinery | 11.0 | 10.8 | 12.6 | 15.0 | 15.5 | . . |
| Transport machinery | 13.7 | 17.5 | 17.3 | 20.3 | 23.9 | . . |
| **Multinational manufacturers** | 16.7 | 17.3 | 18.3 | 22.0 | 25.1 | 27.5 |
| *Memorandum item* | | | | | | |
| Employment in overseas plants | | | | | | |
| Absolute number, millions | 1.3 | 1.1 | 1.5 | 1.8 | 1.8 | . . |
| Per cent of domestic employment | 8.3 | 7.2 | 10.8 | 12.0 | 12.8 | . . |

*Source:* MITI, *Basic Survey of Business Activities Abroad.*

investment abroad. Indeed, according to an EPA survey in the spring of 1997,[1] about one-third of manufacturing firms were still preparing for an expansion of their overseas activities.

The same survey found that companies are able to obtain a normal profit on exports at an exchange rate of Y 106 to the dollar. This is above the current market rate, suggesting that the high profitability of exports will further stimulate their growth. On the other hand, the increased overseas activities of Japanese firms may contribute to restraining manufacturing exports.[2] The experience of the past two decades suggests a close linkage between the loss of market share by Japanese exporters and the deterioration in international competitiveness (Figure 3).

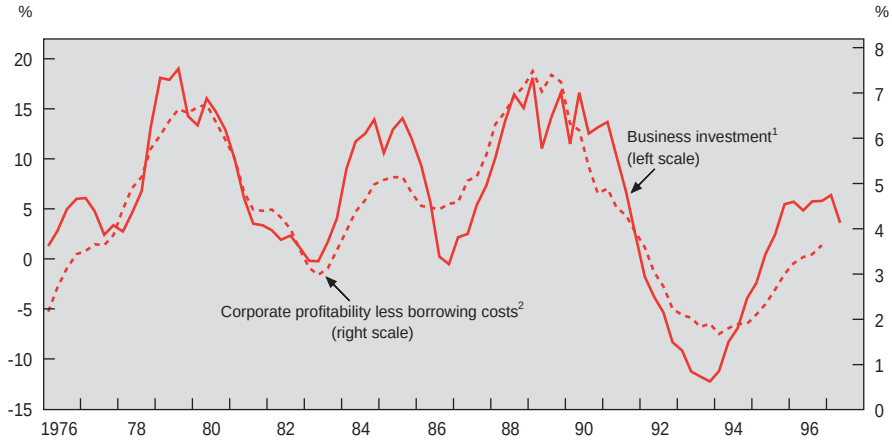
## ... and, until recently, by private investment

The steady expansion of *business investment* was, until the end of 1996, the most sustained factor behind the growth of private demand, even though capital expenditure by firms has risen at a much more moderate pace than in previous recoveries. Its pickup started initially in the semiconductor industry, spread to other large manufacturing firms in 1995 and finally reached non-manufacturing and medium-sized companies in 1996, when overall business investment grew by over 6 per cent. The share of gross business investment in GDP thus recovered

and, by the first half of 1997, was close to the average of the past two decades. However, given the apparent rise in the scrapping rate over the past decade, the share of net business investment in GDP is still considerably below its average level in the past. Consequently, the growth of the capital stock is still relatively slow.

The rebound in investment has been driven mainly by improved profits (Figure 4) and low interest rates. In the second half of 1996, corporate profits rose strongly, especially for large manufacturing companies. Overall, in FY 1996, the profits of quoted companies increased by 17 per cent, significantly more than expected by market analysts, reflecting lower long-term interest rates, an increase in domestic sales and continuing restraint on labour costs. In addition, higher export prices, made possible by the depreciation of the yen, added to corporate profitability. The improved performance of large manufacturing companies has gradually spread to small and medium-sized firms and even to some non-manufacturing enterprises, as orders from large corporations rose. However, the profitability and hence investment of small suppliers has been restrained by the impact of increased imports and higher overseas production.

Figure 4. **BUSINESS INVESTMENT AND CORPORATE PROFITABILITY**




1. Nominal investment, year-on-year increase.
2. Rate of return on physical assets minus long-term interest rates, 4-quarter moving average.

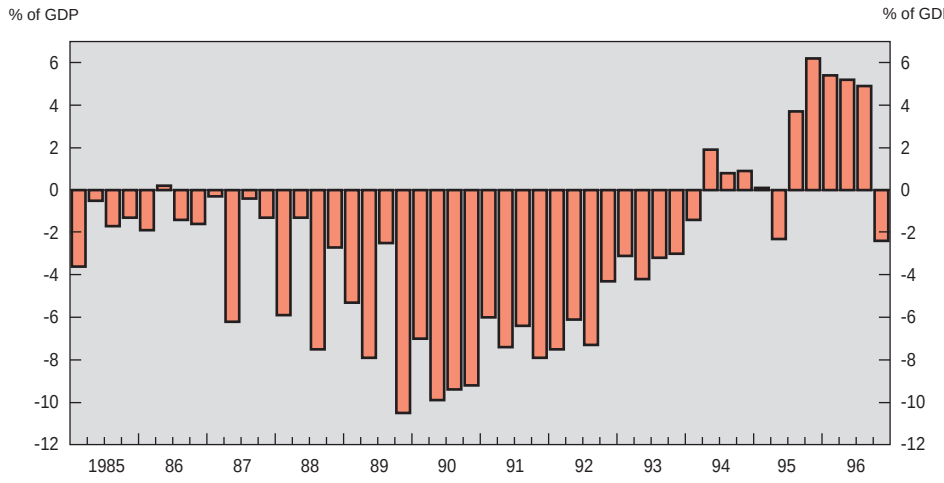
*Source:* EPA, *Report on National Accounts,* Ministry of Finance, *Financial Statements of Incorporated Businesses.*

Higher profits have also strengthened the financial positions of companies. The financial surplus of the business sector increased (Figure 5), as retained earnings and sales of land were more than sufficient to finance net business investment.[3] Large firms have continued to use some of their surplus to reduce their long-term bank borrowing. Consequently, the long-term debt of both manufacturing and non-manufacturing companies (excluding those in real restate) has continued to fall relative to sales (Figure 6). However, there is an exception with small firms, especially in the real estate sector, whose lower profitability has kept their long-term debt at a high level relative both to assets and sales, thereby restraining these firms' investment and risk-taking capacities.

In the first half of 1997, there was a pause in the growth of corporate capital formation, with the first fall in investment since 1994 being registered in the second quarter of the year. Such a development may be linked to an ending of the boom in the mobile telecommunication sector in particular. Although such spending is likely to moderate in 1997, other information-related investment may increase as demands for further upgrading of office automation, communication

Figure 5. **FINANCIAL BALANCE OF THE CORPORATE SECTOR**

Deficit (–), surplus (+)




*Source:* Bank of Japan, *Economic Statistics Monthly.*

Figure 6. **CORPORATE INDEBTEDNESS**[1]

Index 1985 Q1 = 100

1. Corporate debt includes long-term borrowing and bonds.

*Source:* Ministry of Finance, *Financial Statements of Incorporated Businesses*; Bank of Japan, *Flow of Funds Accounts*; EPA, *Annual Report on National Accounts*.

Figure 7. **HOUSING STARTS AND CREDIT CONDITIONS**

1. Seasonally adjusted at annual rate.
2. Housing Loan Corporation.

*Source:* EPA and Bank of Japan.

and computer facilities are rising in response to rapid developments in related technology. At the same time, although construction activity has remained weak, commercial building picked up, as higher demand reduced vacancy rates and stabilised rents in metropolitan areas.

Recent business surveys suggest that corporate investment may continue to expand during the rest of 1997, though at a slower pace, as investment plans, by the middle of the year, appeared somewhat weaker than those of a year before. Nevertheless, manufacturing investment, boosted by the weaker yen, may continue to increase more than non-manufacturing, which may be held back by the lower level of public spending.

As for *housing investment*, there was a surge in the fourth quarter of 1996[4] in anticipation of the consumption tax rate hike. Housing starts collapsed after hitting their highest monthly level in the past two decades in October 1996 and, by the second quarter of 1997, they were 30 per cent below this record high. Such a fall came despite low interest rates (Figure 7) and the stabilisation of housing prices, which, in the Tokyo area, has been reflected in the ratio of annual payments on a new apartment to income falling to nearly 60 per cent of its level in 1990.

## In contrast, private consumption remained subdued...

The weakness of the recovery has dampened the growth of personal income. The increase in compensation per employee has remained sluggish, reflecting the slow improvement in the labour market due to continued corporate restructuring. Moreover, household property income has fallen, in line with the fall in interest rates (Table 4).

Despite sluggish wage growth (see below) and the fall in household property income, the growth of personal consumption picked up in 1996, generating a fall in the savings ratio. Changing demand patterns have increased purchases in some areas, such as personal computers and mobile telephones. In addition, replacement demand for autos has been strong since the autumn of 1996, reflecting the scrapping of cars purchased during the sales boom of the late 1980s.[5] The surge in the consumption of durables was further strengthened by anticipatory spending

Table 4. **Household income**

Percentage change from previous year

| | Per cent of total 1995 | 1992 | 1993 | 1994 | 1995 | 1996[1] | 1997[1] |
|---|---|---|---|---|---|---|---|
| Employee compensation | 62.6 | 3.5 | 2.3 | 2.5 | 2.1 | 2.1 | 3.3 |
| Business income[2] | 12.3 | 6.8 | –3.3 | 3.3 | –7.0 | –2.8 | 1.1 |
| Investment income[3] | 4.1 | –9.5 | –1.8 | –18.8 | –3.5 | . . | . . |
| Transfers | 21.0 | 6.3 | 4.1 | 4.5 | 7.1 | 5.3 | 5.3 |
| Total income | 100.0 | 3.6 | 1.6 | 1.9 | 1.7 | 2.0 | 3.3 |
| Tax and social security payments | –18.3 | 2.0 | –1.0 | –1.6 | 2.7 | 3.0 | 5.9 |
| Other pension payments | –6.7 | 5.3 | –0.4 | 2.0 | 3.1 | 2.5 | 2.5 |
| Disposable income | 75.0 | 3.9 | 2.5 | 2.7 | 1.3 | 1.7 | 2.8 |
| Real disposable income | .. | 2.0 | 1.3 | 2.0 | 1.7 | 1.5 | 1.1 |
| *Memorandum* | | | | | | | |
| Real consumption | | 2.1 | 1.2 | 1.9 | 2.0 | 2.8 | 1.5 |
| Saving rate[4] | .. | 13.1 | 13.4 | 13.3 | 13.1 | 11.9 | 11.5 |
| Employment growth[5] | .. | 2.3 | 1.6 | 0.6 | 0.5 | 1.1 | 1.6 |

1. OECD estimates.
2. Business and investment income in 1996 and 1997.
3. After deducting interest on consumer debt and other interest paid.
4. Net of depreciation.
5. Employees only.

*Source:* Ministry of Labour, *Monthly Labour Survey*; Economic Planning Agency and OECD.

in the months prior to the consumption tax hike in April 1997, especially for items such as furniture and consumer electronics.

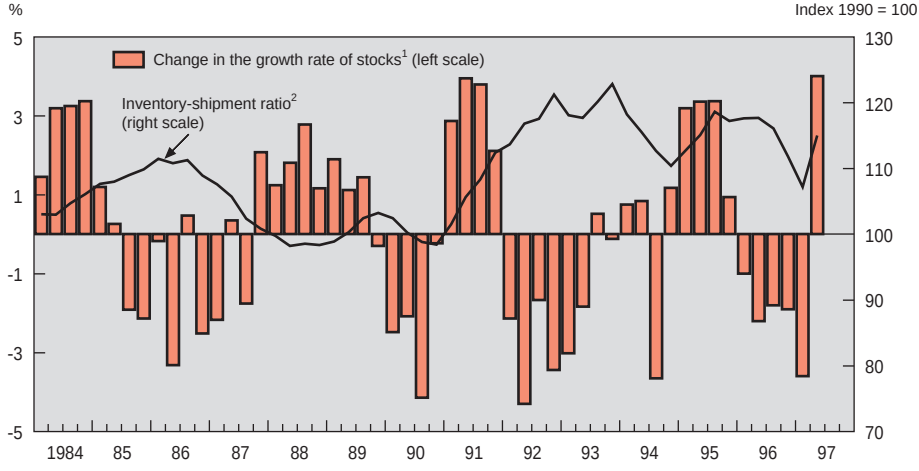
The increase in spending that occurred in the first quarter of 1997 was completely reversed in the second quarter. For the year as a whole, the increase in real personal disposable income is likely to be less than in 1996, due to the hike in indirect taxation, the expiration of the income tax rebate, a rise in social security contributions at the end of last year and increased payments for healthcare. In a full year, these changes will reduce real disposable income by perhaps 3 per cent, with as much as 2 per cent affecting personal income in 1997. Nevertheless, personal spending by the end of the year may still rise, albeit slowly, thanks to employment increases and hikes in bonus payments as well as some possible further fall in the savings ratio as consumer confidence increases.

## ... while both public investment and stockbuilding depressed demand

In contrast to private expenditure which, as indicated above, remained buoyant until the first half of 1997, public-sector demand has fallen since the first half of 1996. This is primarily due to the drop in public investment as the impact of the September 1995 economic stimulus package began to diminish. Indeed, public investment fell by almost 2¼ per cent of GDP between the second quarter of 1996 and the first quarter of 1997. Some rebound in public spending occurred, however, in the second quarter of 1997 and orders for public works suggest that this increase may continue in the remainder of the year as the effect of the supplementary budget for FY 1996 is felt. Nonetheless, such a movement is unlikely to offset fully the low public works outlays included in the initial budget plans of both central and local governments for FY 1997.

A further negative factor has been the slackening in the growth of stocks of finished manufactured products (Figure 8). By the end of 1996, they were falling

Figure 8. **STOCKS IN MANUFACTURING INDUSTRY**




1. Change in the year-on-year growth rate of stocks of finished products in manufacturing industry.
2. The inventory-shipment ratio covers finished products of manufacturing industry.

*Source:* MITI.

and this movement accelerated further in the first quarter of 1997, holding back industrial output growth. While this movement was reversed in the second quarter, the marked increase in stocks in that period presages a further period of falling inventories before the stock output ratio returns to its trend level.

## With import growth slackening...

With the level of stocks of finished products starting to fall from the second half of 1996 onwards, real import growth decelerated sharply (Table 5). The slowdown was accentuated by the rising trend of import prices following the decline in the yen. Nevertheless, import penetration has continued to increase due

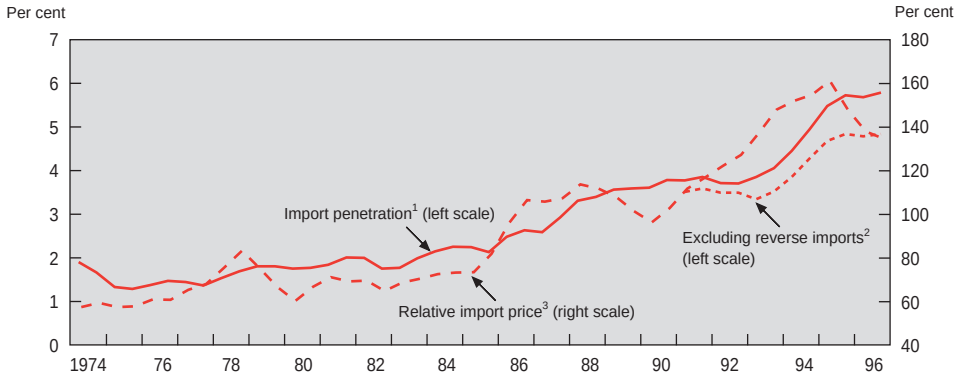
Table 5. **External balances**

Trillion yen

| | 1993 | 1994 | 1995 | 1996 | Seasonally-adjusted annual rates | | |
|---|---|---|---|---|---|---|---|
| | | | | | 1996 | | 1997 |
| | | | | | 1st half | 2nd half | 1st half |
| **Trade** | **15.5** | **14.7** | **12.3** | **9.1** | **9.0** | **9.1** | **11.0** |
| Exports | 39.2 | 39.3 | 40.3 | 43.6 | 42.7 | 44.4 | 49.3 |
| Imports | 23.7 | 24.6 | 27.9 | 34.5 | 33.7 | 35.3 | 38.3 |
| **Services** | **–4.8** | **–4.9** | **–5.4** | **–6.8** | **–6.5** | **–7.0** | **–6.5** |
| **Investment income** | **4.5** | **4.1** | **4.2** | **5.8** | **5.6** | **6.1** | **7.0** |
| **Transfers** | **–0.6** | **–0.6** | **–0.7** | **–1.0** | **–1.1** | **–0.9** | **–1.3** |
| **Current account** | **14.7** | **13.3** | **10.4** | **7.2** | **7.0** | **7.3** | **10.3** |
| (per cent of GDP) | 3.1 | 2.8 | 2.2 | 1.4 | 1.4 | 1.5 | 2.0 |
| *Memorandum* | Percentage change from previous period | | | | | | |
| **Trade in goods** | | | | | | | |
| Volume [1] | | | | | | | |
| Exports | –1.7 | 1.7 | 3.8 | –0.7 | –2.8 | 3.3 | 10.7 |
| Imports | 4.2 | 13.5 | 12.5 | 2.3 | 4.2 | 1.5 | 3.3 |
| Manufactures | 7.1 | 19.6 | 22.1 | 4.1 | 9.5 | 4.5 | 4.9 |
| Unit value [1] | | | | | | | |
| Exports | –4.9 | –0.9 | –1.2 | 8.5 | 9.8 | 5.0 | 4.1 |
| Imports | –12.8 | –7.7 | –0.2 | 17.7 | 18.3 | 16.1 | 9.3 |

1. Custom clearance basis (service trade is not included), year-on-year.
*Source:* Bank of Japan, *Balance of Payments Monthly*; OECD.

Figure 9. **IMPORT PENETRATION AND RELATIVE IMPORT PRICES**




1. Import penetration is measured as the ratio of manufactured imports to total domestic demand.
2. Reverse imports are manufactured products made outside Japan by Japanese companies and sent to Japan.
3. The relative price of imports is measured by the ratio of the import price of manufactured products to the domestic demand deflator.

*Source:* OECD.

to structural factors boosting purchases from abroad (Figure 9). In particular, a significant part of overseas production in other Asian countries is exported back to Japan, either as finished products sold under Japanese brand names or as semi-finished goods for incorporation in other products. These so-called ''reverse'' imports have risen steadily, reaching 15 per cent of total imports in FY 1995.

## ... the external surplus has started to rise

The combination of slower growth in imports and rising exports was reflected in an increase in the goods and services surplus to 0.9 per cent of GDP in the first half of 1997. By the second quarter of the year this surplus had reached almost 1½ per cent of GDP, up from virtual balance a year previously, despite import prices increasing faster than export prices due to the yen's fall (the so-called ''J-curve effect''). This development was accentuated by a marked rise in net investment income (Table 5), which has been boosted by the continued accumulation of net foreign assets and the decline in the value of the yen, since

many of these assets are denominated in foreign currencies. Overall, the current account surplus rose to 2.5 per cent of GDP in the second quarter of 1997.

## Given the slow improvement in the labour market...

The increase in output in 1996 and at the beginning of 1997 has generated a slow improvement in the labour market (Table 6). Unemployment remained high, only slightly below its record post-war high of 3.5 per cent of the labour force in

Table 6. **Labour market developments**

| | Million persons/hours | Percentage change from previous period | | | | | |
|---|---|---|---|---|---|---|---|
| | 1996 | 1994 | 1995 | 1996 | 1996 | | 1997 |
| | | | | | 1st half | 2nd half | 1st half |
| **Working-age population** [1] | 86.9 | 0.0 | –0.0 | –0.1 | –0.2 | –0.1 | –0.2 |
| **Labour force** | 67.1 | 0.5 | 0.3 | 0.7 | 0.5 | 0.8 | 1.5 |
| **Unemployment rate** [2] | 2.3 | 2.9 | 3.2 | 3.4 | 3.4 | 3.3 | 3.4 |
| **Employment by status** | 64.9 | 0.0 | 0.0 | 0.4 | 0.1 | 0.8 | 1.5 |
| Employees | 53.2 | 0.7 | 0.5 | 1.1 | 1.2 | 1.1 | 1.6 |
| Full-time [3] | 42.2 | 0.0 | 2.3 | –1.4 | –1.3 | –1.6 | 0.4 |
| Part-time [4] | 10.9 | 3.5 | –6.7 | 12.4 | 12.1 | 12.7 | 6.6 |
| Self-employed and family workers | 11.5 | –2.4 | –1.8 | –2.9 | –4.5 | –1.2 | 0.7 |
| **Employment by industry** | | | | | | | |
| Private sector | 62.7 | 0.0 | 0.0 | 0.5 | 0.3 | 0.8 | 1.5 |
| Agriculture and forestry | 3.3 | –1.4 | –1.4 | –2.9 | –5.1 | –0.9 | –0.6 |
| Manufacturing | 14.5 | –2.2 | –2.7 | –0.8 | –1.0 | –0.5 | 0.9 |
| Construction | 6.7 | 2.3 | 1.2 | 1.1 | 0.3 | 1.6 | 3.5 |
| Services | 37.7 | 0.7 | 1.2 | 1.2 | 1.3 | 1.0 | 1.4 |
| Government | 2.1 | 2.9 | 1.4 | –1.8 | –4.0 | –0.5 | 2.3 |
| **Hours worked** | | | | | | | |
| Total | 1 919 | –0.4 | 0.1 | 0.1 | 0.6 | 0.2 | –0.9 |
| Standard | 1 796 | –0.3 | 0.0 | –0.3 | 0.3 | –0.3 | –1.3 |
| Overtime | 123 | –1.1 | 2.0 | 6.2 | 5.2 | 8.3 | 3.8 |

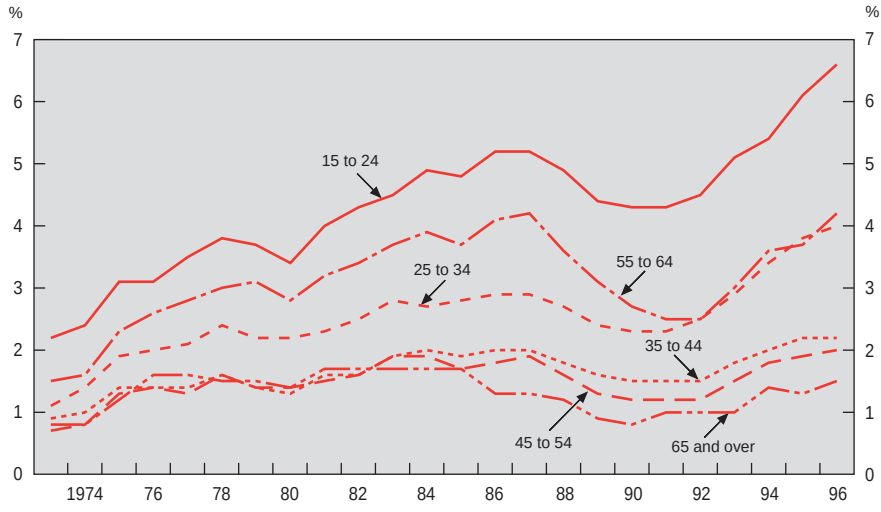
1. Age 15 to 64.
2. Seasonally adjusted.
3. More than 35 hours.
4. Between 0 and 35 hours.

*Source:* Management and Coordination Agency, *Labour Force Survey*; Ministry of Labour, *Report on Employment Service*.

the spring of 1996. It has been concentrated amongst the young and old, for whom the rate of unemployment has moved well above that seen at the last cyclical peak in 1987 (Figure 10). In contrast, middle-age people have been much less affected by unemployment. This reflects the fact that firms have adjusted employment by reducing their hiring, especially of new graduates, rather than by laying-off employees. Indeed, the percentage of workers leaving firms has hardly changed throughout this business cycle. Nonetheless, there has been significant job creation, with private-sector firms adding almost 1 million employees in the year to the first half of 1997.

In addition, there may have been some increase in structural unemployment amongst young people, as job preferences have become more diversified, leading to more job search activities and voluntary unemployment. Indeed, the number of job vacancies increased by 10 per cent in the year to April 1997 suggesting a mismatch between supply and demand. However, preliminary reports suggest that hiring of new high school and college graduates rose significantly in 1997, reflecting an end to the decline in the number of employees as employers became

Figure 10. **UNEMPLOYMENT BY AGE**




*Source:* Management and Coordination Agency, *Labour Force Survey.*

more confident of the durability of the upswing. Employment growth has continued to be boosted by a strong rise in part-time jobs. These have mainly been filled by women whose participation rate in the labour force has thus increased, more than compensating for the fall in the working-age population that started in 1996.

## ... wage and price increases stay low

The growth of basic wage rates is still moderate, reflecting the efforts of companies to reduce the fixed costs of labour as much as possible. The slowdown over the past three years has been greater than might be expected from the annual central spring wage round, due to competitive pressures at the firm level (Table 7). Employers have preferred to restrain the growth of basic wages and emphasise the flexible components of pay. Overtime and bonus payments have picked up somewhat, in line with rising production and profitability, leading to a faster increase in total earnings. The emphasis on raising the share of flexible payments in the total wage bill has continued in this year's wage settlements.

Table 7. **Wages and earnings**

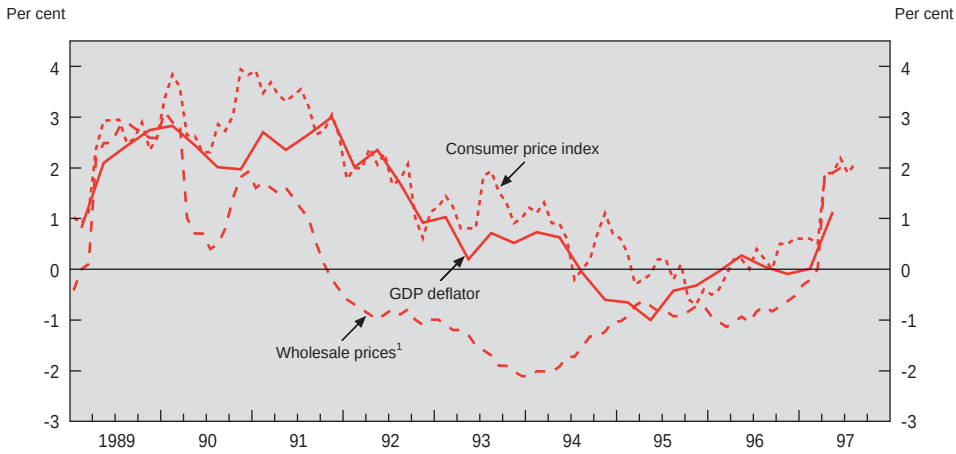
Per person employed

| | Per cent of total 1996 | Percentage change from previous period | | | | |
|---|---|---|---|---|---|---|
| | | 1993 | 1994 | 1995 | 1996 | 1997[1] |
| **Wage survey basis** | | | | | | |
| Spring wage round | .. | 3.9 | 3.1 | 2.8 | 2.9 | 2.8 |
| Basic earnings | 70.2 | 2.7 | 2.5 | 1.8 | 1.6 | 1.5 |
| Overtime | 6.2 | –6.4 | 0.7 | 6.5 | 5.6 | 5.1 |
| Bonus payments | 24.8 | –3.3 | 0.2 | 0.7 | 0.8 | 4.1 |
| Total earnings | 100.0 | 0.6 | 1.8 | 1.8 | 1.6 | 1.7 |
| **National accounts basis** | | | | | | |
| Total earnings | .. | 0.6 | 2.0 | 1.0 | 0.8 | 1.7 |
| Social security contributions | .. | 1.8 | 0.4 | 5.3 | 1.5 | 1.6 |
| Compensation per employee | .. | 0.7 | 1.8 | 1.6 | 0.9 | 1.7 |

1. OECD estimates.
*Source:* Ministry of Labour, *Monthly Labour Survey*; Economic Planning Agency and OECD.

Figure 11. **INFLATION INDICATORS**

Percentage changes from previous year




1. Producer prices for the domestic market.

*Source:* OECD.

The weakness in wage growth has helped keep underlying domestic prices stable, with the GDP deflator showing no increase in the first quarter of 1997. Falling unit labour costs have meant that the impact of the weak yen on domestic prices has been limited to ending the decline in prices recorded in the course of 1996 (Figure 11). However, the increase in the consumption tax boosted the rate of inflation, as measured by retail prices, from April 1997 onwards. The impact of the hike on the price level, though, was less than the 2 percentage point increase in the tax rate as some goods are exempt from the tax.

## Short-term outlook: the recovery could regain some momentum

Output growth is projected to be only ¾ per cent in 1997 as a whole – the lowest annual rate since 1994 – after the sharp drop in activity in the second quarter of the year. Short-term indicators suggest that there will be some rebound in private consumption in the rest of the year, following the earlier erratic

movement of household spending due to the increase in the rate of indirect taxation (Table 12). Moreover, business surveys indicate that corporate investment should resume its upward trend after its fall in the second quarter of the year. At the same time, some increase in public works spending stemming from the supplementary budget at the beginning of 1997 is likely. Nonetheless, given that the second quarter level of private spending was very low, domestic demand will probably be flat in the second half of the year and only a continued, though smaller, boost from net exports may generate a slow expansion in activity.

In 1998, growth is projected to accelerate to just over 2 per cent, assuming that the pace of fiscal tightening will be slowed in line with government guidelines for the FY 1998 budget. With no additional tax hikes foreseen, the increase in real personal disposable income should accelerate. The rise in private-sector employment and some increase in overtime working should indeed more than offset the negative effects, on personal incomes, of the restraint being exercised over spending on public-sector wages by the government in the 1998 budget. As a result, the growth of private consumption should pick up moderately. Elsewhere, with margins on exports remaining high and low financing costs continuing to boost corporate profits, business investment is projected to renew its upswing. Renewed growth of private demand, through, might be partially offset by the continued weakness of public demand. In addition, the contribution of the foreign balance to output growth may slacken as the effects of the weak yen on trade will diminish and as exports to South East Asia are expected to slow as the result of financial turmoil in that region. However, the overall current account surplus may rise moderately to around 2½ per cent of GDP, from 2¼ per cent in 1997, in part due to the weaker yen boosting investment income receipts.

Although the economic recovery is projected to continue, the estimated output gap is expected to remain above its level in the second half of 1993 when the upswing first started (Table 8). With the labour force still rising despite the decline in the working-age population, the unemployment rate may stay high by Japanese standards, at slightly above 3 per cent. As a result, the increase in basic wage rates could remain below 2 per cent in 1998. In this context, the rise in the price level in 1997, following the increase in indirect taxation, should not generate a permanent upward movement in the inflation rate. Indeed, the growth of the private consumption deflator should fall below 1 per cent in the course of 1998.

Table 8. **Short-term prospects**

Percentage change from previous year

| | Average 1983-1993 | 1994 | 1995 | 1996 | 1997[1] | 1998[1] |
|---|---|---|---|---|---|---|
| **Demand and output (volume)** | | | | | | |
| Consumption | 3.2 | 2.0 | 2.2 | 2.8 | 1.4 | 1.6 |
| Private consumption | 3.4 | 1.9 | 2.0 | 2.8 | 1.5 | 1.9 |
| Government consumption | 2.1 | 2.4 | 3.5 | 2.3 | 0.8 | –0.2 |
| Gross fixed investment | 5.0 | –0.8 | 1.1 | 8.7 | –3.6 | 1.9 |
| Public[2] | 4.2 | 2.8 | 0.7 | 9.9 | –14.1 | –1.3 |
| Private residential | 3.2 | 8.5 | –6.4 | 13.6 | –8.7 | –1.9 |
| Private non-residential | 6.2 | –5.3 | 3.9 | 6.6 | 3.9 | 4.5 |
| Stockbuilding[3] | .. | –0.2 | 0.3 | –0.1 | –0.1 | 0.0 |
| Total domestic demand | 3.7 | 1.0 | 2.2 | 4.5 | –0.3 | 1.7 |
| Exports of goods and services | 4.6 | 4.6 | 5.4 | 2.3 | 11.1 | 8.8 |
| Total demand | 3.8 | 1.3 | 2.5 | 4.2 | 1.0 | 2.5 |
| Imports of goods and services | 6.1 | 8.9 | 14.3 | 10.5 | 2.0 | 6.3 |
| GDP | 3.6 | 0.6 | 1.4 | 3.5 | 0.8 | 2.1 |
| **Inflation and capacity utilisation** | | | | | | |
| GDP deflator | 1.7 | 0.3 | –0.6 | 0.0 | 1.0 | 0.9 |
| Private consumption deflator | 1.7 | 0.7 | –0.5 | 0.2 | 1.6 | 1.0 |
| Import deflator | –6.0 | –5.3 | –2.6 | 11.2 | 6.5 | 0.5 |
| Export deflator | –3.3 | –3.9 | –3.1 | 7.0 | 2.8 | 0.2 |
| Total employment | 1.2 | 0.1 | 0.1 | 0.5 | 1.5 | 1.3 |
| Unemployment rate | 2.5 | 2.9 | 3.1 | 3.4 | 3.4 | 3.3 |
| Output gap | 0.1 | –2.7 | –3.9 | –1.6 | –2.7 | –2.6 |
| **Current account balance** | | | | | | |
| Per cent of GDP | 2.7 | 2.8 | 2.2 | 1.4 | 2.2 | 2.6 |
| Net exports[3] | .. | –0.3 | –0.8 | –0.9 | 1.1 | 0.4 |
| *Memorandum* | | | | | | |
| Net government lending (per cent of GDP) | 0.2 | –2.3 | –3.7 | –4.4 | –2.8 | –2.3 |
| Short-term interest rates | 5.6 | 2.3 | 1.2 | 0.6 | 0.6 | 0.9 |

1. OECD estimates.
2. Including public corporations.
3. Contributions to GDP growth.

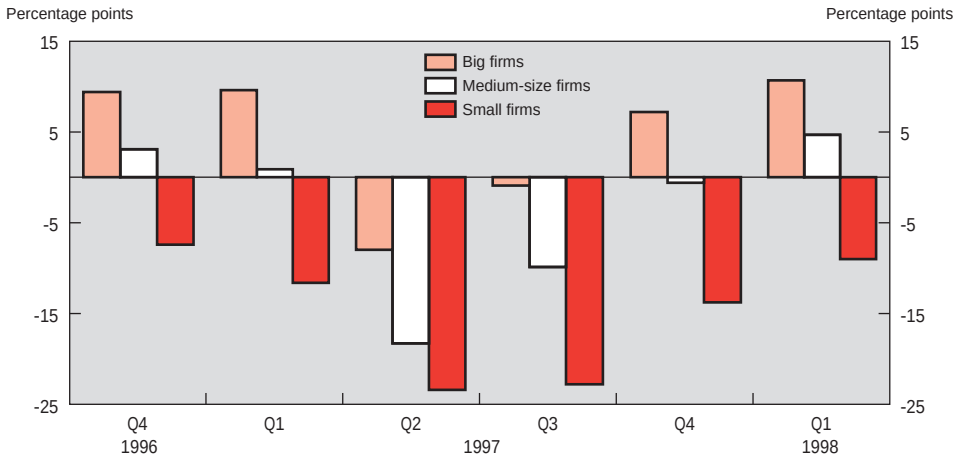
*Source:* OECD.

The projections are based on the following assumptions:

– An unchanged exchange rate of ¥120.9 to the dollar.
– Short-term interest rates will increase to 1 per cent by the second half of 1998, while long-term rates rise, but remain below 3 per cent.

Figure 12. **CHANGE IN THE BUSINESS CLIMATE**

Companies' judgement on business conditions, August 1997




1. The judgement about business conditions reflects the difference between those companies expecting an upturn in the economy and those expecting a downturn.

*Source:* Ministry of Finance, *Business Survey.*

– Fiscal policy will tighten markedly in 1997, with the estimated general government structural budget deficit falling by slightly over 2 per cent of GDP. The following year the structural deficit is projected to fall by a further ½ per cent of GDP.
– Oil prices will remain stable at around $18¼ per barrel from the second half of 1997 onwards.
– Export market growth for manufactures will rise to 11¼ per cent in 1997 and slow to 9 per cent in 1998.

There are uncertainties surrounding such projections, however. While there is some possibility that the rebound in the economy may be faster than expected in the short term, given the extreme volatility of recent movements in GDP, the consequences of the financial difficulties in some ASEAN countries could have a more marked effect on Japanese business sentiment and export growth to that region. The deeper and longer these difficulties last, the more severe would be the impact on Japan's growth prospects, particularly if the economic slowdown spreads to the rest of Asia.

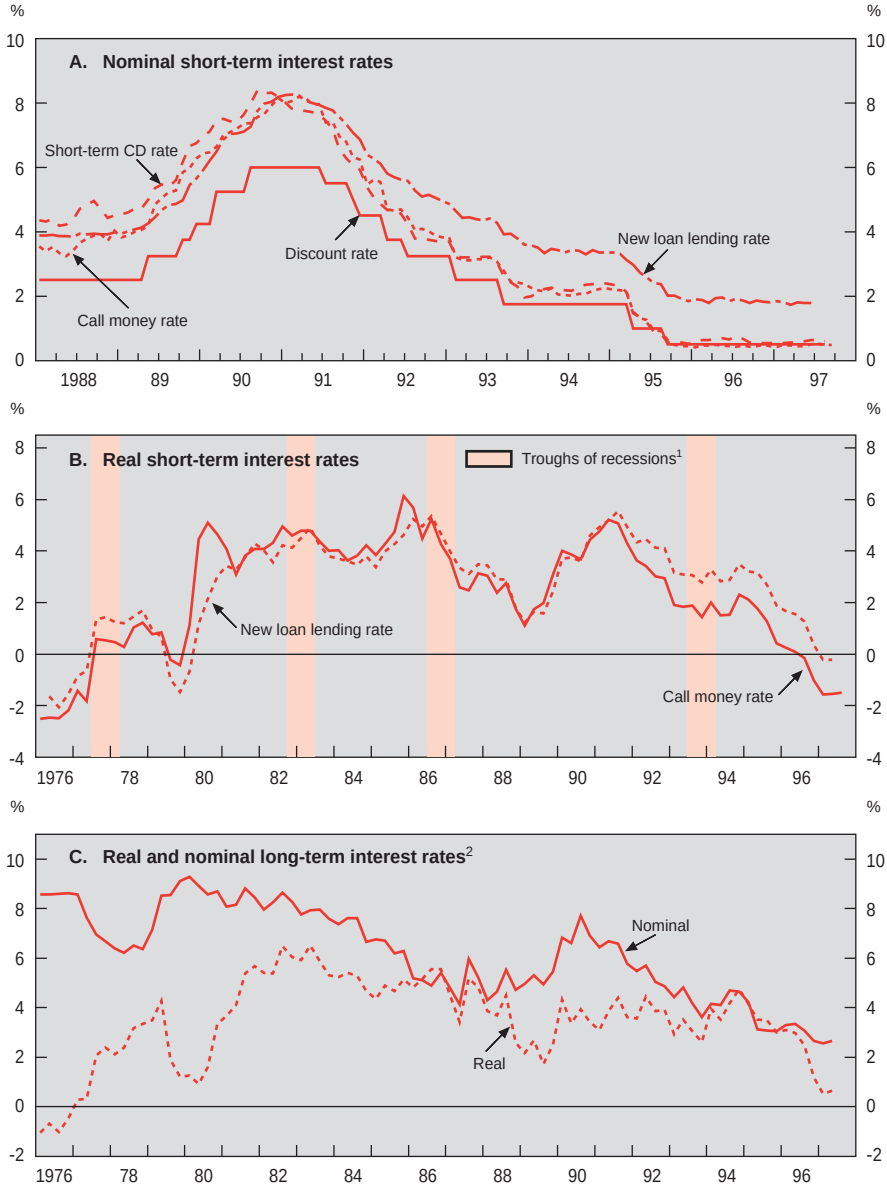
# II. Monetary policy and the financial system

Monetary policy has continued to support activity over the past year. Interest rates have remained at historical lows and, with the substantial depreciation of the exchange rate, overall monetary conditions have become exceptionally easy. Such a policy stance has been justified given the extent of the fall in domestic prices and the weakness of business activity which has aggravated the persistent difficulties of the financial system. Although, as indicated above, the economic recovery is not yet fully established, this stance has shown some positive results with asset prices stabilising and the domestic price level no longer falling – factors which have contributed to support banks' balance sheets. At the same time, the framework for ensuring financial stability has been further improved with the introduction of a new set of more transparent rules for financial regulation and supervision. Moreover, the possibility of additional indirect injections of public money to safeguard the depositors of some financial institutions, such as credit co-operatives, now exists through government-guaranteed borrowing by the Deposit Insurance Corporation. The following paragraphs review these policy developments.

## With interest rates extremely low...

Short-term interest rates have remained at the historically low level of 0.5 per cent since the autumn of 1995. The Official Discount Rate (ODR) has been unchanged at that level for over 20 months, the second longest period of stability for a decade. In addition, call money market rates have been kept slightly below the ODR (Figure 13, Panel A). The rates of interest on certificates of deposit have also been low, reflecting the dissipation of market expectations that official interest rates might be slightly increased. Bank lending rates have

Figure 13. **INTEREST RATE DEVELOPMENTS**




1. The dating of the phases of the business cycle is as defined by the EPA.
2. Ten-year central government bonds.

*Source:* OECD.

remained well above money market rates, though since the middle of 1996 they have tended to fall somewhat as the demand for loans stayed weak.

The movements of long-term interest rates over the past year have been driven by the changing outlook for fiscal policy and growth (Figure 13, Panel C). After rising during the first two quarters of the year, interest rates on ten-year government bonds dropped by nearly 100 basis points to 2½ per cent as the plans for a tighter budget in 1997 reduced expectations about future growth. However, the continued strength of private domestic demand and exports at the beginning of 1997 appear to have improved market sentiments, thereby stabilising long rates. Yields even edged up somewhat in the late spring, though remaining well below the level of a year ago. However, once the pause in economic activity became evident in the summer, long-term interest rates dropped once again to only slightly below 2 per cent. The combination of falling bond yields and a pickup in inflation led to a marked decline in real long-term interest rates, as conventionally calculated, though the extent of the fall has been somewhat exaggerated by the impact of the increase in indirect taxation on inflation.

In 1996, the Bank of Japan changed its operating procedures for implementing monetary policy.[6] While in the past, the Bank used to influence market sentiment by changing the amount it lent to commercial banks at the official rate, it has now stopped such lending, putting greater emphasis on open-market operations, in part through the short end of the government bond market. The Bank has also played a role in improving settlement procedures such as those for government bonds that had proved to be a barrier to the smooth functioning of repurchase agreements. Consequently, market forces now have a greater influence on the movement of interest rates. More importantly, new legislation has been passed which gives the Bank of Japan greater independence (see Box 1).

## ... capital outflows accelerated, pushing down the exchange rate...

Over the past eighteen months, the differential between US and Japanese long rates has widened considerably. For ten-year bonds, the gap has increased from around 250 basis points at the beginning of 1996 to over 410 points by the autumn of 1997 (Figure 14, Panel A). The widening in the real interest rate differential has been much larger, especially at the long end of the market (Figure 14, Panel B). These developments have increased Japanese investors'

### Box 1. **The new Bank of Japan law**

A new legal structure governing the Bank of Japan will come into effect in April 1998, increasing the independence of the Bank of Japan and the transparency of the policy decision process. Under the new law, the independence of the Bank must be respected and existing laws that give the government the authority to order actions by the Bank, including changes in monetary policy, will be repealed.

Under the new legislation, the objectives of the Bank of Japan are twofold:

*i)* The Bank shall pursue price stability, thereby contributing to the sound development of the economy.

*ii)* It shall ensure smooth settlement of funds among banks and other financial institutions thereby contributing to the maintenance of an orderly financial system.

In order to achieve these objectives, the new law has changed the structure of the Bank. There will now be only one decision making body – the Policy Board, consisting of the Governor, two deputy governors and six other members. Two representatives of the government can attend the board meetings for discussions of monetary policy as necessary. They can make propositions, including a demand for the deferral of the implementation of board decisions until the next meeting, but cannot vote.

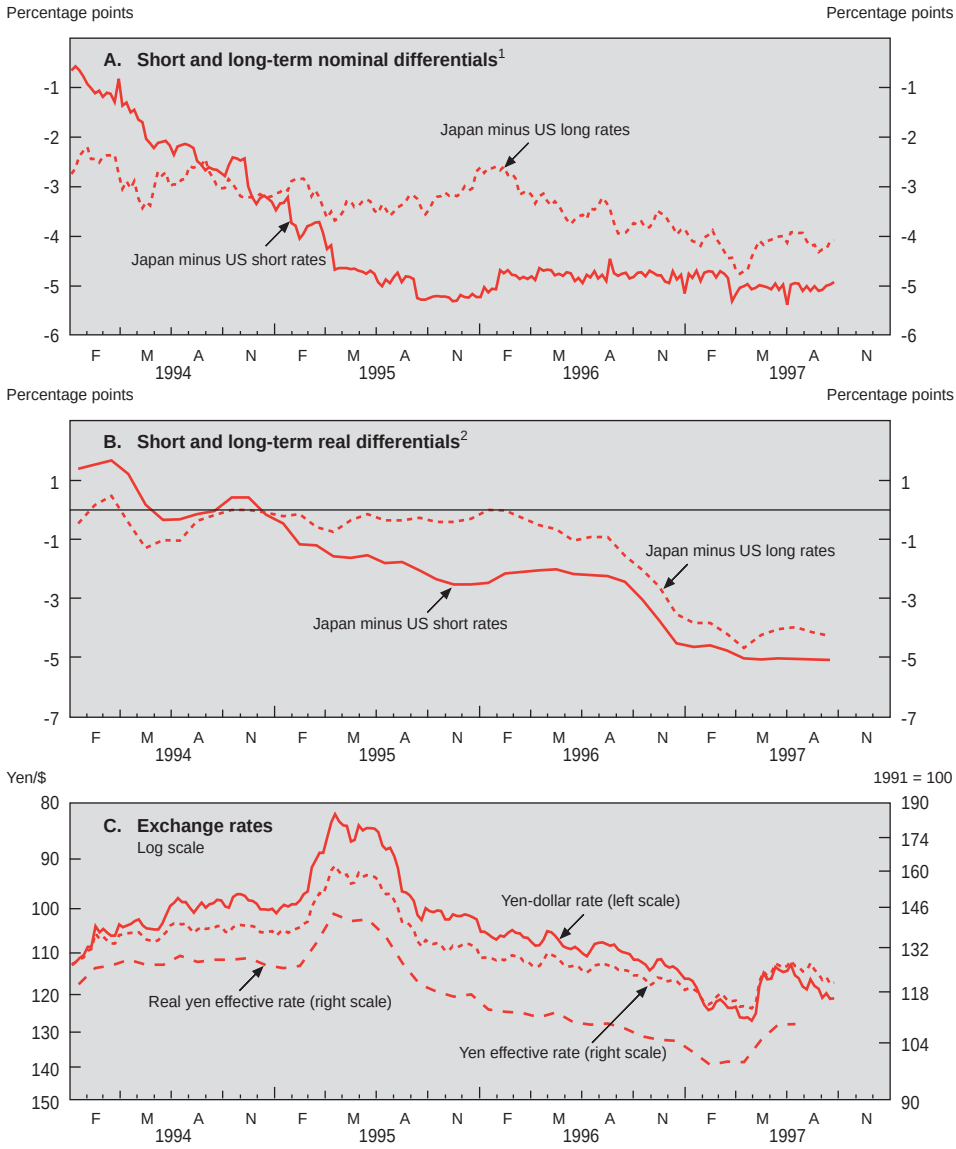
All the governors and board members are appointed by the government for a duration of five years, and their appointment is subject to confirmation by the two houses of the Diet. Up to six executive directors and a few policy advisors of the Bank are appointed by the Minister of Finance on the basis of the recommendation of the Policy Board.

The members of the new Policy Board will be chosen on a different basis to the past. So far, four appointed members were expected to have expertise in the areas of finance, industry and agriculture while the other two came form the Ministry of Finance and the Economic Planning Agency. Under the new system, the members of the Board other than the governor and two deputy governors will be selected on the basis of their academic experience and/or their expertise in the area of the economy or finance.

In order to improve transparency and democratic control, the Board will have to publish a summary of the discussion of its meetings quickly and disclose publicly the full record of the meeting at some point. Every six months, the Bank must report on its activities to the Diet, through the Minister of Finance.

incentive to invest abroad, an attitude strengthened by the perception, which prevailed for much of 1996, that the authorities in Japan and the United States favoured a reversal of the earlier significant appreciation of the yen.

Figure 14. **EXCHANGE RATE BEHAVIOUR AND INTEREST RATE DIFFERENTIALS**




1. Call money for short-term rate and central government bonds for long-term.
2. Measured by the difference between the interest rate and the three-month moving average of the inflation rate measured by the consumer price index.

*Source:* OECD.

The possibility of higher returns abroad led to extensive capital outflows since the beginning of 1996. By the first quarter of 1997, net purchases of foreign bonds reached about 2 per cent of GDP (Table 9). At the same time, capital inflows dropped off, especially foreign purchases of Japanese equities. The pick-

Table 9. **International capital flows**

Yen billion

| | 1995 | 1996 | 1996 | | | | 1997 | |
|---|---|---|---|---|---|---|---|---|
| | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| **Financial account** | **–6 061** | **–2 993** | **1 263** | **–764** | **–945** | **–2 548** | **–3 866** | **–2 292** |
| *Assets* | –19 922 | –14 606 | 1 787 | –5 180 | –1 524 | –9 688 | –4 916 | –12 245 |
| Direct investment | –2 129 | –2 549 | –587 | –688 | –680 | –593 | –660 | –951 |
| Equities | 7 | –905 | 451 | –923 | –73 | –359 | 332 | –1 571 |
| Bonds | –8 479 | –8 798 | –1 523 | –1 523 | –2 919 | –2 834 | –1 125 | –4 347 |
| Money markets | 582 | –1 303 | –59 | –510 | –527 | –207 | –438 | –315 |
| Derivatives | –113 | –1 517 | –441 | –304 | –402 | –370 | –618 | –789 |
| Loans | –15 634 | –906 | 2 093 | –55 | 3 785 | –6 728 | –1 460 | –2 800 |
| Trade credits | 232 | 110 | –49 | 168 | –52 | 43 | 40 | 124 |
| Deposits and currency | 3 755 | 5 080 | 1 811 | 885 | 178 | 2 205 | 138 | –788 |
| Others | 1 858 | –3 818 | 91 | –2 230 | –835 | –844 | –1 127 | –727 |
| *Liabilities* | 13 861 | 11 612 | –524 | 4 416 | 580 | 7 140 | –1 051 | 9 953 |
| Direct investment | 4 | 25 | –7 | –93 | 100 | 24 | 103 | 67 |
| Equities | 4 805 | 5 298 | 2 629 | 2 085 | 129 | 455 | 659 | 2 396 |
| Bonds | –920 | 1 931 | –1 340 | 1 872 | –138 | 1 537 | –1 223 | –547 |
| Money market instruments | 1 893 | 68 | –2 067 | 2 156 | –230 | 209 | –2 156 | 3 923 |
| Derivatives | –852 | 712 | 316 | 197 | 206 | –8 | 463 | 838 |
| Loans | 9 758 | 2 157 | –588 | –1 213 | 515 | 3 442 | 3 447 | 1 782 |
| Trade credits | –28 | 52 | 112 | –25 | –86 | 51 | –2 | –31 |
| Deposits and currency | –26 | 807 | –99 | –5 | 28 | 883 | –19 | 260 |
| Others | –774 | 563 | 520 | –559 | 56 | 547 | –222 | 1 265 |
| Capital transfers | –214 | –388 | –171 | –72 | –87 | –58 | –148 | –101 |
| **Total capital flows** | **–6 275** | **–3 347** | **1 095** | **–828** | **–1 025** | **–2 589** | **–3 995** | **–2 370** |
| Errors and omissions | 1 313 | 132 | –799 | 7 | –174 | 1 097 | 2 411 | –200 |
| Current account | 10 386 | 7 158 | 2 024 | 1 576 | 1 729 | 1 829 | 2 037 | 3 017 |
| **Reserve assets** | –5 424 | –3 942 | –2 320 | –755 | –530 | –337 | –454 | –448 |

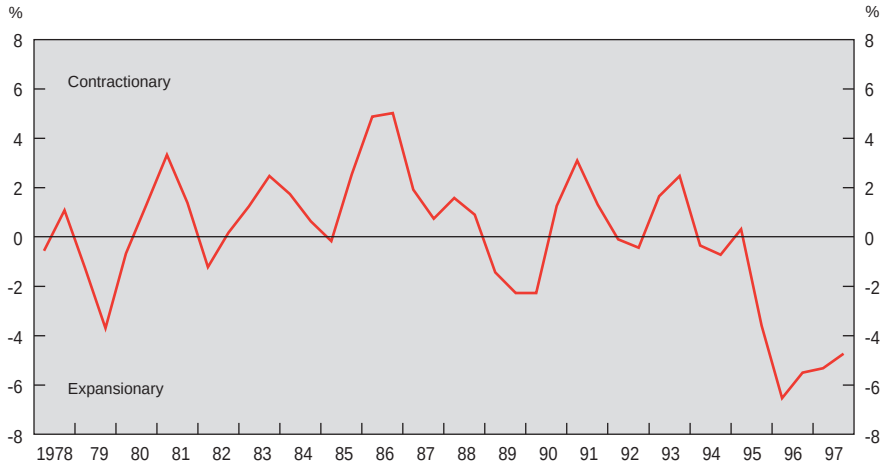
*Source:* Bank of Japan.

up in private net capital outflows allowed the authorities to slow down the rate of accumulation of official reserves.[7] Nevertheless, the yen dropped markedly against the dollar, especially at the beginning of 1997 (Figure 14, Panel C), before rebounding in the spring as buyers returned to the Tokyo stock market and domestic bond yields increased. By June 1997, the yen had regained its level of the final quarter of 1996, both in effective terms and against the dollar, but was still below the average for 1996 as a whole. In line with the weakness of bond yields and the stock market, the yen fell back again in the autumn.

## ... and leaving monetary conditions extremely easy

The depreciation of the exchange rate, together with the low level of interest rates, have resulted in a marked easing in overall monetary conditions as illustrated by the composite index shown in Figure 15.[8] The fact that, in contrast to some previous episodes of monetary easing, both the domestic (interest rates) and external (real exchange rate) components of the index have moved in the same

Figure 15. **MONETARY CONDITIONS INDICATOR**[1]




1. Real short-term interest rate plus 0.2 times change in real exchange rate. The interest and exchange rates are measured as deviations from period averages.

*Source:* OECD.

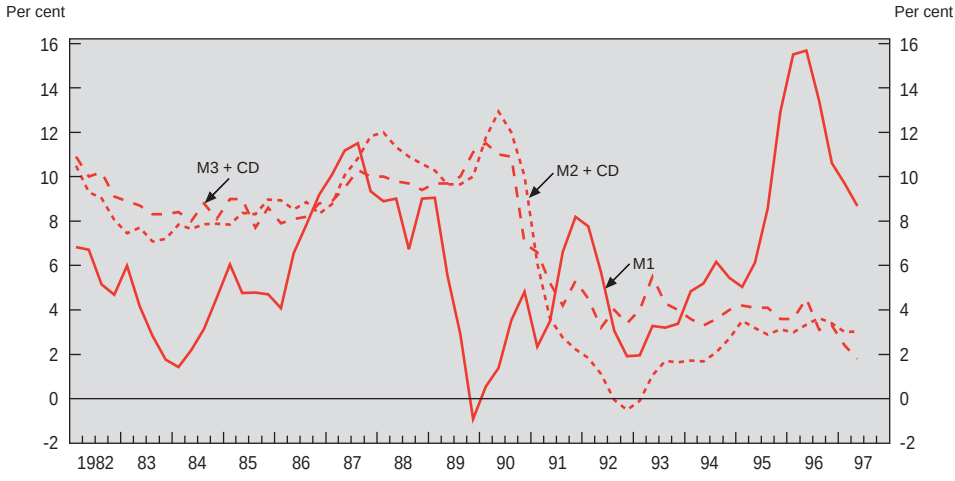
direction witness the exceptionally accommodating recent monetary conditions. However, with the exchange rate appreciating somewhat since May 1997, monetary conditions have become slightly less expansionary than in 1996.

## However, money and credit growth has remained weak

The relatively easy monetary conditions have not led to an acceleration in the demand for money or credit. Rather, since the middle of 1996, there has been a slight deceleration of broad money growth, as measured by the sum of M2 and certificates of deposit, to an annual growth rate of just over 3 per cent (Figure 16). This has been associated with a slackening of transfers from savings to cash deposits, as depositors had achieved a large part of their adjustment to low interest rates, which had boosted the growth of the narrow money supply M1 during the middle of 1996. Nonetheless, the money stock has become even more liquid, with the total of outstanding currency rising at annual rate of 8 per cent.[9]

Figure 16. **GROWTH OF THE MONEY STOCK**

Percentage changes over a year earlier




*Source:* Bank of Japan and OECD.

Table 10. **Contributions to the growth of the money stock**

Percentage points

| | 1994 | 1995 | 1996 | 1996 | | | | 1997 | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 |
| Money stock growth | 2.8 | 3.3 | 3.2 | 3.6 | 4.6 | 2.6 | 3.2 | 2.0 | 1.8 |
| Credit counterparts | 0.1 | 3.6 | 1.4 | 2.0 | 2.2 | 0.3 | 1.4 | 1.9 | 1.6 |
| Foreign credit | 0.5 | 2.1 | –0.7 | 0.7 | 0.3 | –0.7 | –0.7 | –0.1 | 0.4 |
| Domestic credit | –0.4 | 1.5 | 2.1 | 1.3 | 1.9 | 1.0 | 2.1 | 2.0 | 1.3 |
| Public sector | –0.7 | –0.2 | 0.9 | –0.6 | –0.5 | 0.3 | 0.9 | 1.9 | 1.2 |
| Private lending | 0.3 | 1.7 | 1.2 | 1.8 | 2.5 | 0.7 | 1.2 | 0.1 | –0.0 |
| Non-monetary assets | 2.6 | –0.3 | 1.8 | 1.6 | 2.3 | 2.3 | 1.8 | 0.1 | 0.2 |
| *Memorandum* | | | | | | | | | |
| Lending growth | | | | | | | | | |
| Public sector | –4.0 | 2.4 | 2.7 | 0.4 | –1.3 | 1.1 | 2.7 | 9.7 | 6.2 |
| Private sector | –0.5 | 1.3 | 0.8 | 1.3 | 2.0 | 0.0 | 0.8 | 0.1 | 0.0 |

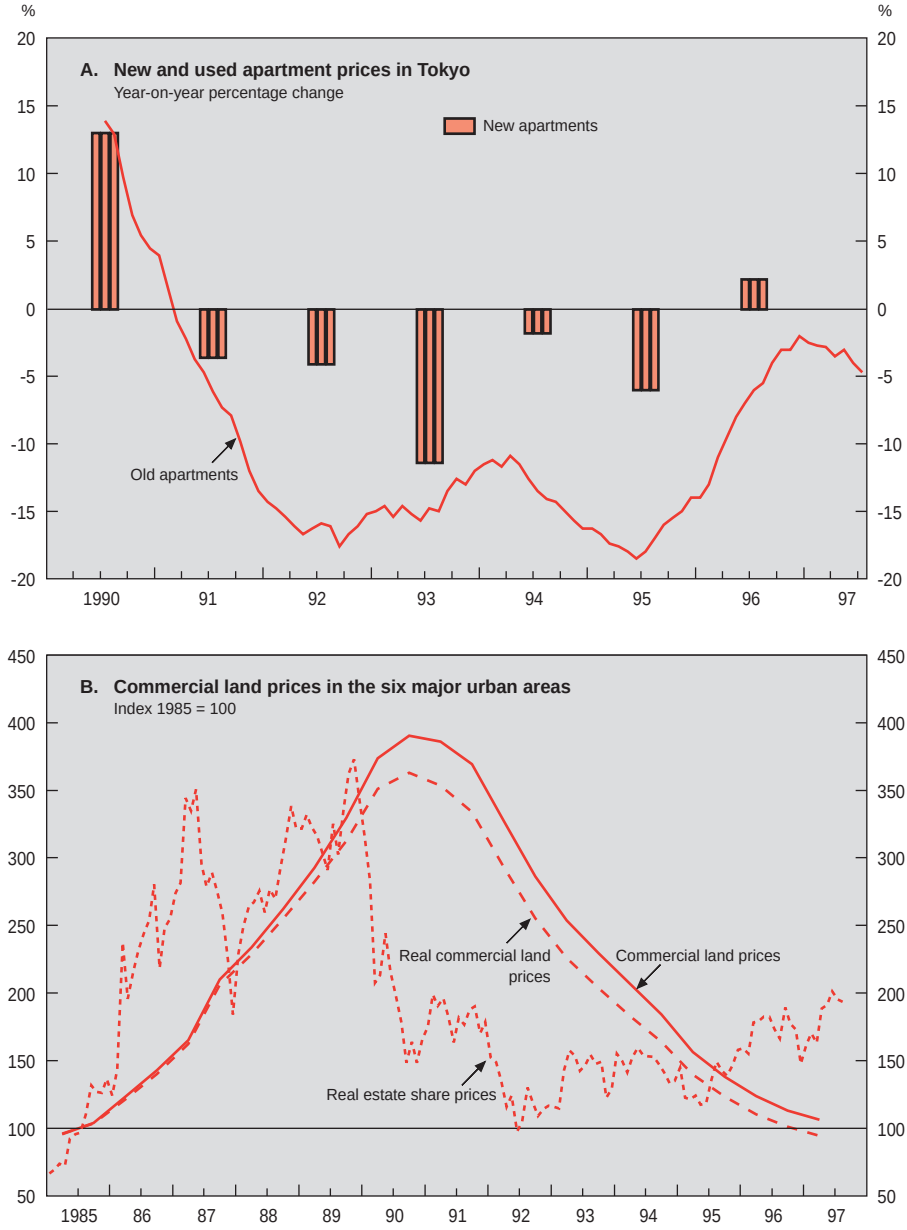
*Source:* Bank of Japan.

To a large extent, the weak monetary expansion reflects the sluggish growth of bank lending to the private sector (Table 10). In 1996, bank loans to households and companies contributed only about one-third of the increase in the money stock and represented a declining share of domestic credit expansion. This movement became more marked in the first quarter of 1997, when the growth in domestic credit was entirely accounted for by increased lending to the public sector, reflecting a rise in holdings of government securities by the Bank of Japan. Over FY 1996, the increase in the central bank's holdings of government debt represented almost half of the growth in the outstanding stock of government bonds.

## Despite some asset prices stabilising...

The easier monetary conditions appear to have contributed to stabilising domestic asset markets. As for real assets, for the first time in six years, the price of new apartments in the Tokyo metropolitan area increased in 1996 (Figure 17, Panel A). This tendency was confirmed in the second-hand market, where the rate of decline in housing prices eased sharply during 1996. By the summer of 1997,

Figure 17. **REAL ESTATE PRICES**

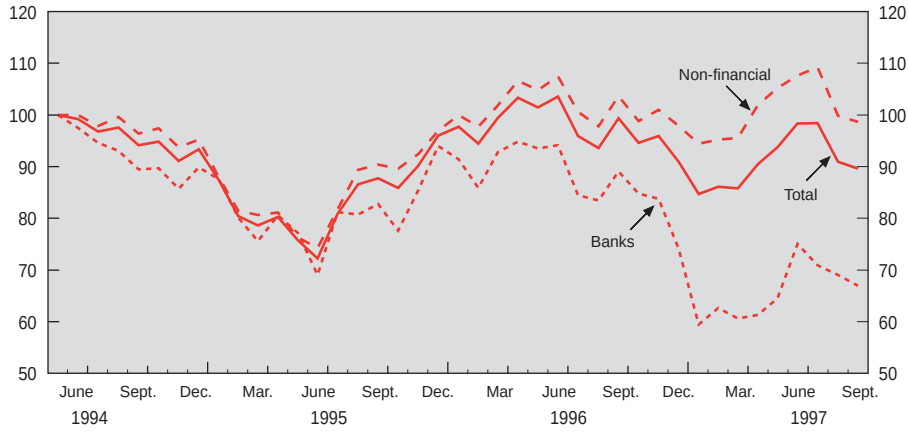



*Source:* Recruit Real Estate Company, Japanese Real Estate Research Institute, Nikko Research Centre.

the year-on-year decline in these prices had fallen to only 4 per cent. In contrast, published price indices for commercial land in the major metropolitan areas continued to show falls which have reduced land prices to their 1985 level (Figure 17, Panel B). However, these indices are based on assessed values and so tend to lag behind transaction prices. The share prices of real estate companies, a more timely indicator of market developments, have increased significantly in the past year. This reflects a stabilisation of office rents in central business districts, thus boosting yields for this type of investment. Some estimates put the return on new office construction well above the yield on government securities. For financial assets, stock market prices in the non-financial sector of the economy have been rising since the beginning of 1997 (Figure 18). The prices of non-financial companies are now more than 30 per cent above their 1995 low. The overall market average has, however, been held back by the poor performance of financial firms.

Figure 18. **SHARE PRICES BY SECTOR**[1]

Index, May 1994 = 100




1. Prices include reinvested dividends.

*Source:* Nikko Research Centre.

## ... balance sheet difficulties persist for certain financial institutions

### *The extent of bad loans*

Although substantial progress has been made in addressing balance sheet difficulties, problems persist for some financial institutions. Despite somewhat improved conditions in the property market, deposit-taking institutions had non-performing loans (NPLs) of Y 27.9 trillion at end-March 1997[10] (Table 11). After deducting loan loss reserves of Y 15.0 trillion, unprovisioned loans represented just under 1 per cent of the total assets of these institutions (just over 2½ per cent of GDP). However, the situation of different types of institutions varied markedly in this regard (Table 11). In particular, credit co-operatives, with unprovisioned bad loans representing over 7 per cent of their total assets, and a few banks are in a weak financial position.[11]

Other studies suggest that the total amount of bad loans may be greater than disclosed in banks' balance sheets. For example, calculating the balance sheets of two major banks according to US accounting principles for the six-month period

Table 11. **Non-performing loans at deposit-taking institutions**

March 1997

| | Assets | Loans | Operating profits[1] | Bad loans | Reserves | Net | Unprovisioned loans | | Workout Years[1] |
|---|---|---|---|---|---|---|---|---|---|
| | Yen trillion | | | | | | Per cent of GDP | Per cent of loans | |
| Agricultural co-operatives | 97.5 | 23.4 | 0.65 | 0.29 | 0.49 | –0.20 | (0.0) | –0.9 | –0.3 |
| Regional banks | 201.0 | 137.3 | 1.33 | 3.34 | 1.97 | 1.37 | (0.3) | 1.0 | 1.0 |
| Trust banks | 246.1 | 58.3 | 1.31 | 3.48 | 1.71 | 1.77 | (0.4) | 3.0 | 1.4 |
| City banks | 441.9 | 284.4 | 2.69 | 10.12 | 6.03 | 4.09 | (0.8) | 1.4 | 1.5 |
| Other credit co-operatives | 26.6 | 17.7 | 0.12 | 0.47 | 0.25 | 0.22 | (0.0) | 1.2 | 1.8 |
| Shinkin banks | 110.6 | 70.2 | 0.86 | 3.24 | 1.62 | 1.62 | (0.3) | 2.3 | 1.9 |
| Regional banks 2nd Tier | 69.9 | 52.9 | 0.51 | 2.01 | 0.98 | 1.03 | (0.2) | 1.9 | 2.0 |
| Long-term credit banks | 87.3 | 52.7 | 0.57 | 2.84 | 1.64 | 1.19 | (0.2) | 2.3 | 2.1 |
| Credit co-operatives | 25.9 | 17.2 | 0.19 | 2.12 | 0.30 | 1.82 | (0.4) | 10.6 | 9.7 |
| All deposit-taking institutions | 1306.8 | 714.1 | 8.22 | 27.90 | 14.99 | 12.91 | (2.6) | 1.8 | 1.6 |

1. Unprovisioned loans divided by FY 1997 operating profits. No account has been taken of residual loan values.
*Source:* Ministry of Finance.

ending September 1996 showed their bad loans to be more than 10 per cent higher than reported on the basis of Japanese standards.[12] Detailed inspection of the assets of liquidated institutions in Japan has also shown bad loans to be much larger than previously declared as has often occurred in bank liquidations in other countries.[13]

Most problem loans have occurred in the construction, real estate and finance industries. Between 1985 and 1996, lending by the major banks to these sectors rose from Y 40.9 trillion to Y 100.8 trillion. Some analysts argue that this increase in lending may turn into bad loans, as in the period considered, there has been no increase in commercial land prices (Figure 17). As a result, the burden on the borrowing companies of such loans is real and imposes a drag on activity, especially in the real estate industry. For example, the development of profitable projects by companies with large debt burdens may not be financially attractive since all the gain would go to the banking system.

The ultimate cost to the financial system of the bad loans will depend on the extent to which collateral can be recovered from borrowers. The experience of the Co-operative Credit Purchasing Company (CCPC) provides some guidance in this area. Established in 1993, the CCPC is a vehicle for bad loans that allows banks to set losses on a loan against tax before the collateral on that loan is realised.[14] By September 1996, the CCPC had acquired 9 075 loans backed by 16 150 properties. On average, the CCPC paid the lending banks just 40 per cent of the initial value of the properties,[15] which it sold for three-quarters of their acquisition prices leaving further substantial losses for banks to absorb on their loans to the CCPC – *i.e.* for about 30 per cent of the original value of the non-performing loan. So far, the CCPC has sold just 20 per cent of the properties thus acquired. Despite the fact that the rate of sales has increased recently, complete disposal of such properties would take some time.

One reason for the slow pace of sales is the fact that foreclosure procedures are rarely used in Japan. Most lending is on a recourse basis and is secured by the cash flow and all the properties of a borrower rather than being secured by an individual project or property. As a result, the transfer of a loan to the CCPC and sale of the collateral requires the voluntary consent of the owner, who remains responsible for selling the property. In such circumstances, using securitisation to speed up sales of collateral, as has been done in several other countries, would be difficult.

### *The position of various categories of banks and non-banks*

The extent of remaining balance-sheet difficulties differs across the various categories of financial institutions. The *major banks'* balance sheets improved for the year ending March 1997. Operating profits[16] remained high, buoyed by the still large, by Japanese standards, spread between the cost of funds and the return on new short-term loans (Table 12). In addition, banks made significant profits on trading government bonds. However, with the stock market still lower than in March 1996, the extent of equity portfolio revaluations fell from the previous year. As a result, total profits dropped slightly.

Major banks continued to use most of this profit flow to both provision and write-off non-performing loans. As a result, the stock of bad loans dropped some 25 per cent and reserves for potential losses now amount to over half of the remaining NPLs. Consequently, unprovisioned bad loans currently represent just over one and a half years of operating profits. Given that total bad loan charges were only slightly greater than profits, the overall BIS capital ratio of the major banks fell only a little to slightly under 9 per cent. However, a broader measure of the banks' capital, which allows for eventual losses on NPLs and loans to the CCPC, as well as unrealised capital gains on their share portfolio, showed a more marked decline. However, although the fall in the value of the banks' shareholdings was partially reversed by a stock market rally and despite prospects of future restructuring, the share prices of banks remain depressed (Figure 18).

The financial position of the individual major banks varies considerably. In another year, many of them will have accumulated reserves equivalent to 70 per cent of their non-performing loans. If such a level of provisioning were coupled with the eventual recovery of 30 per cent of the collateral backing their NPLs – in line with the experience of the CCPC – then these banks would be close to solving their non-performing loan problems (Figure 19). However, a limited number of the major banks face a much worse situation. Indeed, steps have been taken to ensure the survival of the worst-placed of them in each of the different categories of institutions (city banks, long-term credit banks and trust banks). The measures announced in April 1997 to support two banks included a merger, though its timing is not yet certain, and a capital injection. Public money was involved to a certain extent, through the use of funds from the Bank of Japan.

Certain mutual life insurance companies also face a difficult environment. The returns earned by such companies have fallen progressively in the current

Table 12. **Balance sheets of major banks**[1]

Yen trillion, March of each year

| | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|
| ***Profit and loss account*** | | | | | | |
| **Net operating profits**[2] | 2.6 | 3.1 | 2.7 | 2.7 | 3.8 | 4.2 |
| **Security gains** | 0.6 | 0.1 | 2.2 | 3.1 | 4.1 | 1.5 |
| Equity gains (net) | 2.1 | 0.5 | 2.2 | 4.0 | 3.3 | 3.4 |
| Bond trading gains | –0.2 | 0.1 | 0.4 | 0.1 | 1.0 | 0.4 |
| Equity valuation losses | 1.3 | 0.5 | 0.4 | 1.0 | 0.2 | 2.3 |
| **Total bad loan charges** | –0.9 | –1.7 | –4.2 | –5.4 | –11.1 | –5.7 |
| Specific loan provisions | –0.5 | –0.9 | –1.4 | –1.6 | –5.6 | –2.5 |
| Write-offs | –0.1 | –0.2 | –0.2 | –0.7 | –1.6 | –0.5 |
| Loss on debt sales to CCPC | – | –0.2 | –1.8 | –1.5 | –1.9 | –0.7 |
| General provisions[3] | –0.2 | –0.5 | –0.8 | –1.8 | –2.1 | –2.0 |
| **Recurring profits** | 2.0 | 1.3 | 0.8 | 0.2 | –3.3 | –0.2 |
| Extraordinary profits | 0.0 | 0.0 | 0.0 | 0.1 | 0.2 | 0.1 |
| Taxation | –1.1 | –0.8 | –0.3 | –0.2 | –0.5 | –0.1 |
| **After tax profit** | 0.9 | 0.5 | 0.5 | –0.1 | –3.6 | –0.1 |
| ***Balance sheet data*** | | | | | | |
| **Non-performing loans** | n.a. | n.a. | n.a. | n.a | 21.9 | 16.4 |
| Excluding restructured loans | 8.0 | 12.8 | 13.6 | 12.5 | 13.1 | 13.2 |
| To legally bankrupt companies | n.a. | 2.0 | 2.3 | 2.6 | 3.0 | 3.4 |
| Loans six months overdue | n.a. | 10.8 | 11.3 | 9.9 | 10.1 | 9.8 |
| Restructured loans | n.a. | n.a. | n.a. | n.a | 8.8 | 3.2 |
| Supported loans[4] | n.a. | n.a. | n.a. | n.a. | 3.8 | 2.9 |
| **Loans to CCPC** | n.a. | 0.5 | 2.2 | 3.8 | 5.0 | 5.4 |
| Loan provision reserves[5] | 3.0 | 3.7 | 4.5 | 5.5 | 10.3 | 9.4 |
| Total loans | 398.2 | 398.2 | 390.1 | 383.3 | 393.7 | 397.7 |
| Stock holdings | 47.9 | 47.9 | 52.5 | 44.8 | 55.3 | 47.3 |
| Book value | 30.9 | 31.0 | 32.8 | 36.4 | 39.3 | 39.3 |
| Unrealised profits (gross) | 17.0 | 16.9 | 19.7 | 8.4 | 16.0 | 8.0 |
| Risk-weighted assets | 479.5 | 459.9 | 446.6 | 441.9 | 452.3 | 456.9 |
| **Capital** | 39.6 | 42.8 | 43.3 | 39.5 | 41.2 | 40.9 |
| Tier One | 22.5 | 22.9 | 23.1 | 22.8 | 21.0 | 21.4 |
| Tier Two | 17.1 | 20.0 | 20.2 | 16.6 | 20.2 | 19.5 |
| Unrealised stock profits (net) | 7.8 | 8.0 | 9.2 | 4.1 | 7.5 | 3.8 |
| Subordinated debt | 7.2 | 10.6 | 10.3 | 11.3 | 11.3 | 14.5 |
| Other | 2.2 | 2.2 | 1.3 | 1.2 | 1.4 | 1.2 |
| Remaining stock profits | 9.2 | 8.9 | 10.5 | 4.3 | 8.5 | 4.0 |

1. City banks, long-term credit banks and trust banks.
2. Net operating profits represent net interest incomes plus trading income and commissions less operating costs such as staff. They are calculated before any provision for bad loans and exclude income from bond trading.
3. Mainly general provisions against non-performing loans which are set at 0.3 per cent of outstanding loans.
4. Total lending to borrowers to which the lending bank is extending help, including debt forgiveness.
5. Special and general loan-loss reserves.

*Source:* Federation of Bankers' Associations of Japan, *Analysis of Financial Statements of All Banks;* Bank of Japan and Ministry of Finance.

context of low long-term interest rates, while many policyholders are still entitled to high projected yields of 4½ or 5½ per cent, depending on the product. New policyholders are only paid returns of 2¾ per cent. Companies that had large reserves were able to continue to cross-subsidise investors but one company exhausted such reserves and became insolvent. One-third of the deficit will be borne by policyholders through a reduction in yields. The insurance industries' Policyholders Protection Fund has agreed to pay part of the sum necessary to facilitate the transfer of the contracts to other companies. One notable feature of the rescue of the insurance company was that companies which had established long-term business links to the insurer were reluctant to contribute to the rescue operation, perhaps because of the possibility of shareholder law suits. In the wake

Table 13. **Key bank profitability and solvency ratios**

Per cent, March each year

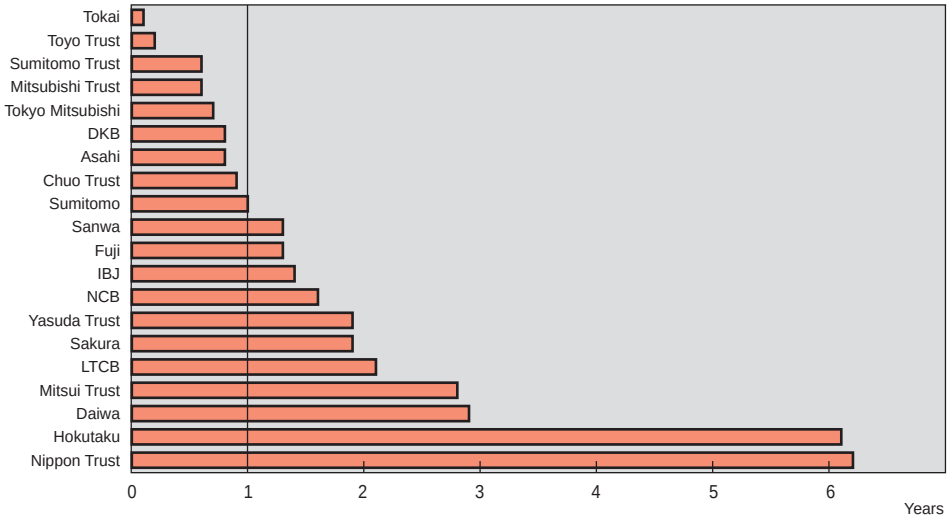
| | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|
| Operating profits/Tier One capital | 11.5 | 13.6 | 11.7 | 11.8 | 18.1 | 19.6 |
| Non-performing loans/Total loans [1] | 2.0 | 3.2 | 3.5 | 3.3 | 5.6 [1] | 4.1 |
| Provisions/Non-performing loans [1] | 37.5 | 28.9 | 33.1 | 44.0 | 47.3 [1] | 57.3 |
| Unprovisioned loans/Tier One capital | 22.2 | 39.8 | 39.3 | 30.7 | 54.9 [1] | 32.7 |
| Loans to CCPC/Tier One capital | – | – | 8.6 | 16.7 | 23.8 | 25.1 |
| BIS capital ratio | 8.3 | 9.3 | 9.7 | 8.9 | 9.1 | 9.0 |
| Tier One | 4.7 | 5.0 | 5.2 | 5.2 | 4.6 | 4.7 |
| Tier Two | 3.6 | 4.3 | 4.5 | 3.8 | 4.5 | 4.3 |
| Unrealised stock profits (net) | 1.6 | 1.7 | 2.1 | 0.9 | 1.6 | 0.8 |
| Subordinated debt | 1.5 | 2.3 | 2.3 | 2.6 | 2.6 | 3.2 |
| Other | 0.5 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| Unprovisioned bad loan loss/Tier One capital [2] | 9.8 | 20.2 | 18.8 | 11.5 | 18.5 [1] | 9.7 |
| Loss on CCPC loans/Tier One capital [3] | 0.0 | 0.0 | 1.9 | 3.6 | 3.6 | 6.9 |
| Estimated market price net worth ratio [4] | 7.8 | 7.6 | 8.5 | 6.3 | 7.1 | 5.7 |

1. From 1996, restructured loans are included in total non-performing loans.
2. The unprovisioned loss is equal to total non-performing loans less provisions less the assumed residual values of loans. The latter value is calculated as 35 per cent of the face value of the loan by the Ministry of Finance estimates.
3. The loss on loans to the CCPC is estimated as the difference between the original discount given to the CCPC and the most recent discount.
4. Tier one capital plus gross unrealised stock gains less estimates of unprovisioned bad loan loss and loss on loans to the CCPC. The figure excludes unrealised gains on property assets.

*Source:* OECD and Federation of Bankers' Associations of Japan.

Figure 19. **THE VARIABILITY OF BAD LOANS BY BANKS**

Expected bad loan losses relative to annual operating profits[1]




1. Expected bad loan losses are calculated as the difference between the stock of bad loans and the sum of provisions for losses and the residual value of the bad loans which is assumed to be 30 per cent.

*Source:* Company results and OECD calculations.

of this failure, new inflows to some mid-sized mutual insurance companies declined, some policies were cancelled and the reserves of the Protection Fund were exhausted.

## To address these problems a new regulatory environment is being introduced

Two laws were enacted by the Diet in June 1996 to establish a new legal framework for resolving the difficulties met by deposit-taking financial institutions and ensure a smooth transition by the time this framework is fully operational in 2001. The first law removed the limit on the amount of money that can be used by the Deposit Insurance Corporation (DIC) in a bank resolution. In

order to finance possible additional payments, the deposit insurance rate was increased sevenfold and the DIC was allowed to borrow with a government guarantee in the case of the failure of a credit co-operative. Consequently, the total resources of the DIC in the period to 2001 should amount to Y 2.7 trillion. The second law allows the supervisory authority to request the reorganisation or liquidation of financial institutions.

A third law, also passed in 1996, established a framework for ''prompt corrective action'' by the authorities in response to a deterioration of a bank's capital ratio. New regulatory standards were set, designed to increase transparency, strengthen risk management, enhance external audits and introduce improved supervisory methods. Under the new system, if the bank's capital ratio falls below certain standards, the authorities request the bank to submit a management improvement plan to take specific action to remedy its situation. Banks will be required to classify their loan portfolios into five risk categories, subject to external audits. In addition, a limited amount of mark-to-market accounting will be introduced and there will be some change to the rules governing the consolidation of subsidiaries.

Finally, the supervision of deposit-taking institutions and other private financial companies will be undertaken by a new organisation, the Supervision Agency for Financial Entities, which will be independent of the Ministry of Finance and under the general authority of the Prime Minister. The Securities Exchange Surveillance Commission will be transferred to this new agency and continue to have responsibility for the oversight of securities markets. In addition to supervising banks, the new supervisory agency will jointly regulate agricultural-related financial institutions with the Ministry of Agriculture, Forestry and Fisheries, other non-bank financial institutions with the Ministry of International Trade and Industry while most credit co-operatives will be supervised by local governments. The Ministry of Finance, meanwhile, will continue to exercise responsibility for financial policy planning and legislation.

The objective of this rule-based system of bank supervision is to move away from case-by-case regulation and resolution of problem institutions. In the past, such a system was, in part, supported by a significant range of regulations which affected banks' profitability and by banks maintaining close links with the regulators.[17] Such supervisory arrangements now appear to be inappropriate as they

constituted one of the factors which resulted in reduced competition and efficiency in the banking industry. With deregulation of the financial sector moving ahead (see Chapter IV), a more transparent system supervising that industry is therefore needed. The introduction of the ‘‘prompt corrective action’’ mechanism with an objective capital adequacy standard is an essential step in this direction.

Improved supervision of banks will be particularly important in view of the planned programme of financial deregulation. While the direct impact of deregulation on the core banking business will be limited, since most regulations in this field have already been abandoned, banks will be given more freedom to enter new business areas, notably the equity markets. Such a move may well proceed through the creation of bank holding companies, though under the Banking Law such companies cannot own subsidiaries engaged in non-financial activities.[18]

The programme of financial deregulation will give banks the possibility of improving profits through diversification of their activity. This is important because the crisis in the Japanese banking industry has been partly linked to poor profitability. Indeed, compared with other countries, Japanese banks earn less than one-third the rate of return on risk assets, leaving them with a smaller cushion to absorb loan losses. For example, despite an increase in the past few years, the net interest margin of Japanese banks is only one-third of the US level. This low profitability does not appear to be related to inefficiency. Although operating costs per employee are significantly higher in Japan, this reflects the higher level of Japanese salaries. In fact, operating costs per dollar of assets are only one-sixth of those in American banks. Reductions in costs, though, could boost profitability; each 10 per cent decline in operating costs would raise the rate of return on equity by 1 percentage point. However, boosting profitability by reducing costs is unlikely to solve the problem of low returns since even if the major banks were to eliminate all costs, their post-tax return on equity would only rise from its current 9 per cent level to 18 per cent, still below the rates of return obtained by many international banks. Thus, the major problem for the Japanese banking system is not high costs and inefficiency, but rather low revenues. This can only be boosted by giving the banks the opportunities to enter new business areas and the freedom to develop better quality financial products. In such a way, households might readjust their portfolios away from bank deposits, so allowing banks to run down their loan portfolios and improve lending margins.

## Overall assessment

The stance of monetary policy has been extremely easy over the past year. According to OECD estimates, real short-term interest rates are negative, while the low level of the exchange rate is boosting demand. Nevertheless, prices are expected to be stable in the near future, though the inflation rate has been boosted by the temporary impact on the price level of the increase in indirect taxation. Moreover, with the recent pause in economic growth, the output gap remains high and the recovery has not yet significantly reduced unemployment. Such considerations suggest that an early tightening of monetary policy does not seem warranted. However, once an increase in private demand is clearly seen to outweigh the ongoing reduction in public demand and the output gap is falling, there would be a case for a progressive reduction in the extent of monetary ease.

Low interest rates, by supporting the economy, have already helped some of the major problems facing the banking industry to be addressed. Banks at the core of the financial system have absorbed the losses on the *jusen* loans and further increased their loan reserves, helped by high operating profits; some of them have almost restored their financial health. A new system for dealing with the resolution of failing credit institutions has been put in place, with the deposit insurance system being given an income of Y 2.3 trillion over the next five years and a borrowing facility backed by government guarantees in the case that a credit co-operative fails. While this should be sufficient to deal with most of the difficulties experienced by credit co-operatives, the resolution of the problems still facing some banks may have to wait for some time. By and large, this may be one factor explaining why capital markets remain uncertain over the full extent of bad loans and the share prices of banks have fallen significantly. Moreover, low interest rates have impaired the profitability of the mutual life insurance industry. Improved disclosure and transparency would lower the cost of capital for financial companies, a move that would be aided by using market prices in the balance sheets of financial entities. For the banks, such an environment may be generated by the new supervisory system based on prompt corrective action.

# III. Fiscal policy in a medium-term perspective

The prolonged slowdown of the Japanese economy in the first part of the 1990s, and the numerous fiscal stimulus packages adopted to combat it, resulted in a sharp deterioration in Japanese public finances. Improving the government's financial position has thus become a priority for policy makers, as reflected in the FY 1997 budget and the introduction of a medium-term plan to reduce the central and local government budget deficit to 3 per cent of GDP by 2003. This chapter first reviews the extent of the problem from an international perspective. It then examines the government's fiscal plan and considers its implications for long-term debt sustainability and short-term economic growth. The final section discusses possible measures to rationalise public spending and improve budgetary transparency.

## How serious is Japan's public finance situation?

### *The deterioration in the government financial position up to 1995*

Between 1990 and 1995, the health of public finances in Japan deteriorated markedly. From a surplus of nearly 3 per cent of GDP in 1990, the general government financial balance had moved into a deficit of almost 4 per cent by 1995 (Table 14), larger than both the OECD average and that of the other six major economies. The increasing stock of government debt also indicates a marked worsening in the health of public finances. Gross general government debt rose from 64 per cent of GDP in 1990 to 81 per cent by 1995, which was 8 percentage points higher than the average in the other major six economies and 5 percentage points above the average of the nineteen OECD countries for which data were available in 1995 (Figure 20, Panel A). On the other hand, despite increasing, net debt (the balance of government liabilities and assets), at 11 per

Table 14. **General government financial balances**

Deficit (–), surplus (+) as percentage of GDP

| | Actual | | | | | Estimates and projections | | |
|---|---|---|---|---|---|---|---|---|
| | 1990 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 |
| **Fiscal year basis** | | | | | | | | |
| Central government | 0.3 | –2.1 | –2.8 | –3.7 | –4.2 | –4.3 | –3.3 | –2.9 |
| Local government | –0.3 | –1.1 | –1.6 | –2.0 | –2.7 | –2.5 | –2.0 | –1.7 |
| Total of above | 0.0 | –3.2 | –4.5 | –5.7 | –6.9 | –6.8 | –5.3 | –4.6 |
| Social security | 3.5 | 3.4 | 3.1 | 2.7 | 2.9 | 2.9 | 2.7 | 2.4 |
| General government | 3.5 | 0.1 | –1.4 | –3.0 | –4.0 | –4.0 | –2.6 | –2.2 |
| **Calendar year basis** | | | | | | | | |
| General government | 2.9 | 1.5 | –1.6 | –2.3 | –3.7 | –4.4 | –2.8 | –2.3 |
| *of which:* | | | | | | | | |
| Cyclical component | 1.1 | 0.5 | –0.4 | –0.9 | –1.3 | –0.7 | –1.2 | –1.2 |
| Structural component | 1.8 | 1.0 | –1.2 | –1.4 | –2.4 | –3.7 | –1.6 | –1.1 |
| Gross debt | 65.1 | 62.4 | 63.6 | 73.1 | 80.6 | 86.4 | 90.6 | 93.3 |

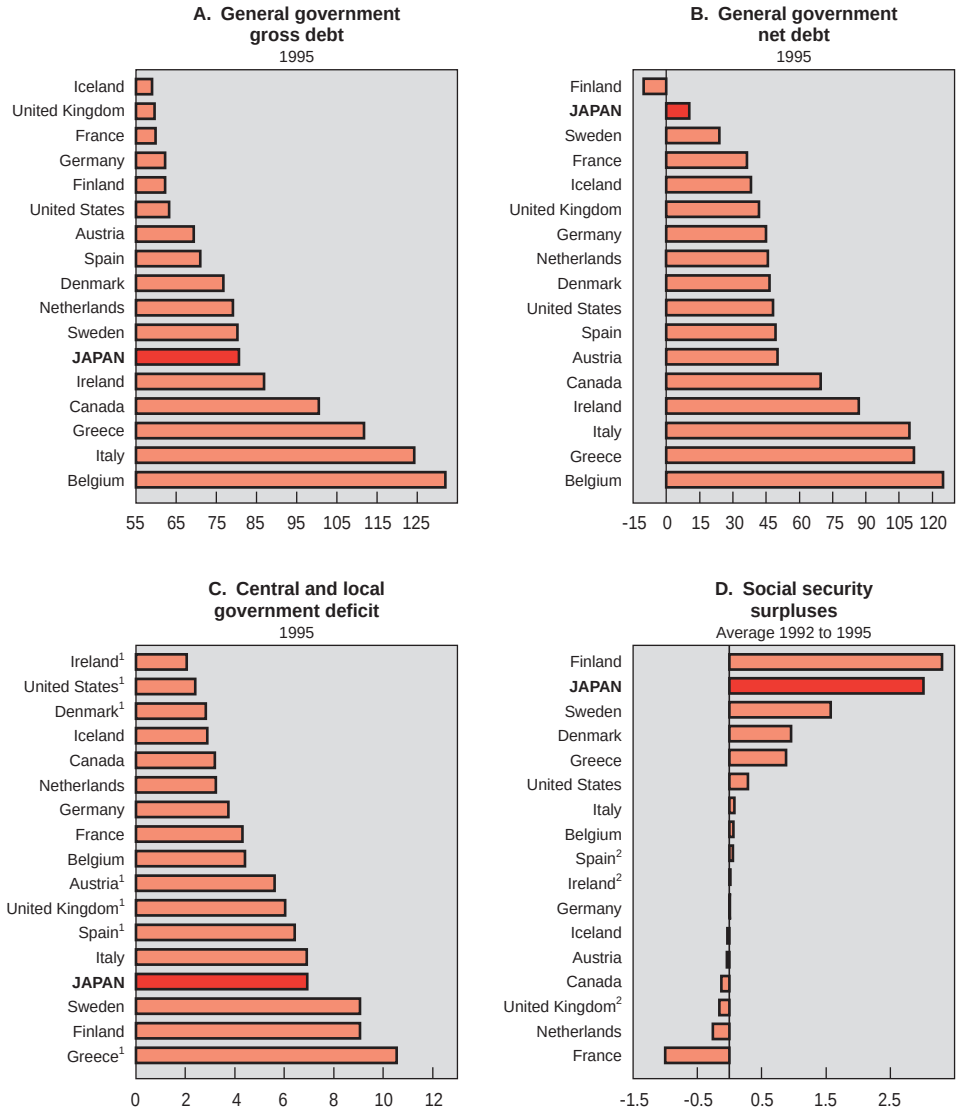
*Source:* OECD and EPA, *Annual Report on National Accounts*.

cent of GDP, was still markedly below that in most other OECD countries (Figure 20, Panel B).

The difference between gross debt and net debt is due to the size of the accumulated social security surpluses which, in Japan, averaged 2¾ per cent of GDP between 1992 and 1995 (Figure 20, Panel D) bringing the assets of the system to 45 per cent of GDP. Only three other OECD countries had a surplus of more than ½ per cent of GDP. However, this does not necessarily indicate that the Japanese economy is better placed than others to finance increasing pension expenditures. To make such a judgement would require an accounting system that includes the accrual of pension liabilities in the government budget position as well as the increase in pension fund assets. Such a measure would likely show that pension benefits are currently being earned by contributors at a faster rate than contributions are being paid. Thus, despite the social security surplus currently reported, the finances of the public pension system may be deteriorating at the moment, requiring a rapid increase in contributions at a later date (see Chapter V).

For the same reason, the use of the general government balance (which includes social security) masks the full extent of the deterioration in the fiscal

Figure 20. **FISCAL INDICATORS**
Per cent of GDP




1. Estimated on the basis of the general government deficit less the social security deficit shown in panel D.
2. 1994 is the last available year.

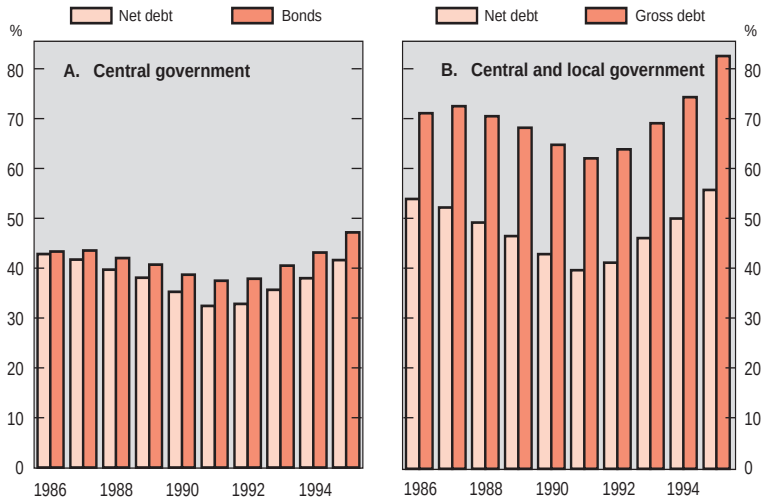
*Source:* OECD, *Annual National Accounts* and OECD.

## Box 2. **Central and local government debt**

The extent of public indebtedness in Japan is often measured by the gross debt of central and local public authorities. As both central and local governments hold considerable financial assets, the return on these assets can, in principle, be used to pay borrowing costs. To the extent that these assets have been acquired at market rates of interest and are of good quality, net debt (liabilities minus assets) can then be regarded as a better measure of the financial position of the government sector. Overall, central and local government net debt in Japan is estimated to have risen to 65 per cent of GDP at the end of 1997, with gross debt amounting to almost 90 per cent of GDP.

Figure 21. **MEASURES OF CENTRAL AND LOCAL GOVERNMENT INDEBTEDNESS**

Per cent of GDP




*Source:* Economic Planning Agency, *Annual Report on National Accounts.*

For central government, net debt has, in practice, moved closely in line with the gross stock of outstanding central government bonds (Figure 21). This is because other forms of government borrowing (through the issuance of short-term bills or borrowing from public sector financial institutions) are generally used to finance the acquisition of financial assets, such as loans and foreign exchange reserves, with, therefore, no implica-

*(continued on next page)*

*(continued)*

tions for net indebtedness (Table 15). For example, in March 1995, central government net debt was 38 per cent of GDP, while the stock of central government bonds amounted to 43 per cent.

Table 15. **Assets and liabilities of central and local governments**

Trillion yen, end March 1995

| | Central government | | Central government non-profit institutions | | Local government | |
|---|---|---|---|---|---|---|
| | Assets | Liabilities | Assets | Liabilities | Assets | Liabilities |
| Short-term government securities | 5.0 | 23.0 | .. | .. | .. | .. |
| Long-term government bonds | 1.0 | 207.1 | 0.9 | .. | .. | .. |
| Deposits with Trust Fund Bureau | 8.1 | .. | 0.5 | .. | .. | .. |
| Public sector loans | 8.5 | 22.5 | 13.0 | 43.9 | 7.4 | 63.4 |
| Public sector investments | 21.5 | .. | .. | 9.4 | 3.7 | 0.0 |
| Foreign claims | 11.5 | 1.3 | .. | .. | .. | .. |
| Other | 15.8 | 3.9 | 0.9 | .. | .. | .. |
| Corporate shares | 0.5 | .. | .. | .. | 0.1 | .. |
| Transfers from general government | 3.5 | .. | .. | .. | .. | .. |
| Public corporation bonds | .. | .. | 0.7 | 0.6 | .. | .. |
| Local authority bonds | .. | .. | 0.2 | .. | .. | 30.0 |
| Other bonds | .. | .. | 3.8 | .. | .. | .. |
| Investments with banks and insurance companies | .. | .. | 36.3 | .. | 30.4.. | .. |
| Private sector loans | .. | .. | .. | 0.3 | .. | 7.4 |
| Assets less liabilities of non-profit institutions | 2.1 | .. | .. | .. | .. | .. |
| **Total** | **77.4** | **257.8** | **56.3** | **54.2** | **41.7** | **100.8** |
| Net position | | –180.3 | | 2.1 | | –59.1 |

*Source:* Economic Planning Agency, *Annual Report on the National Accounts.*

Local government also holds considerable financial assets that moderate the burden of its gross indebtedness. Most of these assets are held as deposits with trust banks. Almost 75 per cent of gross local authority debt is either held or guaranteed by public financial institutions. In March 1995, local authority net debt was 12 per cent of GDP against gross debt of 21 per cent.

position in Japan. Thus, the central and local government deficit reached 7 per cent of GDP in 1995, a level not seen since the late 1970s. Only four countries (Greece, Sweden, Finland and Italy) had larger deficits for this level of government (Figure 20, Panel C). As to debt, the financial position of central and local government has worsened sharply. Its net debt rose to 56 per cent of GDP (see Box 2), a level much higher than that of the general government sector.[19]

The deterioration in public finances was due to the marked slowdown in economic activity in the first half of the 1990s and the series of economic packages then adopted to stimulate domestic demand. These packages accounted for almost half of the 6.9 per cent rise in the central and local government deficit between 1990 and 1995. During this period, the fiscal stimulus boosted public works spending by 2 per cent of GDP, while temporary tax cuts lowered government revenues by slightly over 1 per cent – a total discretionary swing of 3 per cent of GDP. In addition, the cyclical downturn also reduced tax revenues and increased expenditures by almost 3 per cent of GDP, according to OECD estimates (Table 14). Consequently, the general government structural budget balance, which abstracts from cyclical effects on spending and revenue, swung from an estimated surplus of almost 2 per cent of GDP in 1990 to a deficit of similar size in 1995. However, as the economy is unlikely to maintain permanently the high level of capacity utilisation seen in 1990, only about half of this cyclical loss seems to be reversible on a sustainable basis. Finally, there was a fall in tax receipts that represented the unwinding of the unusual tax buoyancy seen during the last expansion, when revenues rose more than might be expected on the basis of the standard estimates of tax elasticities. This factor accounted for further government borrowing of the order of ¾ per cent of GDP. In sum, 45 per cent of the deterioration in the budget balance between 1990 and 1995 was related to discretionary changes and 25 per cent to reversible cyclical factors, with the remaining 30 per cent accounted for by normalisation of tax revenues from an exceptional starting point.

### *The fiscal correction in 1996 and 1997*

Although the initial FY 1996 budget attempted to end the upward movement of the deficit, this objective was thwarted by a supplementary budget (Table 16). The initial budget had envisaged a Y 18.4 trillion (3.7 per cent of GDP)[20] increase in the stock of outstanding bonds, slightly down from the amount issued

Table 16. **Central government budget**

| | Trillion yen | | | | | | Percentage increase from previous period | |
|---|---|---|---|---|---|---|---|---|
| | FY 1994 | FY 1995 | FY 1996 | | | 1997 | 1996 | 1997 |
| | Settled | Revised | Initial budget | First supplementary budget | Total budget | Initial budget | Total relative to revised | Initial relative to total |
| **Total expenditure** | 73.6 | 75.9 | 75.1 | 2.7 | 77.8 | 77.4 | 2.5 | –0.5 |
| General spending | 43.0 | 47.9 | 43.1 | 2.6 | 45.7 | 43.8 | –4.6 | –4.2 |
| *of which:* | | | | | | | | |
| Public works | 9.8 | 11.7 | 8.5 | 1.6 | 10.1 | 8.6 | –13.7 | –14.9 |
| Other | 33.2 | 36.2 | 34.6 | 1.0 | 35.6 | 35.4 | –1.7 | –0.6 |
| Debt servicing[1] | 13.4 | 12.8 | 16.4 | –0.3 | 16.1 | 16.8 | 25.8 | 4.3 |
| Local government | 12.1 | 12.3 | 13.6 | 0.3 | 13.9 | 15.5 | 13.0 | 11.5 |
| Other spending | 5.1 | 1.8 | 2.0 | 0.0 | 2.0 | 1.3 | –31.0 | –35.0 |
| **Total revenue** | 57.1 | 59.3 | 54.1 | 1.3 | 55.4 | 60.7 | –6.6 | 9.6 |
| Taxes | 51.0 | 51.9 | 51.3 | 0.4 | 51.7 | 57.8 | –0.4 | 11.8 |
| Non-tax revenue | 6.1 | 7.4 | 2.7 | 0.9 | 3.6 | 2.9 | –51.4 | –19.4 |
| **Deficit**[2] | 16.5 | 21.2 | 21.0 | 1.3 | 22.4 | 16.7 | 5.3 | –25.3 |
| Construction bonds | 12.3 | 16.4 | 9.0 | 1.7 | 10.7 | 9.2 | – | – |
| Deficit bonds | 4.2 | 4.8 | 12.0 | –0.3 | 11.7 | 7.5 | – | – |
| *Memorandum* | | | | | | | | |
| Net bond issuance[3] | 14.1 | 18.6 | – | – | 19.5 | – | 4.3 | –33.0 |
| (Per cent of GDP) | 3.0 | 3.8 | – | – | 3.9 | – | – | – |

1. Including an amortisation payment on existing debt.
2. The deficit is equal to the gross issuance of government debt.
3. Increase in outsanding stock of ordinary government bonds.

*Source:* Ministry of Finance, Bank of Japan.

in the previous year. It projected a marked fall in public works spending that was to be partially offset by higher debt servicing costs, due to increased payments to the debt settlement account, and by a fall in non-tax revenues. However, as a result of the extra spending in the supplementary budget, the final increase in the outstanding stock of government debt was Y 19.5 trillion (3.9 per cent of GDP).

The government increased its efforts to reduce the budget deficit in FY 1997 with the enactment of the final stage of the phased tax reform. This included an increase in the rate of the consumption tax from 3 to 5 per cent and the removal of the temporary income tax reduction. In total, these changes are projected to generate a significant increase in tax revenues in 1997, so reversing earlier tax reductions. At the same time, public works spending is set to fall by 15 per cent from the level implied by the combined FY 1996 initial and supplementary budgets (Table 16). As a result of these reinforcing movements in taxes and spending, the authorities project that the deficit of the general account of the central government will fall to about 3¼ per cent of GDP, a decline of 1¼ per cent of GDP from the FY 1996 outcome. As for the local authorities, the initial budget plans for FY 1997 suggest that their deficit will decline slightly to 2¼ per cent of GDP. On a national accounts basis, the government estimates the central and local government deficit at 5.4 per cent of GDP, in line with OECD estimates. With the social security system remaining in surplus, OECD projections suggest a fall in the general government deficit to 2.6 per cent of GDP, the first decline in six years. Nevertheless, gross general government debt would rise further to 91 per cent of GDP.

## The need for medium-term fiscal consolidation

### *The government's programme*

Given the continuing increase in public debt, the government decided, at the beginning of 1997, to reduce the central and local government budget deficit (on a national accounts basis) to 3 per cent of GDP by the year 2003 from an expected 5½ per cent in 1997. It is intended to eliminate the issue of so-called deficit bonds by the central government,[21] which are expected to amount to 1.5 per cent of GDP in FY 1997. By June 1997, the government had introduced a medium-term fiscal plan in which the period 1998 to 2000 was designated as an

‘‘intensive reform period’’ during which targets have been set for all spending areas in order to restrain the growth of expenditure (Table 17). Most notably, the government expects to lengthen by three years the period over which the current ten-year basic plan for public investment plan is achieved. The OECD estimates

Table 17. **Public expenditure plans for the period FY 1998 to FY 2000**

| Expenditure area | Budgetary objectives | |
|---|---|---|
| | FY 1998 | FY 1999-FY 2000 |
| Social transfers | Reduce expenditure increase to Y 0.3 trillion | Keep growth below that consistent with the expansion of the elderly population (*i.e.* less than 2 per cent annually).<br>State medical expenditure kept in line with national income. Unemployment benefits for elderly job seekers will be reviewed. |
| Public investment | Cut by 7 per cent | Completion of the 1995 ten-year basic public investment plan delayed 3 years. Spending will fall in FY 1998 and FY 1999. |
| Education | | Plans to raise number of teachers will be extended by two years. |
| Defence | No increase | Equipment spending cut 10 per cent. |
| Energy | No increase | No increase. |
| Small firms | No increase | No increase. Subsidies to Chambers of Commerce to be reviewed. |
| Science and technology | Increase by less than 5 per cent | Expenditure to be restrained. |
| Development assistance | Cut by 10 per cent | No new quantitative medium-term targets. |
| Local finance | Strict restraints on spending | Spending to grow no more than nominal GDP. Debt issue to be restrained. |
| Subsidies | | The laws governing subsidies to businesses will be reviewed. Other subsidies reduced by 10 per cent. All new subsidies will be limited to a five-year period. |
| Agriculture | | Uruguay Round plan delayed two years.<br>Policies to be developed to handle forestry debt. |
| Transport | – | New means to finance JNR settlement debt will be evaluated. |
| Staff and personal costs | – | Strengthen the implementation of the permanent staff reduction programme and conduct a further review. |

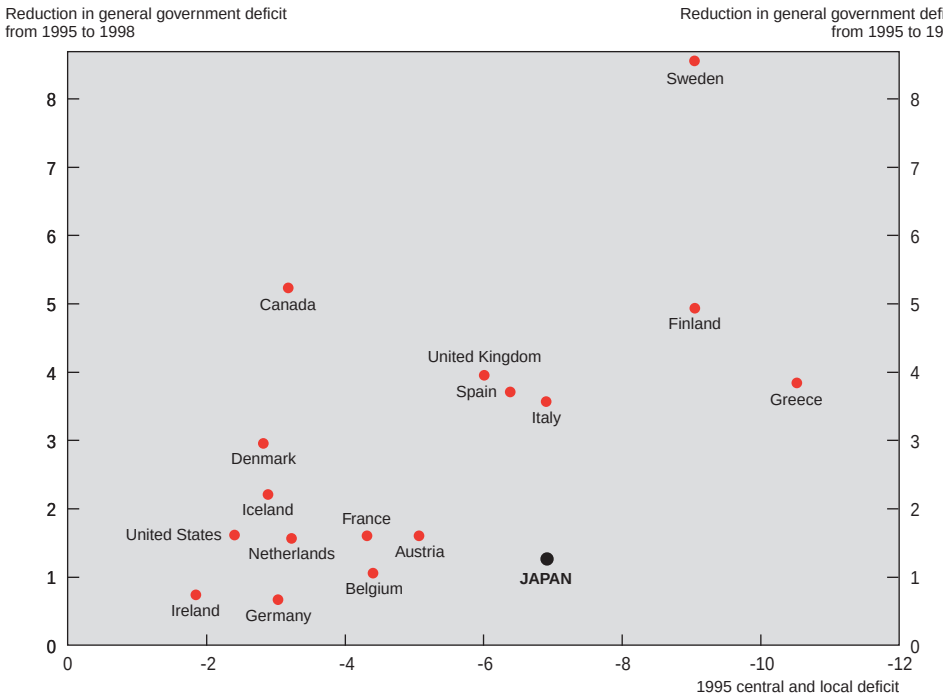
*Source:* Translation of the cabinet decision concerning ‘‘The Promotion of Fiscal Structural Reform’’ 3 June 1997.

that drawing out the implementation of the programme in this way would reduce the annual rate of public investment spending by 15 per cent between FY 1997 and FY 2000, equivalent to 1½ per cent of GDP. As other forms of spending are also being restrained, the fall in total public expenditure may be close to 2 per cent of GDP, with the decline in general government spending being somewhat less. For the longer term, the government intends to gear fiscal policy towards stabilising the level of gross public debt. It also reaffirmed its objective of keeping the total of all taxes, social security contributions and the fiscal deficit below 50 per cent of national income (currently equivalent to 43 per cent of GDP).

In this broad perspective, the government has announced a restrictive budget for FY 1998 in line with its medium-term plan. Public works expenditures are projected to fall by 7 per cent, while other forms of spending are subject to restrictive targets. Overall, the OECD estimates that there will be a ½ per cent decline in nominal central government spending,[22] excluding debt interest payments and obligatory tax transfers to local authorities. This implies that the central government deficit falls by ⅓ per cent of GDP between FY 1997 and FY 1998. Local authority investment spending may also fall by ¼ per cent of GDP. With fiscal drag adding to revenues, the OECD projects that the central and local government deficit would decline to 4.6 per cent, a drop of almost 2½ per cent of GDP in three years. Such a decline would be somewhat less than the fiscal consolidation expected on average in other OECD countries with a similar initial level of deficit (Figure 22). Moreover, Japan's gross general government debt would still increase further to 93 per cent of GDP, up from 81 per cent at the end of 1995.

Additional cuts in expenditure will be required to reduce the deficit from 4.6 per cent of GDP in 1998 to the government's 3 per cent target by 2003. During this period, public investment will be reduced and other spending restrained. In addition, leaving nominal tax brackets unchanged would boost tax revenue (Table 18). Overall, assuming that the current cyclical shortfall in revenue is gradually eliminated over the medium term, the government could reach its deficit target by continuing with a slight fall in nominal expenditures each year, which should generate an annual decline in the deficit of around ⅓ per cent of GDP per year. Nevertheless, general government gross debt would continue rising to 102 per cent of GDP in 2003, up from 91 per cent in 1997. Moreover, a

Figure 22. **DEFICIT REDUCTION COMPARED TO INITIAL DEFICIT**
1995 to 1998, per cent of GDP




*Source:* OECD.

deficit held at the government's 3 per cent target would continue to raise the debt-to-GDP ratio for many years after 2003. Indeed, gross general government debt would eventually stabilise at about 120 per cent of GDP,[23] assuming that inflation rises to 2 per cent and output grows in line with the 1½ per cent trend expected for the first two decades of the next century (see Chapter V). On the other hand, under similar assumptions, a central and local government deficit of 2¼ per cent of GDP, equivalent to a slight general government surplus, would prevent gross debt rising. If, alternatively, prices were to remain stable, the deficit required to stabilise debt would drop to 1 per cent only.

Table 18. **Fiscal tightening required to reach the deficit target of the government**

Fiscal years, per cent of GDP

| | 1998 | 1999 | 2000 | 2001 | 2002 | 2003 |
|---|---|---|---|---|---|---|
| **Central and local government spending assumptions** | | | | | | |
| Baseline structural primary surplus | –2.4 | –2.4 | –2.4 | –2.4 | –2.4 | –2.4 |
| Spending cuts | 0.3 | 0.5 | 0.8 | 1.0 | 1.3 | 1.6 |
| Fiscal drag | 0.0 | 0.1 | 0.2 | 0.2 | 0.2 | 0.3 |
| Projected structural primary surplus | –1.7 | –1.5 | –1.1 | –0.7 | –0.4 | 0.0 |
| Property income | –2.1 | –2.1 | –2.3 | –2.6 | –2.9 | –3.0 |
| Projected structural surplus | –3.8 | –3.6 | –3.4 | –3.3 | –3.3 | –3.0 |
| Cyclical component of deficit | –0.8 | –0.7 | –0.5 | –0.2 | 0.0 | 0.0 |
| **Outcomes** | | | | | | |
| Central and local government deficit | –4.6 | –4.3 | –3.9 | –3.5 | –3.3 | –3.0 |
| Gross general government debt | 93.3 | 95.9 | 98.0 | 99.8 | 101.3 | 102.4 |
| **Economic assumptions** | | | | | | |
| Trend nominal income growth | 2.6 | 2.6 | 2.6 | 2.6 | 2.6 | 2.6 |
| Average interest rate on debt | 3.2 | 3.1 | 3.3 | 3.6 | 4.0 | 4.0 |
| *Memorandum* | | | | | | |
| Central and local government surplus | –4.6 | –4.3 | –3.9 | –3.5 | –3.3 | –3.0 |
| Social security surplus | 2.4 | 2.4 | 2.5 | 2.4 | 2.3 | 2.1 |
| General government surplus | –2.2 | –1.9 | –1.4 | –1.1 | –1.0 | –0.9 |

*Source:* OECD.

## *The impact of the planned deficit reduction on the economy*

Under the government's medium-term plan, the pace of fiscal consolidation is scheduled to slow considerably. In 1998, the fall in the structural balance, as estimated by the OECD, is to be only one-quarter that expected in 1997 (Table 14). Public investment will continue to decline but in the absence of increases in taxation, the growth of real personal disposable income will rise, pushing up household consumption. As a result, the negative impact of fiscal tightening on economic activity will be considerably reduced, so helping growth to accelerate in 1998 (see Chapter I). On the basis of the government's plan, it would appear that the pace of fiscal tightening may slow further in 1999 and 2000. While such a slowdown will ease the drag on the economy coming from fiscal policy, the government will need to take into account any unexpected shocks that might reduce the prospective growth of domestic demand when implementing the medium-term plan.

## The need for some rationalisation of public spending

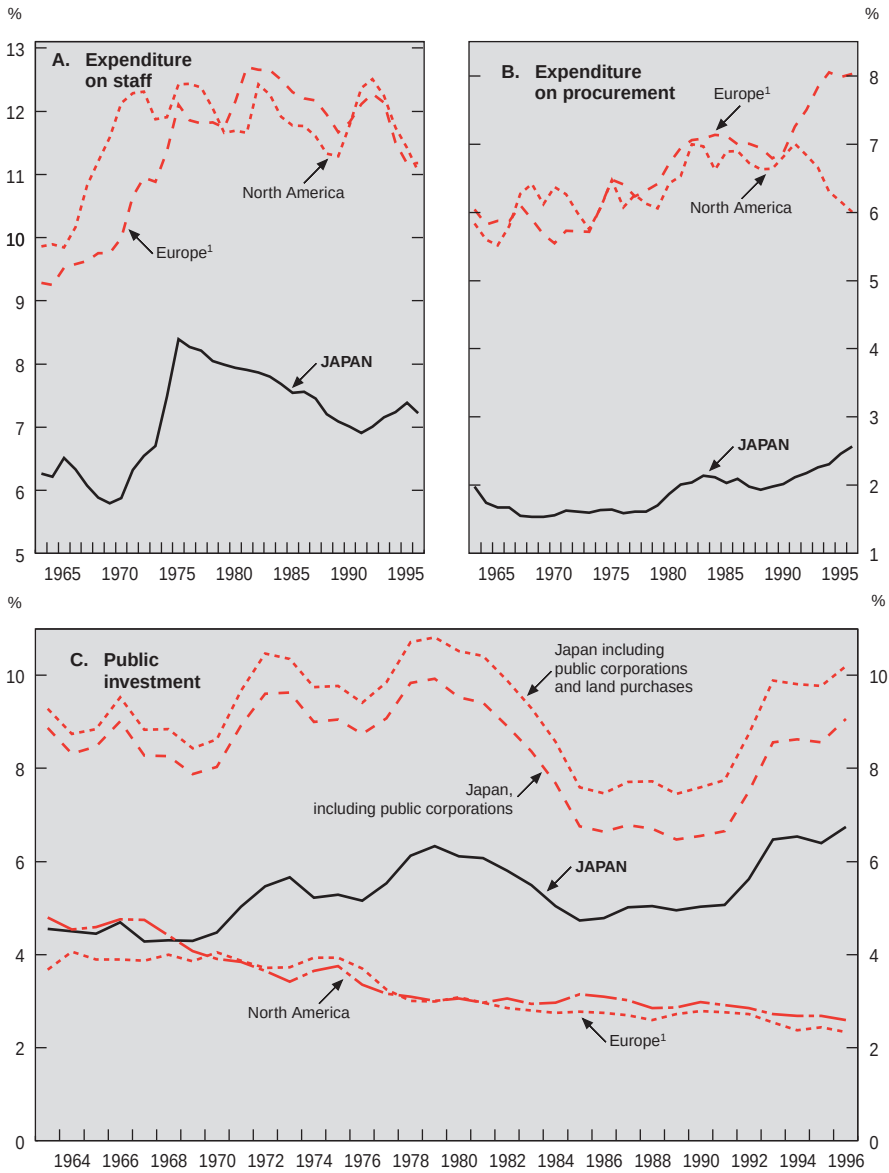
In addition to the need to reduce the deficit, it is important to improve the quality of public spending so that the return from expenditure can be maximised. Given the extensive control over public spending on staff and procurement, efficiency gains may be difficult to achieve in these areas. Outlays on these items have always been well below the levels found in Europe and North America (Figure 23, Panels A and B). In this area, an official study has indicated that it may be possible to substantially reduce the number of ministries and agencies of the central government. Moreover, there are several other areas where the quality of outlays might be improved: investment, including that by public corporations, and lending included in the fiscal investment and loan programme and government financial intermediaries.

### *Improving the quality of public investment*

At nearly 7 per cent of GDP in 1996, total public spending on investment is particularly high in Japan. It is almost three times higher than the average of other major OECD countries (Figure 22, Panel C) and is even above the level found in many developing countries.[24] Moreover, when land purchases and investment by public corporations, mainly in the area of roads and housing, are included, the overall share of public investment in GDP rises to nearly 10 per cent. Cumulative public investment since 1963 amounts to 110 per cent of 1996 GDP, compared to 75 per cent in the United States and 67 per cent in Germany. However, the high level of public investment in Japan reflects, in part, the low initial level of infrastructure as well as the higher cost of construction (Table 19). In 1990, the cost of civil engineering projects was 80 per cent greater than in the United States and almost 60 per cent above that in the European Union.[25] Even after allowing for the fact that the overall price level was some 25 per cent higher in Japan than in other countries, the difference in construction prices was still 40 per cent at that time. Other government surveys suggest that price differences may be smaller.[26] It would appear, nevertheless, that the high level of public investment in Japan does not necessarily generate a large physical public-sector capital stock. Such emphasis on public investment in Japan appears to reflect, in part, a budgetary process that makes it difficult to change the allocation of spending and has favoured across-the-board expenditure limits which have left the share of different ministries in total spending quite stable (see 1996 Survey).

Figure 23. **AN INTERNATIONAL COMPARISON OF PUBLIC SPENDING**

Per cent of GDP




1. Germany, France, Italy and United Kingdom.

*Source:* OECD, Analytical and Annual National Accounts Database.

Table 19. **An international comparison of construction prices**

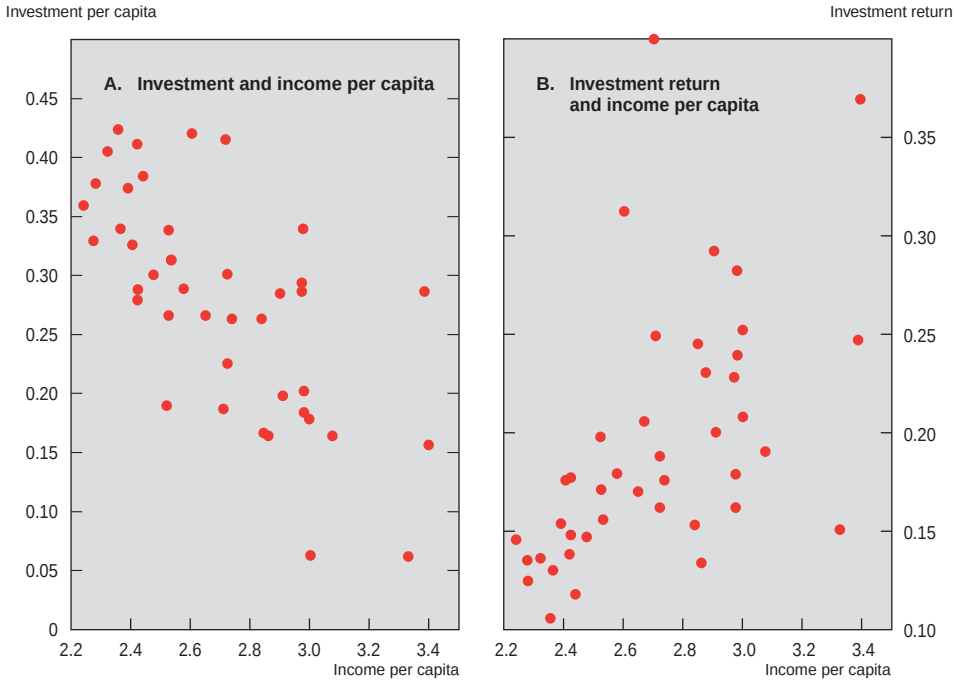
Price levels measured in dollars, 1990
OECD average = 100

| | Civil engineering | All ouput | Relative price |
|---|---|---|---|
| Japan | 146 | 116 | 126 |
| United States | 82 | 90 | 91 |
| Canada | 71 | 99 | 72 |
| European Union | 92 | 104 | 88 |
| Australia | 76 | 95 | 80 |

*Source:* *OECD Purchasing Power Parities Report*, 1990.

Figure 24. **THE LEVEL AND RETURN ON GOVERNMENT INVESTMENT**

47 prefectures of Japan, FY 1993




*Source:* Economic Planning Agency, *Annual Report on Prefectoral Accounts.*

The allocation of public investment in Japan, which is closely related to regional differences in income levels, appears to be rather economically inefficient. For example, several econometric studies found that the rate of return on public investment is positively related to prefectoral income and is twice as large in high as in low-income prefectures[27] (Figure 24). Nonetheless, the lower the level of per capita income, the higher is the level of public investment per capita (Figure 24). In lower-income prefectures, public investment per resident is almost four times greater than in higher-income areas.[28] As a result, economic activity in poorer areas is highly dependent on public-sector construction activity, which accounts for as much as 15 per cent of output in some prefectures. Moreover, the construction sector represents 13½ per cent of non-agricultural employment in the ten lowest-income prefectures compared with less than 10 per cent in the ten highest-income prefectures. While public investment could be seen as a large regional aid programme and so be justified on equity grounds, it remains that the concentration of investment in low-income areas tends to reduce the economic efficiency gains from such outlays. Nonetheless, part of such investment can be justified on non-economic grounds, as it represents the provision of a public good – such as an improved quality of life, a better environment or enhanced public security.

## *Reviewing the role of public corporations and the FILP programme*

### *Non-financial institutions*

In addition to public works spending by the government, investment by non-financial public corporations[29] averaged 2 per cent of GDP in the period 1990 to 1996, suggesting another possible area for rationalisation. These companies, which in 1996 accounted for one-third of total public-sector fixed capital formation, take a variety of forms:

- Special budget accounts managed as enterprises (*e.g.* the National Forestry Service).
- Corporations owned by the central government (*e.g.* Japan Highway Corporation).
- Local government-owned corporations and enterprises.

The enterprises owned by the central government operate principally in the area of transport infrastructure (Table 20). Based on the outstanding debt of these

Table 20. **Debt and investment of principal non-financial public corporations and government enterprises**

FY 1995

| | Per cent of total | Billion yen | | | Thousands |
|---|---|---|---|---|---|
| | | Outstanding debt from FILP | Capital expenditure | Subsidies, transfers and government investment | Employment |
| **Highways** | | | | | |
| Japan Highways | 22.5 | 19 609 | 1 598 | 242 | 8.6 |
| Metropolitan Expressways | 3.8 | 3 289 | 320 | 35 | 1.4 |
| Hanshin Expressways | 3.0 | 2 645 | | | 0.9 |
| Honshu-Shikoku Bridge | 2.2 | 1 919 | | | 0.7 |
| Total | **31.5** | **27 462** | **1 918** | **277** | **11.7** |
| (Per cent of GDP) | | **(5.6)** | **(0.4)** | **(0.1)** | |
| **Railways** | | | | | |
| Japanese National Railways Settlement | 17.6 | 15 399 | 40 | 6 350 | 2.5 |
| Railway Development | 5.3 | 4 631 | 0 | 182 | 0.1 |
| Japan Railway Construction | 1.8 | 1 572 | | | 1.9 |
| Private JR companies | 1.4 | 1 202 | | | |
| Teito Rapid Transit | 0.6 | 553 | | | 11.0 |
| Four other transport corporations | 1.4 | 1 225 | | | 1.4 |
| Total | **28.1** | **24 582** | **40** | **6 532** | **5.9** |
| (Per cent of GDP) | | **(5.0)** | **(0.0)** | **(1.3)** | |
| **Environment** | | | | | |
| Housing and Urban Development | 14.8 | 12 948 | 1 178 | 280 | 4.9 |
| National Forestry Service | 3.8 | 3 330 | 43 | 57 | |
| Sewage, Water, Environment | 2.3 | 1 965 | | | 2.2 |
| Total | **20.9** | **18 243** | **1 221** | **337** | **7.1** |
| (Per cent of GDP) | | **(3.7)** | **(0.2)** | **(0.1)** | |
| **Municipal enterprises** | | | | | |
| Total | **14.3** | **12 447** | 0 | 0 | |
| (Per cent of GDP) | | **(2.5)** | **(0.0)** | **(0.0)** | |
| **Others** | | | | | |
| Total | **5.2** | **4 549** | 0 | 0 | |
| (Per cent of GDP) | | **(0.9)** | **(0.0)** | **(0.0)** | |
| **Total** | **100.0** | **87 283** | **3 179** | **7 146** | **24.6** |
| (Per cent of GDP) | | **(17.9)** | **(0.7)** | **(1.4)** | **(0.05)** [1] |

1. Per cent of the labour force.

*Source:* FILP report 1996, Financial Bureau Ministry of Finance.

corporations to government financial institutions, about 60 per cent of their investments have been either in roads or railways. Housing supply for middle-income families in metropolitan areas accounts for a further 15 per cent of

investment, while a broad range of local enterprises account for a similar amount. The remaining 10 per cent of investment is spread across a wide variety of activities. Although such companies make significant investments in physical assets, the construction work is usually undertaken by private-sector firms. As a result, only 25 000 people are employed by non-financial public corporations (though the number rises to 36 000 if the employees of the last state-owned railway operating company are included).

Overall, the investments made by these corporations have not been profitable on a commercial basis. The government has made considerable transfers to them, averaging 0.7 per cent of GDP between 1992 and 1995, to keep these companies afloat and to fund their investment programmes (Table 21). In addi-

Table 21. **Public corporations: income and outlay accounts**

Per cent of GDP, fiscal years

| | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|
| **Non-financial corporations** | | | | |
| Total income | 0.82 | 0.93 | 0.86 | n.a. |
| Operating surplus | 0.75 | 0.86 | 0.80 | n.a. |
| Property income | 0.07 | 0.07 | 0.06 | n.a. |
| Interest payments | 1.12 | 1.15 | 1.13 | n.a. |
| Taxation | 0.00 | 0.00 | 0.00 | n.a. |
| Profit | –0.30 | –0.21 | –0.28 | n.a. |
| Dividend payments | 0.13 | 0.13 | 0.14 | n.a. |
| Retained earnings | –0.48 | –0.41 | –0.47 | n.a. |
| **Financial corporations** | | | | |
| Profit | 0.46 | 0.70 | 0.91 | n.a. |
| Dividend payments | 0.01 | 0.55 | 0.65 | n.a. |
| Retained earnings | 0.45 | 0.15 | 0.26 | n.a. |
| **All public corporations** | | | | |
| Profit | 0.16 | 0.48 | 0.64 | n.a. |
| Dividend payments | 0.19 | 0.74 | 0.85 | n.a. |
| Retained earnings | –0.03 | –0.26 | –0.21 | n.a. |
| Capital transfers | 0.24 | 0.21 | 0.19 | 0.21 |
| Net capital formation | 1.37 | 1.59 | 1.49 | 1.57 |
| Other transactions | 0.10 | 0.15 | –0.11 | 0.20 |
| Net lending | –1.06 | –1.49 | –1.61 | –1.17 |
| *Memorandum* | | | | |
| Gross capital formation | 1.91 | 2.14 | 2.08 | 2.21 |

*Source:* EPA, *Annual Report on National Accounts*, and government submission.

tion, the companies have continued to borrow from other sources in order to finance their cash-flow deficit, which averaged 1.3 per cent of GDP in the period 1992 to 1995. By March 1995, their debt amounted to 20 per cent of GDP.

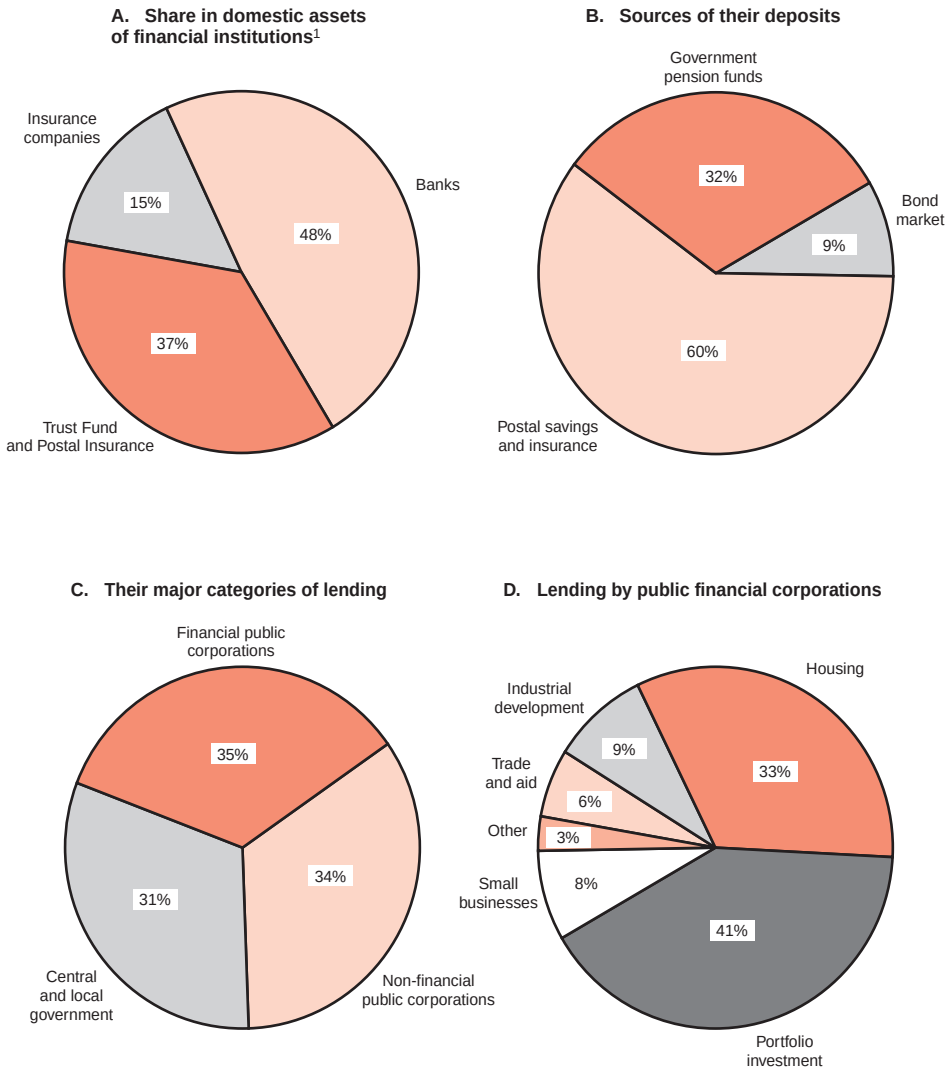
These transfers paid by the government to non-financial corporations are concentrated on a few enterprises. The largest transfer is made to the Japanese National Railways Settlement Corporation,[30] which has an outstanding debt of Y 28 trillion (5.6 per cent of GDP). Total transfers from the government to this corporation (including new lending) amounted to 1.5 per cent of GDP in FY 1995. In addition, the National Forest Service receives interest subsidies on part of its government loans, which financed its low-yielding investments. While other public corporations do not require large subsidies to break even, their investment decisions are often made on non-commercial grounds or use profits from past operations to fund new loss-making projects.

### *Financial institutions*

Public financial institutions are less important as a source of public-sector spending than non-financial corporations, though they do play an exceptionally large role in the flow of capital in the economy. In aggregate, the two key institutions, the Trust Fund Bureau and Postal Life Insurance, control 37 per cent of the total domestic assets of financial institutions in Japan[31] (Figure 25), only 10 percentage points less than that of the banking sector. The assets of the Postal Life Insurance alone are half the size of all mutual life insurance companies in Japan. The Trust Fund Bureau draws its funds largely from postal savings and the social security system, while the postal insurance system lends to the Postal Life Insurance Fund.[32]

The Trust Fund Bureau and Postal Life Insurance are the principal sources of funds for the Fiscal Investment and Loan Programme (FILP), which is presented to the Diet each year along with the regular budget.[33] These two funds lend for a number of purposes, including financing the budget deficit of central and local government, non-financial and financial public corporations. About one-third of their portfolio is borrowed by other central and local government accounts in the form of government bonds, local bonds or short-term financing loans (Figure 25). A significant part of their funds are lent to the unprofitable non-financial public corporation sector (see above), the rest of the loans going to other government financial institutions (in particular to various investment funds

Figure 25. **THE ROLE OF GOVERNMENT FINANCIAL INTERMEDIARIES**
Trust Fund Bureau Special Account
Postal Life Insurance Special Account




1. Banks, insurance companies, Trust Fund Bureau, and Postal Life Insurance.
*Source:* Ministry of Finance, *FILP Report 1996*, Bank of Japan, *Monthly Economic Statistics* and OECD calculations.

that are part of central government[34]). The annual FILP budget includes repayments of past lending by the two key special accounts as well as the allocation of additional funds.

When the public financial institutions were established, the banking sector was highly regulated and interest rate competition was limited. As a result, these public institutions were able to lend money at cheaper rates than the banking system, while keeping a sufficient margin to pay their costs. Moreover, they could direct money towards areas favoured by the government, as there was excess demand for their relatively inexpensive loans.

However, since financial deregulation has reduced the demand for government loans, the government's ability to set priorities through the FILP budget has been limited. Demand often does not even match the planned supply of funds, especially when the yield curve is steep, as the public institutions always lend at long-term rates. With most of the borrowing and lending rates being conducted at market rates, the government has limited scope to make off-budget subsidies to selected sectors through these public financial institutions. Consequently, subsidies resulting from lending rates that are set below borrowing rates are not large. Nevertheless, subsidies still occur because lending rates do not reflect risk, a problem that has become more evident due to the narrowing of margins since interest rates were deregulated.

In fact, public financial institutions face both credit and interest rate risk. As far as credit risk is concerned, the extent of bad loans has been low, with an average of only about 1 per cent of their loans classified as non-performing. However, certain institutions have experienced high default rates, notably those lending to small businesses and for foreign aid purposes. Public financial institutions have also been exposed to interest rate risk. For instance, the regulation requiring the Housing Loan Corporation (HLC) to allow households to pre-pay mortgages without any penalty has led to as much as 15 per cent of their loans being repaid in one year.[35] Prepayments of loans to small businesses had also been allowed in some cases, again with no penalty. However, from 1 July 1996, prepayment of those loans, contracted with the Japan Finance Corporation for Small Businesses, was conditional on the lender's approval and payment of a predefined penalty. Overall, subsidies paid to government financial institutions amount to 0.1 per cent of GDP.

As a group, public financial institutions are profitable, and paid dividends averaging 0.4 per cent of GDP each year between 1994 and 1996. Moreover, these institutions appear to operate efficiently, with their non-interest operating costs representing a proportion of their assets similar to those of the main banks. However, according to a recent MITI report, if the loans of the HLC were securitised and sold on the market, they would have to yield slightly more than 1 per cent above government bonds of a similar maturity. The margins would be even larger for the loans from certain other public institutions, as the HLC has the highest quality asset portfolio, in which case they represent the cost to the taxpayers of inadequate payment for the *ex ante* risks involved in such lending operations.

## The need for increased budget transparency

In Japan, there is a high degree of control over the budget under the present system. Only the general account has the power to issue new government bonds, which are included as revenues, although the debt consolidation account can issue re-financing bonds. Since the Diet approves both revenues and expenditures, it also controls the extent of bond issues during a fiscal year. Consequently, if the deficit is larger than expected, the government has to submit a supplementary budget to the Diet to obtain authority to issue additional bonds.

However, while the extent of control is high, the level of transparency is low. As noted above, the key element in the official fiscal restructuring plan is the 3 per cent target for the deficit of central and local governments, calculated on a national accounts basis. Although extensive information is available concerning the general account, 38 special accounts, the local authority financing plan, the budgets of public corporations and the lending activities of government financial institutions through the FILP programme, it is not possible to monitor progress towards the fiscal objective using the published budget documents. Consequently, as noted in previous surveys, there is a need for a more analytical presentation of government accounts and consolidated statements, as well as returns based on existing legal and public accounting principles. Such statements should split transactions between current expenditures and revenues, capital spending and revenues and financial transactions, following the practice of several other OECD countries. Each category of spending should be broken down by economic

category, with transfers to and receipts from other levels of government clearly identified.

This information would need to cover special accounts as well as the general account. Ideally, the government special accounts should be split into four groups dealing with central government activities, social security and financial and non-financial government enterprises. Such a presentation is already used in the national accounts (Table 22). Most of the fifteen special accounts relating to the central government have been set up to ensure that earmarked taxes are used for the intended purpose. These taxes are supplemented with transfers from the general account and borrowing on a short-term basis. Two of the accounts, Auto Accident Reinsurance and the Bond Consolidation, run a surplus (for the latter account, the surplus reflects the rule requiring a certain proportion of government bonds to be redeemed each year). The inclusion of the Bond Consolidation Account in the general account into a central government account would lower the size of the budget deficit. In addition, the scope of the central government account should be widened to include the various government-owned non-financial public corporations, since there is little difference between their activities and those of government agencies operating in the same field. This could be done either through the provision of a separate sub-account for government non-financial corporations, covering both income and expenditure and financial transactions, with enough details to allow consolidation, or by direct consolidation with central government accounts.

The other special accounts cover the social security system, local authorities and financial and non-financial enterprises. Although the government's fiscal target excludes social security, the potential problems in this area (see Chapter V) suggest that the same analytical approach should be applied. Thus, budget statements should be based on the relevant special accounts and the accounts of the various pension funds and mutual associations. Finally, the same principles used for the central government and social security accounts should be applied to the local government financing plan to permit aggregation and consolidation with the central government accounts. It would then be possible to judge the current year's budget against the medium-term deficit target.

The transparency of the budget process would also be enhanced by the publication of regular quarterly statements of budgetary outcomes within one month of the end of the quarter. At present, monitoring information is limited to

Table 22. **Revenues and surpluses of government special accounts by national accounts sector**

Yen billion FY 1995

| | Total revenue | Transfer from general account | Other revenue | Surplus of account |
|---|---|---|---|---|
| **National accounts sector** | | | | |
| **Central government** | | | | |
| Road Improvement | 5 184 | 2 554 | 2 630 | 0 |
| National Schools | 2 536 | 1 558 | 978 | 0 |
| Flood Control | 1 660 | 1 176 | 484 | |
| National Hospitals | 1 050 | 248 | 802 | 0 |
| Coal and oil account | 701 | 515 | 186 | 0 |
| Harbour improvement | 573 | 348 | 225 | 0 |
| Airport improvement | 547 | 131 | 416 | |
| National Land improvement | 508 | 276 | 232 | 0 |
| Electric Power development | 452 | 0 | 452 | 0 |
| Nation property consolidation | 207 | 0 | 207 | 0 |
| Registration | 167 | 72 | 95 | 1 |
| Patents | 98 | 0 | 98 | 21 |
| Auto inspections | 54 | 3 | 51 | 7 |
| Agricultural Management | 51 | 11 | 40 | 0 |
| **Total of above** | **13 788** | **6 892** | **6 878** | **29** |
| Government bond consolidation | 57 882 | 13 221 | 44 667 | 4 193 |
| Local tax transfer | 26 553 | 13 215 | 13 318 | 463 |
| **Total of above** | **84 435** | **26 436** | **57 986** | **4 656** |
| Foreign Exchange | 1 295 | 0 | 1 295 | 155 |
| **Social security** | | | | |
| Welfare insurance | 59 331 | 3 940 | 55 391 | 10 825 |
| National Pension Fund | 20 993 | 1 518 | 19 475 | 543 |
| Workers Insurance | 8 798 | 256 | 8 542 | 674 |
| Seamens Insurance | 112 | 7 | 105 | 1 |
| **Total of above** | **89 234** | **5 721** | **83 513** | **12 043** |
| **Government financial enterprises** | | | | |
| Postal Savings | 19 947 | 0 | 19 947 | 3 159 |
| Postal Life Insurance | 19 273 | 0 | 19 273 | 7 022 |
| Trust Fund Bureau | 19 265 | 0 | 19 265 | 0 |
| Industrial Investment | 2 497 | 1 128 | 1 369 | 0 |
| Auto Accident Reinsurance | 2 767 | 0 | 2 767 | 1 749 |
| Urban development | 2 445 | 7 | 2 438 | 0 |
| Agricultural Mutual Aid Re-insurance | 286 | 100 | 186 | 19 |
| Fishery Mutural Aid | 51 | 16 | 35 | 12 |
| Earthquake Reinsurance | 34 | 0 | 34 | 0 |
| Forest Insurance | 13 | 0 | 13 | 7 |
| **Total of above** | **66 578** | **1 251** | **65 327** | **11 968** |
| **Government non-financial enterprises** | | | | |
| Postal services | 7 361 | 4 | 7 357 | 0 |
| Food control | 4 738 | 183 | 4 555 | 0 |
| National Forest Service | 787 | 241 | 546 | |
| Printing Bureau | 99 | 0 | 99 | 0 |
| Mint Bureau | 28 | 0 | 28 | 0 |
| Alcohol monopoly | | | | |
| **Total of above** | **13 013** | **428** | **12 585** | **0** |

*Source:* Ministry of Finance Budget Documents.

data on monthly tax receipts, the treasury accounts of the Bank of Japan and the outstanding balance of government bonds. There may be some difficulty in producing comprehensive monitoring figures for all social security and local authority accounts, given the number of small organisations concerned. In this case, a sampling process might be required for these levels of government.

## Summary

The fiscal position of the Japanese government has deteriorated markedly since 1990, mainly as the result of an extensive attempt to offset a downturn in private demand. This action, together with a cyclical loss of tax revenues, led to gross general government debt rising at one of the most rapid rates in the OECD area to a level above the OECD average. The net debt of the general government, in contrast, is still relatively low due to the accumulated surpluses of the social security system. However, the rate of increase of these surpluses is already declining as the rapid growth in the number of elderly people pushes up spending on pensions. This will tend to push the general government deficit towards that, much higher, of central and local governments (which in 1996 represented nearly 7 per cent of GDP). It will also raise the net debt of general government more rapidly than gross debt.

The government expects to reduce the deficit of central and local government – which is responsible for the growth of gross general government debt – to 3 per cent of GDP by 2003. This would be equivalent to a general government deficit of just under 1 per cent of GDP at that time. The major part of the reduction occurred in 1997, so that the speed with which the deficit is being reduced will slacken markedly in 1998, producing a smaller drag on economic growth than in 1997. Nonetheless, flexibility will be needed when implementing future budget consolidation so as to ensure that actions taken to achieve long-run fiscal sustainability are compatible with cyclical economic developments. Over the longer term, lower budget deficit targets than those envisaged in the current plan would probably be needed to achieve the government's aim of stabilising the level of gross public debt.

In addition to the need to reduce the deficit, the quality of public spending should be improved. Public investment, which is relatively high compared to other countries, appears to be allocated on the basis of not only obtaining the

highest return on investment but also supporting local economic activity. Thus, a reallocation of public investment could maintain the benefits of spending even while the total amount is being reduced, as envisaged in the government's medium-term fiscal plan. In order to provide a global picture of public investment spending, the remaining public corporations should be moved onto the budget, so as to lessen the possibility of the recurrence of problems such as that of the loss-making Japan National Railway Settlement Corporation, whose debts – backed by the government – amount to 5 per cent of GDP.

The extensive involvement of the government in the financial sector also suggests that there may be room for improving the allocation of financial resources by introducing a much greater market orientation in this area. Government financial institutions account for over one-third of total lending. While the bad loan experience of these institutions has been limited, they have been exposed to interest rate risks – which the private sector would normally avoid – necessitating increased subsidies in recent years. In addition, credit is allocated at below market interest rates, favouring certain sectors such as housing and small businesses which may generate costs to the taxpayer if lending margins are inadequate to cover the *ex ante* risks involved in such lending operations.

Greater transparency in the production of the budget and the government accounts would also help in assisting the process of fiscal consolidation. At present, it is not possible to monitor progress towards the government's deficit target from the existing budget documents. The existing 39 government accounts and the local authority financing plan need to be presented in a way that permits consolidation and allows the calculation of central government, local government and social security spending and deficits. In addition, it is important to distinguish expenditure and revenue flows from financing items in order to assess accurately the economic implications of the budget.

# IV. Regulatory reform

Progress in deregulating certain sectors of the Japanese economy, notably distribution and telecommunications, has resulted in large economic benefits. Such gains have strengthened the government's commitment to more ambitious reforms to expand business opportunities, increase domestic demand and facilitate imports in order to further improve the standard of living in Japan. Regulatory reform is recognised as a key to boosting labour productivity and meeting the challenges posed by an ageing population (see Chapter V). Moreover, deregulation, by stimulating the development of new business areas, is essential to create jobs, as noted in the chapter of last year's Survey on *Implementing the OECD Jobs Strategy*. Given the scope for further gains, deregulation should be a top priority for economic policy in Japan. Improving the economic structure through deregulation is indeed one of the major objectives established for the year 2000 by the Prime Minister, along with fiscal consolidation and reform of the financial system, the public sector, education and the social security system. Consequently, the pace of reform, which had been excessively cautious, appears to have accelerated during 1996. This chapter first reviews the impact of deregulation in recent years. It then discusses recently announced measures and the scope for further action.

## The scope of deregulation to date

The Economic Planning Agency estimates that deregulation has resulted in an annual average of almost Y 8 trillion (1.6 per cent of GDP) of additional demand during the first half of the 1990s (Table 23). Moreover, it has reduced prices, resulting in annual gains to consumers estimated at 1 per cent of GDP over the same period. One of the most important regulatory changes was the easing of the *Large-Scale Retail Store Law* in 1990, 1992 and 1994, which

Table 23. **Impact of deregulation on demand**

Annual average in billions of yen[1]

| Sector | Period (fiscal years) | Consumption | Investment | Housing | Total increase in GDP |
|---|---|---|---|---|---|
| Information and telecommunications | | 1 489 | 774 | 0 | 2 263 |
| Telecommunications | 1990-95 | 1 285 | 707 | 0 | 1 992 |
| Cable television | 1990-95 | 58 | 54 | 0 | 113 |
| Satellite television | 1990-95 | 146 | 13 | 0 | 159 |
| Large-Scale Retail Law Store | 1990-95 | 3 617 | 917 | 0 | 4 534 |
| Housing and land | 1992-96 | 0 | 0 | 200 | 186[2] |
| Aviation | 1994-95 | 72 | 0 | 0 | 72 |
| Convertible bonds | 1990-95 | 0 | 701 | 0 | 701 |
| Temporary workers | 1990-94 | 137 | 0 | 0 | 137 |
| Small-scale breweries | 1994-96 | 13 | 4 | 0 | 16 |
| Total | | 5 328 | 2 396 | 200 | 7 909 |
| Negative impact on small stores[3] | | 627 | 0 | 0 | 627 |
| Net total | | 4 701 | 2 396 | 200 | 7 282 |
| Per cent of GDP | | 1.0 | 0.5 | 0.1 | 1.6 |

1. The impact on demand is calculated by extending the pre-deregulation trend growth rate in retail sales and investment in these industries into the period after regulatory reforms were implemented. The impact of deregulation is calculated as the difference between this trend and the actual increase in sales and investment recorded following the regulatory change.
2. The total is reduced by a Y 14 billion rise in imports.
3. The negative impact of the increase in large stores is estimated to have reduced consumption demand by Y 627 billion.

*Source:* Economy Planning Agency.

significantly reduced distribution costs (see 1995 Survey). In principle, these reforms exempted stores with a planned retail space of less than 1 000 square metres from the deliberation procedure and put a time limit on the process, which reduced its average length from 34 months to about half a year (Table 24). This helped boost the number of notifications to build large stores from an annual average of 610 in the second half of the 1980s to 1 617 in the first half of the 1990s, while the declining trend in the number of smaller stores accelerated. The opening of larger, more efficient retail outlets is estimated to have increased annual demand by about Y 4.5 trillion (0.8 per cent of GDP) per year between 1990 and 1995. In addition, other changes in the distribution system, such as the easing of regulations on sales of cosmetics, medicines and liquor, have benefited consumers by reducing prices.[36]

Table 24. **Regulation of large stores**

| Regulation | Before May 1990 | After May 1994[1] |
|---|---|---|
| Duration of application procedure | Average of 34 months<br>No upper limit | Average of 6.2 months<br>Upper limit of 1 year |
| Store size | more than 500 $m^2$ | 1 000 $m^2$ or more[2] |
| Store closing time | Before 6 p.m. | Before 8 p.m.[3] |
| Number of days store must be closed | At least 48 days a year | At least 24 days a year[3] |

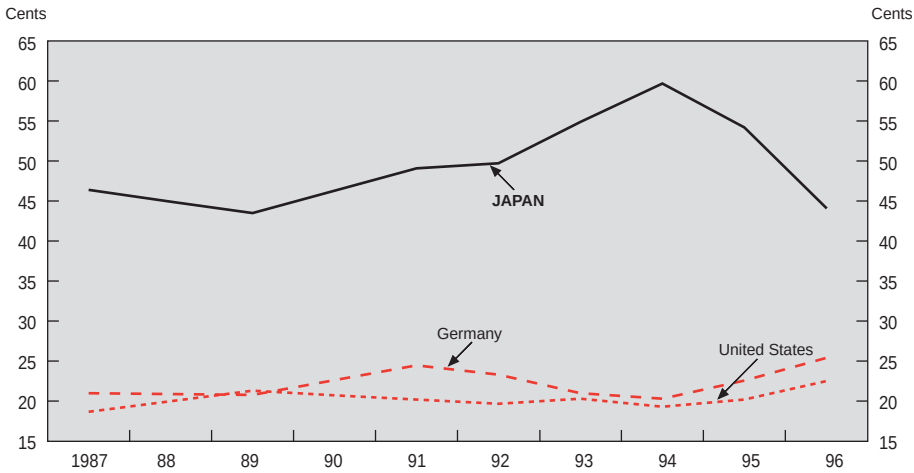
1. The Large-Scale Retail Store Law was revised in May 1990, January 1992 and May 1994.
2. In principle, no deliberation procedures are required for stores from 500 $m^2$ to 1 000 $m^2$.
3. A store must notify the Minister for International Trade and Industry or the heads of local governments when it wants to set its closing time after 8 p.m. or close less than 24 days a year.

*Source:* Ministry of International Trade and Industry.

There have also been significant changes in the *telecommunications* sector since the introduction of competition in 1985 (see 1996 Survey). Deregulation, combined with technological progress, has expanded the size of the market, while reducing prices and improving services. For example, the cost of a three-minute telephone call between Tokyo and Osaka has been reduced from Y 400 in 1985 to Y 100 in 1997 while subscription fees in the mobile telephone market have fallen by 96 per cent over the same period. Moreover, the number of mobile telephone subscribers increased from 40 000 in 1985 to 23 million in 1996. According to the EPA, the deregulation of telecommunications has generated an annual average of almost Y 2 trillion (0.4 per cent of GDP) in additional demand between 1990 and 1995 (Table 23). Meanwhile, changes in regulations governing cable and satellite television broadcasting have added Y 0.3 trillion of extra demand. The number of cable television subscribers increased from 400 000 to 2.8 million during the first half of the 1990s, while the number of firms in this industry tripled to 250 (EPA, 1997). In addition, the relaxation of controls on land and housing, aviation and temporary workers generated another Y 0.4 trillion of demand (0.1 per cent of GDP).

Important steps have also been taken in the area of *petroleum* products. The pre-tax price of gasoline has fallen by 26 per cent since 1994, when the government began to consider removing the ban on imports of refined petroleum products, a measure that was implemented in April 1996 (Figure 26). The decline

Figure 26. **GASOLINE PRICES**
Pre-tax price of regular gasoline in US cents per litre




*Source:* IEA, *Energy Prices and Taxes,* 1997.

occurred despite an increase in crude oil prices, which boosted the cost of gasoline in other countries. These trends reduced the ratio of Japanese to US gasoline prices before taxes from three in 1994 to two in 1996.

Regulatory reforms have also taken place in *financial markets*. In particular, the removal of the limits on issues of convertible bonds has reduced the cost of capital, generating additional business investment (Table 23). The financial conditions that a firm must meet before issuing straight bonds have also been relaxed, thus making it possible for weaker companies to raise capital through this instrument. Moreover, the securities subsidiaries of commercial banks were allowed to underwrite straight bonds in 1994. Increased competition from these firms, which had one-third of the market by 1996, has helped reduce commissions by 10 basis points. Meanwhile, the commercial paper market has been liberalised by relaxing maturity restrictions and allowing additional firms, such as leasing and insurance companies, to use this instrument. The EPA estimates that reforms in the bond market (both convertible and straight) and the commercial

paper market reduced financing costs by an annual average of Y 766 billion (0.2 per cent of GDP) during the first half of the 1990s.

## Recent initiatives

Deregulation is currently proceeding along three tracks in Japan. *First*, a final revision of the three-year deregulation plan (FY 1995 to FY 1997) was announced in March 1997. More than 1 500 deregulation measures had already been implemented, with about 400 unfinished items (Table 25). In addition, the plan includes almost 900 new measures, which draw heavily on proposals from the Administrative Reform Committee, an independent council created in December 1994 for a three-year term to oversee regulatory reform, as well as from foreign governments and private-sector organisations in Japan and overseas. In total, 1 300 items in twelve areas remain to be implemented by April 1998.

Table 25. **The revised ''Deregulation Action Programme''**

March 1997

| Area | New items | Unfinished items from March 1996 programme | Total items to be implemented | Items implemented from March 1996 programme[1] | Other measures already implemented | Total |
|---|---|---|---|---|---|---|
| Housing and land | 55 | 7 | 62 | 116 | 8 | 186 |
| Information and telecommunications | 71 | 19 | 90 | 108 | 15 | 213 |
| Distribution system | 86 | 18 | 104 | 141 | 11 | 256 |
| Transport | 88 | 42 | 130 | 208 | 30 | 368 |
| Standards, certification and imports | 150 | 71 | 221 | 283 | 31 | 535 |
| Finance, securities and insurance | 162 | 23 | 185 | 156 | 68 | 409 |
| Energy | 17 | 7 | 24 | 32 | 10 | 66 |
| Employment and labour | 18 | 16 | 34 | 39 | 5 | 78 |
| Pollution and the environment | 24 | 4 | 28 | 23 | 10 | 61 |
| Disaster relief and public security | 133 | 114 | 247 | 92 | 15 | 354 |
| Education | 27 | 4 | 31 | 20 | 6 | 57 |
| Others | 59 | 77 | 136 | 96 | 18 | 248 |
| Total | 890 | 402 | 1292 | 1304 | 227 | 2823 |

1. Partially or fully.
*Source:* Management and Coordination Agency.

*Second*, the deregulation process was also advanced by the December 1996 recommendations by the Economic Council, an advisory body to the Prime Minister. It proposes far-reaching reforms in six areas – land and housing, medical care and welfare, distribution, employment and labour, the financial system and telecommunications. These recommendations influenced the three-year plan, which also addresses these sectors.

A *third* track for deregulation is the Action Plan for Economic Structural Reform, which was adopted by the government in May 1997. This programme is intended to create an environment that is more favourable to new business activities, in particular by reducing Japan's high-cost structure through deregulation in sectors such as telecommunications, energy and distribution. These sectors are among the fifteen key areas, including as well medical care and the environment, which are projected to grow from a combined output of Y 200 trillion in 1997 to Y 550 trillion in 2010, while creating 7.4 million jobs during that period. The Action Plan, which is being co-ordinated by the Ministry of International Trade and Industry (MITI), contains around 1 000 specific proposals to reform regulations and promote efficiency in the public sector.

### *The ''Big Bang'' planned in the financial system*

Deregulation of the financial sector is a key element in all three deregulation initiatives. In November 1996, the Prime Minister announced a five-year plan to make Japan's domestic financial system ''free, fair and global'' and transform Tokyo into an international financial centre on a par with New York and London (Table 26). The first step of this programme is the reform of the foreign exchange and foreign trade control law, which limits transactions involving foreign currencies to the authorised foreign exchange banks and restricts the netting of transactions. These regulations appear to have been obstacles to swift transactions by Japanese corporations, thus making it difficult to quickly seize business opportunities. In May 1997, the Diet passed legislation eliminating the authorised foreign exchange bank system from April 1998 and allowing other companies and individuals to buy and sell foreign currencies freely. In addition, the new law abolished, in principle, the permission and prior notification requirements on capital transactions. As a result of these changes, new intermediaries, such as securities firms and trading companies, are likely to enter foreign exchange markets, thereby increasing competitive pressures and reducing transaction costs.

Table 26. **Outline of major elements in the ''Big Bang'' deregulation plan**

| Sector/market | Reform | Implementation date |
|---|---|---|
| Foreign exchange | Elimination of the ''authorised foreign exchange bank'' system | April 1998 |
| | Liberalisation of cross-border capital transactions | April 1998 |
| Financial holding companies | Holding companies permitted | FY 1998 |
| Banks | Permit banks to sell mutual funds over the counter | FY 1998 |
| | Loans can be transformed to asset-backed securities | FY 1998 |
| | Allow commercial banks to issue long-term debentures | FY 1999 |
| | Removal of controls on banks' securities subsidiaries | FY 1999 |
| | Permit banks to sell certain insurance products | Around 2001 |
| Securities | Securities firms can establish bank-type accounts for customers | FY 1997 |
| | Allow securities firms to handle unlisted stocks | FY 1997 |
| | Sale of derivatives liberalised | FY 1998 |
| | Free entry to securities business | FY 1998 |
| | End to fixed brokerage commissions | FY 1999 |
| | Securities firms' trust banking units allowed to enter the pension management business | FY 1999 |
| | Securities firms allowed to enter insurance industry | 2001 |
| Insurance | End to uniform premiums on auto, fire and personal accident insurance | July 1998 |
| | Insurance firms can establish banking and securities subsidiaries | FY 1999 |
| | Subsidiaries of life and non-life insurance firms can fully enter the ''third sector'' of the insurance market | 2001 |
| Pensions | End to the 5:3:3:2 rule on the allocation of funds | By 2001 |
| Accounting practices | Financial assets to be assessed at market, rather than book, value | FY 1999 |
| | Practices to be brought up to international standards | |

*Source:* Ministry of Finance.

Moreover, firms will be allowed to handle foreign currency transactions in-house, which should reduce bank fees.[37] In addition, individual Japanese will be able to open bank accounts abroad, either in yen or foreign currencies, thus facilitating cross-border fund movements.

The improved access to international markets beginning in FY 1998 is likely to force further deregulation in Japan to prevent financial activities from moving to overseas markets with lower costs. In particular, fixed commissions on equity purchases make trading relatively expensive in Japan. Since 1993, the volume of Japanese equities traded in London as a percentage of that in Tokyo has almost

doubled from 10 to 18 per cent.[38] Moreover, the trading volume of foreign stocks in the Tokyo Stock Exchange has declined sharply. To avoid a further shift away from Japan, all brokerage commissions are to be liberalised by the end of 1999.[39] Increased freedom to set prices will also be extended to the insurance market, where, at present, a rating organisation consisting of the insurance companies establishes uniform premiums for auto, fire and personal accident coverage at levels that help to make these kinds of insurance available as well as affordable for consumers. This system, which has prevented competition based on price and reduced the link between risk and the level of the premium, is to be abolished in July 1998. This reform, which was promised in the December 1996 insurance agreement with the United States,[40] should attract new entrants, both domestic and foreign. For example, major trading companies in Japan have already announced plans to enter the non-life insurance market.

In addition to granting greater freedom to set prices, the government plans to boost competition by reducing financial market segmentation. Although the Ministry of Finance has allowed institutions to open subsidiaries in other financial fields since 1993, it continues to limit the scope of their activities. For example, the trust banking units set up by securities firms are not allowed to handle pension funds, while the securities subsidiaries established by commercial banks are barred from underwriting and brokering equities. Such regulations on subsidiaries are to be lifted in the latter half of FY 1999. In addition, insurance companies will be allowed to open banking and securities subsidiaries, while securities companies will be permitted to open bank-style accounts through which customers can make payments. Entry into other markets will be facilitated by the partial lifting of the ban on holding companies (see below). Bank holding companies, for example, are likely to be allowed to own other financial firms, including trust banks, securities and insurance firms, and leasing and credit companies.[41] However, bank holding companies will be barred from owning businesses in other fields.

Competition within the banking sector will also be enhanced with the easing of restrictions on the financing methods of commercial banks. In particular, they will be allowed to issue corporate bonds, a fund-raising channel that is to be opened to non-bank lending institutions, such as leasing companies and consumer credit firms. Finally, in July 1997, the Ministry of Finance ended regulations on

the opening and location of bank branches and on the operating hours of banks and cash-dispensing machines.

The deregulation plan will also relax other types of controls. Securities firms, for example, have been able to offer option trading in individual stocks since June 1997 and over-the-counter-derivatives will be allowed by the end of FY 1998. Moreover, the Ministry of Finance will allow banks to sell loan-backed securities to individual investors and corporations, a strategy which may help them to reduce their non-performing loans. Likewise, the freedom to manage pension funds will be increased by the removal in FY 1998 of the so-called 5-3-3-2 rule, which requires that at least half of the funds be placed in investments with guaranteed principal, no more than 30 per cent in domestic equities or foreign assets and less than 20 per cent in real estate. The return to pension assets should also be boosted by the elimination of the rule in April 1999 which allows investment advisory firms to manage only half of pension funds.[42] Finally, tougher accounting rules are to be imposed to bring Japanese practices closer into line with international standards. One government advisory body recommended that companies appraise their financial holdings at the current market value rather than at book value, a reform that would help investors to assess more accurately the position of financial institutions.

With the revision of the foreign exchange and foreign trade control law, significant changes in Japan's financial system appear inevitable. The increased competition resulting from the deregulation plan should benefit consumers and firms by providing a greater variety of products at lower costs. Greater competition may also result in a market restructuring that eliminates inefficient firms. In the securities market, for instance, the abolishment of fixed brokerage commissions may reduce revenues by as much as 10 per cent according to some estimates and encourage mergers among the existing 225 companies. In particular, the major banks, which have already become major competitors in the bond market, are likely to enter the equity broking market as well. These factors could lead to a rationalisation in the securities industry similar to that which occurred following the ''Big Bang'' in London in 1986. Overall, the planned reforms should lead to greater integration of Japanese markets and companies into the global arena, while keeping some transactions, which would have otherwise moved overseas, in Japan.

## *Other areas*

The deregulation plans include significant measures in other sectors, such as transport, telecommunications, energy, housing and land, the distribution system and the labour market. These are summarised below, as well as developments in competition policy and progress in improving administrative procedures.

### *Transport*

The Ministry of Transport has announced that it will ease entry barriers in this sector. The ''supply-demand balancing provision'', which is intended to prevent excess supply, will be abolished by FY 1999 for air transport and charter buses and by FY 2001 for taxis, buses and passenger boats. In the air transport industry, four companies have announced, since November 1996, plans to enter the domestic market. They would be the first new entrants in over forty years. Increased competition in domestic routes, where the market share of the three leading airlines (Japan Airlines, All Nippon Airlines and Japan Air System) is 94 per cent, appears to be the key to reducing prices.[43] For example, one of the new companies has announced plans to halve the fares on the Tokyo to Sapporo route, reportedly the busiest in the world, using leased aircraft and foreign crews. In addition to the high level, there has been little variation in prices between the airlines on the routes where they compete, even after the easing of government controls on fares in June 1996 (see below). At that time, there were 38 domestic routes served by at least two of the major airlines, according to a study by the FTC (Table 27). On thirty of these routes, prices were identical between airlines, while for the other eight, the average difference was less than 2 per cent.

The opportunities for new airlines, though, remain limited by the shortage of take-off and landing slots at airports in Tokyo and Osaka. About four-fifths of domestic air travel in Japan involves these two cities either as a departure or destination. The completion of a new runway at Haneda Airport in Tokyo in March 1997 added an additional forty daily slots (for both landings and take-offs). Although the Ministry of Transport had promised to give priority to the new carriers in allocating the new slots, two new companies, which had requested a total of 21 slots, received only three each. In contrast, the three leading airlines were given the remaining 34. It is uncertain whether such an allocation will encourage new entrants to the industry as three flights a day

Table 27. **Variation in airline ticket prices**

June 1996

| | Number of routes served by more than one airline[1] | Identical prices on all flights | Identical prices on two airlines[2] | Other[3] |
|---|---|---|---|---|
| Routes served by all three airlines | 20 | 14 | 6 | 0 |
| Routes served by ANA and JAL | 11 | 11 | – | 0 |
| Routes served by ANA and JAS | 7 | 5 | – | 2 |
| Total | 38 | 30 | 6 | 2 |

*Note:* JAL refers to Japan Airlines, ANA to All Nippon Airlines and JAS to Japan Air System.
1. Does not include routes where JAL and JAS or ANA and JAS provide joint service.
2. On these six routes, JAL and ANA offered identical fares while JAS was 1.9 per cent less expensive on average.
3. On these two routes, ANA was 1.5 per cent less expensive on average.

*Source:* FTC (1997).

appears to be insufficient to achieve the economies of scale necessary to challenge the established airlines.[44]

Establishing a more transparent system of allocating take-off and landing slots at major airports would enable new airlines to compete with existing carriers. One option would be an auction system. The Ministry of Transport is current considering this approach as well as a lottery system. Exposing the large firms to competitive market forces appears necessary for new carriers to have a chance to succeed. Indeed, experience suggests that greater competition would reduce prices. On the 99 routes reserved for just one airline in 1996, the ticket price was only 4.3 per cent below the government-set standard, compared to a difference of more than 9 per cent on the 38 routes served by two or three airlines (FTC, 1997).

Airline ticket prices remain under government control in Japan, although reforms in recent years have permitted increased diversity in pricing. In December 1994, airlines were allowed to discount standard prices up to 50 per cent, subject to notification of the Ministry of Transport, and, in June 1996, they were permitted to set their regular fares up to 25 per cent below the government standard. Following the introduction of the new system, however, prices were increased on almost half of domestic flights, while less than a third saw a decline (FTC, 1997). The government is now considering a further liberalisation of prices such as a system which only sets maximum prices, to accompany the elimination of the excess supply criteria in FY 1999. At that stage, airlines will be free, to

some extent, to set their own prices as well as to choose their own routes. A first step in this direction was the abolishment in April 1997 of the criteria, based on the number of passengers, for determining which routes can be served by more than one company. The results of the 1996 liberalisation, however, suggest that competition from new carriers may be necessary to reduce ticket prices significantly.

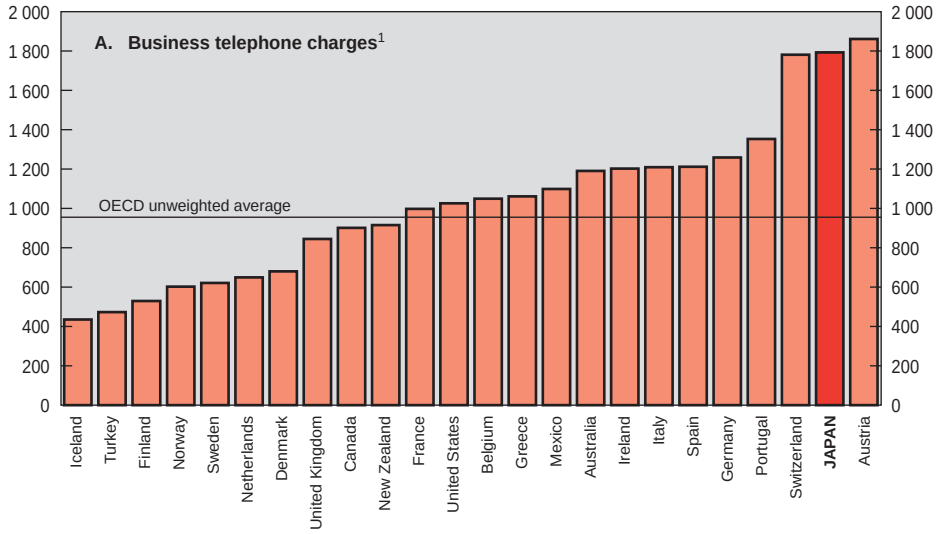
The Ministry of Transport's control over the number of taxis and the fares they charge is also being relaxed. Regulation of this sector may have led to relatively high charges for taxis in Japan and contributed to a decline in the number of passengers from 4.3 billion in 1970 to 2.8 billion in 1996. The Ministry will allow a 10 per cent rise in the number of taxis during FY 1997, while the minimum fleet size of a new taxi company will be reduced to ten cars from the present limit of sixty cars in Tokyo and thirty in Osaka and Nagoya. Over the next three years, the geographic limits on taxi firms will be eased by reducing the number of zones by half. As for fares, taxi firms were given the freedom, beginning in April 1997, to charge up to 10 per cent below the government-set standard. A further liberalisation of fares is planned along with the elimination of the excess supply provision in FY 2001.

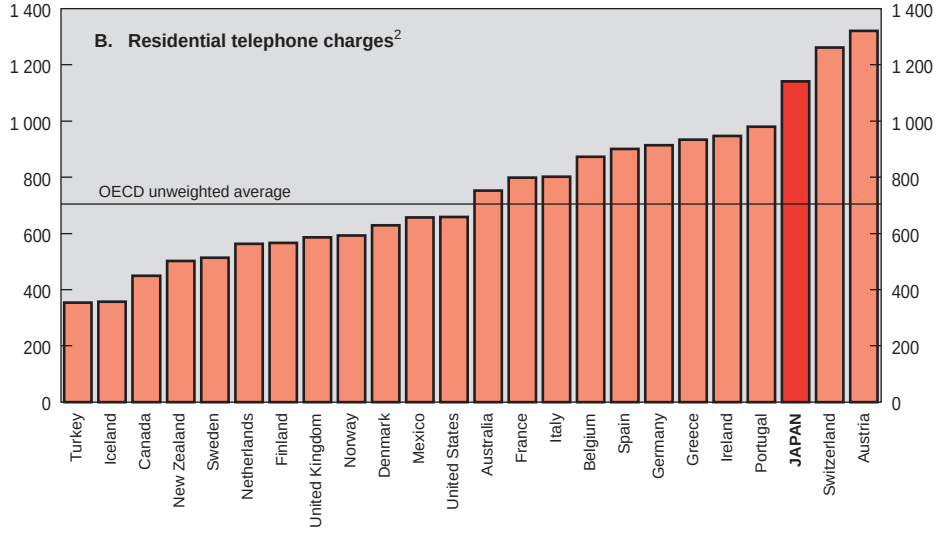
### *Telecommunications*

Despite significant declines in recent years, the prices of telecommunication services in Japan remain relatively high compared to other countries. In 1996, business telephone charges for domestic services were calculated (using market exchange rates) to be 74 per cent above the OECD average, while residential rates were 52 per cent above the average (Figure 27). As for international calls, the price has fallen by almost 40 per cent during the past five years, the second-largest decline in the OECD area.[45] Nevertheless, according to the OECD, international calls from Japan were the most expensive among Member countries in 1996 at 46 per cent above the average (Figure 28). The government plans to promote the reduction of telecommunications charges through measures to enhance competition, such as the restructuring of Nippon Telegraph and Telephone (NTT).

Japan's telecommunications market has been segmented between local, domestic long-distance and international calls. In January 1997, the right to operate in all areas of the market was confirmed for all firms with the exceptions

Figure 27. **TELEPHONE CHARGES IN OECD COUNTRIES**
Expenditure in US dollars in January 1996

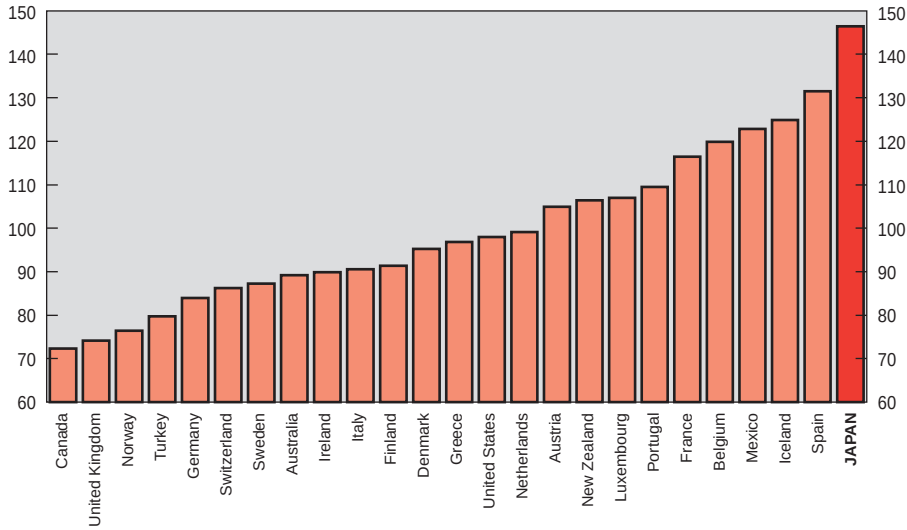



1. Includes 2 911 calls.
2. Includes 1 114 calls.

*Source:* OECD, *Communications Outlook,* 1997.

Figure 28. **THE COST OF INTERNATIONAL TELEPHONE CALLS**[1]
1996, OECD average = 100




1. The average for business and residential use. OECD countries are divided into three regions (Europe, North America and Pacific) and international call destinations are assumed to be distributed equally to each region. The average cost is calculated using current exchange rates.

*Source:* OECD, *Communications Outlook,* 1997.

of NTT and Kokusai Denshin Denwa (KDD). NTT, which is 65 per cent government-owned, is to be restructured into two regional companies and one long-distance provider operating under a holding company, which would be responsible for research and development. The two regional companies, East NTT and West NTT, which are to be special corporations (like the current NTT), will be required to provide universal service in their respective regions. The long-distance provider will be allowed to offer international services as well. Although the three companies will be linked by a holding company, they will report separate accounts and transfers between them will be prohibited. The increased transparency will limit the scope for any cross-subsidisation and put pressure on them to cut costs and improve services. This restructuring, which was approved by the Diet in June 1997, is to be implemented by the end of 1999.[46] Another bill

has been introduced to allow KDD, a private firm which is the major international operator with about 65 per cent of the market, to compete domestically. KDD has already joined an alliance with firms affiliated with electric power utilities to use their networks.

Allowing NTT to offer international services is also intended to maintain its competitiveness in an increasingly global telecommunications market. At the same time, it may strengthen NTT's dominant position in the domestic market, where it accounted for 92 per cent of telephone calls within Japan in FY 1994. The entry of NTT and KDD into each others' markets and the elimination of the barriers separating domestic and international calls are expected to lead to a major restructuring of the telecommunications industry.[47] Moreover, by the end of 1997, domestic carriers will be allowed to connect leased international telephone lines to domestic networks to offer discount international services. In addition, Japan agreed in the World Trade Organisation's negotiations on basic telecommunication services to allow greater foreign investment in domestic firms in this sector, a move which should encourage competition. The 33 per cent limit on foreign ownership will be abolished for the 133 Type I firms, excluding KDD and NTT, where the limit will remain at 20 per cent.

However, these measures alone will not be sufficient to boost competition – it is also necessary to address the ''interconnection'' issue. In Japan, a new firm cannot enter the market without connecting to NTT's local network. For example, the three privately-owned carriers that were allowed to enter the domestic long-distance market in 1987 pay about 40 per cent of their telephone service revenue to NTT to use its local network, a factor that may have helped NTT to maintain a market share of almost 70 per cent.[48] The government recently introduced legislation to ensure that any telecommunications firm can interconnect promptly on fair and transparent conditions. NTT will be required to make public its rates for using its network.[49] If negotiations with NTT fail, the minister is to provide arbitration to reach an agreement.

The entry of new firms into the industry requires a licence from the Ministry of Post and Telecommunications, which judges applications based on certain criteria. The removal of one condition in a bill recently introduced to the Diet – that the proposed investment be ''appropriate in the light of demand'' and not result in excess capacity – should boost competition. Firms will be permitted to enter the telecommunications market subject to the three remaining criteria: the

applicant must have an adequate financial basis and technical capacity, the business plan must be feasible and the planned activities must be appropriate for the sound development of the telecommunications industry. The Economic Council report recommended further measures to promote liberalisation of market entry in this sector. As for price regulations, the prior approval system was abolished, except for services considered to have a ''substantial influence on the daily life of citizens and the national economy'',[50] and replaced by a notification system in October 1995. As a result, the number of prices needing advance approval – a process that can take more than one month – was reduced by half. At the end of 1996, this reform was extended to the rates charged for cellular and most mobile telephones. The Ministry will review the regulation of other rates once the conditions for fair and effective competition are satisfied following the restructuring of NTT in FY 1999.

### *Energy*

The cost of electricity remains high in Japan, at more than double the OECD average for both household and industrial users in 1995, although the gap may have narrowed somewhat since then.[51] The government has set a target of reducing its price to internationally comparable levels by 2001. To achieve this goal, the Electric Utility Council, an advisory body to the government, is considering the introduction of competition in this industry. At present, the nine utilities have regional monopolies to produce and to distribute electricity. One option under consideration is to allow other electricity-producing firms to sell directly to consumers. Since the end of 1995, such firms have been allowed to sell their surplus electricity to the power companies through open tender (see 1996 Survey). A recent study found that the cost of new power sources provided by non-utility firms was 25 per cent cheaper on average than that supplied by the power companies.[52] In FY 1996, the utilities purchased 2 per cent of their total output from non-utility firms, who had offered to sell four times that amount. An April 1997 report by the Fair Trade Commission stated that it is necessary to reform the current system in which the nine regional electric utilities engage in power generation, transmission and distribution.[53] A first step in this direction would be to separate the accounting statements for these three activities and to increase the number of auctions in which the non-utility firms sell surplus electricity to the utilities. Such an approach would encourage entry into the power generation industry by allowing companies to select the most efficient

suppliers of power. These reforms would need to be accompanied by other measures, such as the elimination of the supply-demand balancing clause used to control entry while taking into account other objectives such as the security of supply, environmental protection and universal service. The Electric Industry Utility Council is currently considering these proposals.

The latest deregulation plan includes further measures in the market for gasoline, which, as noted, remains substantially more expensive in Japan than in other countries (Figure 26). *First*, the requirement imposed in 1977 that service stations possess ''Gasoline Purchase Certificates'' from oil refiners is to be abolished by the end of FY 1997. Some refiners have reportedly refused to grant such certificates to supermarkets and other retailers that want to start selling gasoline, fearing that the increased competition would drive down prices. *Second*, service stations' opportunities to purchase oil products from more than one supplier will be enhanced. At present, *keiretsu* networks prevent service stations from buying from other suppliers, a system that protects the interests of oil wholesalers under the guise of ensuring a stable supply. *Third*, the government will study the safety aspects of self-service gasoline stations, a step which could eventually lead to allowing such stations.

### *Housing and land*

The three-year plan includes measures to promote efficient use of land and to lower construction costs. Most importantly, a new zoning system to facilitate the supply of high-rise condominiums in major cities was introduced in June 1997. Under this system, the regulations on the ratio of floor space to land can be raised from four to six in the centres of major cities in FY 1997. These regulations are one of the factors that have limited the number of high-rise buildings: for example, the average building height in the central 23 wards of Tokyo is less than three storeys (see 1994 Survey). The government expects this reform to boost the supply of condominiums in the three major urban areas. In addition, the criteria in the Building Standards Law are to be replaced with clear performance standards, a reform that should encourage flexibility in design and reduce construction costs.

### *The distribution system*

MITI is reviewing the law on the establishment of large stores, which, despite significant liberalisation, continues to limit their size and operating hours. A survey[54] of nine regional offices in FY 1996 reported that, in the course of obtaining authorisation to build, new retail outlets had to reduce their planned floor space by an average of 16.2 per cent, up from 15 per cent the previous year. In addition, 42 per cent of the applicants who wished to stay open past 8 p.m. had to trim their plans, compared with 37 per cent the year before. MITI's review of the law is to be completed by the end of 1997. The government is also taking measures to ensure that the benefits realised by liberalising the nationwide regulations is not offset by local authorities. In addition, beginning in 1998, convenience stores will be allowed to sell certain basic medical products, such as vitamins, which have been restricted to pharmacies. Finally, the monopoly in the salt trade was ended in April 1997.

### *The labour market*

Reforms to ensure flexibility in the labour market are essential in the face of a changing industrial structure and an ageing population. As recommended in the chapter of last year's Survey on *Implementing the OECD Jobs Strategy* (Table 28), the regulations that had limited the activities of private-sector job placement agencies to 29 job categories were relaxed in 1997[55] while the commissions they charge have been set freely, subject to the approval of the Labour Minister, since April 1997. Allowing such firms to play a greater role is particularly important in the context of rising job turnover, especially among younger workers (see 1996 Survey). As for temporary (dispatched) worker agencies, the number of job categories where such workers are allowed was increased from sixteen to 26 in December 1996. Other measures are under consideration, such as extending the maximum length of employment for such workers and giving dispatching agencies freedom to operate in all job categories with certain exceptions. Finally, regulations aimed at protecting female employees by limiting their number of overtime working hours as well as restricting holiday and night-time work are to be abolished in 1999.[56] This is expected to encourage the employment of women. In contrast to the measures noted above, no steps have been taken in implementing the other recommendations made in the *OECD Jobs Strategy*.

Table 28. **Implementing the OECD Jobs Strategy – an overview of progress**

| Jobs Strategy proposal | Actions taken | OECD assessment/recommendations |
|---|---|---|
| **I. Increase the flexibility of employment** | | |
| – allow temporary employment agencies to play a larger role | – The number of job categories allowed to use temporary workers was expended from 16 to 26, with a decision on further measures expected at the end of 1997. | – Further reducing the restrictions on these firms would be beneficial. |
| – make corporate retirement allowances portable | – No action. | – Such a measure would help workers change companies in mid-career. |
| **II. Taxes and transfers** | | |
| – relax the earnings test for pension recipients | – No action. | – The earning test discourages persons over age 60 from working. |
| **III. Active labour market policies** | | |
| – permit private job placement firms to play a more important role | – The number of job categories where these firms are allowed to operate was expanded in 1997 and controls on their commissions were liberalised. | – The number of job categories should be further expanded to help match firms with workers possessing needed skills. |
| – improve the functioning of the Public Employment Security Office | – No action. | |
| – evaluate the effectiveness of active policies | – No action. | – Evaluation of policies in place could identify needed changes to improve their effectiveness. |
| **IV. Enhance product market competition** | | |
| – accelerate deregulation | – Most recent plans appear to speed up the pace of deregulation. | – Removal of entry barriers and price controls would promote the growth of new businesses and employment. |
| – further strengthen competition policy | – The FTC has been given more resources and has begun to widen the scope of its investigations. | – Effective enforcement of competition policy will help realise the benefits of deregulation. |

Table 28. **Implementing the OECD Jobs Strategy – an overview of progress** *(cont.)*

| Jobs Strategy proposal | Actions taken | OECD assessment/recommendations |
|---|---|---|
| **V. Increase labour force skills** | | |
| – improve the quality and availability of training outside of firms | – No action. | – Improvements in education and training are important in the context of more workers changing companies and less on-the-job training. |
| – establish a systematic recognition of qualifications | – No action. | |
| – improve the education system | – A measure to allow parents more choice in selecting schools for their children is under consideration. | |
| **VI. Technology and innovation** | | |
| – Increase the emphasis on basic research and improve the efficiency of public spending on technology | – No action. | |

*Source:* OECD.

### *Competition policy*

Enforcement of the Anti-Monopoly Act is to be strengthened as part of the revised deregulation plan. This commitment is backed by the provision of additional resources to the Fair Trade Commission (FTC) at a time of tighter controls on public expenditure. The FTC's budget is scheduled to increase 3.3 per cent in the context of a small 1.5 per cent increase in government spending, while the size of the investigation staff will expand by 12 to 248. The increase in the size of the FTC in recent years has been associated with a growing volume of surcharges imposed on violators of the Anti-Monopoly Act (Table 29).

One recent initiative in this area is the FTC's decision to conduct a fact-finding survey of so-called ''private regulations'', such as certification and standardisation systems. Such regulations, which are often handled by trade associations consisting of private firms, may impede free competition by preventing newcomers from entering markets.[57] The FTC will seek to correct such practices that restrict competition and will co-ordinate with other ministries in cases where anti-competitive administrative guidance is involved. In May 1996,

Table 29. **Activities of the Fair Trade Commission**

| | 1992 | 1993 | 1994 | 1995 | 1996 | Total |
|---|---|---|---|---|---|---|
| **A. Legal actions** | | | | | | |
| Private monopolisation | 0 | 0 | 0 | 0 | 1 | 1 |
| Cartels | 30 | 24 | 21 | 24 | 15 | 114 |
| *of which:* | | | | | | |
| Price cartels | 9 | 9 | 1 | 4 | 10 | 33 |
| Bidding cartels | 20 | 14 | 19 | 20 | 5 | 78 |
| Unfair trade practices | 4 | 5 | 1 | 4 | 2 | 16 |
| *of which:* | | | | | | |
| Retail price maintenance | 0 | 4 | 1 | 1 | 1 | 7 |
| Non-price constraints | 4 | 0 | 0 | 2 | 0 | 6 |
| Other | 0 | 2 | 2 | 3 | 3 | 10 |
| Total | 34 | 31 | 24 | 31 | 21 | 141 |
| Criminal cases[1] | 1 | 0 | 1 | 0 | 1 | 3 |
| **B. Surcharges ordered by the FTC** | | | | | | |
| Number of enterprises | 135 | 406 | 512 | 741 | 368 | 2162 |
| Amount of surcharges (billions of yen) | 2.7 | 3.6 | 5.7 | 6.4 | 7.5 | 25.8 |

1. The FY 1996 criminal case involved accusations filed against 25 companies and 34 individuals in a bid-rigging case regarding water meters ordered by the Tokyo Metropolitan Government.

*Source:* Fair Trade Commission.

for example, the FTC decided against a health industry organisation in charge of certification, which was found to prevent new firms from entering their business. In April 1997, the Commission also ended its exemptions which had allowed the retail price maintenance system for 28 cosmetic products and non-prescription medicines. The remaining exemptions for copy-righted materials will be limited and clarified by March 1998.

Under the three-year plan, the FTC is also taking legal steps to reduce the types of business cartels that are exempted from the Anti-Monopoly Act. Such exemptions have allowed, for example, export cartels, which have been used to limit overseas shipments in line with voluntary restraint agreements, and anti-depression cartels in declining industries.[58] In addition, the FTC is planning to introduce a notification limit for mergers in 1998. The threshold will be Y 10 billion in assets for the first company and Y 1 billion for the second. Under this criterion, most of the mergers in Japan will no longer need advance notification. This reform should facilitate the restructuring of Japanese firms while allowing the FTC to focus on those mergers likely to impinge on the degree of competition.

In June 1997, the Diet repealed the ban on holding companies, which was imposed 50 years ago to prevent a re-emergence of the pre-war conglomerates known as *zaibatsu*. The decision to permit such companies[59] is intended to make the Japanese market more open and to facilitate the business activities by making it easier to create or buy firms. Moreover, it may encourage smaller, better-focused and more independent business activities. Another advantage is that it will give corporations more flexibility in adjusting employment and wage levels in different parts of their operations, depending on their performance. The legislation, however, includes several restrictions on the formation of holding companies. First, holding companies with firms in at least five different industries and with combined assets of more than Y 15 trillion will not be permitted.[60] Second, holding companies which include a financial institution with total assets of more than Y 15 trillion will not be able to own enterprises outside the financial sector. Third, holding companies with at least five different subsidiaries will not be allowed if their business areas are inter-related and they hold substantial market shares in these areas.

To be fully effective, the legislation permitting holding companies should be accompanied by a well-constructed consolidated taxation system, which would

allow a holding company to offset the losses of one subsidiary against the tax liabilities of another. Otherwise, the introduction of holding companies might actually increase the corporate tax burden compared to the present system, which allows profits and losses to be consolidated within an enterprise. The Tax Commission, an advisory body to the Prime Minister, urged the government to examine the issue cautiously, taking into account the problem of tax evasion and the impact on tax revenue, among other issues. In addition to tax issues, there are labour market concerns. Nikkeiren, which represents Japanese employers, has agreed to hold consultations with Rengo, the largest labour union, to discuss protection for workers under the new system. Unions want the right to negotiate wages and other issues directly with the holding company rather than with its subsidiaries.

### *The Administrative Procedures Law*

This law, which was implemented in October 1994 (see 1995 Survey), was intended to make the administrative decision-making process more fair and transparent. A 1996 survey by the Management and Coordination Agency on the impact of this law found 24 instances since 1994 where the content of administrative guidance had been made available to the public.[61] The survey also examined the government's handling of applications for regulatory permits from the private sector. According to the law, administrative agencies should, in principle, estab-

Table 30. **The central government's handling of applications for regulatory permits**
March 1996

| Ministry responsable | Number of regulatory permits surveyed[1] | No concrete criteria available (per cent) | No standard timetable for responding to application (per cent) |
|---|---|---|---|
| International Trade and Industry | 792 | 7.4 | 21.0 |
| Health and Welfare | 559 | 20.6 | 39.4 |
| Agriculture, Forestry and Fishery | 554 | 4.9 | 7.8 |
| Transportation | 514 | 7.6 | 22.6 |
| Finance | 414 | 24.4 | 33.3 |
| Other | 915 | 12.0 | 29.8 |
| Total | 3748 | 12.0 | 25.5 |

1. Applications handled by local offices of the ministries are not included. Of the 865 surveyed, 5 per cent had no concrete criteria and 12 per cent had no standard timetable.
*Source:* Management and Co-ordination Agency.

lish clear criteria and a standard timeframe for completing the process. It found that no definite criteria for evaluating applications existed in 12 per cent of the cases and no standard timetable in 25 per cent (Table 30). The results varied widely by ministry, with the applications administered by the Ministry of Finance least likely to be subject to clear criteria and timetables.

### *The links between the private and public sectors*

The decline in the number of retiring civil servants taking private-sector jobs in 1995 and 1996 may also suggest a changing relationship between the government and the private sector. Although the law officially bans civil servants from taking jobs within two years after retirement at firms under the supervision of their ministry, the National Personnel Authority can waive this waiting period at the request of the ministry concerned. *Amakudari* – the employment of senior-level retired civil servants – may have been a way of strengthening links between government ministries and the firms that they regulate. The number of *amakudari* retirements, which peaked at 318 in 1985, remained about 200 between 1992 and 1994. In 1995, though, it fell to 189 before dropping to 134 in 1996 (Table 31). However, these figures do not include retirees who take up posts at government-owned or affiliated corporations. According to one survey, as many as three-fourths of all executives at the approximately 90 government-backed organisations were retired civil servants.

Table 31. **Number of retiring civil servants taking private-sector jobs**

| Ministry or Agency | 1995 | 1996 |
|---|---|---|
| Finance | 59 | 27 |
| International Trade and Industry | 17 | 17 |
| Agriculture, Forestry and Fisheries | 16 | 7 |
| Construction | 16 | 13 |
| Posts and Telecommunications | 13 | 5 |
| Transport | 11 | 14 |
| Education | 9 | 15 |
| Maritime Safety | 9 | 18 |
| Health and Welfare | 4 | 0 |
| Other | 35 | 18 |
| Total | 189 | 134 |

*Source:* National Personnel Authority.

## Scope for further action

The deregulation process in Japan has been widely criticised for its slow pace. Reform plans have included large numbers of minor technical measures, while failing to address the key entry and price restrictions. During the past year, however, the pace of deregulation appears to have picked up with the announcement of important steps, such as the ''Big Bang'' programme in financial markets, the reform of telecommunications, including the restructuring of NTT, and the Ministry of Transport's commitment to abolish excess supply criteria for allowing entry. Moreover, there appears to be greater recognition of the effect of regulations in raising prices in Japan. Progress could be further accelerated by establishing clear objectives and timetables for removing remaining regulations that limit competition. In addition, the continuation of the Administrative Reform Committee, or the creation of a similar independent group, to oversee the deregulation process would help increase the pace of reform. Moreover, as noted in the *OECD Report on Regulatory Reform*, deregulation should be accompanied by stronger enforcement of competition policy.

The potential productivity gains from further deregulation appear significant in Japan. An OECD study of the long-term effect of liberalisation of five sectors – electricity, air and road transport, telecommunications and distribution – estimated that output might rise by 5.9 per cent (Table 32). Such an increase would be larger than the 3½ to 5 per cent gains estimated for liberalisation of the same sectors in three major European countries. Studies by the EPA and MITI on the benefits of regulatory reform in Japan also project output gains of about 6 per cent. These studies, which include sectors not covered in the OECD report, generally assume that deregulation allows productivity and prices in key non-manufacturing sectors in Japan to converge to international levels. Implementing far-reaching reforms, such as eliminating entry barriers and price regulations, as recommended by the Economic Council, would help Japan obtain these potential benefits.

Such output gains would probably be accompanied by large shifts of labour between sectors. According to the EPA study, employment could decline significantly in the distribution and financial sectors. This would be offset, though, by a greater number of jobs in the service, manufacturing and construction sectors. Increased labour market flexibility would be essential to help accommodate such

Table 32. **Estimated economic effects of deregulation**

| | OECD[1] | MITI[2] | EPA[3] |
|---|---|---|---|
| Real output | 5.9 | 6.0 | 5.8 |
| Consumer prices | .. | –3.4 | –7.3 |
| Change in unemployment | 0.0 | .. | 0.0 |

1. OECD (1997) estimates the long-run effect of changes in five sectors: electricity, air and road transport, telecommunications and distribution.
2. This study by MITI calculates the impact that deregulation of transportation, energy, telecommunications, finance and distribution would have over the period 1995 to 2001.
3. The EPA estimates the effect over the period of 1998 to 2003 of deregulation in eight fields: finance, telecommunications, transportation, distribution, energy, land and housing, health and employment and labour. It projects a decline in net exports would reduce Japan's current account surplus by 0.9 per cent of GDP.

*Source:* OECD (1997), *The OECD Report on Regulatory Reform*, Vol. II, MITI and EPA.

shifts. The EPA study notes the importance of reducing controls on private employment and temporary worker agencies to increase information flows and thus tend to minimise any increase in unemployment. Such a reform would need to be based on a consensus between employers and labour unions. Restructuring may also be aided by the recent decision to lift the ban on holding companies. It is important, though, to introduce a well-designed consolidated taxation system in order to make this reform really effective.

Perhaps the most ambitious reforms are planned in the financial sector, which will be profoundly affected by liberalisation of foreign exchange operations. Given the greater opportunities to conduct financial business overseas, prompt implementation of the ''Big Bang'' programme of reform will be necessary to maintain the competitiveness of Tokyo as a financial centre. However, the plan does not address key issues in several areas. One of the most important is taxation, most notably with respect to equity transactions, which is one of the factors which makes Japan more expensive. This issue is currently under review by the government. Second, the role of public financial institutions, such as the postal savings and the FILP institutions (see Chapter III), has not been addressed.

In telecommunications, the plan to divide NTT into three firms headed by a holding company may help reduce its dominant role in the domestic market. Moreover, the entry of international and domestic long-distance providers, including NTT and KDD, into each others' markets should increase competition and lower the cost of telephone service in Japan. It is also important to implement a transparent and fair interconnection framework to allow new firms to connect to

NTT's local network. Moreover, the remaining criteria for licensing new firms should not be used to generate entry barriers.

The removal of the excess supply criteria for entry to the transport sector over the next three to five years should also be beneficial. However, this should be accompanied by the removal of regulations, such as those on prices and area of service, which discourage potential entrants, notably in the taxi and trucking businesses. Moreover, the minimum fleet size requirements on these two industries should be phased out (see 1996 Survey). In air transport, the entry of new carriers could help reduce the high price of air travel. However, given the shortage of additional take-off and landing slots at airports in Tokyo and Osaka, a more transparent approach to allocate these slots is needed to encourage the entry of additional airlines. Reforms in the gasoline market will probably allow service stations more opportunities to purchase oil products from more than one supplier. Permitting self-service stations could also help lower the relatively high price of gasoline in Japan. Measures to increase competition in the electricity industry, such as ending regional monopolies, might reduce prices, which remain high compared to other OECD countries. In addition, reforming the current system, in which electric utilities engage in the generation, transmission and distribution of power should help reduce prices. In the distribution sector, the Large-Scale Retail Store Law continues to restrict the size and operating hours of new large stores. Phasing out this law would help continue the rapid increase in the number of larger, more efficient retail outlets seen during the past five years. Finally, allowing the incorporation of businesses in agriculture and health care, a proposal that was not included in the most recent deregulation plan, would help increase competition in those sectors.

Finally, government intervention in the economy should be limited. Moreover, improving the transparency of government regulations would reduce their burden on the private sector. In particular, administrative agencies should endeavour to establish clear criteria and publicly-announced deadlines in handling private-sector applications. Moreover, discouraging the employment of retiring civil servants in the firms they once regulated would help improve the transparency of relations between the government and the private sector.

# V. The sustainability of the social security system

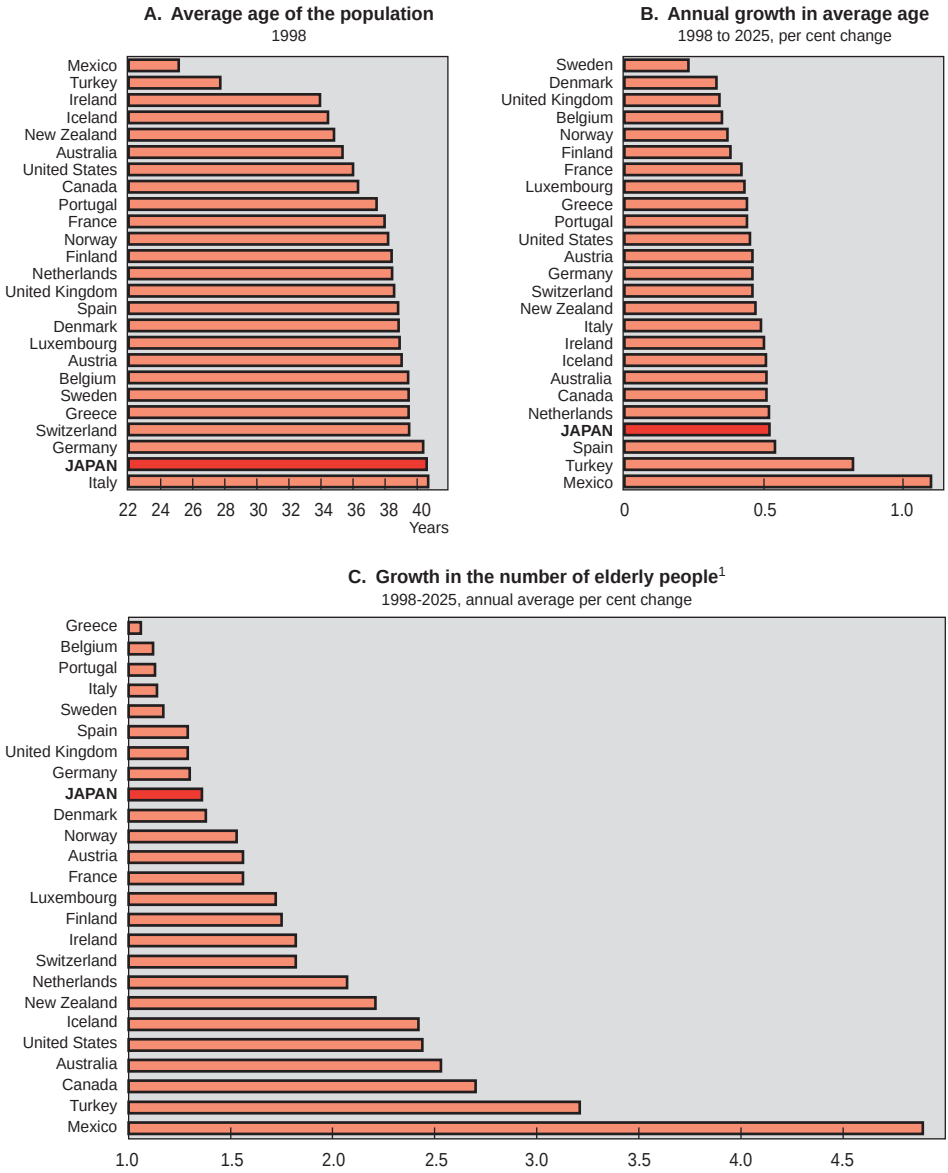
Population ageing is a problem facing Japan along with many other OECD countries. In the past decade, the Japanese population has already become the second most elderly in the OECD area and will continue to age, though less rapidly. Moreover, Japan's birth rate is too low to sustain the working-age population at its present level. By raising government spending and lowering social contributions, such developments will put additional pressure on public finances and call into question the sustainability of the social security system. This chapter first reviews the scale of the problem in the light of Japan's demographic trends and their impact on saving, investment and economic growth. It then considers the implications for the pension, medical insurance and welfare systems. Policy options are examined in the concluding section.

## The scale of the problem

During the last three decades Japan has become a more elderly society, with the average age of its population increasing from 31 years in 1970 to about 40 years currently. At present, only one OECD country – Italy – has a more elderly population than Japan (Figure 29, Panel A). In the period to 2025, the Japanese population will continue to age, though at a slightly slower pace (Panel B). The absolute increase of 5¼ years in the age of the Japanese population to an average of 45 years in this period is likely to be somewhat above the OECD average reflecting its already high average age.[62] By 2025, the average age of the population in Japan will be the highest in the OECD area. On average, though, the number of elderly people will increase somewhat less rapidly than in other OECD countries (Figure 29, Panel C).

Defining elderly as people over 60 and the working-age population as those between 20 and 59,[63] Japan's dependency ratio (*i.e.* the ratio between these two

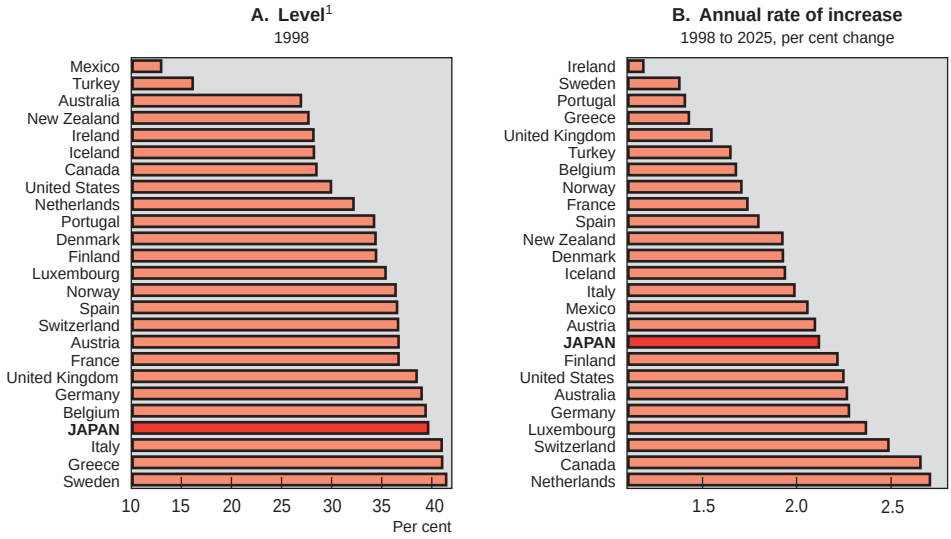
Figure 29. **POPULATION AGEING: AN INTERNATIONAL COMPARISON**




1. People aged 60 and over.

*Source:* Ministry of Health and Welfare and Bos (1994).

Figure 30. **RATIO OF ELDERLY TO YOUNGER PEOPLE**




1. Ratio of population aged 60 and over to that aged 20 to 59.
*Source:* Ministry of Health and Welfare and Bos (1994).

variables) is the fourth highest in the OECD area in 1998 (Figure 30, Panel A). While the elderly population is increasing, the number of younger people is declining and this has already led to a reduction in the working-age population, which tends to reduce income growth. In the period to 2025, this ratio will increase by 2 per cent per year, though less rapidly than in some other OECD countries, including the United States. However, given the high initial level, the ratio in Japan will be the third highest in the OECD area by then.

Official Japanese projections (Table 33),[64] which assume a fertility rate below that required to stabilise the population, show a slowdown of the ageing process in the period 2025 to 2050, with the number of elderly people expected to fall slightly. However, the working-age population is projected to decline by 1 per cent annually. As a result, the average age of the population may rise as high as 48. Meanwhile, the total population, which was 126 million in 1996, is projected to start declining after 2007 and to fall to 100 million by 2050, a decline of more than 20 per cent in half a century.

Table 33. **Main assumptions underlying the official population projections**

| | Old projections (1992) | New projections (1997) | Date at which assumption is reached |
|---|---|---|---|
| Fertility rate | 1.8 | 1.6 | 2025/2050 |
| Age at first marriage | 27.4 | 27.4 | 2050 |
| Children per marriage | 2.1 | 1.6 | – |
| Women never marrying | 11.0 | 13.8 | – |

*Source:* Ministry of Health and Welfare.

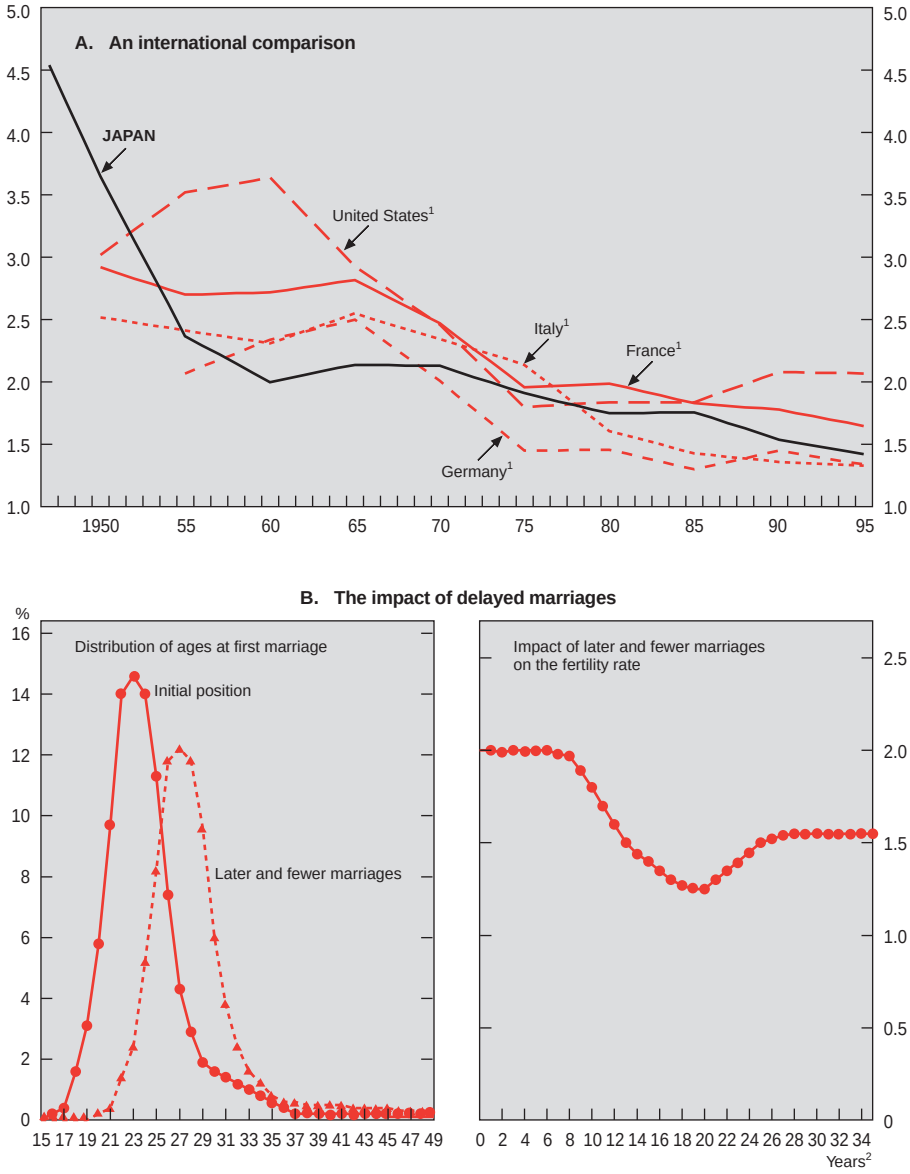
The expected decline in the population is due to the considerable fall in the birth rate in the post-war period. The fertility rate[65] came down from 4.5 in 1947 to 2.1 in the 1960s, a level sufficient to maintain a stable population. However, by 1995, it had fallen to only 1.4, a rate similar to those in continental Europe (Figure 31, Panel A). In contrast, the fertility rate in North America has been rising for fifteen years and went above 2 in the early 1990s.

The initial post-war decline in the fertility rate was mainly due to a drop in the average number of children per family, reflecting a shift of the population from agriculture to the service sector, and from rural to urban areas. During the past 25 years, though, the decline has been mainly due to later marriage. As the number of children born to unmarried couples or to single mothers is negligible in Japan,[66] the fall in the proportion of married couples has gone together with a corresponding decline in births. Although the fertility rate for married women remains slightly above 2, about 30 per cent of women in the 15 to 49 age group are not married.

The major reasons for delayed marriages include the rising enrolment of women in higher education and their expanding job opportunities in urban areas. In the Tokyo metropolitan area, the average age of a woman's first marriage is now more than two years above the national average, while that for university graduates is almost three years above the average.[67] Given the sharp rise in the proportion of women attending university and the growing share of the urban population, the trend towards later marriage is likely to continue in the future.

According to the official population projections, the fall in the fertility rate is, to some extent, temporary and will be partially reversed by a rise in the birth rate once the marriage age has increased to a new equilibrium level (Figure 31,

Figure 31. **FERTILITY RATES**

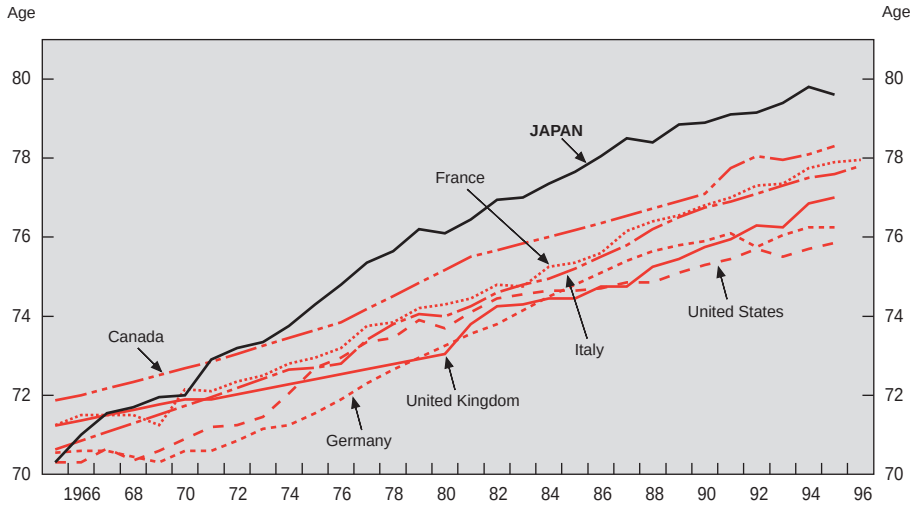



1. Last available date: 1991 for USA, 1992 for Italy and 1994 for Germany and France.
2. Number of years after the initial delay in marriage.

*Source:* Institute of Population Problems, Ministry of Health and Welfare.

Panel B). However, the projections also assume that only 86 per cent of women born in 1980 and after will ever get married and that consequently the fertility rate will remain below the replacement level at 1.61 by 2050. In addition to the fall in fertility, there has been a sharp decline in mortality rates in Japan which now has the longest life expectancy of all major industrial countries (Figure 32). Average life expectancy for males has increased from 50.1 years in 1947 to 76.4 years in 1995 and for females from 54.0 years to 82.8 years during the same period.[68] Initially, increased life expectancy was due to a sharp fall in the mortality rate among children under the age of one, thanks to improvements in nutrition and sanitary conditions. More recently, better health care for the elderly has also significantly increased life expectancy at retirement age and beyond. In 1994, the average remaining life expectancy at age 65 was 16.7 years for men and 21 years for women, compared with an average of 14.9 and 18.6 years, respectively, in the remainder of the OECD area.

Figure 32. **AVERAGE LIFE EXPECTANCY IN OECD COUNTRIES**




*Source:* OECD, Health Database.

## Macroeconomic consequences

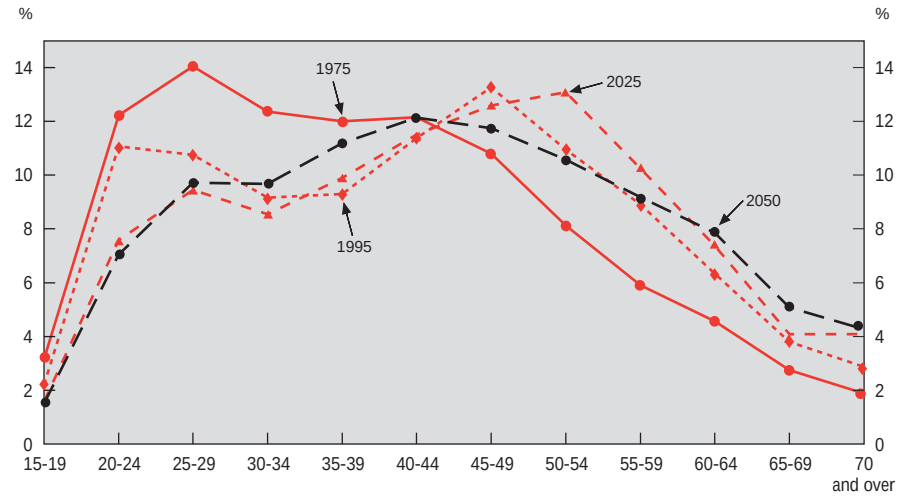
The ageing of the population will have a number of far-reaching implications for the economy. Most notably, a low birth rate will reduce the growth of the working-age population and so will tend to lower the growth of the economy and the level of investment. An increasing proportion of elderly in the total population is also likely to affect saving. While the full interaction of all of these forces is difficult to assess, their main implications are discussed below.

### *Impact on the labour supply*

By 1995, the decline in fertility rates had already started to reduce the working-age population.[69] Over the period 1995 to 2025, the working-age population will fall at an annual rate of 0.7 per cent, with the decline accelerating to 1.0 per cent in the following 25 years.[70] Consequently, the average age of the working population, which increased from 39.8 in 1975 to 42.9 in 1995, will rise further to 45 by 2025 before stabilising. The increase in the proportion of workers over age 50 in the labour force during the past 20 years (Figure 33) has already put strains on the seniority pay system, leading to a less-pronounced rise in wages as age increases. The rise in the proportion of the workforce over age 50 should slow in the period to 2025 and then begin to decline back towards its 1995 level, as the impact of the assumed rise in the fertility rate from 1.4 to 1.6 is felt.

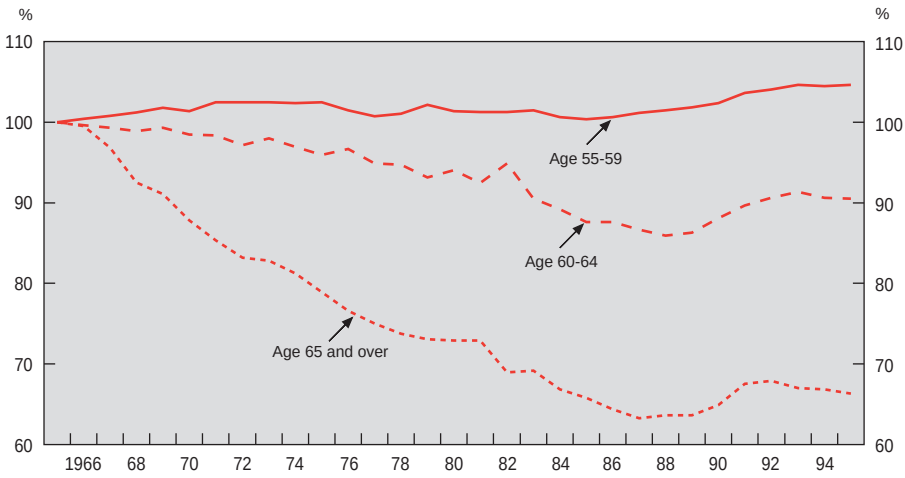
Thus far, the fall in the working-age population has been offset by increasing labour force participation by women and older workers. The participation rate of men over 60, which had fallen until the mid-1980s in line with the trend in other OECD countries, has been reversed in Japan in recent years (Figure 34). Between 1987 and 1995, the participation rate of men aged between 60 and 64 rose from 71.7 to 74.9 per cent, while for those over age 65, it increased from 35.6 to 37.3 per cent. However, these aggregate figures[71] mask the full extent of the change that has occurred. Excluding the self-employed, the participation rate of male employees in the age group 60 to 64 rose from 38 to 46 per cent, while that for those above 65 increased from 13 to 17 per cent. While the trend of delayed retirement was initially attributable to cyclical factors, it has continued during the period of sluggish growth in the 1990s, suggesting that it represents strong supply-side pressure. Overall, the OECD estimates that the rising number of women and elderly people in the labour force has raised the trend participation

Figure 33. **THE CHANGING AGE STRUCTURE OF THE LABOUR FORCE**
Per cent of the total




*Source:* Management and Coordination Agency.

Figure 34. **THE CHANGE IN THE LABOUR FORCE PARTICIPATION RATES OF OLDER MEN**
Index, 1965 = 100




*Source:* Management and Coordination Agency, *Labour Force Survey.*

rate by about 0.8 per cent per year over the past decade. Although this trend may slow somewhat in the future, it may reduce the fall in the labour force to as little as 0.3 per cent annually in the period to 2025. However, the decline is likely to accelerate in the following quarter century.

### *Impact on future growth prospects*

The prospects for future growth depend on the balance between the expansion of the labour force and the increase in labour productivity. The growth of the latter will depend on the balance between the rise in total factor productivity and the increase in capital intensity. Overall productivity growth, after accelerating from the mid-1980s, has now dropped back to the rate seen in the 1970s and early 1980s. On the other hand, the increasing trend in capital intensity has been more stable. If the increase in total factor productivity and capital intensity continues at its current pace, the growth of real incomes of the employed would remain unchanged. Nonetheless, the overall pace of expansion of the economy will slow down, in line with the fall in the labour force. Indeed, it seems likely that economic growth may decline from just under 3 per cent during the past decade to only 1.5 per cent in the decades beyond 2000 (Table 34), reflecting expected developments on the supply side. The slowdown is primarily due to a decline in the labour force, which leads to a marked slowing in the growth of the capital stock. Capital per worker and technical progress, though, continue to increase steadily. In addition, the projections include a further decline in the

Table 34. **Factors contributing to economic growth**

Annual change, per cent

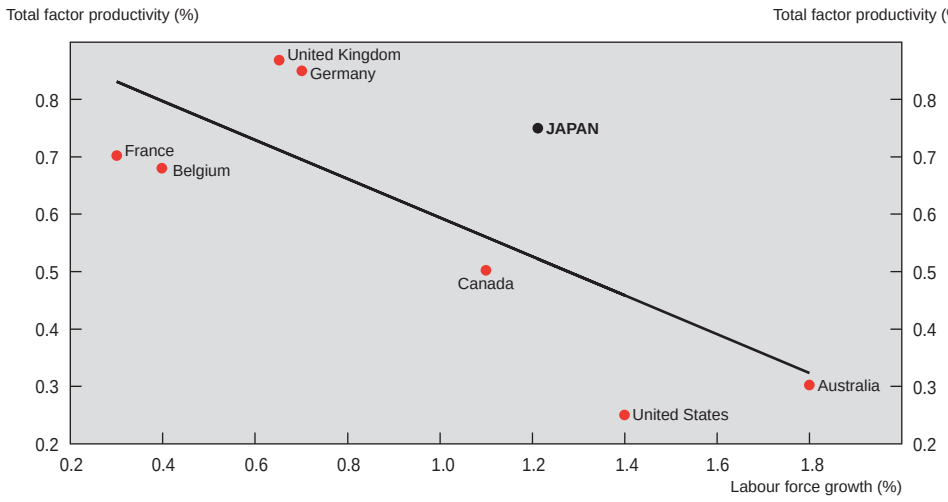
| | | 1987-1997 | 1998 | 2000-2025 | 2025-2050 |
|---|---|---|---|---|---|
| Capital stock | | 4.8 | 4.3 | 2.4 | 1.6 |
| Capital per worker hour | | 4.6 | 2.9 | 3.0 | 3.0 |
| Technical progress | | 1.2 | 0.9 | 0.9 | 0.9 |
| Hours worked | | –0.8 | 0.4 | –0.3 | –0.3 |
| Employment | | 1.0 | 1.0 | –0.3 | –1.1 |
| GDP | | 2.9 | 2.8 | 1.5 | 1.0 |
| Capital output ratio | (level) | 2.1 | 2.3 | 2.6 | 2.8 |
| Net business investment | (per cent of GDP) | 10.2 | 10.1 | 6.2 | 4.5 |

*Source:* OECD.

average hours worked per year, as changing life patterns generate a demand for shorter working weeks and longer holidays.

The projections assume that total factor productivity growth remains constant. However, there are two factors which could accelerate the rise in productivity and so offset the impact of a falling labour force. *First*, deregulation and other policies to stimulate competition (see Chapter IV) should improve efficiency, particularly in Japan's large non-manufacturing sector, which has a low productivity level compared to the Japanese manufacturing sector and the service sectors in other countries. *Second*, technical progress that boosts labour productivity may be stimulated by the fall in the labour supply. The experience of major OECD countries shows a correlation between high labour force growth (in countries such as the United States and Australia) and slow increases in total factor productivity, while the opposite was true in major European countries (Figure 35).[72] On the other hand, to the extent that technical progress only occurs through new investment, a slowdown in output growth might reduce the increase in productivity.

Figure 35. **GROWTH OF TOTAL FACTOR PRODUCTIVITY AND THE LABOUR FORCE**




*Source:* Yashiro and Oishi (1993).

### *Impact on investment*

The future path of domestic investment will depend on the extent of capital formation needed to raise capital intensity and the investment necessary to match the change in the labour force. On balance, slower economic growth will be associated with a significant deceleration in the growth of the business capital stock. If business sector fixed assets per unit of labour input continue to increase at a stable rate while the labour force declines, the growth of the capital stock would slacken from around 4¾ per cent in the past ten years to 2½ per cent, according to OECD estimates. Given the current level of the capital-output ratio, the share of net investment in GDP might well fall by more than 4 percentage points. The decline in the population might also lead to a stabilisation in the housing stock, though continued reductions in the average household size might partially offset this effect. In such circumstances, net housing investment could fall to zero and land prices would likely decline.

A major uncertainty about the macroeconomic effect of ageing on investment relates to the quality of labour. The average age of workers will continue to rise in the next 30 years as more elderly people remain in the workforce. In addition, with more women restarting work with markedly less on-the-job training, the labour force may become less homogeneous. This may stimulate labour-saving investment and increased use of new information-technologies in order to utilise better these new segments of the labour force.

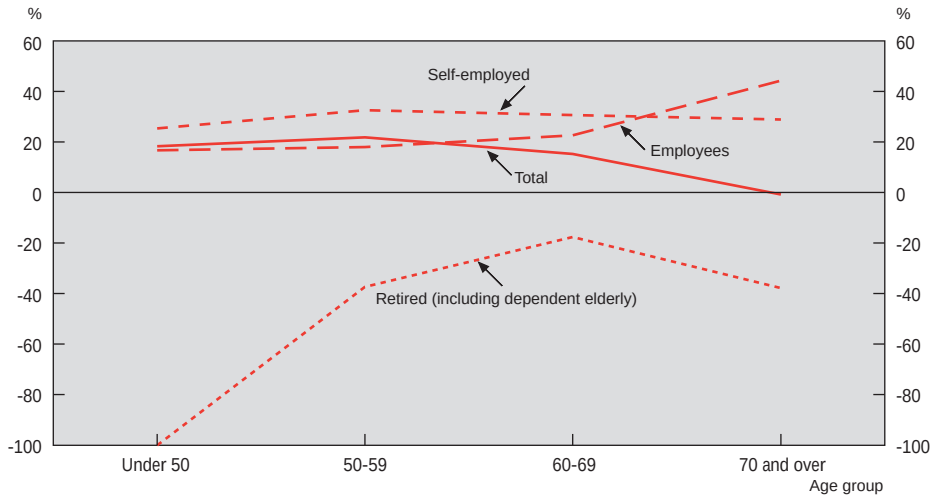
### *Impact on saving*

#### *Households*

One of the most important economic effects of population ageing will be on the household savings rate. A simple life-cycle model of consumer behaviour assumes significant dissavings by the elderly, who run down their stock of assets following retirement. Such a model would predict a large fall in the household saving rate as the population ages. Indeed, one analysis[73] suggested that the rate would become negative in Japan after 2005, although other studies[74] predict a smaller decline.

Microeconomic data on household savings behaviour in Japan, however, appear to be inconsistent with the simple life-cycle hypothesis. Several studies have found that elderly people continue to save until they are very old and leave

Figure 36. **SAVING RATES OF DIFFERENT CATEGORIES OF HOUSEHOLDS**
Per cent, 1989




*Source:* EPA, *Economic Survey of Japan,* 1994-95.

a considerable portion of their accumulated wealth as bequests.[75] For example, in the annual household survey, the elderly have savings rates as high as 50 per cent, well above the level of younger households (Figure 36). However, this source only includes the elderly head of households who continue to work – a relatively high-income group. To obtain a global picture, it is necessary to consider the behaviour of the elderly who are self-employed[76] or retired and those who live with their children. The latter group are likely to have a high dissaving rate as they are relatively poor.[77]

It is possible to calculate an aggregate saving ratio for the elderly by combining the estimated rates for those who live with their children with data on those who are self-employed, retired or working as employees. Such a calculation shows that the saving rate for those over 60 was about 8 per cent of income (Table 35), less than half the figure for those under 60.[78] Moreover, the rate was negative for those over 70. Consequently, the growth in the proportion of elderly households should lower the overall savings rate. However, the impact will be

Table 35. **Factors influencing the savings rate, 1995-2005**

| Age of household head | Age-specific saving rate | Household structure | | Relative incomes | Age-specific saving rate |
|---|---|---|---|---|---|
| | 1995 | 1995 | 2025 | 1995 | 2025 |
| 25-49 | 18.3 | 50.7 | 36.2 | 109 | 18.3 |
| 40-59 | 21.8 | 18.7 | 18.8 | 138 | 21.8 |
| 60-69 | 15.3 | 17.1 | 16.9 | 80 | 15.3 |
| 70 plus | –0.8 | 13.5 | 28.1 | 39 | –0.8 |
| Overall saving rate | 17.8 | 100.0 | 100.0 | 100 | 16.5 |

*Source:* OECD, EPA, World Bank.

moderated by the fact that the elderly's share of income will rise less rapidly than their share in the total number of households.[79] Overall, if the saving rate of each age group were to remain constant from 1995 to 2025, population ageing would reduce the aggregate household savings rate by 1.3 per cent of income, an amount similar in magnitude to the standard deviation of the annual change in the household savings rate over the past 34 years.

In reality, the saving rate of each age group is likely to change as the population ages. A continued decline in the prevalence of three-generation families (with negative saving rates) and a rise in the labour force participation of the elderly (with high saving rates) will tend to raise overall household savings formation. More fundamentally, younger people may save more given the uncertainty about the future of the public pension scheme in the context of increasing life expectancy. This concern may also motivate people to save more to provide security in old age.[80] However, these factors, which tend to increase saving, may be offset to some extent by falling family size. Given the importance of intergenerational transfers and bequests, it is possible that saving by children may decline as inheritances per child increase.[81] On the other hand, the impact of lower family size on the saving of the parents is not clear.[82]

### *Corporate pension funds*

The build-up of assets in occupational pension schemes may partially compensate for the decline in household saving rates. The development of funded schemes did not begin until the 1960s in Japan. Moreover, their coverage expanded rapidly during the 1980s. Consequently, these schemes are far from

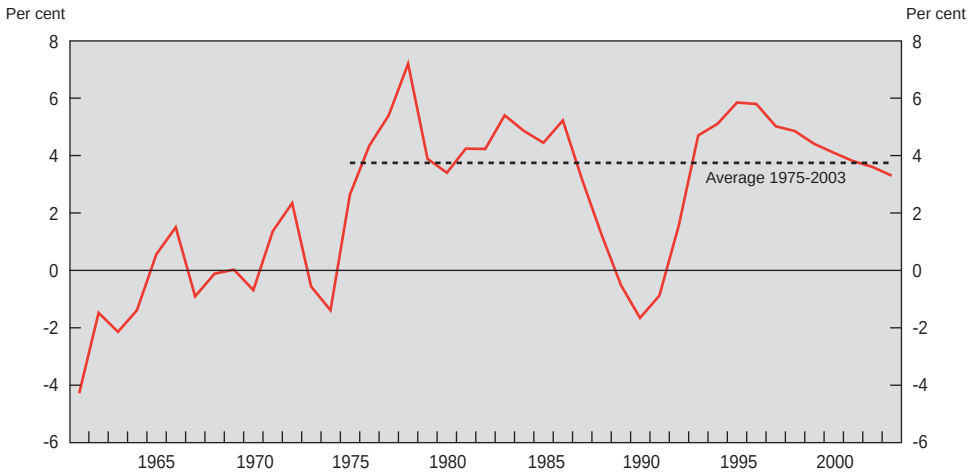
reaching maturity and are likely to accumulate assets at an accelerating pace until 2015. Overall, corporate pension fund assets may rise from 8 per cent of GDP to 24 per cent by 2025,[83] representing an annual growth of about ½ per cent of GDP. Such an increase might offset a significant part of the decline that may occur in the household saving rate, on the assumption that households have already incorporated future asset accumulation by private pension funds into their current saving plans.

In sum, ageing seems likely to lower the household savings rate. However, the magnitude of the decline will be limited by a number of factors, such as increased uncertainty over the future of the public pension scheme, which could quite easily offset the impact of the change in the population structure.

### *Impact on the external balance*

Developments in the current account will depend on movements in both the public-sector deficit and the private-sector savings-investment balance. The latter rose markedly at the end of the high growth era, showing on average a surplus of about 3¾ per cent of GDP since 1975, with considerable cyclical swings (Figure 37). At present, this surplus is even higher due to the weakness of the current recovery. Medium-term projections by the OECD suggest that it will gradually fall to around 3½ per cent of GDP as the economy recovers and the government deficit is reduced. The movement of this balance beyond the year 2000 is less clear as it will be influenced by a marked slowdown in economic growth, a decline in the size of the working population and an increase in the number of elderly. The discussion above suggests that investment may fall more than saving, a result that appears to be consistent with past movements in the private-sector savings-investment balance. However, small changes in assumptions about future movements in saving ratios for different age groups or about movements in capital intensity would result in markedly different projections of the private-sector savings-investment imbalance and so such extrapolations are subject to wide margins of uncertainty. Indeed, a number of studies suggest a marked drop in this balance, mainly because they assume a marked impact of ageing on the aggregate saving rate. Moreover, future movements in the public-sector deficit, which are discussed below, will also influence the size of the current account balance.

Figure 37. **PRIVATE-SECTOR SAVING AND INVESTMENT BALANCE**
Per cent of GDP




*Source:* OECD.

## The effects on public finances

The major fiscal impact of population ageing will come through substantial increases in public pension benefits for retired persons. In addition, public health expenditure is also likely to rise. The following sections examine the consequences for public expenditure and taxation of the increasing proportion of the elderly in the population.

### *Social security pensions*

#### *Design*

The state pension scheme in Japan is a two-tier system, a third tier being provided by the private sector and run by companies. The first tier is a flat-rate benefit while the second tier is earnings-related. These schemes (see Box 3) are designed to generate a considerable amount of income redistribution from high to low-income families, as contributions to the employees' scheme are entirely

## Box 3. **Pension provision in Japan**

Pensions in Japan are provided from a mixture of public and private sources. Although the first public pensions were introduced in 1890, the current system was effectively started in 1961 and did not take its present form until 1985. The private pension system acts both as a complement and as a substitute for the public system.

The public system has two tiers. The *first*, known as the national pension, is a flat-rate system, which covers in principle, all residents between the ages of 20 and 60. The *second* tier has two elements: the Employees' Pension Insurance (KNH) for private-sector employees and a voluntary scheme, the National Pension Fund, for the self-employed. For the first of these, both contributions and benefits are related to income, while for the second, benefits are proportional to contributions.

In addition, there is a third tier supplied by almost 90 per cent of private enterprises, which provides some form of occupational benefits at retirement. About half of enterprises pay just a lump-sum benefit, while the others pay pensions as well.

There are three main methods for funding these private-sector pensions. *First*, the employer can create a contracted-out plan, known as an Employees' Pension Fund (KNK). These plans are responsible for the payment of second-tier social security benefits, although the government must pay all pre and post-retirement increases in the pension due to the legal indexation of past earnings. The second-tier contributions of the employer and the employee that would normally be paid to the government are partly diverted to the plan. The establishment of such a plan requires the company to provide a complementary third-level pension which the employee can opt to take as a lump sum on retirement. Such plans have to be funded externally through a trust or insurance account. *Second*, there are tax-qualified plans (TQPs) which also have to be funded externally. Employer contributions are a tax-deductible corporate expense. However, the assets of these plans are taxed at the rate of 1.173 per cent, though the fund pays no income tax. *Finally*, some companies run book-reserve schemes whereby a notional fund is created and held as a liability on the balance sheet of the company and so offer less protection to pensioners than fully funded schemes. This option has been made less attractive by changes in the tax system: at present, only 40 per cent of contributions are tax-deductible against 100 per cent when this system was established in 1952.

Private pension plans are supervised by the government. The Ministry of Health and Welfare monitors KNK funds while the Ministry of Finance oversees TQPs. In the event that a private scheme is unable to pay its liabilities, the Federation of Employees' Pension Funds takes over the payment of part of the benefits and this cost is shared amongst KNK funds. This organisation also takes over the payment of benefits for former members of KNK funds, with less than ten years service in a company, and is compensated by a transfer payment from these funds.

Public-sector employees are covered through a number of plans, known as mutual aid schemes, which pay benefits similar to those in the private sector.

income related while benefits are partly flat rate, and, perhaps not intentionally, from married women who work to those who do not. Pension payments depend on family characteristics. For instance, the average male worker with a non-working wife and 40 years of contributions would receive a pension amounting to 54 per cent of gross average total earnings. This ratio would fall to 52 per cent for a couple where both have worked for 40 years and to 42 per cent for a single person. It is higher than in most Anglo-Saxon countries, reflecting the provision of pensions for women who have not been employed.

Pensions are paid at a relatively early age despite the higher life expectancy in Japan (Box 4). Although the 1994 reform will raise the eligibility age for the first-tier pension to 65, the second-tier pension, which accounts for over half of the average benefit for retired people, will still be payable at age 60. This structure is equivalent to an average retirement age of 61.6 years, well below the average in other countries. Given that the life expectancy for Japanese people at age 65 is two years longer than in other OECD countries, pension payments will last almost one-third longer on average in Japan.

Table 36. **A comparison of state pension schemes**

| | Retirement age | | | | | | Date |
|---|---|---|---|---|---|---|---|
| | 1994 | | | Future | | | |
| | Men | Women | Average | Men | Women | Average | |
| Australia | 65 | 60 | 62.5 | 65 | 65 | 65 | 2014 |
| Canada | 65 | 65 | 65 | 65 | 65 | 65 | – |
| France | 60 | 60 | 60 | 60 | 60 | 60 | – |
| Germany | 63 | 60 | 61.5 | 65 | 65 | 65 | 2018 |
| Italy | 60 | 55 | 57.5 | 65 | 60 | 62.5 | 2001 |
| Spain | 65 | 65 | 65 | 65 | 65 | 65 | – |
| Sweden | 65 | 65 | 65 | 65 | 65 | 65 | – |
| United Kingdom | 65 | 60 | 62.5 | 65 | 65 | 65 | 2020 |
| United States | 65 | 65 | 65 | 67 | 67 | 67 | 2020 |
| Average of above | 63.7 | 61.7 | 62.7 | 64.7 | 64.1 | 64.4 | |
| Japan [1] | 60 | 60 | 60 | 60/65 | 60/65 | 60/65 | 2014 |

1. For Japan, the age of payment for pensions will be 65 for the basic pension and 60 for the earnings related pension. Given the expenditure on the two schemes this is equivalent to an average retirement age of 61.6.

*Source:* OECD.

## Box 4. **Benefits and contributions under the state pension system**

### Contributions

The first-tier contribution is a flat rate and payable by all residents between the ages of 20 and 60. Self-employed persons and students (19 million people) make direct flat-rate contributions. Although these payments are compulsory in theory, the absence of an enforcement mechanism makes them voluntary in practice.[1] For employees (38.5 million people), contributions, which are compulsory and income-related, are combined with payments to the second-tier system. This system then transfers the required flat-rate payment to the first-tier scheme. The employees' contributions also finance the flat-rate pensions granted to non-working wives (12 million people).

A minimum of 25 years of contributions is required to receive a pension under the first-tier system. A maximum of 40 years of contributions can be made but must be paid by the age of 60, with some exceptions. A contribution credit is granted for certain categories, with very low incomes, but each year of credited contributions has only one-third the value of a paid contribution year. In order to help finance the first-tier pension, tax revenues, equivalent to one-third of the actual benefits received, are transferred to this scheme by the central government, except for local government employees where the transfer comes from this level of government.

The second-tier contribution rate has been 17.35 per cent of basic earnings (excluding bonuses) since October 1996, split equally between employees and employers and, since April 1995, one per cent of bonuses. As noted, this includes the payment for the first-tier benefit. In this scheme, contributions can be made up to the age of 65.

### Benefits

The actual pension paid under the first-tier scheme is proportional to the number of years of contributions that have been paid. In 1995, the maximum benefit to those with 40 years of contributions would have been Y 65 500 (18 per cent of monthly average earnings). However, since the system only began in 1961, the average benefit paid in 1994 was only 70 per cent of the maximum. The pension is currently payable at age 60 for employees[2] but by 2014, this age will be raised to 65.[3] For the self-employed, the pension is already payable at age 65.

The second-tier earning-related pension benefits are proportional both to the number of years of contributions and the average level of basic earnings. Benefits accrue at the rate of 0.75 per cent of basic earnings per year,[4] giving a pension of 30 per cent of earnings after 40 years' contributions. There is no ceiling on the number of years of contributions, but these must be paid by age 65. Past earnings are revalued every five years to reflect the growth in post-tax earnings. Between revaluations, the amount of the

*(continued on next page)*

*(continued)*

benefit is indexed to the increase in the CPI. After retirement, the same indexation rules apply to benefits as apply to the re-rating of past earnings.

A full second-tier pension benefit, including the flat-rate component, is currently payable to persons who retire at age 60. For those who continue to work between 60 and 65, benefits are reduced by between 20 and 100 per cent, depending on the level of earnings. From age 65, full pension benefits are paid, regardless of whether the person is working. An additional flat rate benefit of Y 18 833 per month is paid to retired male employees if they are married, if the wife receives no benefit.

1. Contributors must purchase a pension stamp monthly from their local municipal office and keep them to prove their contribution record.
2. Between age 60 and 64, the flat-rate pension is paid entirely by the KNH without any government subsidy.
3. The flat-rate basic pension can be paid before age 65. If the pension is taken at age 60, it is reduced by 42 per cent. On the other hand, if it is taken at age 70, it is increased by 88 per cent. These rates are, however, not actuarially fair, as they are based on relatively short life expectancies.
4. There is a special provision for those who were born before 1946 for whom the more favourable rates of up to 1 per cent are applied.

The Japanese pension system allows participation in the labour force by people over the age of 60. Those between 60 and 65 can receive full benefits while working part time with income below certain limits. In the past, employment could generate a reduction in the overall total of income and pension payments at certain income levels. Bonus payments, though, were disregarded in this calculation, thus encouraging employers to pay a higher proportion of income in the form of bonuses.[84] Employees were sensitive to the implied marginal reduction rates that were more than the increase of income at certain earnings levels. As a result, there was a marked clustering of earnings by the retired at levels that were just below the points where such high rates of benefit reduction occured. In view of this, the maximum reduction rate was limited to 100 per cent from 1994 onwards. Pension recipients over age 65 are allowed to work without any deduction in benefits. Moreover, they do not pay any contributions to the pension scheme. On the other hand, the size of their benefits is not increased by working past age 65.

The redistributive impact of the public pension system is partially offset by the fact that social security insurance and some private pension contributions are tax-deductible.[85] Given the top marginal tax rate of around 65 per cent, high-earners effectively pay low premia to the social insurance system. Tax deductibility of pension contributions could be justified if pension benefits were fully taxed as normal income at the same marginal rate as paid during a working life. However, pension income is subject to a special schedule which gives very high deductions for public pensions. For example, a married couple over 65 can deduct Y 3.35 million of their pension income in calculating their taxable income base. As the average married employee receives a pension of just under Y 3 million per year, state pensions are effectively free of income taxation. Even if the special allowances for pension income were ended, many high-income earners would still benefit from the tax deductibility of social security contributions since that income is taxed at a lower marginal rate once they are retired. Moreover, a private pension can be converted to an actuarially equivalent lump-sum cash payment, which is taxed differently to income from pension benefits.

### *Cost: outlook and risks*

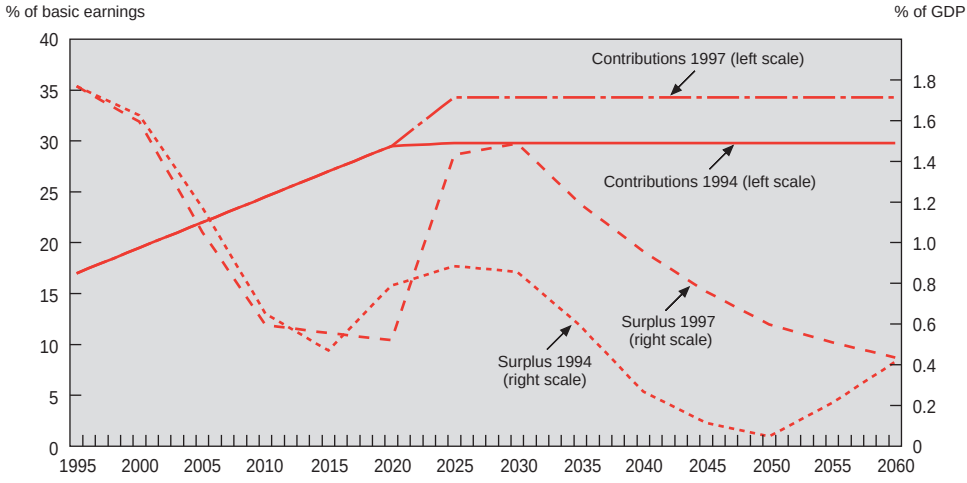
Pension expenditures in all OECD countries will be increased by the rising number of elderly people. In the case of Japan, from now on the rise in the number of elderly is no more marked than in other countries. However, a second factor boosting pension expenditure in Japan is the fact that the system is not fully mature. The average period for which new retirees have contributed is rising by almost 1½ per cent annually, at the moment. As a result, average pension entitlements could rise by almost 10 per cent during the next 25 years under the current system.[86]

Taking these factors into account, the government estimated in 1994 that the contribution rate for both the first and second-tier pensions would have to rise to 29.8 per cent of basic pay (equivalent to 23 per cent of total earnings) by 2025, split equally between employees and employers. Under such a scenario, the annual surplus of the employees' pension insurance fund[87] would become a constant proportion of spending after that date. However, official population projections, made in 1997, have lowered the estimated size of the working-age population in 2050 by 12 per cent while increasing the number of elderly people

by 3 per cent. Consequently, the dependency ratio would rise by 18 per cent by 2050. Such a deterioration implies that, as the pension system is largely managed as a pay-as-you-go system, the contribution rate would need to be raised even more than planned, so balancing future revenues and expenditures. Indeed, the government has estimated that the rate would rise to 34.3 per cent of basic pay by 2025, a 5 percentage point increase from the level planned for this date.[88] As the rise in the contribution rate occurs prior to the decrease in the number of contributors, the financial position of the fund should improve (Figure 38). Rather than moving towards balance after 2025 as originally expected, the fund would continue to run a surplus amounting to 1 per cent of GDP, falling to about 0.6 per cent by 2050.[89] By then, total expenditure of the system would amount to just under 10 per cent of GDP. At this level of contributions, the financial health of the social security system would be assured for the following 50 years.

In addition to the employees' pension insurance, there is a further scheme which, since 1986, provides first-tier pensions. Since more than 6 million of the self-employed have not joined the system, and given that the flat-rate pensions which had started to be paid in 1986 or earlier continue to be the responsibility of

Figure 38. **OFFICIAL PROJECTIONS OF THE SURPLUS AND CONTRIBUTIONS OF THE EMPLOYEES RETIREMENT INSURANCE FUND**




*Source:* Ministry of Health and Welfare and OECD estimates on the basis of national projections.

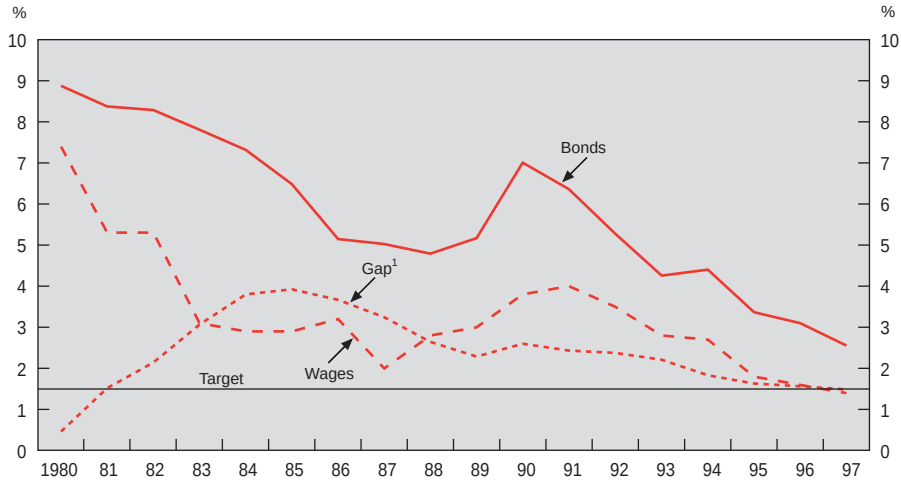
the employees fund, expenditure by this scheme is relatively small. By 2025, spending on this part of first-tier pensions is projected to amount to less than 1 per cent of GDP, with a surplus of 0.1 per cent and assets of 3.1 per cent of GDP. The new government projections suggest that the real value of the flat-rate contributions for non-employees need to rise by 3½ per cent annually, on average, in the period to 2015, rather than the 3 per cent increase foreseen in the 1994 reform.

Taking both the employees' insurance fund and the national pension scheme together, the assets of the social security system will stabilise as a share of GDP from the beginning of the next century. However, an increasing share will be held and managed by the private sector. In 1966, the government introduced the possibility for companies to contract-out of part of the second tier of the state system by establishing employee pension funds that are actuarially-sound and supervised by the government (see Box 3). Employers who establish such funds pay lower contributions to the state system. Although these funds are in the private sector, the portion that relates to the contracted-out part of the state system is counted in the state's reserves. As these private funds were only established recently, they are rapidly increasing in size as they reach maturity. The proportion of the state system's assets that are held in company pension funds is projected to rise from 17 per cent in 1995 to 55 per cent in 2025.[90] The asset portfolio managed by the state will fall from 23 per cent of GDP to 12 per cent over the same period.

Projections of the balance between future pension spending and contributions are surrounded by considerable uncertainties. One *downside risk* is that future demographic developments might be worse than expected. As noted above, the official projections assume that the fertility rate will increase from its current level of 1.4. If, on the other hand, it were to remain unchanged, the number of contributors to the pension system would eventually be reduced by 12 per cent, lowering revenue by about 1.2 per cent of GDP and bringing the employees' insurance fund into a slight deficit. However, such a deficit could be maintained for a considerable time as the fund's assets represent 30 per cent of GDP.

Another risk stems from the fact that the current rate of return on pension funds is well below the official assumption of 5.5 per cent. However, what matters for the financial soundness of both the state and corporate systems is not the absolute level of the rate of return but the difference between the nominal rate

Figure 39. **BOND YIELDS AND WAGE GROWTH**




1. Four-year moving average.
*Source:* OECD.

of interest and the nominal growth of basic earnings (which determines the increase in pension expenditure). In the official projections, this differential is set at 1.5 per cent, an assumption consistent with the current situation (Figure 39). Although nominal interest rates have fallen to 2.7 per cent, the growth of basic earnings has slowed to 1.2 per cent.[91] Nonetheless, the gap has declined from the early 1980s, putting pressure on companies to increase contributions to schemes.

There are also some *upside risks*, such as a further rise in the labour force participation of women. Official projections assume that female participation rates remain constant, despite the marked increase recorded in recent years.[92] Increases in the labour force participation of married women would lower the dependency rate and produce a long-term saving under the current pension system, since married women who work pay contributions but only receive the earnings-related part of benefits since they already receive the basic pension benefits even if they do not work and do not pay contributions.[93] To encourage women to participate in the labour force, it will be necessary to remove aspects of the social security systems that tend to limit the incentives of married women to work, as well as improving othe aspects of social and labour market policy.

Another upside risk comes from the demographic projections. *First*, a large share of single persons may eventually marry. The proportion of Japanese women married by the age of 40 has never fallen below 95 per cent in contrast to the assumption of the current population projections (see above). *Second*, despite the fall in the fertility ratio, there is no evidence thus far that the completed family size has fallen below the replacement rate. Women born in the period 1928-32 had a completed family size of 2.33 by 1977. It fell to 2.18 children for women born between 1943 and 1947. If these patterns continue, the fertility rate would eventually increase further than expected. However, given the number of young women not being married at age 30, the higher age of marriage may lead to a decline in the completed family size.

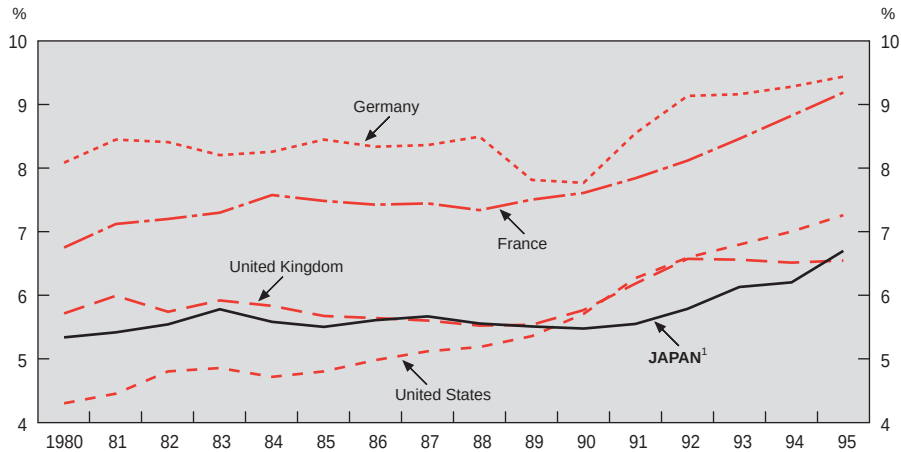
A further upside risk comes from the change in the formula to be used in the indexation of benefits. Since 1994, benefits have been indexed to the growth of earnings after the deduction of tax and social insurance contributions. The official projections appear to allow only a 0.2 per cent annual difference between the growth of gross and net pay. However, the current provisional projected increase in contributions will generate a gap of 0.3 per cent annually.

Another uncertainty related to the level of public spending on the elderly is the new in-work benefit to be paid to those between the ages of 60 and 64. This scheme, which in practice, is similar to a partial pension system, was introduced in 1994, when the government decided to gradually raise the eligibility age for first-tier pensions to 65. It is designed to assist workers following their mandatory retirement, which is set at 60 in most companies. Although most employees subsequently find new jobs, either with the same company or with a smaller enterprise, pay levels are generally about 40 per cent below pre-retirement earnings. Under the new scheme, the government pays a benefit of 25 per cent of the basic wage to employees whose earnings fall by more than 15 per cent.[94] This benefit can be paid concurrently with a retirement benefit subject to the earnings limits described above.[95] As yet, there is no information on the cost of this scheme, which will be financed from employment insurance contributions.

### *Health insurance for the elderly*

Health care is the second largest form of public social expenditure in Japan, accounting for over one-third of the total in this category. Relative to output,

Figure 40. **PUBLIC HEALTH EXPENDITURE**
Per cent of national income




1. Estimate for 1995.
*Source:* OECD, Health Database and *Annual National Accounts.*

though, government expenditures in this area are lower than in most OECD countries. Medical spending was relatively stable in the 1980s at around 5½ per cent of GDP before rising to 6½ per cent in the 1990s (Figure 40).

Health insurance in Japan is universal and financed largely by compulsory social security contributions or general taxation. Around 88 per cent of total medical expenditure is covered by the state, either through insurance premiums (57 per cent) or grants from central (24 per cent) and local government (7 per cent). Private expenditure is limited to 12 per cent of the total. The system of financing public medical care is provided by two main sectors: the first is ‘‘medical insurance for the working population and their dependants’’, while the second, known as the Health Service System for the Elderly (HSSE), is for elderly people. Within the medical insurance sector, there are a large number of schemes, which are linked by complicated revenue sharing agreements and also contribute to the HSSE (see Box 5).

The share of the elderly in total medical expenses has risen continuously, reaching 32 per cent in 1995. The cost of medical care for the over-70 age group

## Box 5. **Health insurance in Japan**

Universal health insurance was introduced in Japan in 1961. All citizens under the age of 70 belong either to a workplace or community-based system. In addition, there is a third programme for the elderly. As noted, a little more than half of total medical expenditure is funded by insurance premiums, with government grants covering about a third. The remainder is financed by private expenditure.

Table 37. **Outline of Japan's health insurance system**

**A. Coverage**

| Programme name | Category | Insurer | Number insured (millions) | Percentage of expenses covered |
|---|---|---|---|---|
| Employee's Insurance | Employees at SMEs | Central government | 37.6 | Insured: 80 |
| | Employees at large companies | 1815 private-sector health insurance societies | 32.5 | Dependent, Inpatient: 80 Outpatient: 70 |
| | Mutual aid association [1] | 82 private-sector health insurance societies | 12.0 | |
| | Seamen's Insurance | Central government | 0.3 | |
| National Health Insurance | Farmers and self-employed | Local government [2] | 26.9 | 70 |
| | | 166 non-profit associations | 4.6 | |
| | Retirees up to age 69 | Local government [2] | 6.7 | |
| National Health and Medical Service | Persons over age 70 | Local government | 11.6 | 100 [3] |

**B. Financing**

| Programme name | Share of expenditure | Government subsidies | | | | |
|---|---|---|---|---|---|---|
| | | Premiums | Central | Local | Patients | Transfers of premiums |
| Employee's Insurance | 44.0 | 81.5 | 5.4 | 0 | 13.5 | 0 |
| National Health Insurance | 22.2 | 35.7 | 33.9 | 7.1 | 23.2 | 0 |
| National Health and Medical Service [4] | 33.7 | 0 | 35.3 | 11.8 | 4.7 | 48.2 |
| Total | 100 | 43.7 | 21.8 | 5.6 | 12.7 | 16.3 |

1. Covers national and local public employees and private-school teachers.
2. There are 3 251 units of local government involved in health care.
3. Complete coverage following co-payment of Y 500 per day for outpatient service (maximum of four times a month), and Y 1 000 per day for inpatient care.
4. About a third of government subsidies are channelled via the other two systems. About four-fifths of the premia income transferred to the elderly is from the employee Insurance system.

*Source:* Ministry of Health and Welfare.

*(continued on next page)*

*(continued)*

Employee health insurance, which covered about 82 million workers and their dependants as of March 1995, is very decentralised (Table 37). Employees at small and medium-sized companies are covered by a government-run scheme, primarily financed by payroll taxes set at 8 per cent of basic earnings (shared equally by employers and employees). In addition, the government provides a subsidy equivalent to 13 per cent of total costs. For those working at large companies, almost 2 000 private-sector firms provide insurance, which is almost entirely financed by employers and employees, with only minimal subsidies from the government. Finally, civil servants and employees of public corporations are insured by mutual aid associations.

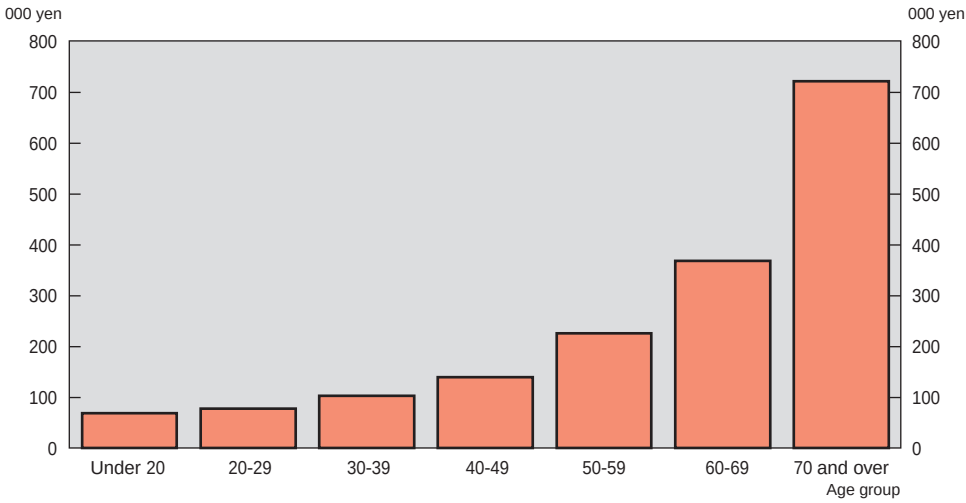
The National Health Insurance system, a community-based scheme, provides coverage to persons who are not included in employer-based programmes. This group, which consists primarily of the self-employed, farmers and retirees up to age 70, includes a larger proportion of low-income households. Consequently, the government covers one-half of the costs. The premium paid by the insured varies according to income and family size.

The largest subsidies are provided for the Health Service System for the Elderly, which covers persons over the age of 70. There are individual contributions by the elderly and co-payments. However, these payments, which were introduced in 1983, remain quite low: Y 1 020 per month for outpatient services and Y 710 per day for inpatient care and from April 1996 will be indexed to prices.

has grown twice as rapidly as that of the rest of the population. In 1994, such patients utilised more than five times more medical services than those below age 70 (Figure 41). This ratio is one of the largest amongst the seven major OECD countries, reflecting Japan's health insurance system, which provides unrationed, almost free medical services to the elderly above the age of 70.[96]

Terminal medical care is concentrated in hospitals, even in cases where intensive nursing, rather than intensive medical care, is required. As a result, the average stay in hospital in Japan is about triple that in other OECD countries. In practice, care for the elderly, which is covered by health insurance, is substituted for nursing home care, which is not covered. One factor accounting for the prevalence of hospital care is that it is reimbursed on a fee-for-service basis by the social security system with no limit on payments. Consequently, hospital

Figure 41. **MEDICAL COSTS OF THE AVERAGE PATIENT BY AGE GROUP**
Thousand yen per year




*Source:* Ministry of Health and Welfare.

services are in abundant supply. On the other hand, cheaper nursing care services are provided by the government but their provision is limited for fiscal reasons.

Future demographic developments are likely to put severe pressure on health and nursing expenditure. Given that medical spending is concentrated in the last year of people's life, with most of this expenditure concentrated in the last month, when average expenses are four times higher than in the previous year, an ageing of the population due to increased longevity need not raise such spending. However, the age structure of the population in Japan means that the annual number of deaths will increase very rapidly in the next half century from 0.91 million in 1995 to a projected 1.66 million in 2050, so lifting health spending on the elderly four times as rapidly as in the rest of the OECD area (Table 38). Moreover, about 6 per cent of the elderly are bed-ridden or suffering from senility. With increasing longevity, the proportion of the very old to the overall elderly population will rise, so raising the proportion of the elderly requiring intensive care. Indeed, the number of such elderly people is projected to rise from 1.4 to 3.75 per cent of the working-age population by 2025.[97]

Table 38. **Projections of public health care spending: an international comparison**

Per cent of GDP

| | 1995 | Projected spending in 2030 Growth in line with number of: | | Change in spending 2030 – 1995 | |
|---|---|---|---|---|---|
| | | elderly | deaths | elderly | deaths |
| United States | 6.4 | 8.2 | 6.9 | 1.8 | 0.5 |
| **Japan** | **4.9** | **6.3** | **7.2** | **1.4** | **2.3** |
| Germany | 6.2 | 7.8 | 7.4 | 1.6 | 1.2 |
| France | 7.0 | 8.9 | 8.3 | 1.9 | 1.3 |
| Italy | 6.4 | 8.1 | 7.7 | 1.7 | 1.3 |
| United Kingdom | 6.0 | 7.0 | 6.2 | 1.0 | 0.2 |
| Canada | 7.4 | 10.3 | 9.3 | 2.9 | 1.9 |
| Australia | 5.8 | 7.6 | 6.2 | 1.8 | 0.4 |
| Austria | 7.4 | 10.3 | 8.0 | 2.9 | 0.6 |
| Belgium | 7.4 | 9.5 | 7.8 | 2.1 | 0.4 |
| Denmark | 5.6 | 7.0 | 5.8 | 1.4 | 0.2 |
| Finland | 6.9 | 9.4 | 7.4 | 2.5 | 0.5 |
| Iceland | 7.4 | 9.1 | 7.1 | 1.7 | –0.3 |
| Ireland | 5.1 | 5.6 | 4.7 | 0.5 | –0.4 |
| Netherlands | 6.7 | 9.8 | 7.4 | 3.1 | 0.7 |
| Norway | 7.6 | 9.2 | 7.4 | 1.6 | –0.2 |
| Portugal | 4.1 | 5.0 | 4.2 | 0.9 | 0.1 |
| Spain | 5.7 | 7.4 | 6.1 | 1.7 | 0.4 |
| Sweden | 6.2 | 7.9 | 6.7 | 1.7 | 0.5 |
| Average | 6.3 | 8.1 | 6.9 | 1.8 | 0.6 |

*Source:* OECD, 1996.

## *Overall fiscal impact*

On the basis of unchanged policies, government expenditure in Japan will rise significantly during the next quarter century. The increase in social security pension spending will be more rapid than in other OECD countries, rising from an estimated 5.8 per cent of GDP in 1998 to 10.2 per cent in 2025 (Table 39). Given the expected increase in other pension benefits for government and public corporation employees, total pension expenditure might be twice the level of spending in North America, the United Kingdom and Australia, although still less than in continental Europe (Table 40).[98] Health spending on the elderly will also

Table 39. **Projections of government spending in Japan**

| | Per cent of GDP | | | Per cent of national income | | |
|---|---|---|---|---|---|---|
| | 1994 | 1998 | 2025 | 1994 | 1998 | 2025 |
| **Current services** | | | | | | |
| Education | 3.2 | . . | . . | 3.7 | . . | . . |
| General public services | 2.6 | . . | . . | 3.1 | . . | . . |
| Defence | 0.9 | . . | . . | 1.0 | . . | . . |
| Other services | 1.9 | . . | . . | 2.2 | . . | . . |
| Total | 8.6 | 8.6 | 8.6 | 10.1 | 10.1 | 10.1 |
| **Total spending on the elderly** | | | | | | |
| Social security pensions | 4.5 | 6.2 | 10.2 | 5.3 | 6.9 | 11.9 |
| Other pensions | 1.9 | 2.1 | 3.3 | 2.2 | 2.9 | 3.9 |
| Health care for the elderly | 1.9 | 2.1 | 4.2 | 2.0 | 2.2 | 4.9 |
| **Other social security spending** | | | | | | |
| Health care for others | 3.4 | 3.4 | 3.4 | 4.0 | 4.0 | 4.0 |
| Family benefits | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 |
| Other transfers and administration | 2.7 | 2.7 | 2.7 | 3.1 | 3.2 | 3.2 |
| **Investment** | | | | | | |
| Capital formation | 6.4 | 5.8 | 5.8 | 7.5 | 6.8 | 6.8 |
| Land purchases | 1.1 | 1.0 | 1.0 | 1.3 | 1.2 | 1.2 |
| **Other** | | | | | | |
| Interest payments | 3.7 | 3.6 | 3.6 | 4.4 | 4.2 | 4.2 |
| Subsidies | 0.7 | 0.7 | 0.7 | 0.9 | 0.8 | 0.8 |
| **Total spending in Japan** | 35.0 | 36.3 | 43.6 | 41.0 | 42.6 | 51.1 |
| **Total spending, OECD average** | 46.4 | 42.9 | 44.2 | . . | . . | . . |
| *Memorandum* | | | | | | |
| Social security spending Japan | 14.4 | 16.6 | 23.8 | 16.9 | 19.5 | 28.0 |
| Other current spending | 13.1 | 12.9 | 12.9 | 15.3 | 15.1 | 15.1 |
| Investment | 7.5 | 6.8 | 6.8 | 8.8 | 8.0 | 8.0 |

*Note:* The projections of debt interest assume a stable debt to GDP ratio. The projections of spending in the remainder of the OECD area assume stable non-pension spending and pension spending that rises in line with the estimates in OECD Economics Department Working Paper No. 168.
*Source:* OECD and national projections.

grow. If it were to increase in line with the number of deaths, its share of GDP might almost double between 1995 and 2025. Expenditure on long-term care might also rise. If other forms of government spending are held constant as a proportion of GDP, total expenditure would rise from 36 per cent of GDP in 1995 to just over 43 per cent in 2025,[99] a figure comparable to the overall average of the OECD area projected for that year.

Table 40. **Projected pension expenditure relative to GDP**

| | 1995 | 2020 | 2050 |
|---|---|---|---|
| United States | 4.1 | 5.2 | 7.0 |
| Canada | 5.2 | 6.9 | 8.7 |
| Australia | 2.6 | 2.9 | 4.5 |
| United Kingdom | 4.5 | 5.1 | 4.1 |
| Average of above | 4.1 | 5.0 | 6.1 |
| Germany | 11.1 | 12.3 | 17.5 |
| France | 10.6 | 11.6 | 14.4 |
| Italy | 13.3 | 15.3 | 20.3 |
| Sweden | 11.8 | 13.9 | 14.5 |
| Spain | 10.0 | 11.3 | 19.1 |
| Average of above | 11.4 | 12.9 | 17.2 |
| Japan | 6.4 | 14.3 | 14.4 |

*Note:* In 1998, spending on pensions in Japan is estimated to be 8.3 per cent of GDP.
*Source:* OECD Economics Department Working Paper No. 168 for countries except Japan. For Japan, figures are OECD estimates based on national projections.

Although part of the rise in spending due to ageing will be met by higher social security contributions, central and local government will be exposed to increased expenditure in three ways (Table 41). *First*, the government is legally committed to pay part of the expenditures of the National Pension Plan. According to the 1994 reform plan, the central government contribution would rise by 2.4 per cent annually in real terms between 1994 and 2025, although this estimate has been revised up to 3 per cent since the new population projections add somewhat to the number of elderly. With real GDP projected to average around 1.7 per cent, central government transfers to the pension plan would have to rise from 0.9 per cent of GDP in 1994 to 1.3 per cent by 2025. *Second*, the government is required to pay the pensions of government and public corporation employees and war veterans. Overall, these pension payments amounted to about 1.6 per cent of GDP in 1994, with staff contributions and transfers from the KNH accounting for some of the expenditure. The demographics of these funds are relatively favourable. Pensions to veterans and retired Japan Railway staff should decline gradually.[100] Such factors will limit the increase in budgetary transfers to these areas to 1.0 per cent of GDP. *Third*, central and local government must

Table 41. **General government spending on the elderly by level of government**

Fiscal year 1994

| | Central government | Local government | Central and local government | Social security | Contribution transfers | Age-related spending |
|---|---|---|---|---|---|---|
| | Billion yen | | | | | |
| Health care for 65-70 year old | 0 | 0 | 0 | 352 | 842 | 1 161 |
| Health care for the over 70s | 3 300 | 819 | 4 119 | 0 | 3 600 | 7 816 |
| Employees Pension Insurance | 0 | 0 | 0 | 11 855 | 0 | 10 326 |
| National Pension | 4 460 | 0 | 4 460 | 4 473 | 0 | 10 256 |
| Employees Pension Funds | 0 | 0 | 0 | 2 962 | 0 | 903 |
| Government employees | 1 468 | 3 728 | 5 155 | 2 063 | 0 | 5 413 |
| Public corporations | 545 | 0 | 545 | 578 | 500 | 1 263 |
| War victims | 1 752 | 0 | 1 752 | 0 | 0 | 1 645 |
| Others | 212 | 9 | 221 | 0 | 0 | 676 |
| Total | 11 698 | 4 554 | 16 252 | 22 283 | 4 942 | 39 459 |
| | Per cent of GDP | | | | | |
| Health care for 65-70 year old | – | – | – | 0.07 | 0.18 | 0.24 |
| Health care for the over 70s | 0.69 | 0.17 | 0.86 | 0.00 | 0.75 | 1.64 |
| Employees Pension Insurance | – | – | – | 2.48 | – | 2.16 |
| National Pension | 0.93 | – | 0.93 | 0.94 | – | 2.15 |
| Employees Pension Funds | – | – | – | 0.62 | – | 0.19 |
| Government employees | 0.31 | 0.78 | 1.08 | 0.43 | – | 1.13 |
| Public corporations | 0.11 | – | 0.11 | 0.12 | 0.10 | 0.26 |
| War victims | 0.37 | – | 0.37 | – | – | 0.34 |
| Others | 0.04 | – | 0.05 | – | – | 0.14 |
| Total | 2.45 | 0.95 | 3.40 | 4.66 | 1.03 | 8.26 |

*Source:* ''The Cost of Social Security in Japan'', Social Development Research Institute, 1996.

ensure that the expenditure and revenue of the health care insurance schemes for retired people are balanced. In 1994, government transfers to these funds amounted to 0.9 per cent of GDP. If there were no increase in health insurance contributions, the additional government expenditure would be 2.3 per cent of GDP. Overall, the central government might need to raise taxation or reduce other forms of expenditure by 3¾ percentage points of GDP mostly due to rising health expenditure. In addition, contributions to the social security system are planned to increase by 5½ per cent of GDP between 1994 and 2025. This movement will progressively erode the income tax base as such payments are tax deductible, so adding to central government's financing problems.

## The policy response

The likely combination of a marked slackening in growth in the coming decades and a substantial increase in the size of the public sector poses difficult challenges. Although there is little evidence on the optimum size of the public sector or on the dead-weight cost of taxation, an abrupt increase in the size of the government might be best avoided. This is especially true as it would significantly increase the gap between the cost of employing a worker and the income received by the employee. Such a movement, if not reflected in lower wages, might adversely affect the demand for labour and, if it were reflected in lower wages, might lead to a lower labour supply. A range of policy reforms should be considered in order to minimise the costs of ageing.

### *Reforming public pensions*

#### *Sustainability*

On the basis that economic growth slackens and the fall in the labour force continues to accelerate, as in the latest official population projections, contributions would have to rise substantially. Indeed, the government has calculated that a further increase in the overall pension contribution rate to 34.4 per cent of basic earnings would be sufficient to ensure long-run stability. However, under this scenario, the pension fund would continually accumulate assets, with the surplus representing about one-fifth of contributions. One option to lower the burden, in the short term, would be to stop accumulating surpluses and aim for the fund to break even. In this case, according to OECD calculations, a lower contribution rate of around 28 per cent would be sufficient from 2025. Given that the dependency ratio will tend to decline after 2050, as the number of elderly starts to fall, there might indeed be some advantage in a variant of this option which would allow a slight rundown in assets so as to smooth over the increase in contributions that could otherwise occur in the middle of the century. However, over the longer term, either of these cases would imply that contributions would have to offset lower interest income. Moreover, the impact of such a change on intergenerational equity would need to be considered.

Another option that might be considered would be to raise the eligibility age for the second-tier pension to 65, the same as for the first-tier system. As noted above, many OECD countries have already decided to raise their retirement ages

Table 42. **Number of persons over the retirement age relative to the working-age population**

2025

Percentage points

| | 1995 | Projection for 2025 | | | | |
|---|---|---|---|---|---|---|
| | | | | | Per cent | |
| | Current retirement age | Current retirement age | Planned retirement age | Retirement age of 65 | Impact of the planned increase on the proportion of retired persons | Further impact of a retirement age of 65 |
| Japan | 35.3 | 66.7 | 60.2 | 47.9 | –9.8 | –20.4 |
| France | 36.0 | 58.1 | 58.1 | 39.5 | 0.0 | –32.0 |
| Italy | 46.9 | 85.9 | 56.3 | 45.0 | –34.5 | –20.0 |
| Germany | 30.8 | 59.2 | 44.0 | 0.0 | –29.0 | 0.0 |
| Sweden | 30.1 | 41.7 | 41.7 | 0.0 | 0.0 | 0.0 |
| Spain | 24.1 | 37.9 | 37.9 | 0.0 | 0.0 | 0.0 |
| United Kingdom | 32.1 | 47.0 | 37.5 | 0.0 | –20.2 | 0.0 |
| Canada | 19.4 | 37.2 | 37.2 | 0.0 | 0.0 | 0.0 |
| Australia | 22.5 | 40.1 | 32.2 | 0.0 | –19.8 | 0.0 |
| United States | 21.4 | 36.2 | 30.3 | 0.0 | –16.2 | 0.0 |

*Source:* OECD.

to 65. Such policies will markedly reduce the increase in the dependency ratios in these countries (Table 42). In Japan, a gradual increase in the retirement age to 65 over the next fifteen years might eventually reduce expenditure on pensions by 14 per cent and increase the working population by up to 6 per cent. This reform alone could reduce the break-even contribution rate to 27.7 per cent of basic earnings. If it were coupled with a policy of no longer accumulating assets, the contribution rate could be lowered to 22.5 per cent (equivalent to just under 17 per cent of total earnings).

Other approaches could be considered as possible alternatives to the above suggested options of raising the retirement age or stopping the accumulation of assets. One would be to reduce the extent of the post-retirement indexation of benefits, perhaps linking the growth of pensions to prices rather than to earnings net of tax and social security contributions. If it were considered desirable to go further in order to limit the role of the public sector, then it would probably be necessary to envisage a matched reduction in contributions and the amount of benefits earned. Such an option, though, might need to be accompanied by an increase in the private provision of pensions.

At the same time, it is important to improve the rate of return on the assets of the social security system. For those assets that are managed by companies in KNK pension funds, one option would be to end the current regulations on asset allocation and competition in the fund management industry. These rules have limited the proportion of assets invested in equities to 30 per cent, while requiring at least half to be held in bonds or bank deposits. Experience in the United States over the post-war period suggests that such an asset portfolio (high on bonds and low on equities) would not only have reduced the average return on pension fund assets, but would have also increased the riskiness of the portfolio. One analysis suggests that the optimal portfolio in the United States, generating the best combination of risk and reward, has been consistently found to have between a 50 and 75 per cent equity share depending on whether a risk-free asset existed.[101] In the future, asset allocation would need to take into account the results of a similar analysis for Japanese markets, in order to achieve the optimal balance between the volatility and long-term performance of different financial investments. One problem for Japanese pension fund investors is that it is difficult to determine the performance of fund managers. Full performance disclosure and accounting based on market prices should also be introduced, so that investors can better judge the performance of insurance company and trust bank fund managers and put them in competition with the other independent fund mangers. Many of these elements are being considered as part of the new financial deregulation programme (see Chapter IV).

To raise the return on pension assets held by the government, one option would be to end the system of depositing annual surpluses with the Trust Fund Bureau. Such funds only earn a return equivalent to that on government bonds, which is below that earned on equities in the long term. Moreover, for society as a whole, the rate of return on money invested in the public sector is even lower than the government bond rate due to the subsidies that are required for FILP institutions to break even, those projects undertaken by public corporations which have low returns and any inefficient investments that are financed by the government construction bonds purchased by the Trust Fund Bureau. All of the assets of the pension fund system should be managed privately in a way that balances the risks and rewards of different financial instruments, while noting that the return on equity investment is uncertain. Given the current stock of assets and the future path of contributions and benefits, an illustrative simulation

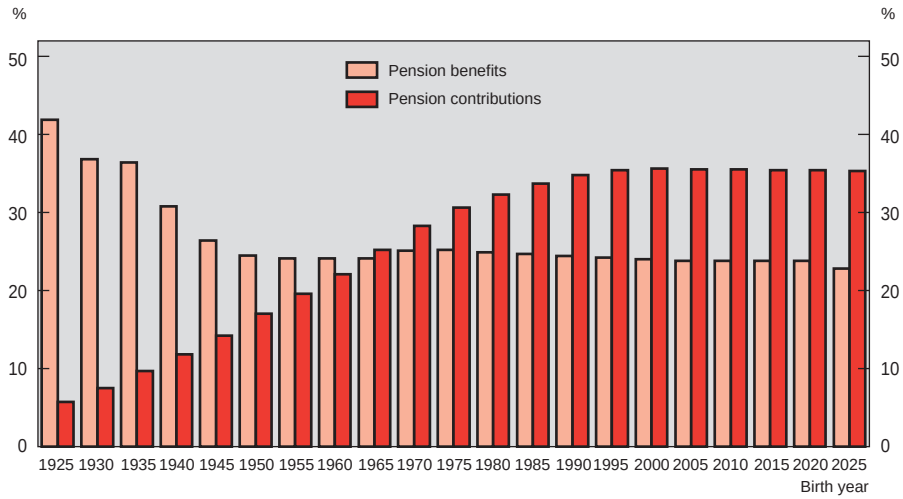
suggests that if the rate of return on assets were raised by perhaps the large amount of 1 percentage point, contributions could be reduced by 12 per cent from the year 2025 onwards which would be equivalent to lowering the contribution rate by 4 percentage points or raising the retirement age by two years.[102]

### *Intergenerational equity*

In addition to the question of financial sustainability, there are equity issues that require reform. The system has never directly linked individual contributions and benefit, so that expected benefits and contributions were not equalised. The initial participants received immediate pensions without having to establish a full contribution record. In addition, life expectancy has increased to a greater extent than was expected when the schemes were established. According to estimates by the Economic Planning Agency, people born in 1925 and who retired in 1985 will receive benefits equivalent to more than eight times their contributions, in present value terms. Although contributions have been progressively increased, generations born prior to 1965 will be net gainers from the system (Figure 42). However, there are many other forms of intergeneration transfers that may be affected by ageing. For instance, an increase in per capita bequests may to some extent offset transfers brought about by the social security system. Nonetheless, the present value of benefits that are currently being paid or have been promised exceed the assets of the state pension system by an amount estimated to be almost equivalent to one year's GDP.[103] As a result, future contribution rates will have to be above the inter-generationally fair contribution rate that EPA estimates put at around 22 per cent of basic earnings.

The financing of such a liability gives rise to issues of inter-generational equity which would occur even without an ageing population. The government plan boosts the contribution rate to a level that exceeds the actuarially fair rate where future benefits and contributions are equalised in present value terms, thus meaning that future contributors will have to finance this deficit. However, other possibilities could be envisaged. For instance, if the contribution rate were set at the actuarially fair level, the revenue shortfall could be financed by gradually increasing direct and indirect taxation. This approach would improve fairness by ensuring that current retired generations participate to a greater extent in the financing of their own pensions. Alternatively, a one-time increase in general

Figure 42. **STATE PENSION BENEFITS AND CONTRIBUTIONS BY YEAR OF BIRTH**
Per cent of lifetime income




*Source:* Economic Planning Agency, *White Paper on Japan's Economy,* 1996.

taxation could be introduced. However, such an increase would have to be as large as 2.5 per cent of GDP. Such an increase in taxation would be the equivalent of issuing a pension liability bond and financing the interest from general tax receipts.

Such reforms could theoretically permit the development of a partially privatised three-tier pension system in Japan. The first tier would be the current flat-rate pension scheme, which has a redistributional aspect since contributions are proportional to income but benefits are flat rate. The redistributive element of this tier could be heightened by completely financing it from general taxation. The required increase in taxation would be large and might result in difficulty in maintaining the real value of benefits. Once a pension liability bond had been issued or the redistributive element transferred to the tax system, the second-tier system, based on an actuarially fair contribution and private-sector management of the assets, would eventually generate a large stock of assets. From time to time, such a system might be faced with difficulties in maintaining the real value of its benefits if its investments became unprofitable. However, viewed over the

long term, such a system would be welfare-enhancing if the rate of return on capital were greater than the rate of growth of wages,[104] as has usually been the case.

### *Reforming health care*

The growing number of very old persons will expand the demand for medical, welfare and other public services. The degree to which the government must increase social transfers to the elderly will depend on their ability to help themselves and on the extent of family support, which traditionally has been large in Japan. Even today, 57 per cent of elderly Japanese males live with their relatives, compared with only 7 per cent in the United States.

However, the Japanese family structure has been changing rapidly and elderly people are less likely to live with their children than in the past. The changing industrial structure and the associated concentration of the population in urban areas have geographically separated parents from their adult children. In addition, the share of self-employed families – which are characterised by implicit contracts that children will inherit their parents' household assets in return for providing care for them – has declined.[105]

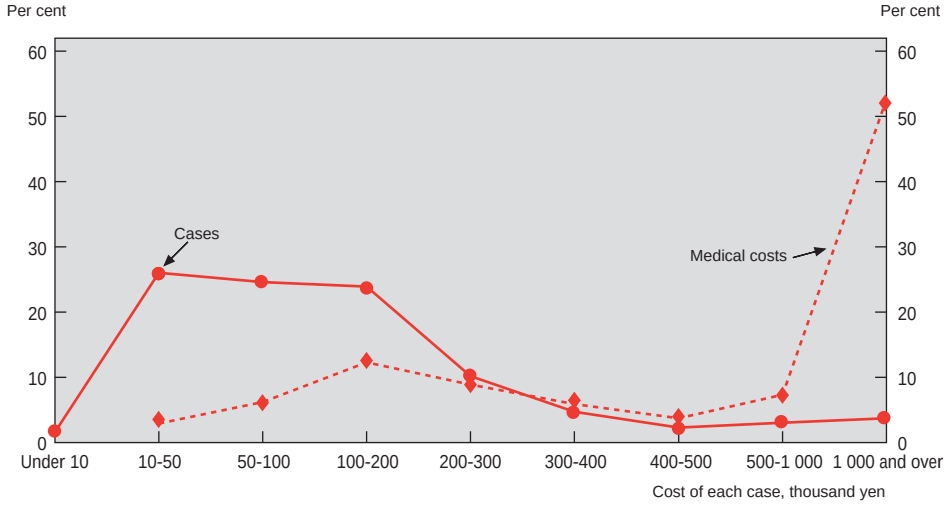
One possible area for development would be to encourage an increased role for the market in supplying welfare services. Most elderly people in Japan are no longer poor. Their average income is approaching the national average, while their assets are double those of younger households. This reflects the elderly's high rate of home-ownership, which exceeds 80 per cent, and the high value of land on which those homes are located. Given their incomes and assets, elderly people could afford to contribute to their medical and nursing expenses to a greater degree than they actually do at present, bringing their position into line with that of younger people with the same income and assets. In the public sector, co-payments for the elderly are practically zero and those in long-stay care continue to receive state pensions while being free to transfer their assets to their children.

Recent government proposals to reform the health insurance scheme suggest doubling the patient's share of medical costs from 10 to 20 per cent when the person is an employee – the rate already paid by the dependants of employees. However, for the elderly, there are only small co-payments for hospital treatment.[106] For the elderly, hospital fees will be increased and they will be

charged a portion of drug costs (which are currently not separated from other fees). However, the emphasis of these reforms is on increasing the share of the patient's direct payments – and, even then, will not introduce co-payments by the elderly in contrast to the position of younger people – rather than reducing costs.

Structural reform in Japan's health care system is also required in order to reduce costs. It could be made more efficient by shifting medical resources from minor treatments to serious diseases. Currently, 85 per cent of health insurance claims are for minor medical treatments costing less than Y 30 000. These services, which account for 20 per cent of total costs, could be directly purchased by households in the market (Figure 43). At the same time, the basis for payment of doctors could be shifted to an annual capitation fee per patient. Under such schemes, the incentives of the suppliers would be reversed from cost-maximising to cost-minimising treatment. Recent empirical studies, based on limited experience, suggest that managed care systems do reduce expenditure in Japan.[107] Finally, it will be necessary to separate long-term care from medical treatment of the elderly. According to the official estimates, shifting the frail elderly from

Figure 43. **DISTRIBUTION OF AVERAGE MEDICAL COSTS PER CASE**
Per cent of total cases and expenditure




*Source:* Ministry of Health and Welfare.

hospitals to nursing homes and providing better non-medical support services might reduce hospital costs by Y 1.2 trillion in 2000.[108]

To facilitate such a shift, the government has proposed introducing long-term care insurance. Under the new system, the costs are to be financed equally by contributions from the working population and the elderly and by a grant from the central and local government. For employees, only people aged between 40 and 64 will contribute, with payments shared by both the employee and employer and made through individual medical insurance schemes. Payments by the elderly will be taken, basically, at source from their public pension benefits. Contributions will rise with income, but within the five income bands chosen by the government will be flat rate. In addition, there will be a user charge of 10 per cent of the costs up to a certain ceiling. The beneficiaries are free to purchase services from either the private or the public sectors.

For the present, such a system may be adequate. However, in the future, demand for such services may increase rapidly so that two further steps seem likely to be required. First, all of the working population should contribute rather than just those over 40. Second, there may be some case for increasing the provision of aid for the frail elderly at home, in order to minimise their stays in specialised institutions.

### *Raising participation and birth rates*

The recent revisions to the official population projections have highlighted the fact that the major difficulty facing the economy in the period after 2025 will be the fall in the population caused by the current low birth rate rather than the increase in the number of elderly people.

Policies that aim to increase saving, raise the capital stock through extensive public investment and foreign asset accumulation and may boost income per capita; but, if the number of births stays at the low level seen in recent years, the working population would continue to fall, pulling down the growth of output. Such considerations suggest that a priority for government policy should be to take steps to raise the labour force in the short term by raising participation rates and in the long term by encouraging, if possible, the birth of more children.

Increasing the labour force participation of elderly workers would be doubly effective as their share in the total population is growing. Avoiding excessive taxes and high social security and benefit reduction rates should help improve

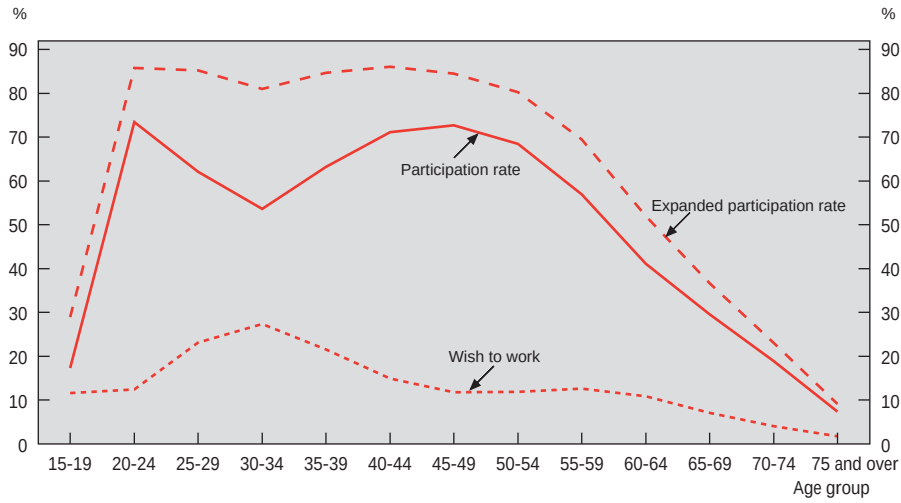
work incentives. Although the 1994 reform made progress in some of these areas, a significant tax on working remains.[109] To increase labour force participation, demand-side measures are also important since well-paid employment opportunities for older workers are limited. One reason may be that the skills of older workers are firm-specific and so their human capital cannot be fully utilised elsewhere. Another problem is the linkage of pay to seniority, which may encourage firms to impose mandatory retirement. Consideration may have to be given to introducing more flexible arrangements to promote the utilisation of older workers.

Offsetting the decline in the domestic labour force by increasing the use of immigrants, a theoretical possibility, poses problems more of a social than economic nature. At the moment, the government is allowing some increase in the immigration of skilled foreign workers, while tightly restricting the inflow of unskilled foreigners.[110] According to official estimates, foreign workers (including illegal workers) accounted for over 1 per cent of total employees in 1996.

Reforms may also be required to increase the labour force participation of women. The pension rights of married women should be improved while separate income taxation for married couples should be considered. At present, married women lose significant pension advantages when their earnings exceed a relatively low limit. There is evidence to suggest that modifying these constraints would generate a considerable supply-side response. Recent empirical work has indicated that the labour supply elasticity for married women is around 0.33[111] and that average annual earnings of many married women are just below the cut-off point for losing these benefit advantages.[112] Moreover, the limits encourage women to work part time, which tends to depress their hourly wage rates.

Despite a considerable rise in the past 20 years in Japan, only 50 per cent of women are in the labour force compared with around 56 per cent in Scandinavia and North America.[113] Participation falls markedly in the 30 to 45 age group before recovering for older women (Figure 44). Moreover, while only 40 per cent of married women in the 30 to 34 age group actually work, 60 per cent of those who are not working state that they would like to have a job. If all the women who would like to work were able to, their participation rate would increase to 65 per cent, boosting the labour force by 11 per cent. Although this would be a significant gain, it would only serve to offset by one decade the reduction in the labour force caused by the low fertility rate.

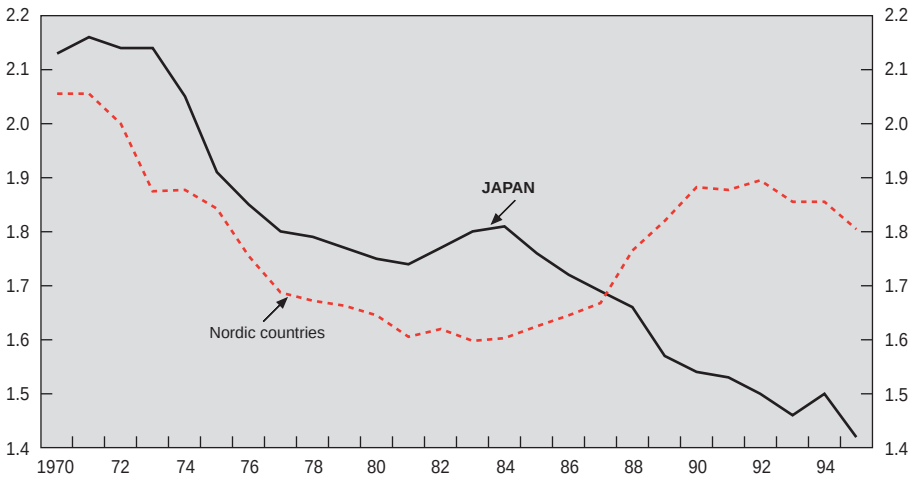
Figure 44. **LABOUR FORCE PARTICIPATION RATES OF WOMEN BY AGE**
Per cent, 1992




*Source:* Management and Coordination Agency, *Employment Status Survey.*

Child allowances are low (government spending is only 0.1 per cent of GDP) and the tax system is biased against married couples while the state pension scheme is biased against married working women. Child-care leave is allowed for one year after the birth of a child and the employer allows the person on leave to return to work if they so wish. Since April 1995, an allowance equivalent to one-quarter of previous income is paid by the social security system to a person on such child-care leave. In addition, maternity leave is allowed for a certain period prior to the birth of the child. The existing family policies of the government are quite limited. The experience in OECD countries of these policies is mixed. In Sweden there were quite large responses to certain incentives, suggesting that a pro-family policy can increase the birth rate. However, such policies have to be introduced in a way that does not generate unsustainable public spending, otherwise there may be pressure to reverse policies and the fertility rate may fall back. In general, Nordic countries have a developed family policy and that is favourable to the integration of women in the labour force and birth rates are markedly higher there than in Japan (Figure 45).

Figure 45. **FERTILITY RATES: COMPARISON WITH NORDIC COUNTRIES**




*Source:* Ministry of Health and Welfare, OECD, Health Database.

In the case of Japan, an improved family policy should aim to lessen the opportunity costs to women of having children. It should aim to reduce the preference on the part of employers in many sectors for married women to withdraw from the labour force. Given the changing expectations of women in modern society, such attitudes may force them to choose between pursuing a career and remaining single and childless and the traditional pattern of marriage and children. Such an approach would encourage mothers who wish to work to do so, thereby perhaps reducing the number of women choosing neither to marry nor to have children. Lowering the costs of child-rearing for families by raising child allowances and improving child-care leave and benefits should also be considered. Improved child-care facilities might also be necessary. Indeed, some research suggests that the trend away from establishing families is so strong that improvements in child care may be necessary just to stabilise birth rates.[114]

In addition, deregulation of key sectors of the economy would boost labour productivity and help offset the negative impact of a shrinking workforce on overall economic growth. As noted in previous Surveys, the level of productivity

in Japan's non-manufacturing sector is low compared to other major OECD countries. This is largely due to regulations which limit market entry and price competition. Regulatory reforms in such industries as telecommunications, transport, electricity and distribution could boost output by about 6 per cent according to estimates by both the OECD and the Japanese Government (see Chapter IV). However, as long as pensions are linked to the growth of wages, policies that increase productivity – and hence wages – will also raise pension benefits. Thus, however desirable in its own right, such policies will not ease the financing problem of the social security system. The boost to real incomes brought about by deregulation might, though, make the cost of higher pension contributions more bearable to the working population.

## Concluding remarks

The Japanese population is already the second most elderly in the OECD area. Over the next 25 years, the number of elderly people will grow at about 1¾ per cent annually, a somewhat slower pace than in the rest of the OECD area. Nevertheless, by 2025, the Japanese population will be the most elderly in the OECD area and only two other OECD countries will have a higher ratio of elderly to working-age people. Although the number of elderly should stabilise after 2025, the current low birth rate will cause both the active and the total population to fall quite rapidly. By 2050, the total population will be almost one-fifth below its current level.

These demographic factors may result in a slowing of economic growth as the labour force declines. Both saving and investment may fall, with the latter perhaps dropping by more than the former as there is little evidence that elderly persons dissave. In addition, with the total population declining, there is likely to be reduced demand for housing, pushing down residential investment and lowering land prices. Overall, there is a possibility that the private-sector saving-investment balance may increase, suggesting that the current account surplus is unlikely to decline unless there is a marked worsening in public finances. Such a result appears to be consistent with past movements in the private-sector savings-investment balance. However, small changes in any of the assumptions could result in markedly different projections and so such extrapolations linking ageing

and the movement of the current account surplus are subject to a wide measure of uncertainty.

Government expenditure will be under considerable pressure from both rising pension benefits and increasing medical costs for the elderly. Under present policies, the growth in pension spending will be the most rapid in the OECD area. This is due not only to a rise in the number of retired people, but also to a considerable increase in the average benefit paid as the pension system matures. If other forms of public expenditure remain constant as a share of GDP, general government spending may rise to around 44 per cent of GDP and over 50 per cent of national income, bringing Japan in line with the average in the OECD area. Present government policy is to partially accommodate such increased spending while limiting the share of tax, social security contributions and the fiscal deficit to 50 per cent of national income.

To achieve this objective, reform of the public pension system will be necessary. One move would be to raise the eligibility age to 65 for the earnings-related pension, bringing it into line with that for the flat-rate pension. Such a reform, which has already been adopted in many OECD countries, should result in a considerable reduction in expenditure. Another option would be to improve the management of the assets held by the state pension funds in order to raise the rate of return. The management should be privatised, ending the system of placing surpluses with the Trust Fund Bureau. At the same time, the management of state-system reserves held in company pension funds should be improved by the planned abolition of regulatory control over the allocation of assets, and allowing assets to be invested in a way that optimised risks and returns. Raising the rate of return on pension funds by 1 percentage point would generate funds equivalent to 10 per cent of contribution income by 2050. Further options would include changing the post-retirement indexation of benefits and, if the objective were to limit contributions to their current level, some reduction in the rate at which pension benefits accrue might be required.

More significant reform of the pension system could be envisaged to reduce the extent of the imbalance in the inter-generation transfers that occurs in the current framework. Essentially, this could be accomplished by reducing contributions to the state system to the actuarially-fair level and then funding the remaining cost by general taxation. An eventual target could be a redistributive flat-rate pension financed by general taxation and a second-tier system based on actuari-

ally-fair contributions that accumulates a full pension fund under private management. However, such a system would need large increases in taxation and bond issue. Moreover, private-sector financial investment might – in some circumstances – have difficulty generating a sufficiently rapid increase in real benefits. A number of other reforms seem necessary to improve the equity of the system. The tax deductibility of pension contributions should be limited or the tax treatment of pensions should be aligned with that of income from employment. In addition, the elderly could contribute much more to their health and nursing care expenses, given their high levels of income and wealth. In the absence of control over the extent of public spending on health, central and local governments will likely be constrained to either cut other forms of spending or raise revenues. Such potential pressures add to the need to reduce the current level of the central and local government budget deficit.

The social security system would become more sustainable if the labour force participation of women and the elderly were to increase and if the birth rate were to rise. To accomplish this, social policy should be more oriented to helping families and reducing the cost to women of working and having families. In addition, tax and social benefit policies that discourage women and the elderly from working should be revised.

## Notes

1. Source: EPA, *The Survey of Corporate Activities*, 1997.
2. See for instance, Nomura Research Institute, *Quarterly Economic Outlook*, April 1997.
3. The relatively large cash flow in cars and other export-related manufacturing industries was sufficient to finance their strong demand for investment.
4. For housing purchases, there was an exemption from higher tax rates planned in April 1997 for housing which was contracted by the end of September, even if it was not available for use until after April.
5. The large increase in auto sales also reflected the introduction of new models, including recreational vehicles to match consumers' preferences, and sales promotion, backed by increased corporate profitability.
6. For a detailed discussion of these procedures, see K. Ishida, ''Recent Developments in the Implementation of Monetary Policy in Japan and its Operating Procedure'', *Proceedings of the Conference of Central Bank Economists''*, BIS, Basle, 1997.
7. The growth of reserves appears to have been limited to the re-investment abroad of the interest earned on the existing stock of reserves (about 0.3 per cent of GDP annually).
8. There are different ways of measuring overall monetary conditions. For a detailed discussion see Ishida (*op. cit.*).
9. This was partially offset by strong growth of certificates of deposits. Such instruments are not covered by deposit insurance, and so, with the sevenfold increase in the deposit insurance premium, offer a marginally higher interest rate than equivalent time deposits.
10. This figure excludes ''supported loans'', a new category of loan defined by the Japanese Bankers Federation. The introduction of such a category reflects a progressive widening of the definition of a bad loan. From March 1993, major banks disclosed overdue loans and loans to borrowers in legal bankruptcy. Regional banks only declared the latter category. From March 1996, regional banks disclosed overdue loans and all banks disclosed restructured loans. From March 1997, all banks disclosed supported loans. These amounted to Y 2.9 trillion for major banks and Y 3.4 trillion for all banks.
11. Excluding the two resolved credit co-operatives.
12. The Bank of Tokyo Mitsubishi and the Industrial Bank of Japan publish balance sheets using both US and Japanese accounting standards. After allowing for the fact that US accounting standards require the consolidation of losses at subsidiaries, bad loans are reported to be 12 per cent greater than those calculated using Japanese standards.

13. Such results were also found in the United States when credit institutions were liquidated following the savings and loans crisis.
14. Most of the eventual losses made by the CCPC when the collateral is sold are passed back to the originating bank which, in the meantime, has been accruing interest on its loan to the CCPC.
15. The portfolio is equally split between commercial and residential property, with undeveloped land accounting for 40 per cent of the total. The remainder is equally split between rented and empty property. Nearly all the properties are located in Greater Tokyo or Osaka.
16. Operating profits, as defined here, exclude earnings from bond and securities trading and valuation losses.
17. In the past, this was facilitated, in part, by the practice of *amakudari* – the employment of retired, high-ranking officials from the regulatory agencies. For an analysis of this issue, see Horiuchi (1996). It must be noted that similar practices, though different in nature and scope, exist elsewhere.
18. In addition, such a holding company cannot have more than five sizeable subsidiaries in separate principal fields of business.
19. International comparisons are not available for this debt indicator as data on central government net debt is not collected by the OECD. However, in countries with pay-as-you-go social security systems which have been balanced on an annual basis, the net debt of the social security system is close to zero in line with cumulated deficits and hence central and local government net debt will be the same as general government net debt. Restricting the comparison to such countries, the net debt of central and local government averaged 47 per cent of GDP, somewhat less than in Japan. Only three countries – Belgium, Italy and Canada – had a higher ratio than Japan.
20. This is equivalent to the issuance of Y 21 trillion of new government bonds. Each year a certain proportion of outstanding bonds is paid to an off-budget account for debt redemption, which is financed by the issuance of new bonds.
21. The basic budget law only allows the issuance of bonds to finance certain capital account activities. Such debt is known as ‘‘construction bonds’’. Since 1975, the government has received annual authorisation, in addition, to issue deficit bonds to cover other forms of expenditure.
22. In *Economic Outlook 61*, it was assumed that general government consumption and investment would be held constant in nominal terms, broadly equivalent to the planned budget.
23. The deficit which stabilises net central and local government debt can be calculated from the formula $D = d(1 + 1/g)$ where $D$ is the eventual stable net debt ratio associated with a given initial deficit $d$ and a growth of nominal incomes of $g$. Thus, to stabilise debt at its projected level of 67 per cent at end-1997, and assuming $3\frac{1}{2}$ per cent nominal income growth, the government deficit would have to be $67/(1 + 1/.035)$, *i.e.* $2\frac{1}{4}$ per cent of GDP. If the deficit remains at 3 per cent, net central and local government debt would almost double from its current level of 67 per cent to about 90 per cent of GDP before stabilising. As the assets of central and local governments amount to 29 per cent of GDP, gross debt (both of general government and central and local government) would reach 120 per cent of GDP at that point.

24. In such countries, public investment is typically close to 5 per cent.

25. This estimate refers solely to the building cost and excludes differences in land prices. See *OECD Purchasing Power Parities Report*, 1990.

26. For instance, a study by the Ministry of Construction found in 1993 that the cost of public works was between 13 and 45 per cent greater than in the United States, while the overall price level was 41 per cent higher.

27. Several papers provide evidence on the extent of the productivity-enhancing effects of public infrastructure investment in different regions of Japan [Asarco *et al.* (1994), Mitsui *et al.* (1995) and Ohgawara (1996)]. The research presents cross-sectional or time-series estimates of the marginal productivity of infrastructure capital by estimating regional aggregate production functions (most commonly of the Cobb-Douglas type). The estimated functions incorporate the public capital stock as a production factor, as well as the private capital stock and the labour force. They are similar to the work of Aschauer (1989) and Munnell (1990) for the US economy. All the Japanese studies conclude that the marginal productivity of infrastructure is relatively higher in metropolitan areas such as Tokyo, Osaka and Tokai. There are, though, some differences in the estimates of the levels of marginal productivity in the different studies which stem from differences in estimation periods, the statistical definition of infrastructure and the precise functional forms used in the estimation.

28. A similar but less marked relationship is found for public consumption.

29. Public corporations are defined here according to the methodology of the Japanese National Accounts.

30. This company was established in 1987 to take over the debt of the Japanese National Railways before its privatisation. The debt reflects the past losses of the JNR. Initially, the settlement corporation was endowed with land from JNR, which was estimated to have a value equal to 30 per cent of the debt. Since then, however, in the face of falling land and stock market prices, the profit on asset sales has not been sufficient to meet interest payments. As a result, the imbalance between long-term liabilities and assets has increased further.

31. Deposit-taking institutions, life insurance companies and non-life insurance companies.

32. The Trust Fund Bureau relends about one-third of the social security pension deposits to the Pension Welfare Corporation. This company is owned by the government and managed by the Ministry of Health and Welfare. It functions as a portfolio investment company.

33. The Industrial Investment Special Account Fund and government-guaranteed bonds are additional sources of funds.

34. These are administered by the ministries that originate the flow of funds to the Ministry of Finance. Thus, the Pension Welfare Corporation is run by the Ministry of Health and Welfare and the Postal Insurance and Annuities Welfare Corporation by the Ministry of Posts and Telecommunications, which is also in charge of the investments of the Postal Savings Special Account. It should be noted that these institutions are not officially regarded as financial institutions.

35. Prepayment is all the more likely since most HLC loans are sold by commercial banks who keep information on the borrowing of their clients from the HLC.

36. The gain to consumers from these types of changes has been estimated at Y 145 billion per year during the first half of the 1990s (EPA, 1997).

37. Matsushita Electric, for example, expects to save Y 600 million a year by integrating foreign currency management within the firm.

38. In contrast, the share of US equities traded in London has remained constant at just under 1 per cent.

39. As a first step, transactions above Y 50 million will be liberalised as early as April 1998. Commissions on deals of more than Y 1 billion were liberalised in 1995, leading to a significant decline in rates on these transactions, which account for about 10 per cent of trading in equities.

40. Under this agreement, Japanese life and non-life insurance companies will be able to enter the so-called third sector (personal accident, sickness, injury and nursing care) through subsidiaries two and a half years after the deregulation of the primary sectors. Foreign companies have about 35 per cent of the third sector, which accounts for almost 5 per cent of Japan's insurance market.

41. Securities holding companies that include a bank will be limited to the same financial fields. However, those without a bank will be allowed to hold industrial and commercial companies.

42. The pension fund market was partially opened to investment advisory firms, both domestic and foreign, in 1996. In the year following this reform, the amount of corporate pension funds managed by life insurers declined for the first time in the more than thirty years for which data are available.

43. The average price per passenger kilometre was 13 per cent above the OECD average in 1993, while the average cost was 20 per cent higher (see 1996 Survey).

44. In the allocation of slots at congested airports in the United States and the EU, new entrants are given priority up to a certain point (twelve slots in the United States and four in the EU).

45. This measure refers to the average cost of a one-minute call to OECD countries at peak rates, calculated in national currencies (OECD, *Communications Outlook*, 1997).

46. NTT will be permitted to enter the international market through its affiliates prior to its reorganisation.

47. For example, one company providing international services recently announced that it would merge in October with Japan Telecom, one of the three private firms in the domestic long-distance market. The merger will create the third largest telecommunications firm in Japan, although its current sales would only amount to 5 per cent of NTTs.

48. One long-distance firm, Japan Telecom, has announced plans to build a local network to connect large corporate customers to its long-distance network and thus reduce its dependence on NTT.

49. NTT must also keep separate accounts for interconnection-related revenues and expenses.

50. From a submission by the Japanese government that is included in the OECD's *Communications Outlook*, 1997.

51. An April 1997 survey by MITI found that the price of electricity was only 21 per cent cheaper in the United Kingdom than in Japan and 17 per cent cheaper in Germany.

52. One company, Nissan Steel, sold electricity at a price 35 per cent below that produced by the power companies.

53. This approach is also recommended in ''Regulatory Reform in the Electricity Sector'', in *The OECD Report on Regulatory Reform*, Vol. I (1997).

54. By *Nihon Keizai Shimbun*, reported on 30 May 1997.

55. In April 1997, the ''positive list'' regulation, which had allowed them to operate in 29 categories, was replaced by a ''negative list'', which prohibits them from activities in five areas. The size of the negative list is to be discussed.

56. In the manufacturing sector, for example, female employees have been limited to six hours of overtime per week in principle, with an annual maximum of 150 hours. In addition, there are restrictions on women working on holidays and between the hours of 10 p.m. and 5 a.m.

57. A discussion of these issues is included in ''Product Standards, Conformity Assessment and Regulatory Reform'' in *The OECD Report on Regulatory Reform*, Vol. I (1997).

58. Such cartels have been allowed under the Anti-Monopoly Act itself, the Exemption Act and various individual laws. Measures to reform or repeal 41 exemption systems allowed under individual laws have been taken. In addition, the status of an additional 40 exemption systems allowed under the Anti-Monopoly Act or the Exemption Act are under review.

59. The legal definition is a company in which the total book value of the stockholdings in subsidiaries constitutes more than half of the value of the total assets of the company.

60. If the five companies have a substantial position in their respective industries, such a holding company is prohibited regardless of the size of its total assets.

61. The use of administrative guidance by local government may also have become less burdensome according to a 1997 survey by *Nihon Keizai Shimbun.*

62. The latest national demographic projections have been used here. Projections for other countries are taken from Bos (1994): they assume that the fertility rate will recover to slightly over 2.0 by the year 2030. While this assumption differs from that used for Japan, it does not greatly affect the results, provided that the analysis does not go beyond the year 2025 and the focus remains on the population above age 20.

63. The working-age population is often defined as the age range 15 to 64. In this case, 20 has been used as the lower limit in view of the high and growing proportion of the 15 to 19 year-olds enrolled in full-time education. The upper limit is set at 60, the age at which social security retirement pensions are paid.

64. The projection of the Japanese population is based on the *1997 Estimates by the Population and Social Security Research Institute*, Ministry of Health and Welfare, published in January 1997.

65. Defined as the average number of children born to all women between the age of 15 and 49 in a given year.

66. In 1994, children born to unmarried women accounted for only 1.2 per cent of total births in Japan, compared to 30 per cent in the United States, 28 per cent in France and 51 per cent in Sweden.

67. The data are based on the White Paper by the Ministry of Health and Welfare in 1996. Similar patterns are observed for men, though to a much smaller extent.

68. This refers to the average life expectancy at birth.

69. Ages 15 to 65, though in Japan many people over 65 work.

70. According to the latest national projections.

71. The rise in the participation of employees has been masked by a significant fall in the participation rates of self-employed older workers, many of whom are in the agricultural sector. As the labour force participation of the self-employed is higher, and their share in the total labour force is now shrinking in size, the rising trend in participation of older people may be more evident in the future.

72. The negative association between lower labour force growth and higher productivity growth was pointed out by Culter *et al*. (1990), and the similar relationship between labour force growth and total factor productivity growth was shown by Yashiro and Oishi, 1993.

73. Horioka, 1991. This estimate was based on a single equation regression model linking the savings rate to economic growth and the dependency rate.

74. Yashiro and Oishi, 1993. This study includes other structural variables, such as the proportion of women who work.

75. See Hayashi (1992) and Horioka (1993) for a survey.

76. ''National Survey of Family Income and Expenditure'', conducted every five years, provides information on consumption, saving and income for employees, the retired and the self-employed (in part).

77. It is not possible to directly observe the savings rate of the elderly who live with their children. The method adopted here, and in other research, is to compare households, with and without dependent parents, and to attribute differences in income and spending to the presence of the dependent families. Such a methodology assumes that nuclear and extended families are identical except for the presence or absence of cohabiting parents.

78. EPA, 1995 (also see Yashiro and Maeda, 1994).

79. These calculations assume that the relative incomes of each age group remain constant. However, as the pension system matures, the average income of the elderly will rise relative to that of people of working age. This will push down saving rates, other things being equal.

80. Levhari and Mirman, 1977.

81. Weil, 1994.

82. In the presence of intergenerational altruism (Barro, 1974), an elderly person would decrease his saving, since the marginal utility of bequests for any given size of the parents' wealth are reduced by an increase in the average bequest per child. Altruism, though, may not be the only motive for bequests. Genetic survival may also be a factor. However, a parent who has a strategic bequest motive (Bernheim, Shleifer and Summers, 1985) would increase saving, since a decline in the number of children may reduce the credibility of a threat to disinherit

any one heir and thus raise the amount of bequest that he must pay in order to receive attention from a child. However, forced inheritance laws in Japan lessen this threat in any case. In the case of Japan, Horioka (1993) surveyed existing literature and concluded that ''bequests appear to be relatively prevalent in Japan, but the bulk of these bequests appear to be unintended or accidental, or intended bequests motivated by implicit annuity contracts between aged and their children or by a strategic bequest motive. Only a relatively small proportion of bequests appear to be motivated by intergenerational altruism''.

83. Government Actuary, Ministry of Health and Welfare, cited in Murakami (1997). This figure excludes that part of the government's pension fund system which is contracted out to corporations. These funds are included in the assets of the state system.

84. Moreover, public-sector employees who start to work in the private sector are exempt from the limits on earnings.

85. While social security contributions are completely deductible, the situation for private pension contributions varies.

86. For example, a KNH pension benefit in 1994 for the average male was about Y 189 000 per month or 43 per cent of average earnings (including bonus for male employees in companies with five or more employees). The average pension benefit for the most recent beneficiaries was Y 214 000 or 48 per cent of average earnings. The maximum benefit represents 53 per cent of earnings.

87. Although the first and second-tier pension schemes are conceptually separate in accounting terms, the the Employees Pension Insurance pays the second-tier benefit and transfers the cost of the first-tier scheme to the Basic Pension Account of the government. More than 80 per cent of social security pensions are, in fact, paid by the employees' scheme, as all basic pensions for employees in payment before 1986 continue to be paid by the schemes for employees. In addition, expenditure on benefits for the self-employed is limited by the fact that many self-employed do not pay contributions and so receive no benefits.

88. These official projections are based on economic growth of 1½ per cent, real wage growth of 2 per cent and real long-term interest rates of 3½ per cent.

89. This estimate uses both the government assumptions for the surplus of the fund and for inflation and real economic growth.

90. According to projections by the Government Actuary.

91. This figure for the growth of earnings is based on the average standard earnings of employees in large, medium and small companies.

92. According to the official projections, the ratio of female contributors in KNH to the female working-age population will be basically stable between 2000 and 2025, despite the sharp increase in the 1975-95 period from 19 to 26 per cent.

93. The discrimination occurs because a married couple with one worker receive two flat-rate national pensions and one earnings-related pension. A married couple, both of whom work, pay twice the contributions but only receive an extra earnings-related pension. Given that the flat-rate pension is typically equal to the earnings-related pension, when a married woman starts to work, contributions double but benefits rise by only one-third.

94. Given that bonus and overtime payments account for 25 per cent of total earnings, such a level of subsidy should stabilise – or slightly increase – total earnings.

95. There is also a ceiling on the subsidy and pension benefit which cannot exceed Y 380 000 per month. From 1998, the pension benefits of those who receive the wage subsidies are to be reduced by 10 per cent when the average salary is under 64 per cent of the wage of the person at 60. This makes an already complicated system even more complex and does not stop some people receiving a total income above their income at 60.

96. Recently, nominal user charges of Y 1 020 (roughly $9) per month and Y 710 ($6) per day for hospitalisation were imposed on those aged 70 and above (excluding the poor).

97. The total number of frail people is officially projected to rise from 2.8 per cent of the working population in 1995 to 7.5 per cent in 2025. Half of these people are likely to be bedridden or senile.

98. For details of the projections of pension expenditure in countries other than Japan, see OECD (1996*b*).

99. In Japan, a commonly-used measure of government spending is the national burden ratio. This is defined as the sum of tax, social security contributions and the central and local government fiscal deficit. This definition is equivalent to public expenditure plus the social security surplus less the property income of general government.

100. When JR was privatised, the government took responsibility for the accumulated pension liability of the railway system.

101. Canner, Mankiw and Weil (1997).

102. The calculus would be less favourable if companies retained the benefit of higher returns on KNK funds.

103. Hatta and Oguchi (1993).

104. Assuming that the rate of time preference and wage growth was not negative (Feldstein, 1995).

105. ‘‘Three-generation families’’ in which parents, children, and grandchildren live together and share the family budget accounted for nearly 50 per cent of agricultural households in 1990. The equivalent ratios for non-agricultural households and employee households were 16 and 11 per cent, respectively (Management and Coordination Agency, 1990).

106. For a detailed study of the possibilities for reform and control over public health expenditure see OECD (1995*b*).

107. Ogura and Oishi, 1995, indicated that the costs of medical treatment are lower in the hospitals where certain ceilings are set on the total expenses than in others.

108. However, shifting the elderly from hospitals to nursing homes might require additional expenditures for welfare services. For example, the standard per capita living space in public nursing homes is about three times larger than in hospitals, which were originally designed for short-term stays.

109. The negative impact of the earnings test on older persons’ labour supply will be reduced by the scheduled increase of the eligibility age for the flat pension from 60 to 65.

110. The exceptions for accepting unskilled foreigners are trainees and descendants of the Japanese emigrants to other countries, primarily in South America.

111. Seike, 1993.

112. Higuchi, 1995.

113. These participation rates are calculated as the total number of women working relative to the total population aged over 15. The usual measure used in Japan – the working population to the total population aged between 15 and 65 – overstates the participation rate due to the large number of workers between 60 and 65.

114. This is based on simultaneous equation models on determining fertility, women's labour force participation, and provision of day-care facilities (Yashiro *et al*., 1997).

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*Annex*

# Chronology of main economic events

## 1996

### November

The Prime Minister announces a five-year plan, referred to as the ''Big Bang'', for financial deregulation.

The Ministry of Finance orders Hanwa Bank to suspend operations.

### December

The Economic Council releases its report, ''Economic Structural Reforms in Six Areas''.

A sub-committee of the Administrative Reform Committee announces their proposed deregulation measures.

The Fiscal System Council submits its final report on fiscal structural reform to the Minister of Finance.

The US – Japan framework talks on insurance come to a conclusion.

The Cabinet approves the ''Action Programme for Structural Economic Reforms''.

The Cabinet announces its official economic projection of 1.9 per cent growth in FY 1997.

The Cabinet approves a medium-term target of reducing central and local governments' deficits to 3 per cent by 2003.

The Cabinet approves the initial FY 1997 draft budget, containing total expenditure of Y 77.3 trillion, a 3 per cent increase over the initial budget for the previous year.

## 1997

### January

The Cabinet approves the FY 1997 tax changes, including a phase-out of the temporary income tax cut and a hike in the consumption tax rate from 3 to 5 per cent.

The Diet adopts the first supplementary budget of FY 1996.

### March

The Diet adopts the FY 1997 initial budget.

The Cabinet approves a second revision of the ''Deregulation Action Programme'' of March 1995.

The government announces an overall package for stimulating liquidity in the real estate market, aiming at resolving bad loan problems and promoting urban development.

The merger of Hokkaido Takushoku Bank and Hokkaido Bank is announced.

Japan Credit Bank presents a substantial restructuring plan.

### April

The Ministry of Finance orders Nissan Life Insurance to suspend operations.

### May

The Diet adopts changes in the Commercial Law, which will enable listed companies to introduce the stock option system.

The Diets adopts changes in the Foreign Exchange Law, which will fully liberalise foreign exchange transactions from April 1998.

### June

The Fiscal Reform Conference announces its final report on plans for medium-term spending cuts, which is also approved by the Cabinet.

The Diet adopts changes in the Bank of Japan Law, which strengthen its independence and the transparency of the policy decision-making process.

The Diet passes changes in the Anti-Monopoly Law to lift the ban on pure holding companies.

The Diet adopts changes in the NTT Law, which will spilt it into three companies operating under a holding company.

The final reports on reforms in the areas of banking, securities and insurance, as part of the financial ''Big Bang'', are submitted to the Minister of Finance.

A law establishing the Supervisory Agency for Financial Entities is adopted by the Diet.

**September**

The Administrative Reform Committee announces its interim report on a blue-print for administrative reform, which would trim the government to one cabinet office plus 12 ministries and agencies, down from a present count of 22.

Legislation to implement medium-term spending cuts are approved by the Cabinet.

*STATISTICAL ANNEX AND STRUCTURAL INDICATORS*

Table A. **Selected background statistics**

| | Average 1987-96 | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **A. Percentage change in constant 1990 prices** | | | | | | | | | | | |
| Private consumption | 3.1 | 4.2 | 5.3 | 4.8 | 4.4 | 2.5 | 2.1 | 1.2 | 1.9 | 2.0 | 2.8 |
| Gross fixed capital formation | 4.6 | 9.1 | 11.5 | 8.2 | 8.5 | 3.3 | –1.5 | –2.0 | –0.8 | 1.1 | 8.7 |
| Public | 6.4 | 6.7 | 4.3 | –0.4 | 4.9 | 4.9 | 14.5 | 15.7 | 2.8 | 0.7 | 9.9 |
| Private residential | 4.2 | 22.4 | 11.4 | 0.9 | 4.8 | –8.5 | –6.5 | 2.4 | 8.5 | –6.4 | 13.6 |
| Private non-residential | 4.2 | 5.9 | 14.7 | 14.5 | 10.9 | 6.3 | –5.6 | –10.2 | –5.3 | 3.9 | 6.6 |
| GDP at market prices | 3.1 | 4.2 | 6.2 | 4.8 | 5.1 | 3.8 | 1.0 | 0.3 | 0.6 | 1.4 | 3.5 |
| GDP price deflator | 1.0 | 0.1 | 0.7 | 2.0 | 2.3 | 2.7 | 1.7 | 0.6 | 0.2 | –0.6 | 0 |
| Industrial production | 2.2 | 3.4 | 10.5 | 5.9 | 4.1 | 1.8 | –6.1 | –4.5 | 0.8 | 3.5 | 2.7 |
| Employment | 1.0 | 1.0 | 1.7 | 2.0 | 2.0 | 1.9 | 1.1 | 0.2 | 0 | 0.1 | 0.5 |
| Compensation of employees (current prices) | 4.5 | 3.6 | 5.6 | 7.1 | 8.4 | 7.8 | 3.5 | 2.3 | 2.5 | 2.1 | 2.0 |
| Productivity (GDP/employment) | 2.0 | 3.2 | 4.4 | 2.8 | 3.0 | 1.9 | 0 | 0.1 | 0.6 | 1.3 | 3.1 |
| Unit labour costs (compensation/GDP) | 1.4 | –0.6 | –0.5 | 2.2 | 3.2 | 3.8 | 2.5 | 2.0 | 1.8 | 0.7 | –1.5 |
| **B. Percentage ratios** | | | | | | | | | | | |
| Gross fixed capital formation as per cent of GDP at constant prices | 30.4 | 28.4 | 29.8 | 30.8 | 31.8 | 31.6 | 30.8 | 30.1 | 29.7 | 29.6 | 31.1 |
| Stockbuilding as per cent of GDP at constant prices | 0.4 | 0.2 | 0.7 | 0.8 | 0.6 | 0.8 | 0.3 | 0.2 | 0 | 0.3 | 0.2 |
| Foreign balance as per cent of GDP at current prices | 1.8 | 3.1 | 2.3 | 1.4 | 0.7 | 1.7 | 2.2 | 2.3 | 2.1 | 1.5 | 0.5 |
| Compensation of employees as per cent of GDP at current prices | 54.7 | 53.7 | 53.0 | 53.1 | 53.6 | 54.2 | 54.6 | 55.3 | 56.3 | 57.0 | 56.1 |
| Direct taxes as per cent of household income | 8.1 | 8.1 | 8.0 | 8.0 | 8.9 | 9.3 | 8.9 | 8.2 | 7.6 | 6.9 | 6.9 |
| Household saving as per cent of disposable income | 13.0 | 13.8 | 13.0 | 12.9 | 12.1 | 13.2 | 13.1 | 13.4 | 13.3 | 13.1 | 11.9 |
| Unemployment rate | 2.6 | 2.8 | 2.5 | 2.3 | 2.1 | 2.1 | 2.2 | 2.5 | 2.9 | 3.1 | 3.4 |
| **C. Other indicator** | | | | | | | | | | | |
| Current balance (billion US dollars) | 87.9 | 87.0 | 79.6 | 57.2 | 35.8 | 68.2 | 112.4 | 131.9 | 130.5 | 110.4 | 65.8 |

*Source:* OECD.

## Table B. **Gross domestic product and expenditure**

Billion yen, current prices

| | 1986 | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Private consumption | 196 712 | 205 956 | 217 839 | 232 890 | 249 288 | 261 891 | 272 294 | 278 703 | 286 154 | 290 515 | 299 426 |
| Public consumption | 32 387 | 32 975 | 34 183 | 36 275 | 38 807 | 41 356 | 43 262 | 44 771 | 45 743 | 47 555 | 48 969 |
| Gross fixed capital formation | 91 499 | 99 152 | 110 856 | 122 274 | 136 467 | 143 998 | 143 525 | 140 433 | 137 291 | 136 792 | 148 231 |
| Change in stocks | 1 560 | 662 | 2 676 | 2 976 | 2 430 | 3 453 | 1 489 | 620 | 49 | 947 | 1 010 |
| **Total domestic demand** | 322 159 | 338 745 | 365 555 | 394 415 | 426 992 | 450 698 | 460 571 | 464 527 | 469 237 | 475 809 | 497 636 |
| Exports of goods and services | 38 090 | 36 210 | 37 483 | 42 352 | 45 920 | 46 722 | 47 384 | 44 197 | 44 410 | 45 393 | 49 700 |
| *less:* Imports of goods and services | 24 791 | 25 195 | 29 065 | 36 768 | 42 872 | 39 121 | 36 891 | 33 343 | 34 387 | 38 272 | 47 022 |
| **Gross domestic product at market prices** | 335 457 | 349 760 | 373 973 | 399 998 | 430 040 | 458 299 | 471 064 | 475 381 | 479 260 | 482 930 | 500 314 |
| *less:* Consumption of fixed capital | 46 205 | 48 930 | 52 398 | 58 081 | 62 987 | 68 541 | 72 823 | 74 383 | 75 605 | 77 216 | . . |
| Net indirect taxes | 21 535 | 24 961 | 27 470 | 29 053 | 30 568 | 31 116 | 33 996 | 33 728 | 34 537 | 35 229 | . . |
| **Domestic product at factor cost** | 267 717 | 275 869 | 294 106 | 312 864 | 336 485 | 358 642 | 364 245 | 367 270 | 369 118 | 370 486 | . . |

Quarterly data seasonally adjusted at annual rates
Billion yen, current prices

| | 1994 | 1995 | | | | 1996 | | | | 1997 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | IV | I | II | III | IV | I | II | III | IV | I | II |
| Private consumption | 287 614 | 288 132 | 288 408 | 291 350 | 293 709 | 300 542 | 297 381 | 298 232 | 301 325 | 315 157 | 301 787 |
| Public consumption | 45 420 | 47 337 | 47 866 | 48 230 | 47 055 | 48 600 | 48 912 | 49 805 | 48 782 | 50 224 | 50 253 |
| Gross fixed capital formation | 135 257 | 133 563 | 135 365 | 136 168 | 141 367 | 145 637 | 149 214 | 149 514 | 149 690 | 144 273 | 142 160 |
| Change in stocks | –74 | 1 104 | 928 | 881 | 900 | 2 040 | 1 111 | 777 | 111 | 579 | 1 076 |
| **Total domestic demand** | 468 217 | 470 135 | 472 568 | 476 629 | 483 031 | 496 818 | 496 618 | 498 328 | 499 908 | 510 233 | 495 277 |
| Exports of goods and services | 44 823 | 44 034 | 43 644 | 46 058 | 47 722 | 47 980 | 48 964 | 49 458 | 52 317 | 53 960 | 57 998 |
| *less:* Imports of goods and services | 35 468 | 36 161 | 35 523 | 39 123 | 42 192 | 44 306 | 47 011 | 47 330 | 49 411 | 51 831 | 50 831 |
| **Gross domestic product at market prices** | 477 572 | 478 009 | 480 689 | 483 564 | 488 561 | 500 492 | 498 570 | 500 456 | 502 813 | 512 362 | 502 444 |

*Source:* Economic Planning Agency (EPA), *Annual Report on National Accounts;* OECD.

Table C. **Gross domestic product and expenditure**

Billion yen, 1990 prices

| | 1986 | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Private consumption | 207 669 | 216 413 | 227 860 | 238 703 | 249 139 | 255 410 | 260 708 | 263 744 | 268 801 | 274 169 | 281 950 |
| Public consumption | 35 962 | 36 524 | 37 375 | 38 113 | 38 681 | 39 452 | 40 224 | 41 170 | 42 154 | 43 636 | 44 633 |
| Gross fixed capital formation | 95 707 | 104 453 | 116 428 | 126 006 | 136 685 | 141 163 | 138 981 | 136 269 | 135 243 | 136 733 | 148 642 |
| Public | 24 186 | 25 800 | 26 915 | 26 812 | 28 139 | 29 521 | 33 807 | 39 120 | 40 209 | 40 485 | 44 490 |
| Private residential | 17 668 | 21 623 | 24 088 | 24 297 | 25 465 | 23 304 | 21 789 | 22 309 | 24 196 | 22 636 | 25 710 |
| Private non-residential | 53 853 | 57 030 | 65 425 | 74 896 | 83 081 | 88 338 | 83 385 | 74 840 | 70 839 | 73 612 | 78 442 |
| Change in stocks | 1 467 | 705 | 2 806 | 3 140 | 2 435 | 3 493 | 1 484 | 827 | 16 | 1 403 | 1 144 |
| **Total domestic demand** | 340 804 | 358 095 | 384 469 | 405 962 | 426 940 | 439 518 | 441 397 | 442 010 | 446 213 | 455 941 | 476 368 |
| Exports of goods and services | 37 434 | 37 232 | 39 441 | 43 044 | 46 012 | 48 426 | 50 864 | 51 500 | 53 865 | 56 793 | 58 124 |
| *less:* Imports of goods and services | 25 358 | 27 771 | 33 584 | 39 822 | 42 966 | 41 628 | 41 337 | 41 229 | 44 881 | 51 278 | 56 655 |
| **Gross domestic product at market prices** | 352 880 | 367 556 | 390 325 | 409 184 | 429 986 | 446 315 | 450 924 | 452 282 | 455 197 | 461 456 | 477 837 |

Quarterly data seasonally adjusted at annual rates

Billion yen, 1990 prices

| | 1994 | 1995 | | | | 1996 | | | | 1997 | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | IV | I | II | III | IV | I | II | III | IV | I | II |
| Private consumption | 269 834 | 270 799 | 272 267 | 275 170 | 277 840 | 283 441 | 280 601 | 280 113 | 283 418 | 295 756 | 278 787 |
| Public consumption | 42 526 | 43 351 | 43 499 | 43 845 | 43 891 | 44 242 | 44 284 | 44 879 | 45 163 | 45 186 | 44 615 |
| Gross fixed capital formation | 133 918 | 132 825 | 135 624 | 136 384 | 141 432 | 146 203 | 149 690 | 150 198 | 149 640 | 143 991 | 141 322 |
| Public | 39 246 | 37 774 | 39 682 | 41 053 | 42 662 | 45 456 | 46 401 | 45 486 | 42 117 | 36 900 | 38 395 |
| Private residential | 23 995 | 23 785 | 22 555 | 21 394 | 23 013 | 24 403 | 25 849 | 25 712 | 26 752 | 25 713 | 22 762 |
| Private non-residential | 70 677 | 71 266 | 73 387 | 73 938 | 75 758 | 76 344 | 77 439 | 79 000 | 80 772 | 81 378 | 80 165 |
| Change in stocks | 832 | 1 180 | 1 035 | 1 526 | 1 856 | 2 166 | 1 084 | 894 | 437 | 253 | 1 164 |
| **Total domestic demand** | 447 110 | 448 154 | 452 425 | 456 925 | 465 019 | 476 052 | 475 659 | 476 085 | 478 659 | 485 185 | 465 888 |
| Exports of goods and services | 55 137 | 55 620 | 57 635 | 56 686 | 57 246 | 56 869 | 56 940 | 57 849 | 60 739 | 61 172 | 65 063 |
| *less:* Imports of goods and services | 46 571 | 48 218 | 50 037 | 52 178 | 54 617 | 55 703 | 56 731 | 56 577 | 57 603 | 57 859 | 56 723 |
| **Gross domestic product at market prices** | 455 676 | 455 556 | 460 023 | 461 434 | 467 649 | 477 218 | 475 868 | 477 357 | 481 796 | 488 499 | 474 228 |

*Source:* EPA, *Annual Report on National Accounts;* OECD.

Table D. **Net domestic product by industry of origin**

Billion yen, current prices

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|
| Agriculture, forestry and fishing | 8 115 | 8 139 | 8 596 | 8 932 | 8 782 | 8 633 | 7 940 | 8 287 | 7 500 |
| Mining and quarrying | 718 | 758 | 707 | 882 | 868 | 899 | 843 | 805 | 796 |
| Manufacturing | 84 999 | 91 220 | 96 660 | 103 813 | 109 395 | 106 978 | 101 045 | 97 598 | 99 464 |
| Construction | 28 259 | 32 034 | 35 965 | 39 436 | 42 449 | 43 767 | 45 566 | 45 990 | 44 478 |
| Electricity, gas and water | 7 356 | 7 140 | 6 789 | 7 360 | 8 307 | 8 680 | 8 705 | 9 078 | 8 959 |
| Wholesale and retail trade | 43 837 | 46 879 | 49 564 | 54 523 | 58 536 | 58 788 | 56 603 | 56 432 | 56 621 |
| Bank, insurance and real estate | 47 154 | 51 547 | 56 294 | 58 107 | 60 410 | 62 401 | 64 199 | 67 623 | 66 973 |
| Government services | 26 098 | 26 989 | 28 389 | 30 180 | 31 699 | 33 041 | 34 045 | 34 746 | 35 718 |
| Other services | 69 962 | 73 524 | 79 689 | 85 718 | 92 105 | 97 687 | 101 830 | 103 000 | 105 584 |
| **Total** | 316 499 | 338 229 | 362 652 | 388 951 | 412 550 | 420 875 | 420 773 | 423 557 | 426 092 |
| Import tax | 1 166 | 1 217 | 2 252 | 2 733 | 2 874 | 2 894 | 2 549 | 2 676 | 2 838 |
| Imputed rent | –15 806 | –16 728 | –20 677 | –22 606 | –22 955 | –23 332 | –20 819 | –21 372 | –20 818 |
| Other[1] | . . | . . | –1 458 | –2 204 | –2 308 | –2 047 | –1 865 | –1 749 | –1 799 |
| **Net domestic product at factor cost** | 301 859 | 322 718 | 342 769 | 366 873 | 390 161 | 398 390 | 400 639 | 403 112 | 406 312 |

1. General consumption tax adjustment for investment expenditure.

*Source:* EPA, *Annual Report on National Accounts;* OECD, *National Accounts.*

Table E. **The distribution of national income**

Billion yen

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|
| GDP at market prices | 349 760 | 373 973 | 399 998 | 430 040 | 458 299 | 471 064 | 475 381 | 479 260 | 482 930 |
| Net factor income from abroad | 2 054 | 2 302 | 2 849 | 2 932 | 3 190 | 4 268 | 4 381 | 3 942 | 3 991 |
| GNP | 351 814 | 376 275 | 402 848 | 432 972 | 461 489 | 475 332 | 479 762 | 483 202 | 486 921 |
| Taxes *less* subsidies[1] | 23 930 | 26 327 | 28 203 | 30 747 | 30 713 | 33 847 | 34 088 | 35 080 | 34 630 |
| GNP at factor cost | 327 883 | 349 949 | 374 645 | 402 225 | 430 776 | 441 485 | 445 674 | 448 122 | 452 291 |
| Depreciation | 48 930 | 52 398 | 58 081 | 62 987 | 68 541 | 72 823 | 74 383 | 75 605 | 77 216 |
| NNP at factor cost[2] (A) | 278 953 | 297 551 | 316 564 | 339 237 | 362 235 | 368 662 | 371 291 | 372 517 | 375 075 |
| Compensation of employees (B) | 187 812 | 198 364 | 212 528 | 230 446 | 248 392 | 257 139 | 263 099 | 269 654 | 275 332 |
| Operating surplus [= (A) – (B)] | 91 141 | 99 187 | 104 036 | 108 791 | 113 843 | 111 523 | 108 192 | 102 863 | 99 743 |
| Dividends, interest, rent, etc.[3] | 25 074 | 25 498 | 29 326 | 35 920 | 41 111 | 36 524 | 33 836 | 31 435 | 26 356 |
| Retained income of enterprises[3] | 66 067 | 73 689 | 74 710 | 72 871 | 72 732 | 74 999 | 74 356 | 71 428 | 73 387 |
| Private corporations[4] | 32 646 | 37 819 | 34 555 | 32 990 | 33 515 | 30 814 | 29 831 | 27 658 | 28 608 |
| Public enterprises | 143 | 579 | 3 629 | 4 576 | 2 550 | 2 772 | 3 187 | –918 | 2 933 |
| Self-employment[5] | 33 278 | 35 292 | 36 526 | 35 305 | 36 667 | 41 412 | 41 338 | 44 688 | 41 846 |
| *Memorandum items:* | | | | | | | | | |
| Total property income | 42 848 | 43 797 | 48 298 | 56 506 | 63 231 | 58 676 | 55 971 | 53 664 | 49 043 |
| *of which:* | | | | | | | | | |
| Paid by government and households | 16 844 | 17 374 | 18 009 | 19 386 | 20 711 | 20 786 | 20 905 | 21 136 | 21 659 |

1. Includes the statistical discrepancy.
2. National income.
3. Operating surplus = Dividends, etc. + Income of enterprises.
4. After dividends payments.
5. Personal companies.

*Source:* EPA, *Annual Report on National Accounts;* OECD, *National Accounts.*

Table F. **Income and expenditure of households**

Billion yen

| | 1985 | 1986 | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Compensation of employees | 173 815 | 181 325 | 187 813 | 198 364 | 212 529 | 230 447 | 248 392 | 257 139 | 263 100 | 269 654 | 275 333 |
| Wages and salaries | 151 291 | 157 113 | 161 255 | 170 397 | 182 118 | 196 898 | 212 338 | 221 863 | 226 645 | 232 794 | 236 321 |
| Employers'contribution to Social Security | 13 437 | 14 618 | 15 386 | 16 267 | 17 768 | 20 076 | 21 430 | 22 564 | 23 162 | 23 379 | 25 319 |
| Others | 9 087 | 9 595 | 11 171 | 11 700 | 12 642 | 13 473 | 14 624 | 12 712 | 13 293 | 13 481 | 13 693 |
| Income from property and entrepreneurship | 60 222 | 62 719 | 63 505 | 65 270 | 70 224 | 74 735 | 79 916 | 81 176 | 78 829 | 76 428 | 71 752 |
| Income from independent traders | 40 890 | 42 551 | 44 858 | 47 312 | 49 418 | 51 419 | 54 294 | 57 999 | 56 075 | 57 949 | 53 912 |
| Property income received, net | 19 331 | 20 168 | 18 647 | 17 957 | 20 805 | 23 316 | 25 622 | 23 177 | 22 754 | 18 479 | 17 840 |
| Current transfers from Government | 36 709 | 39 471 | 42 533 | 44 548 | 46 151 | 51 303 | 52 193 | 55 757 | 59 138 | 62 365 | 67 213 |
| Other transfers | 16 655 | 17 288 | 17 705 | 18 995 | 18 917 | 20 306 | 22 457 | 23 559 | 23 441 | 23 932 | 25 230 |
| **Household income** | 287 401 | 300 803 | 311 555 | 327 177 | 347 820 | 376 790 | 402 958 | 417 631 | 424 508 | 432 377 | 439 527 |
| *less:* Direct taxes on households and private non-profit institutions | 21 248 | 22 995 | 25 098 | 26 268 | 27 925 | 33 349 | 37 633 | 37 034 | 34 909 | 32 763 | 30 276 |
| *less:* Current transfers to Government | 26 436 | 28 037 | 30 015 | 31 684 | 33 703 | 39 310 | 41 656 | 43 856 | 45 203 | 46 048 | 50 661 |
| *less:* Other transfers | 19 061 | 19 833 | 20 519 | 22 064 | 22 251 | 23 998 | 26 156 | 27 525 | 27 374 | 27 911 | 28 797 |
| **Disposable income** | 220 656 | 229 939 | 235 924 | 247 160 | 263 940 | 280 133 | 297 513 | 309 217 | 317 022 | 325 656 | 329 793 |
| *less:* Consumption expenditure | 186 235 | 194 051 | 203 342 | 214 992 | 229 831 | 246 154 | 258 332 | 268 677 | 274 696 | 282 354 | 286 532 |
| Food | 41 537 | 41 723 | 41 855 | 43 888 | 43 844 | 46 328 | 48 484 | 49 429 | 49 482 | 49 763 | 48 721 |
| Clothing | 12 491 | 13 196 | 13 970 | 13 549 | 15 469 | 16 392 | 17 279 | 16 895 | 16 267 | 15 162 | 14 909 |
| Rent | 35 082 | 36 126 | 37 951 | 40 036 | 43 273 | 46 886 | 50 600 | 54 316 | 57 624 | 61 303 | 64 638 |
| Other | 97 125 | 103 006 | 109 566 | 117 519 | 127 245 | 136 548 | 141 969 | 148 037 | 151 323 | 156 126 | 158 264 |
| **Household saving** | 34 421 | 35 888 | 32 582 | 32 168 | 34 110 | 33 979 | 39 181 | 40 540 | 42 326 | 43 302 | 43 261 |
| (Per cent of disposable income) | 15.6 | 15.6 | 13.8 | 13.0 | 12.9 | 12.1 | 13.2 | 13.1 | 13.4 | 13.3 | 13.1 |

*Source:* EPA, *Annual Report on National Accounts;* OECD, *National Accounts.*

Table G. **Appropriation account for general government**

Billion yen

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|
| *Receipts:* | | | | | | | | | |
| 1. Direct tax | 44 615 | 48 329 | 53 901 | 58 367 | 62 253 | 59 519 | 54 761 | 50 363 | 48 243 |
| 2. Social security contributions | 29 694 | 31 363 | 33 387 | 38 957 | 41 264 | 43 436 | 44 753 | 45 585 | 50 197 |
| 3. Other current transfers received | 1 026 | 1 091 | 1 173 | 1 298 | 1 419 | 1 533 | 1 566 | 1 600 | 1 628 |
| 4. Indirect taxes | 28 379 | 30 878 | 32 158 | 35 212 | 34 968 | 37 301 | 37 199 | 38 003 | 38 939 |
| 5. Property income | 10 013 | 11 109 | 11 617 | 13 188 | 15 122 | 14 588 | 14 138 | 18 084 | 15 455 |
| 6. Current receipts, total | 113 727 | 122 771 | 132 236 | 147 021 | 155 027 | 156 377 | 152 417 | 153 634 | 154 462 |
| *Disbursements:* | | | | | | | | | |
| 7. Expenditure on goods and services | 32 975 | 34 183 | 36 275 | 38 807 | 41 356 | 43 262 | 44 771 | 45 743 | 47 555 |
| 8. Property income payable (Interest on public debt) | 15 347 | 15 671 | 16 023 | 16 820 | 17 377 | 17 733 | 17 690 | 17 971 | 18 468 |
| 9. Subsidies | 3 419 | 3 409 | 3 104 | 4 644 | 3 852 | 3 304 | 3 471 | 3 466 | 3 711 |
| 10. Social security outlays | 40 420 | 42 235 | 43 766 | 48 823 | 49 587 | 53 142 | 56 590 | 59 770 | 64 597 |
| 11. Other current transfers paid | 1 717 | 1 855 | 2 047 | 2 240 | 2 528 | 2 794 | 2 999 | 3 157 | 3 433 |
| 12. Current disbursements, total | 93 877 | 97 353 | 101 215 | 111 333 | 114 699 | 120 236 | 125 521 | 130 107 | 137 763 |
| 13. Saving (6 – 12) | 19 850 | 25 418 | 31 021 | 35 688 | 40 327 | 36 141 | 26 896 | 23 528 | 16 699 |
| 14. Gross investment | 17 536 | 18 860 | 19 808 | 21 614 | 23 229 | 26 474 | 30 777 | 31 335 | 30 860 |
| 15. Net capital transfers received | 271 | 530 | 216 | 224 | –1 357 | 404 | –216 | –507 | –1 151 |
| 16. Consumption of fixed capital | 2 245 | 2 325 | 2 458 | 2 508 | 2 525 | 2 624 | 2 767 | 2 962 | 3 249 |
| 17. Other capital account items | 3 220 | 3 829 | 3 928 | 4 465 | 5 060 | 5 858 | 6 304 | 5 716 | 5 816 |
| 18. Net lending (13 – 14 + 15 + 16 – 17) | 1 610 | 5 583 | 9 959 | 12 342 | 13 207 | 6 837 | –7 635 | –11 068 | –17 879 |

*Source:* EPA, *Annual Report on National Accounts.*

Table H. **Allocation of funds through the FILP**[1]

| | Fiscal year[2] | | | | | | Proportion of total |
|---|---|---|---|---|---|---|---|
| | 1980 | 1985 | 1990 | 1995 | 1996 | 1997 | |
| | Billion yen | | | | | | Per cent |
| **FILP funding sources** | | | | | | | |
| Employees' pension and national pension fund | 4 660 | 5 325 | 6 803 | 7 380 | 6 920 | 7 300 | 13.0 |
| Postal savings | 9 487 | 8 739 | 4 595 | 10 000 | 10 000 | 11 700 | 20.8 |
| Postal life insurance fund | 1 689 | 2 577 | 6 033 | 8 182 | 8 650 | 7 392 | 13.4 |
| Industrial Investment Special Account | 17 | 31 | 64 | 66 | 65 | 64 | 0.1 |
| Government-guaranteed bonds and borrowing | 1 567 | 3 181 | 1 899 | 3 250 | 3 100 | 3 000 | 5.3 |
| Total of above | 17 420 | 19 853 | 19 394 | 28 878 | 28 735 | 29 806 | 52.7 |
| Redemption and others | 5 792 | 9 578 | 18 420 | 19 313 | 24 990 | 26 551 | 47.3 |
| **Total funding** | 23 211 | 29 432 | 37 814 | 48 190 | 53 725 | 56 157 | 100.0 |
| **Classification by use of funds**[3] | | | | | | | |
| Public-sector financing | 8 381 | 10 516 | 14 143 | 17 855 | 19 631 | 19 397 | 34.5 |
| Public corporations | 4 349 | 5 653 | 8 707 | 9 281 | 9 213 | 8 796 | 15.7 |
| Local authorities | 4 032 | 4 863 | 5 436 | 8 574 | 10 418 | 10 801 | 18.9 |
| Private-sector financing | 9 723 | 9 974 | 14 721 | 22 386 | 20 903 | 19 931 | 35.5 |
| Housing | 4 064 | 4 263 | 6 639 | 12 003 | 12 143 | 11 632 | 20.7 |
| Small businesses | 3 375 | 3 733 | 4 893 | 6 302 | 5 507 | 5 260 | 9.4 |
| Others | 2 284 | 1 978 | 3 189 | 4 081 | 3 253 | 3 039 | 5.4 |
| Portfolio investment | 0 | 0 | 6 950 | 7 950 | 8 591 | 12 030 | 21.4 |
| **FILP total** | 18 104 | 20 490 | 35 814 | 48 190 | 49 125 | 51 359 | 91.5 |
| Other use of funds | 5 107 | 8 942 | 2 000 | 0 | 4 600 | 9 800 | 8.5 |
| **Total lending** | 23 211 | 29 432 | 37 814 | 48 190 | 53 725 | 56 155 | 100.0 |
| | Per cent of GDP | | | | | | |
| FILP total | 7.4 | 6.3 | 8.2 | 10.0 | – | – | |
| FILP net of redemption | n.a | 3.0 | 4.5 | 6.5 | – | – | |

1. Fiscal Investment and Loan Program.
2. The figures of FY 1980, FY 1985 and FY 1990 were actual. The figures from FY 1995 to FY 1997 were from the initial plans.
3. The classification has been developed by the OECD.

*Source:* Ministry of Finance.

Table I. **Appropriation account for public enterprises**

Billion yen

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 |
|---|---|---|---|---|---|---|---|---|---|
| *Receipts:* | | | | | | | | | |
| Operating surplus | 3 447 | 3 566 | 4 396 | 4 385 | 2 901 | 2 323 | 2 662 | 2 682 | 2 531 |
| Property income | 28 514 | 29 494 | 31 179 | 33 460 | 36 673 | 39 229 | 40 854 | 42 294 | 43 297 |
| Current receipts, total | 31 961 | 33 060 | 35 575 | 37 845 | 39 574 | 41 552 | 43 516 | 44 976 | 45 828 |
| *Disbursements:* | | | | | | | | | |
| Direct taxes | 745 | 328 | 335 | 247 | 1 094 | 2 492 | 1 856 | 1 138 | 1 230 |
| Property income payable | 32 217 | 33 195 | 31 946 | 33 269 | 37 024 | 38 780 | 40 329 | 45 894 | 42 896 |
| Other current transfers, net | 413 | 321 | 285 | 319 | 296 | 322 | 355 | 357 | 377 |
| Current disbursements, total | 33 375 | 33 843 | 32 566 | 33 835 | 38 414 | 41 594 | 42 540 | 47 388 | 44 502 |
| Saving | –1 414 | –784 | 3 009 | 4 010 | 1 160 | –42 | 976 | –2 512 | 1 326 |
| *Memorandum items:* | | | | | | | | | |
| Non-residential investment | 5 405 | 5 386 | 5 245 | 5 627 | 6 205 | 7 632 | 8 510 | 8 590 | 8 986 |
| Residential investment | 788 | 801 | 829 | 929 | 1 023 | 1 236 | 1 408 | 1 406 | 1 508 |
| Increase in stocks | –56 | –386 | –268 | 85 | –193 | –57 | –185 | 204 | 109 |

*Source:* EPA, *Annual Report on National Accounts.*

Table J. **Local government budget (initial plan)**

Billion yen

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *General account* | | | | | | | | | | | |
| Total revenue | 54 380 | 57 820 | 62 773 | 67 140 | 70 885 | 74 365 | 76 415 | 80 928 | 82 509 | 85 285 | 87 060 |
| Local tax | 24 223 | 26 501 | 28 646 | 30 791 | 32 678 | 34 024 | 34 555 | 32 581 | 33 764 | 33 782 | 37 014 |
| Tax transfer from central government | 10 557 | 11 126 | 13 922 | 15 600 | 16 615 | 17 563 | 17 386 | 17 428 | 18 139 | 18 840 | 18 201 |
| Treasury disbursements | 9 919 | 9 817 | 10 094 | 10 252 | 10 683 | 11 993 | 12 229 | 14 174 | 12 802 | 13 066 | 13 259 |
| Local government bonds | 5 390 | 6 048 | 5 559 | 5 624 | 5 611 | 5 140 | 6 225 | 10 392 | 11 305 | 12 962 | 12 129 |
| Other | 4 291 | 4 328 | 4 551 | 4 873 | 5 298 | 5 645 | 6 020 | 6 353 | 6 499 | 6 636 | 6 457 |
| Expenditures | 54 380 | 57 820 | 62 773 | 67 140 | 70 885 | 74 365 | 76 415 | 80 928 | 82 509 | 85 285 | 87 060 |
| Salaries and pensions | 16 354 | 16 721 | 17 381 | 18 311 | 19 645 | 20 947 | 21 900 | 22 330 | 22 698 | 22 883 | 23 216 |
| General administration | 11 226 | 11 561 | 12 306 | 12 864 | 13 839 | 14 963 | 15 908 | 16 111 | 16 817 | 17 510 | 17 984 |
| Debt services | 6 125 | 6 185 | 6 169 | 5 902 | 5 842 | 6 070 | 6 555 | 8 921 | 7 694 | 8 862 | 9 640 |
| Investment | 17 594 | 19 527 | 20 554 | 21 355 | 22 735 | 24 466 | 26 792 | 29 072 | 30 362 | 31 065 | 31 069 |
| Transfers to public enterprises | 1 433 | 1 549 | 1 699 | 1 844 | 2 043 | 2 264 | 2 574 | 2 788 | 2 991 | 3 199 | 3 119 |
| Other | 1 684 | 2 276 | 4 664 | 6 865 | 6 781 | 5 656 | 2 687 | 1 705 | 1 947 | 1 765 | 2 032 |

*Source:* Ministry of Finance, *Monthly Financial and Monetary Statistics* (in Japanese).

Table K. **Foreign trade by commodity**

$ million

| | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|
| | Exports, fob | | | | | | | |
| *SITC Sections:* | | | | | | | | |
| Food and live animals | 1 546 | 1 482 | 1 607 | 1 655 | 1 676 | 1 668 | 1 634 | 1 573 |
| Beverages and tobacco | 141 | 164 | 215 | 275 | 331 | 370 | 496 | 402 |
| Crude materials inedible except fuels | 1 829 | 1 839 | 1 903 | 2 137 | 2 137 | 2 483 | 3 021 | 2 978 |
| Mineral fuels lubricants and related materials | 972 | 1 283 | 1 323 | 1 625 | 2 005 | 2 326 | 2 505 | 1 970 |
| Animal and vegetable oils and fats | 83 | 98 | 81 | 67 | 63 | 64 | 68 | 73 |
| Chemical products | 14 782 | 15 879 | 17 483 | 19 125 | 20 234 | 23 669 | 30 198 | 28 855 |
| Other manufactured goods classified chiefly by material | 35 573 | 34 501 | 37 279 | 38 674 | 40 010 | 42 615 | 49 496 | 45 464 |
| Machinery and transport equipment | 192 179 | 201 311 | 221 205 | 240 935 | 257 748 | 282 754 | 310 199 | 285 422 |
| Miscellaneous manufactured articles | 24 188 | 25 900 | 28 513 | 29 921 | 30 639 | 32 021 | 36 009 | 33 879 |
| Other | 3 943 | 4 584 | 5 016 | 5 368 | 6 173 | 7 644 | 9 422 | 10 338 |
| **Total** | 275 236 | 287 040 | 314 625 | 339 781 | 361 017 | 395 614 | 443 047 | 410 955 |
| | Imports, cif | | | | | | | |
| Food and live animals | 28 119 | 28 211 | 30 657 | 33 234 | 35 442 | 41 817 | 46 034 | 46 191 |
| Beverages and tobacco | 2 882 | 3 323 | 3 733 | 3 932 | 3 931 | 4 797 | 5 054 | 4 479 |
| Crude materials inedible except fuels | 30 224 | 27 952 | 26 659 | 25 307 | 26 736 | 27 760 | 32 119 | 29 723 |
| Mineral fuels lubricants and related materials | 43 038 | 56 698 | 54 675 | 52 716 | 48 816 | 47 734 | 53 345 | 60 422 |
| Animal and vegetable oils and fats | 423 | 413 | 477 | 560 | 524 | 627 | 833 | 830 |
| Chemical products | 15 947 | 16 046 | 17 408 | 17 355 | 17 939 | 20 211 | 24 616 | 23 306 |
| Other manufactured goods classified chiefly by material | 30 762 | 30 927 | 31 962 | 26 828 | 28 081 | 32 397 | 41 270 | 39 104 |
| Machinery and transport equipment | 29 894 | 37 859 | 39 445 | 39 321 | 42 752 | 54 646 | 77 922 | 87 752 |
| Miscellaneous manufactured articles | 23 733 | 26 998 | 25 861 | 28 146 | 31 308 | 38 066 | 47 295 | 50 995 |
| Other | 5 765 | 6 165 | 5 494 | 5 352 | 5 072 | 6 346 | 7 476 | 6 359 |
| **Total** | 210 788 | 234 591 | 236 371 | 232 749 | 240 599 | 274 400 | 335 962 | 349 161 |

*Source:* OECD, *Monthly Statistics of Foreign Trade, Series A.*

Table L. **Foreign trade by area**

Monthly averages, $ million

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|
| | Exports, fob | | | | | | | | | |
| Total | 19 273 | 22 079 | 22 931 | 23 912 | 26 210 | 28 304 | 30 076 | 32 967 | 36 915 | 34 275 |
| OECD countries | 13 191 | 14 856 | 15 428 | 15 774 | 16 619 | 17 171 | 17 521 | 19 126 | 20 637 | 19 044 |
| European Union [1] | 3 536 | 4 355 | 4 417 | 4 881 | 5 338 | 5 587 | 4 991 | 5 111 | 5 863 | 5 260 |
| North America | 7 555 | 8 059 | 8 377 | 8 135 | 8 278 | 8 637 | 9 393 | 10 383 | 10 654 | 9 851 |
| Australia | 433 | 557 | 650 | 575 | 541 | 587 | 641 | 727 | 675 | 618 |
| Other | 1 666 | 1 885 | 1 984 | 2 183 | 2 461 | 2 360 | 2 496 | 2 905 | 3 444 | 3 315 |
| Non-OECD countries | 6 081 | 7 223 | 7 503 | 8 138 | 9 592 | 11 133 | 12 555 | 13 841 | 16 278 | 15 231 |
| Non-OECD Europe | 262 | 317 | 317 | 293 | 244 | 162 | 182 | 146 | 153 | 141 |
| OPEC | 957 | 954 | 898 | 1 131 | 1 373 | 1 582 | 1 437 | 1 370 | 1 486 | 1 447 |
| Other | 4 862 | 5 952 | 6 289 | 6 714 | 7 975 | 9 390 | 10 937 | 12 325 | 14 639 | 13 643 |
| *of which:* Asia | 4 070 | 5 133 | 5 487 | 5 983 | 7 099 | 8 272 | 9 705 | 11 130 | 13 473 | 12 632 |
| | Imports, cif | | | | | | | | | |
| Total | 12 585 | 15 614 | 17 571 | 19 567 | 19 728 | 19 418 | 20 056 | 22 895 | 27 997 | 29 133 |
| OECD countries | 6 695 | 8 903 | 10 064 | 10 997 | 10 836 | 10 465 | 10 645 | 12 178 | 14 776 | 14 958 |
| European Union [1] | 1 613 | 2 178 | 2 541 | 3 136 | 2 865 | 2 808 | 2 744 | 3 241 | 4 073 | 4 123 |
| North America | 3 178 | 4 217 | 4 768 | 5 098 | 5 125 | 5 030 | 5 317 | 6 006 | 7 227 | 7 502 |
| Australia | 662 | 857 | 967 | 1 031 | 1 084 | 1 037 | 1 018 | 1 136 | 1 211 | 1 186 |
| Other | 1 242 | 1 651 | 1 787 | 1 731 | 1 761 | 1 589 | 1 565 | 1 796 | 2 265 | 2 147 |
| Non-OECD countries | 5 890 | 6 711 | 7 507 | 8 570 | 8 892 | 8 954 | 9 411 | 10 716 | 13 221 | 14 175 |
| Non-OECD Europe | 221 | 271 | 288 | 308 | 306 | 238 | 267 | 324 | 442 | 374 |
| OPEC | 2 247 | 2 231 | 2 656 | 3 461 | 3 270 | 3 242 | 3 122 | 3 173 | 3 571 | 3 936 |
| Other | 3 422 | 4 209 | 4 562 | 4 800 | 5 316 | 5 474 | 6 022 | 7 219 | 9 208 | 9 864 |
| *of which:* Asia | 3 229 | 3 867 | 4 312 | 4 655 | 5 138 | 5 339 | 5 902 | 6 971 | 8 813 | 9 573 |

1. Including Austria Finland and Sweden from 1995.

*Note:* Detail may not be add due to rounding.

*Source:* OECD, *Monthly Statistics of Foreign Trade.*

Table M. **Exports to the United States and to the European Union**

| | Millions of dollars | | | Annual rate of change | | | Per cent of total | | |
|---|---|---|---|---|---|---|---|---|---|
| | 1994 | 1995 | 1996 [1] | 1994 | 1995 | 1996 [1] | 1994 | 1995 | 1996 [1] |
| **Exports to the United States** | 117 560 | 120 859 | 111 932 | 11.5 | 2.8 | –7.4 | 100 | 100 | 100 |
| Foodstuffs | 312 | 304 | 287 | 2.0 | –2.7 | –5.5 | 0.3 | 0.3 | 0.3 |
| Textiles | 785 | 666 | 629 | –1.4 | –15.2 | –5.5 | 0.7 | 0.6 | 0.6 |
| Chemicals | 4 065 | 4 822 | 4 999 | 14.4 | 18.6 | 3.7 | 3.5 | 4.0 | 4.5 |
| Non-metallic mineral manufactures | 994 | 1 040 | 1 131 | 13.5 | 4.6 | 8.8 | 0.8 | 0.9 | 1.0 |
| Metal products | 4 174 | 4 052 | 3 872 | 18.9 | –2.9 | –4.4 | 3.6 | 3.4 | 3.5 |
| Machinery and equipment | 98 391 | 100 200 | 91 529 | 11.7 | 1.8 | –8.7 | 83.7 | 82.9 | 81.8 |
| Television receivers | 339 | 462 | 516 | 20.4 | 36.2 | 11.8 | 0.3 | 0.4 | 0.5 |
| Motor vehicles | 25 656 | 22 088 | 20 596 | 1 1.2 | –13.9 | –6.8 | 21.8 | 18.3 | 18.4 |
| Vessels | 147 | 5 | 19 | 69.1 | –96.9 | 313.2 | 0.1 | 0.0 | 0.0 |
| Others | 8 839 | 9 776 | 9 486 | 7.2 | 10.6 | –3.0 | 7.5 | 8.1 | 8.5 |
| **Exports to the European Union** | 61 206 | 70 291 | 62 933 | 2.5 | 14.8 | –10.5 | 100 | 100 | 100.0 |
| Foodstuffs | 117 | 86 | 78 | –7.3 | –26.6 | –9.0 | 0.2 | 0.1 | 0.1 |
| Textiles | 848 | 879 | 760 | 2.2 | 3.7 | –13.6 | 1.4 | 1.3 | 1.2 |
| Chemicals | 3 916 | 4 870 | 4 780 | 11.0 | 24.4 | –1.9 | 6.4 | 6.9 | 7.6 |
| Non-metallic mineral manufactures | 4 58 | 5 68 | 5 39 | 1 2.2 | 2 4.0 | –5.1 | 0.7 | 0.8 | 0.9 |
| Metal products | 1 425 | 1 665 | 1 404 | 23.4 | 16.8 | –15.7 | 2.3 | 2.4 | 2.2 |
| Machinery and equipment | 49 271 | 56 309 | 49 600 | 1.8 | 14.3 | –11.9 | 80.5 | 80.1 | 78.8 |
| Television receivers | 195 | 243 | 337 | 7.9 | 25.0 | 38.5 | 0.3 | 0.3 | 0.5 |
| Motor vehicles | 9 362 | 9 530 | 8 601 | –12.3 | 1.8 | –9.7 | 15.3 | 13.6 | 13.7 |
| Vessels | 395 | 565 | 416 | –33.3 | 43.2 | –26.4 | 0.6 | 0.8 | 0.7 |
| Others | 5 172 | 5 913 | 5 773 | –1.8 | 14.3 | –2.4 | 8.4 | 8.4 | 9.2 |

1. OECD estimates.
*Source:* Ministry of Finance, *Summary Report on Trade of Japan.*

Table N. **Balance of payments**

Billion yen

| | 1987 | 1988 | 1989 | 1990 | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|---|---|---|---|---|
| Exports | 325 233 | 334 258 | 373 977 | 406 879 | 414 651 | 420 816 | 391 640 | 393 485 | 402 596 | 435 659 |
| Imports | 192 915 | 216 113 | 263 567 | 306 350 | 285 423 | 263 055 | 236 823 | 246 166 | 279 153 | 344 693 |
| **Trade balance** | 132 319 | 118 144 | 110 412 | 100 529 | 129 231 | 157 764 | 154 816 | 147 322 | 123 445 | 90 966 |
| Services, net | –29 389 | –38 800 | –50 713 | –61 899 | –56 311 | –55 709 | –47 803 | –48 976 | –53 898 | –67 792 |
| Transportation | n.d. | n.d. | n.d. | n.d. | –11 941 | –11 003 | –11 030 | –11 589 | –12 563 | –13 068 |
| Travel | n.d. | n.d. | n.d. | n.d. | –27 608 | –29 413 | –25 821 | –27 763 | –31 595 | –35 880 |
| Other services | n.d. | n.d. | n.d. | n.d. | –16 762 | –15 293 | –10 954 | –9 620 | –9 740 | –18 844 |
| Income | 23 483 | 26 436 | 31 773 | 32 874 | 34 990 | 45 125 | 45 329 | 41 307 | 41 573 | 58 180 |
| Transfers | –4 553 | –4 323 | –4 354 | –6 768 | –16 150 | –4 833 | –5 651 | –6 225 | –7 253 | –9 775 |
| **Current balance** | 121 862 | 101 461 | 87 113 | 64 736 | 91 757 | 142 349 | 146 690 | 133 425 | 103 862 | 71 579 |
| Capital and financial account | –61 511 | –83 420 | –74 651 | –48 679 | –92 662 | –129 165 | –117 035 | –89 924 | –62 754 | –33 472 |
| Direct investment | –27 368 | –46 032 | –65 271 | –70 903 | –40 887 | –18 426 | –15 234 | –17 611 | –21 249 | –25 236 |
| Portfolio investment | –15 4645 | –98 164 | –25 790 | 11 900 | 60 212 | –33 401 | –77 620 | –23 657 | –30 772 | –45 140 |
| Others | 121 637 | 62 074 | 18 287 | 11 857 | –110 371 | –75 697 | –22 533 | –46 738 | –8 585 | 40 442 |
| Change in reserve assets | –55 492 | –21 255 | 18 487 | 13 703 | 11 391 | –753 | –29 973 | –25 854 | –54 235 | –39 424 |
| Errors and omissions | –4 857 | 3 214 | –30 950 | –29 761 | –10 487 | –12 432 | 318 | –17 648 | 13 127 | 1 317 |

*Source:* Bank of Japan, *Balance of Payments Monthly.*

Table O. **Public sector**

**A. Budget indicators: general government accounts** (per cent of GNP[1])

| | 1970 | 1980 | 1985 | 1995 |
|---|---|---|---|---|
| | Fiscal years | | | |
| Non-interest current receipts | 20.2 | 26.1 | 28.3 | 28.5 |
| Non-interest expenditure | 13.7 | 21.9 | 22.1 | 24.4 |
| Primary budget balance | 6.6 | 4.2 | 6.2 | 4.0 |
| Net interest payments[2] | –0.3 | 1.3 | 1.9 | 0.6 |
| General government saving | 6.9 | 2.9 | 4.3 | 3.4 |
| General government net lending | 1.8 | –4.0 | –0.8 | –4.0 |
| *of which:* Central | 0.0 | –5.4 | –3.6 | –4.2 |
| Local | –0.4 | –1.3 | –0.3 | –2.6 |
| Social Security Fund | 2.2 | 2.6 | 3.1 | 2.8 |
| | Calendar years | | | |
| **General government debt** | | | | |
| Gross debt | 12.1 | 52.0 | 68.5 | 89.4 |
| *of which:* Central | 8.0 | 39.2 | 53.7 | 68.8 |
| Net debt[3] | –6.5 | 17.3 | 26.6 | 11.7 |
| *of which:* Central | 2.0 | 27.1 | 41.0 | 39.8 |

**B. The structure of general government expenditure and taxation** (per cent of GNP)

| | 1970 | 1980 | 1985 | 1995 |
|---|---|---|---|---|
| | Fiscal years | | | |
| Total expenditure | | | | |
| Current consumption | 7.5 | 9.8 | 9.5 | 9.7 |
| Transfers to persons | 4.7 | 10.2 | 11.0 | 13.2 |
| Subsidies | 1.2 | 1.5 | 1.1 | 0.8 |
| Net interest payments[2] | –0.3 | 1.3 | 1.9 | 0.6 |
| Capital formation | 5.2 | 7.1 | 5.6 | 6.6 |
| Total expenditure by function | | | | |
| Education | 3.5 | 4.8 | 4.1 | 3.8 |
| Economic services[4] | 4.9 | 5.8 | 4.8 | 5.2 |
| Health | 3.0 | 4.5 | 4.7 | 5.5 |
| Housing | 1.4 | 2.4 | 2.0 | 2.9 |
| Social security | 2.7 | 7.0 | 7.7 | 9.7 |
| Total tax revenue | 20.0 | 25.9 | 28.1 | 28.2 |
| Direct tax | 8.5 | 11.1 | 12.3 | 9.9 |
| *of which:* Personal | 4.1 | 6.3 | 6.8 | 6.1 |
| Corporate | 4.4 | 4.8 | 5.4 | 3.8 |
| Social security | 4.4 | 7.4 | 8.4 | 10.4 |
| Indirect tax | 7.2 | 7.4 | 7.5 | 7.9 |
| Tax rates (per cent) | | | | |
| National personal income tax | | | | |
| Lowest/top rate | 10/75 | 10/75 | 10.5/70 | 10/50 |
| Number of brackets | 19 | 19 | 15 | 5 |
| General consumption tax | – | – | – | 3.0 |

1. National accounts basis.
2. Positive sign means net expenditure.
3. Corporate shares are excluded from financial assets.
4. Includes transportation and infrastructure.

*Source:* Economic Planning Agency, *Annual Report on National Accounts*, Ministry of Finance, OECD, *Revenue Statistics*.

Table P. **Financial markets**

| | 1970 | 1980 | 1985 | 1995 |
|---|---|---|---|---|
| Size of the financial sector (percentages)[1] | | | | |
| Sector employment/total employment | 2.4 | 3.0 | 3.2 | 3.1 |
| Financial assets/GNP | 159.8 | 228.0 | 291.6 | 387.2 |
| Structure of financial assets and liabilities | | | | |
| Financial institutions' share in domestic financial assets (per cent) | 39.5 | 41.9 | 44.0 | 44.9 |
| Net worth (per cent of GNP) | | | | |
| Total | 405.1 | 558.1 | 571.4 | 658.6 |
| Public sector | 58.1 | 71.7 | 57.2 | 88.6 |
| Private sector | 346.9 | 486.4 | 514.2 | 570.0 |
| *of which:* Households | 250.2 | 356.5 | 384.7 | 452.4 |
| Financial assets, net (per cent of GNP)[2] | | | | |
| Total | 40.6 | 53.4 | 89.0 | 114.7 |
| Public sector | –9.9 | –39.4 | –49.2 | –35.6 |
| Private sector | 50.5 | 92.8 | 138.2 | 150.3 |
| *of which:* Households | 60.9 | 87.8 | 113.0 | 158.8 |

1. Financial institutions and insurance.
2. Financial assets (including corporate shares at market prices) less financial liabilities (excluding corporate shares).

*Source:* Economic Planning Agency, *Annual Report on National Accounts.*

Table Q. **Labour market indicators**

**A. Labour force performance**

| | Cyclical peak: 1985 | Cyclical trough: 1986 | 1994 | 1995 | 1996 |
|---|---|---|---|---|---|
| Unemployment rate: | | | | | |
| Total | 2.6 | 2.8 | 2.9 | 3.2 | 3.4 |
| Male | 2.6 | 2.7 | 2.8 | 3.1 | 3.4 |
| Female | 2.7 | 2.8 | 3.0 | 3.2 | 3.3 |
| Youth [1] | 4.8 | 5.2 | 5.4 | 6.1 | 6.6 |
| Share of long-term unemployment in total unemployment [2] | 12.8 | 17.1 | 16.5 | 18.1 | 19.6 |
| Dispersion of regional unemployment rates [3] | 0.83 | 0.74 | 0.54 | 0.56 | 0.58 |

**B. Structural or institutional characteristics**

| | 1970 | 1980 | 1985 | 1996 |
|---|---|---|---|---|
| Participation rate: [4] | | | | |
| Total | 65.4 | 63.3 | 63.0 | 63.5 |
| Male | 81.8 | 79.8 | 78.1 | 77.7 |
| Female | 49.9 | 47.6 | 48.7 | 50.0 |
| Total employment/population (15 years and over) | 64.6 | 62.0 | 61.4 | 61.4 |
| Employees/total employment | 64.9 | 71.7 | 74.3 | 82.1 |
| Part-time employment (as per cent of employees) | 6.7 | 10.1 | 11.2 | 19.5 |
| Non-wage labour costs [5] (as percentage of total compensation) | 8.3 | 11.0 | 13.0 | 13.9 |
| Government unemployment insurance replacement ratio [6] | 60/62 | 60/80 | 60/80 | 60/80 |
| Unionisation rate | 35.4 | 30.8 | 28.9 | 23.2 |
| *of which:* | | | | |
| Private sector | – | 24.7 | 24.4 | 20.2 |
| Public sector | – | 74.5 | 61.7 | 63.4 |
| Annual hours worked per employee | 2 239 | 2 108 | 2 110 | 1 919 |

| Percentage changes (average annual rates) | 1970/1960 | 1980/1970 | 1990/1980 | 1996/1995 |
|---|---|---|---|---|
| Labour force | 1.3 | 0.9 | 1.2 | 0.7 |
| Employment: | | | | |
| Total | 1.4 | 0.8 | 1.2 | 0.4 |
| Primary sector | –4.0 | –4.2 | –2.4 | –3.0 |
| Secondary sector | 3.7 | 0.7 | 0.9 | –0.2 |
| Tertiary sector | 2.7 | 2.3 | 2.0 | 1.0 |

1. Unemployed persons between 15 and 24 years as a percentage of the labour force of the same age group.
2. People looking for a job for one year or more in February.
3. Measured by standard deviation for 10 regions.
4. Labour force as a percentage of relevant population group, aged 15 years and over.
5. Employers' contribution to social security, pension funds and others on National Accounts basis. The latest figure is for 1995.
6. Unemployment benefits per unemployed divided by the compensation per employee. Minimum and maximum ratios are shown.

*Source:* Management and Coordination Agency, *Labour Force Survey, Report on Special Survey of the Labour Force Survey;* Ministry of Labour, *Monthly Labour Survey, Basic Survey on Labour Unions;* Economic Planning Agency, *Annual Report on National Accounts.*

Table R. **Production structure and performance indicators**

**A. Production structure**

| | Per cent share of GDP at current market prices[1] | | | Per cent share of total employment | | |
|---|---|---|---|---|---|---|
| | 1970 | 1980 | 1995 | 1970 | 1980 | 1995 |
| Agriculture | 5.9 | 3.6 | 1.9 | 19.7 | 12.9 | 7.3 |
| Mining and quarrying | 0.8 | 0.5 | 0.2 | 0.5 | 0.3 | 0.1 |
| Manufacturing | 34.8 | 28.2 | 23.7 | 26.7 | 24.0 | 22.5 |
| *of which:* | | | | | | |
| Food[2] | 3.7 | 3.2 | 2.7 | 2.5 | 2.4 | 2.7 |
| Textiles | 1.9 | 1.0 | 0.4 | 3.3 | 2.1 | 1.0 |
| Chemicals and chemical products[3] | 2.9 | 2.2 | 2.0 | 1.0 | 0.8 | 0.8 |
| Primary metal industries | 3.9 | 3.6 | 1.7 | 1.2 | 1.0 | 0.9 |
| Fabricated metal products[4] | 2.1 | 1.3 | 1.4 | 2.3 | 2.0 | 1.7 |
| General machinery | 3.7 | 3.0 | 2.8 | 2.9 | 2.5 | 2.5 |
| Electrical and electronic products | 3.8 | 3.1 | 3.7 | 2.9 | 2.8 | 3.4 |
| Transportation equipment | 3.8 | 3.2 | 2.4 | 2.4 | 2.4 | 2.2 |
| Construction | 7.5 | 9.0 | 9.9 | 8.1 | 10.1 | 10.5 |
| Market services | 43.9 | 48.7 | 54.5 | 37.8 | 44.2 | 50.8 |
| *of which:* | | | | | | |
| Electricity, gas and water | 2.1 | 2.6 | 2.7 | 0.5 | 0.6 | 0.7 |
| Wholesale and retail trade | 13.9 | 14.8 | 12.2 | 16.0 | 17.8 | 16.7 |
| Finance and insurance | 4.1 | 5.0 | 4.7 | 2.4 | 3.0 | 3.1 |
| Real estate | 7.8 | 9.1 | 12.4 | 0.6 | 1.0 | 1.5 |
| Transport and communications | 6.7 | 5.9 | 6.3 | 5.5 | 5.7 | 5.6 |
| Community, business, social and personal services | 9.3 | 11.3 | 16.3 | 12.7 | 16.1 | 23.2 |
| Non-market services | 7.1 | 9.9 | 9.9 | 7.2 | 8.6 | 8.7 |
| Government services | 6.1 | 8.2 | 7.7 | 5.8 | 6.7 | 6.0 |
| Community and personal services | 1.0 | 1.7 | 2.2 | 1.4 | 2.0 | 2.7 |

**B. Manufacturing sector performance**

| | Productivity growth by sector, real GDP/employment (annual rate) | |
|---|---|---|
| | 1980/1970 | 1995/1980 |
| Food[2] | 3.3 | –1.4 |
| Textiles | 5.8 | 2.6 |
| Chemicals and chemical products[3] | 12.9 | 6.9 |
| Primary metal industries | 7.3 | 0.2 |
| Fabricated metal products[4] | 2.9 | 4.4 |
| General machinery | 7.4 | 2.8 |
| Electrical and electronic products | 26.8 | 9.5 |
| Transportation equipment | 5.0 | 2.5 |

1. GDP without the adjustment for import tax, imputed interest and general consumption tax is used.
2. Excluding tobacco.
3. Excluding rubber and plastic products.
4. Excluding machinery.

*Source:* Economic Planning Agency, *Annual Report on National Accounts.*

*BASIC STATISTICS:*

*INTERNATIONAL COMPARISONS*

## BASIC STATISTICS: INTERNATIONAL COMPARISONS

| | Units | Reference period[1] | Australia | Austria | Belgium | Canada | Denmark | Finland | France | Germany | Greece |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | | |
| Total | Thousands | 1994 | 17 840 | 8 031 | 10 124 | 29 251 | 5 206 | 5 088 | 57 960 | 81 407 | 10 430 |
| Inhabitants per sq. km | Number | 1994 | 2 | 96 | 332 | 3 | 121 | 15 | 106 | 228 | 79 |
| Net average annual increase over previous 10 years | % | 1994 | 1.4 | 0.6 | 0.3 | 1.6 | 0.2 | 0.4 | 0.5 | 2.9 | 0.5 |
| **Employment** | | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 7 680 (93) | 3 737 | 3 724 (92) | 13 292 | 2 508 | 2 015 | 21 781 (93) | 35 894 | 3 790 |
| *of which:* Agriculture | % of TCE | 1994 | 5.3 (93) | 7.2 | 2.6 (92) | 4.1 | 5.1 | 8.3 | 5.1 (93) | 3.3 | 20.8 |
| Industry | % of TCE | 1994 | 23.7 (93) | 33.2 | 27.7 (92) | 22.6 | 26.8 | 26.8 | 27.7 (93) | 37.6 | 23.6 |
| Services | % of TCE | 1994 | 71 (93) | 59.6 | 69.7 (92) | 73.3 | 68.1 | 64.9 | 67.2 (93) | 59.1 | 55.5 |
| **Gross domestic product (GDP)** | | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 331.6 | 198.1 | 227.9 | 544 | 146.7 | 97.2 | 1 328.5 | 1 832.3 | 73.1 (93) |
| Per capita | US$ | 1994 | 18 588 | 24 670 | 22 515 | 18 598 | 28 181 | 19 106 | 22 944 | 27 826 | 7 051 (93) |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 327.9 | 162.3 | 204.2 | 596.7 | 107 | 82.5 | 1 111.8 | 1 601.7 | 118 |
| Per capita | US$ | 1994 | 18 382 | 20 210 | 20 166 | 20 401 | 20 546 | 16 208 | 19 201 | 24 325 | 1 450 |
| Average annual volume growth over previous 5 years | % | 1994 | 2.2 | 2.5 | 1.6 | 1.1 | 1.9 | –1.6 | 1.1 | 2.6 | 1.4 (93) |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 21.4 | 24.8 | 17.4 | 18.7 | 14.8 | 14.3 | 18.1 | 18.5 | 17.4 (93) |
| *of which:* Machinery and equipment | % of GDP | 1993 | 9.8 | 8.7 | 7.8 | 6.2 | 7.2 | 5.90 | 8.10 | 7.5 | 7.8 |
| Residential construction | % of GDP | 1993 | 5.2 | 6.3 | 4.6 | 6.1 | 2.9 | 3.7 | 4.9 | 6.5 | 3.6 |
| Average annual volume growth over previous 5 years | % | 1994 | 0.8 | 3.7 | 0.4 | –0.1 | –2.8 | –12.9 | –1 | 0.8 | 2.7 (93) |
| **Gross saving ratio**[4] | % of GDP | 1994 | 16.8 | 25.3 | 22 | 16 | 17 | 16.6 | 19 | 21 | 15.5 (93) |
| **General government** | | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 17.5 | 18.8 | 15 | 20.2 | 25.3 | 22.4 | 19.6 | 17.7 | 19.1 (93) |
| Current disbursements[5] | % of GDP | 1993 | 36.9 | 48.4 | 55.3 | 49 | 61.1 | 58.9 | 51.5 | 45.6 | 51.2 |
| Current receipts | % of GDP | 1993 | 33.5 | 48.6 | 50.1 | 43 | 58.3 | 52.5 | 46.8 | 45.7 | 40.2 |
| **Net official development assistance** | % of GNP | 1993 | 0.34 | 0.31 | 0.41 | 0.46 | 1.03 | 0.76 | 0.66 | 0.44 | .. |
| **Indicators of living standards** | | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 10 803 | 10 546 | 12 090 | 11 863 | 10 042 | 8 814 | 11 395 | 10 733 | 6 367 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 430 | 382 | 387 | 469 | 311 | 386 | 413 | 480[8] | 169 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 464 | 432 | 410 | 586 | 577 | 544 | 511 | 420[8] | 413 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 480 | 478 | 451 | 639 | 536 | 501 | 407 | 556[8] | 197 |
| Doctors, per 1 000 inhabitants | Number | 1993 | 2.2 (91) | 2.3 | 3.7 | 2.2 | 2.8 (92) | 2.6 (92) | 2.8 | 3.2 (92) | 3.8 (92) |
| Infant mortality per 1 000 live births | Number | | 6.1 | 6.5 | 8 | 6.8 | 5.4 | 4.4 | 6.5 | 5.8 | 8.5 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | 3 | 5.5 | 3.7 | 3.3 | 3.5 | 4.8 | 3.5 | 5.2 | 14.6 |
| Consumer prices | % | 1994 | 3 | 3.4 | 2.8 | 2.8 | 2.1 | 3.3 | 2.5 | 3.3 | 16.2 |
| **Foreign trade** | | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 47 363 | 44 881 | 137 259[7] | 165 358 | 41 850 | 29 514 | 235 337 | 422 243 | 8 958 |
| As % of GDP | % | 1994 | 14.3 | 22.7 | 60.2 | 30.4 | 28.5 | 30.4 | 17.7 | 23 | 11.5 (93) |
| Average annual increase over previous 5 years | % | 1994 | 5 | 6.7 | 6.5 | 7.1 | 8.3 | 4.9 | 5.6 | 4.4 | 3.4 |
| Imports of goods, cif* | Mill. US$ | 1994 | 49 731 | 55 071 | 126 006[7] | 148 297 | 35 932 | 23 091 | 220 508 | 376 566 | 21 111 |
| As % of GDP | % | 1994 | 15 | 27.8 | 55.3 | 27.3 | 24.5 | 23.8 | 16.6 | 20.6 | 30.1 (93) |
| Average annual increase over previous 5 years | % | 1994 | 4 | 7.2 | 5 | 5.4 | 6.1 | –1.3 | 3.5 | 6.9 | 5.4 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 7 730 | 11 523 | 9 505[7] | 8 416 | 6 203 | 7 304 | 17 986 | 52 994 | 9 924 |
| As ratio of average monthly imports of goods | Ratio | 1994 | 1.9 | 2.5 | 0.9 | 0.7 | 2.1 | 3.8 | 1 | 1.7 | 5.6 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

## BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

| | Units | Reference period[1] | Iceland | Ireland | Italy | Japan | Luxembourg | Mexico | Netherlands | New Zealand |
|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | |
| Total | Thousands | 1994 | 267 | 3 571 | 57 190 | 124 960 | 398 | 93 010 | 15 382 | 3 526 |
| Inhabitants per sq. km | Number | 1994 | 3 | 51 | 190 | 331 | 153 | 47 | 377 | 13 |
| Net average annual increase over previous 10 years | % | 1994 | 1.1 | 0.1 | 0 | 0.4 | 0.8 | 2 | 0.6 | 0.8 |
| **Employment** | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 138 | 1 168 (93) | 20 152 (93) | 64 530 | 162 (91) | 32 439 | 6 631 | 1 560 |
| *of which:* Agriculture | % of TCE | 1994 | 9.4 | 12.7 (93) | 7.5 (93) | 5.8 | 3.7 (91) | 25.8 | 4 | 10.4 |
| Industry | % of TCE | 1994 | 26.1 | 27.7 (93) | 33 (93) | 34 | 31.5 (91) | 22.2 | 23 | 25 |
| Services | % of TCE | 1994 | 65.2 | 59.7 (93) | 59.6 (93) | 60.2 | 64.8 (91) | 52.1 | 73 | 64.6 |
| **Gross domestic product (GDP)** | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 6.2 | 52 | 1 017.8 | 4 590 | 10.6 (92) | 371.2 | 334.3 | 51.2 |
| Per capita | US$ | 1994 | 23 199 | 14 550 | 17 796 | 36 732 | 27 073 (92) | 3 991 | 21 733 | 14 513 |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 5.1 | 54.3 | 1 068.4 | 2 593.7 | 11.7 | 673.3 | 285.9 | 57.3 |
| Per capita | US$ | 1994 | 19 271 | 15 212 | 18 681 | 20 756 | 29 454 | 7 239 | 18 589 | 16 248 |
| Average annual volume growth over previous 5 years | % | 1994 | 0.6 | 4.7 | 1 | 2.1 | 4.1 (92) | 3 | 2.3 | 2.5 |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 15.2 | 15.1 | 16.4 | 28.6 | 20.4 (93) | 20.7 | 19.3 | 20 |
| *of which:* Machinery and equipment | % of GDP | 1993 | 3.9 | 6.3 | 7.4 | 11.5 | .. | 9.4 | 8.6 | 9.3 |
| Residential construction | % of GDP | 1993 | 4.4 | 4.1 | 5.3 | 5.4 | .. | 4.9 | 5.1 | 4.9 |
| Average annual volume growth over previous 5 years | % | 1994 | –4 | 1 | –2.3 | 1.4 | 6.5 (92) | 7.7 | 0.4 | 5.8 |
| **Gross saving ratio**[4] | % of GDP | 1994 | 16.9 | 19.5 | 18.8 | 31.2 | 60.2 (92) | 15.1 | 24.4 | 20.7 |
| **General government** | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 20.6 | 16 | 17.1 | 9.8 | 17.1 (92) | 11.8[9] | 14.2 | 14.7 |
| Current disbursements[5] | % of GDP | 1993 | 34.9 | .. | 53.2 | 26.9 | .. | .. | 55.4 | .. |
| Current receipts | % of GDP | 1993 | 35.9 | .. | 47.1 | 32.9 | .. | .. | 54.5 | .. |
| **Net official development assistance** | % of GNP | 1993 | .. | 0.15 | 0.42 | 0.27 | 0.34 (92) | .. | 0.88 | 0.22 |
| **Indicators of living standards** | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 11 546 | 7 750 | 11 029 | 11 791 | 15 545 | 4 853 | 10 726 | 9 266 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 464 | 228 | 478 | 282 | 470 | 85 | 356 | 440 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 527 | 300 | 400 | 454 | 511 | 70 | 477 | 436 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 319 | 276 | 421 | 613 | 267 | 148 | 485 | 443 |
| Doctors, per 1 000 inhabitants | Number | 1993 | 3 | 1.7 (92) | 1.7 (91) | 1.7 (92) | 2.1 (92) | 1 | 2.5 (90) | 2 |
| Infant mortality per 1 000 live births | Number | | 4.8 | 5.9 | 7.3 | 4.5 (92) | 8.5 (92) | 18 | 6.3 | 7.3 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | .. | 4.6 | 5.9 | 2.4 | .. | 5.3 | 3.2 | 2.1 |
| Consumer prices | % | 1994 | 6.3 | 2.7 | 5.2 | 2 | 3.1 | 16.1 | 2.8 | 2.5 |
| **Foreign trade** | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 1 628 | 34 125 | 189 802 | 396 149 | .. | 60 882 | 155 084 | 12 169 |
| As % of GDP | % | 1994 | 26.3 | 65.7 | 18.6 | 8.6 | .. | 16.4 | 46.4 | 23.8 |
| Average annual increase over previous 5 years | % | 1994 | 2.7 | 10.5 | 6.2 | 7.6 | .. | 21.7 | 7.6 | 6.5 |
| Imports of goods, cif* | Mill. US$ | 1994 | 1 464 | 25 812 | 167 690 | 274 916 | .. | 79 346 | 139 800 | 11 859 |
| As % of GDP | % | 1994 | 23.6 | 49.7 | 16.5 | 6 | .. | 21.4 | 41.8 | 23.2 |
| Average annual increase over previous 5 years | % | 1994 | 0.7 | 8.1 | 1.9 | 5.5 | .. | 25.5 | 6.1 | 6.1 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 201 | 4 189 | 22 102 | 86 214 | .. | 4 301 | 23 655 | 2 540 |
| As ratio of average monthly imports of goods | Ratio | 1994 | 1.6 | 1.9 | 1.6 | 3.8 | .. | 0.7 | 2 | 2.6 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

## BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

| | Units | Reference period[1] | Norway | Portugal | Spain | Sweden | Switzerland | Turkey | United Kingdom | United States |
|---|---|---|---|---|---|---|---|---|---|---|
| **Population** | | | | | | | | | | |
| Total | Thousands | 1994 | 4 337 | 9 900 | 39 150 | 8 781 | 6 994 | 60 573 | 58 375 | 260 651 |
| Inhabitants per sq. km | Number | 1994 | 13 | 107 | 78 | 20 | 169 | 78 | 238 | 28 |
| Net average annual increase over previous 10 years | % | 1994 | 0.5 | –0.1 | 0.2 | 0.5 | 0.7 | 2.1 | 0.3 | 1 |
| **Employment** | | | | | | | | | | |
| Total civilian employment (TCE)[2] | Thousands | 1994 | 1970 (93) | 4 372 | 11 760 | 3 926 | 3 772 | 19 664 | 25 044 (93) | 123 060 |
| *of which:* Agriculture | % of TCE | 1994 | 5.6 (93) | 11.5 | 9.8 | 3.4 | 4 | 44.8 | 2.2 (93) | 2.9 |
| Industry | % of TCE | 1994 | 23.1 (93) | 32.8 | 30.1 | 25 | 28.8 | 22.2 | 26.2 (93) | 24 |
| Services | % of TCE | 1994 | 71.3 (93) | 55.7 | 60.2 | 71.6 | 67.2 | 33 | 71.6 (93) | 73.1 |
| **Gross domestic product (GDP)** | | | | | | | | | | |
| At current prices and current exchange rates | Bill. US$ | 1994 | 103.4 (93) | 87 | 482.4 | 196.6 | 257.3 | 130.7 | 1 019.5 | 6 649.8 |
| Per capita | US$ | 1994 | 23 984 (93) | 8 792 | 12 321 | 22 389 | 36 790 | 2 157 | 17 468 | 25 512 |
| At current prices using current PPP's[3] | Bill. US$ | 1994 | 95.3 | 122 | 531.7 | 153 | 167.4 | 319.3 | 1 030.2 | 6 649.8 |
| Per capita | US$ | 1994 | 21 968 | 12 335 | 13 581 | 17 422 | 23 942 | 5 271 | 17 650 | 25 512 |
| Average annual volume growth over previous 5 years | % | 1994 | 2.1 (93) | 1.4 | 1.5 | –0.3 | 0.5 | 3.6 | 0.8 | 2.1 |
| **Gross fixed capital formation (GFCF)** | % of GDP | 1994 | 22 (93) | 25.7 | 19.8 | 13.7 | 22.8 | 24.5 | 15 | 17.2 |
| *of which:* Machinery and equipment | % of GDP | 1993 | .. | .. | 5.7 | 5.7 | 7.5 | 10.3 | 7.3 | 7.7 |
| Residential construction | % of GDP | 1993 | .. | .. | 4.1 | 4 | 14.9[10] | 9.1 | 3.1 | 4 |
| Average annual volume growth over previous 5 years | % | 1994 | –3.93 | 2.7 | –1.2 | –7.6 | –0.4 | 5.1 | –2.1 | 4.6 |
| **Gross saving ratio**[4] | % of GDP | 1994 | 21.9 (93) | 24.2 | 18.8 | 13.7 | 29.3 | 22.5 | 13.5 | 16.2 |
| **General government** | | | | | | | | | | |
| Current expenditure on goods and services | % of GDP | 1994 | 22.1 (93) | 17.2 | 16.9 | 27.3 | 14.1 | 11.7 | 21.6 | 16.4 |
| Current disbursements[5] | % of GDP | 1993 | .. | .. | 43.7 | 67.3 | 36.7 | .. | 42.7 | 35.8 |
| Current disbursements[5] | % of GDP | 1993 | .. | .. | 40.1 | 59 | 36 | .. | 36.8 | 31.7 |
| Current receipts | % of GDP | 1993 | 1.23 | 0.36 | 0.32 | 1.33 | 0.49 | .. | 0.34 | 0.19 |
| **Net official development assistance** | % of GNP | 1993 | 9 826 | 7 780 | 8 412 | 9 240 | 13 730 | 3 617 | 10 942 | 16 444 |
| **Indicators of living standards** | | | | | | | | | | |
| Private consumption per capita using current PPP's[3] | US$ | 1993 | 378 | 260 | 307 | 418 | 441 | 29 | 361 | 568 |
| Passenger cars, per 1 000 inhabitants | Number | 1990 | 515 | 273 | 340 | 687 | 603 | 143 | 445 | 553 |
| Telephones, per 1 000 inhabitants | Number | 1991 | 423 | 187 | 400 | 468 | 406 | 175 | 434 | 814 |
| Television sets, per 1 000 inhabitants | Number | 1991 | 3.2 (92) | 2.9 | 4.1 | 3 | 3 | 0.9 | 1.5 (92) | 2.3 (92) |
| Doctors, per 1 000 inhabitants | Number | 1993 | 5 | 8.7 | 7.6 | 4.8 | 5.6 | 52.6 | 6.6 | 8.5 (92) |
| Infant mortality per 1 000 live births | Number | | 4 | .. | 7.2 | 5.4 | .. | .. | 6.7 | 2.8 |
| **Wages and prices** (average annual increase over previous 5 years) | | | | | | | | | | |
| Wages (earnings or rates according to availability) | % | 1994 | 2.7 | 9 | 5.6 | 5.7 | 3.9 | 73 | 4.6 | 3.6 |
| Consumer prices | % | 1994 | 34 645 | 17 072 | 73 129 | 61 122 | 70 467 | 18 456 | 205 170 | 512 627 |
| **Foreign trade** | | | | | | | | | | |
| Exports of goods, fob* | Mill. US$ | 1994 | 30.9 (93) | 19.6 | 15.2 | 31.1 | 27.4 | 14.1 | 20.1 | 7.7 |
| As % of GDP | % | 1994 | 5 | 6.1 | 10.5 | 3.4 | 6.4 | 9.5 | 6.1 | 7.1 |
| Average annual increase over previous 5 years | % | 1994 | 27 345 | 25 967 | 92 182 | 51 730 | 68 126 | 22 976 | 227 026 | 663 256 |
| Imports of goods, cif* | Mill. US$ | 1994 | 23.3 (93) | 29.9 | 19.1 | 26.3 | 26.5 | 17.6 | 22.3 | 10 |
| As % of GDP | % | 1994 | 2.9 | 6.6 | 5.2 | 1 | 3.2 | 37.9 | 2.8 | 7 |
| Average annual increase over previous 5 years | % | 1994 | 13 033 | 10 627 | 28 475 | 15 929 | 23 790 | 4 911 | 28 094 | 43 350 |
| **Total official reserves**[6] | Mill. SDRs | 1994 | 5.7 | 4.9 | 3.7 | 3.7 | 4.2 | 2.6 | 1.5 | 0.8 |

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

*Sources:* Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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12-1996

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 97 03 1 P) ISBN 92-64-15429-9 – No. 49705 1997
ISSN 0376-6438